GEK-83866

DRIVE SYSTEM INSTRUCTIONS

LOAD COMMUTATED INVERTER AC ADJUSTABLE-SPEED DRIVE STATIC POWER CONVERSION AND CONTROL EQUIPMENT

Information contained herein is proprietary information of General Electric Company. The dissemination, use, or duplication of this information for any purpose other than that for which the information is provided is prohibited by General Electric Company except by expressed permission.

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL@ ELECTRIC

LOAD COMMUTATED INVERTER AC ADJUSTABLE-SPEED DRIVE STATIC POWER CONVERSION AND CONTROL -EQUIPMENT

GEK-83866

ISSUE DATE: November 1986

These instructions do not purport to cover all details or variations in equipment, nor **to** *provide* **for every** *possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.*

Copyright 1986
by
General Electric Co., U.S.A.

TABLE OF CONTENTS

No.		Page
	CHAPTER 1	
	SYSTEM DESCRIPTION	
1-1.	Basic Drive Control	1-1
1-2.	Power Converters	1-3
1-2.1.	Basic Operation.	1-3
1-3.	Twelve-Pulse Operation	1-8
	CHAPTER 2	
	Power Equipment	
2-1.	Equipment Options	2-1
2-1.1.	2KV Medium Voltage Bridge	2-1
2-1.2.	4KV Medium Voltage Bridge	2-1
2-1.3.	SILCOMATIC ® Plus Power Conversion Bridge	2-1
2-2.	Thyristor Modules, Medium Voltage Bridge (MVB)	2-1
2-3.	Gate Driver Transformers	2-3
2-4.	Gate Driver Card (HPTK)	2-4
2-5.	Voltage Feedback Attenuator Card (XPTN)	2-5
2-6.	Signal Distribution Board (XTFH)	2-5
2-7.	Exciter Voltage Controller (EVC)	2-5
	CHAPTER 3	
	CONTROL ELECTRONICS PANEL	
3-1.	Introduction	3-1
3-2.	Electronics Module (LCG)	3-1
3-3.	Control Operational Description	3-2
3-3.1.	Basic Principles	3-2
3-3.2.	Phase Locked Loop Operation..	3-3
3-3.3.	Electronics Module (LCG) Printed Wiring Board (PWB) Descriptions	3-4
3-3.4.	Microprocessor - HMPG/DMPB	3-4
3-3.5.	Electrically-alterable ROM = DS3800DMPB	3-4
3-3.6.	LCI Processor Extender Board = DS3800HLCB	3-6
3-3.7.	Analog/Digital Interface Card DS3800HAIA	3-7
3-3.8.	Flux Integrators - DS3800NLIB	3-7
3-3.8.1.	Voltage integrators..	3-7
3-3.8.2.	Voltmeter driver	3-8
3-3.8.3.	Overvoltage protection.	3-8
3-3.8.4.	Undervoltage sensing.	3-8
3-3.8.5.	Suppression..	3-8
3-3.8.6.	Current meter driver	3-9
3-3.9.	Gating Reference Board - DS3800NSFC	3-9
3-3.9.1.	R-bus interface	3-9
3-3.9.2.	Firing mask latch..	3-9
3-3.9.3.	Gate drivers	3-10
3-3.9.4.	Pulse train generation..	3-10
3-3.9.5.	Buffers..	3-10
3-3.9.6.	Line-to-line voltage signal conditioners.	3-10
3-3.9.7.	Cell state sensors	3-13
3-3.9.8.	D-C Link volts	3-13
3-3.10.	Field Excitation Control Board = DS3800NSFE.	3-13
3-3.10.1.	R Bus interface..	3-13
3-3.10.2.	Firing mask latch..	3-13
3-3.10.3.	Gate drivers	3-14
3-3.10.4.	Current feedback conditioning	3-14
3-3.11.	Analog LCI Board = DS3800NLCB	3-14
3-3.11.1.	Speed feedback..	3-15
3-3.11.2.	Test speed reference..	3-15
3-3.11.3.	Torque reference signal conditioning	3-15
3-3.11.4.	Selftest features	3-15
3-3.11.5.	4 - 20 MA signal drivers	3-15
3-3.11.6.	Ground fault detector	3-15
3-3.11.7.	Power applied reset	3-15
3-3.11.8.	Fault relay drivers	3-15
3-3.11.9.	Horsepower meter driver	3-15
3-3.11.10,	Continuity detection circuits	3-15
3-3.11.11.	Test switch	3-15
3-3.11.12.	Restart upon power loss..	3-15
3-4.	Relay driver board = DS3800HRDA	3-15
3-5.	Input signal conditioning = DS3800HISA	3-17
3-6.	Display Module (LDM).	3-18
3-7.	Signal Isolation Module (SIM)	3-18
3-a.	I/O Module (IOM)	3-19
	CHAPTER 4	
	FIELD ADJUSTABLE PARAMETERS AND PUBLIC VARIABLE PARAMETERS	
	Section A - EEPROM (Field) Variables	

TABLE OF CONTENTS-Continued

NO.		Page
	CHAPTER 5	
	SECTION A	
	STRUCTURED LOAD COMMUTATED INVERTER	
	DIAGNOSTIC MONITOR SPECIFICATIONS	
5-1.	INTRODUCTION	5-1
5-2.	OPERATING INSTRUCTIONS..	.5-1
5-2.1.	General Information	5-1
5-2.2.	Editing Functions	5-3
5-2.3.	Command Syntax	.5-3
5-2.4.	Protected Commands	5-4
5-2.5.	Help	.5-4
5-3.	MONITOR COMMANDS	.5-4
5-3.1.	ASSIGN	.5-4
5-3.2.	DISPLAY..	.5-6
5-3.3.	HELP	5-7
5-3.4.	PASSCODE..	.5-7
5-3.5.	QUIT	5-7
5-3.6.	RESTART	.5-7
5-3.7.	SYMBOLS	.5-7
5-3.8.	TERMINAL	.5-10
5-3.9.	VERIFY	5-10
5-4.	EXTENDED MODES	5-11
5-4.1.	ENGINEERING	.5-11
5-4.2.	DISPLAY	5-11
5-4.3.	HELP	5-11
5-4.4.	INPUT	5-11
5-4.5.	PUT	5-12
5-4.6.	MOVE	5-12
5-4.7.	OUTPUT	5-12
5-4.8.	QUIT	5-12
5-4.9.	SUBSTITUTE	5-12
5-4.10.	TUNEUP	.5-12
5-4.11.	APPLY	5-13
5-4.12.	HELP	5-13
5-4.13.	MODIFY	.5-13
5-4.14.	MODIFY REFERENCE	.5-14
5-4.15.	QUIT	.5-14
5-4.16.	REMOVE	5-14
5-4.17.	SET PASSCODE	5-15
5-4.18.	SAVE	5-15
5-4.19.	SAVE D TO A (SDA)	5-15
5-4.20.	TUNEUP SYMBOLS	.5-15
5-4.21.	TUNEUP	5-15
5-4.22.	Monitor Commands -.	.5-16
5-5.	LOADING DISPLAY MESSAGES HEADERS..	.5-16
	CHAPTER 6	
	INSTALLATION AND STARTUP GUIDE (OUTLINE)	
6-1.	INTRODUCTION	6-1
6-1.1.	DETAILED INSTALLATION INSTRUCTIONS	6-1
	CHAPTER 7	
	DIAGNOSTICS and TROUBLESHOOTING	
7-1.	SECTION A - SELFTEST I.	7-1
7-1.1.	BASIC STRUCTURE	7-1
7-1.1.1.	PWB-Specific Tests	7-1
7-1.1.2.	Paired PWB Tests:	7-1
7-1.1.3.	Panel-level Tests	7-1
7-1.2.	EXECUTING THE TESTS	..7- 1
7-1.3.	DIAGNOSING FAILURES..	.7-1
7-1.4.	DESCRIPTION OF TESTS	.7-2
7-2.	SECTION B - GATE-TEST-MODE..	.7-8
7-2.1.	Gate-Test-Mode	.7-8
7-2.2.	Getting Into Gate-Test-Mode..	.7-8
7-2.3.	Gate-Test-Mode Checkout..	.7-8
7-3.	Section C - Software Regulator Block Diagrams	7-13
7-4.	SECTION D - TERMINALS..	7-13
7-4.1.	Serial Monitor and Hand Terminal (or other Diagnositic Terminal)	7-13

LIST OF ILLUSTRATIONS

No.		Page
1-1	Simplified LCI One-Line Diagram..	1-1
1-2	Single Channel LCI Drive	1-2
1-3	Thyristor Bridge Leg Conduction Sequence.	1-3
1-4	Load Converter Voltage and Current..	1-4
1-5	LCI-Driven Synchronous Motor Diagram; Leading Power Factor..	1-5
1-6	Flux and Torque as a Function of Stator Current	1-5
1-7	Typical LCI Drive Excitation, Voltage, and Torque Variations with Speed..	1-6
1-8	Forced Commutation Firing Mode..	1-7
1-9	Motor Voltage and Current in the Self-Commutated Mode	1-9
M2-1	Medium Voltage Bridge Assembly..	2-2
M4-1	Medium Voltage Bridge Assembly, 4KV	2-3
S-1	SILCOMATIC® Plus Power Conversion Bridge Assembly	2-4
s-2	SILCOMATIC® Plus Power Conversion Bridge Attenuator Connections..	2-5
M-2	MVB Thyristor Module	2-6
M-3	MVB Cell Stack Assembly	2-6
M-4	Cell with Gating, Display, Snubber, and Voltage Equalizing Circuits..	2-7
M-S	HPTK Block Diagram	2-7
s-3	SILCOMATIC® Plus Power Conversion Bridge Firing Circuit Elements..	2-8
M-6	Bridge Attenuator Circuits	2-9
CE-1	Synchronous Field Exciter Module (DS3820SFE).	2-9
3-o	Control Electronic Panel..	3-1
3-1	Small Electronics Module (SEM) Card and Connector Layout	3-2
3-2	SEM Input/Output..	3-3
3-3	Electronics Module Block Diagram	3-4
3-4	LCI System Digital Regulator Block Diagram..	3-5
3-5	Flux Wave Zero-Crossings	3-10
3-6	Analog Typical Waveforms	3-12
5A-1	Monitor Command Structure	5-2
6-1	Firing Signals and Current Signals..	6-6
6-2	Phase Current Relationships..	6-8
6-3	IFB during SEGFIRE..	6-10
6-4	CRALPHA1/2 TUNE-UP	6-10
6-5	MOTOR LINE TO LINE VOLTAGE WAVE FORM	6-11
6-6	Ideal Flux Wave	6-12
6-7	Flux Regulator Response	6-13
6-8	Speed Regulator..	6-14
6-9	Load-side FALAP	6-15
6-10	Source-side Firing Angle	6-16
7B-1	Firing Angle Limits..	7-9
7B-2A	Source Firing - Full Advance..	7-10
7B-4A	Typical Gate Pulse Trains..	7-11
7B4B	Typical Firing Pulse (Cell Gate Current Pulse)	7-11
7B-6	Software Interrupts - Full Advance (Source).	7-12
7D-1	Hand Terminal Connection	7-13
7D-2	Connection Diagram Between Silent 700 and HMPG Using HNMB	7-14
7D-3	Connection Diagram Between Silent 700 and HMPG Without Using HNMB	7-15
7D-4	Alternate Connection Diagram Between Silent 700 and HMPG Without Using HNMB	7-16

LIST OF TABLES

No.		Page
3-1A	Microprocessor Board Features - RS232C Connector Signals..	3-6
3-1B	Microprocessor - Berg Jumpers, Testpoints and LED's	3-6
3-1C	Electrically-Alterable ROM - LEDs and Switches	3-7
3-2	LCI Extender - LED's and Testpoints	3-8
3-3	Analog/Digital Input Signals - Berg Jumpers and Testpoints	3-9
3-4	Flux Integrators - Jumpers, Adjustments, LEDs and Testpoints	3-11
3-5	Gating Reference - Berg Jumpers, Adjustments, LEDs and Testpoints	3-13
3-6	Field Excitation - Adjustments and Testpoints	3-14
3-7	Analog LCI Board - Berg Jumpers, Adjustments, Testpoints and LEDs	3-16
3-8	Relay Driver - Testpoints and LEDs	3-17
3-9	Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LEDs	3-17
4-o	Signal Isolation Module (SIM)	3-19
6-1	SLOT 1A	6-3

CHAPTER 1
SYSTEM DESCRIPTION

1-1. Basic Drive Control

The Load Commutated Inverter (LCI) is a static, adjustable frequency drive system which controls a synchronous motor from near zero to rated speed. The basic system consists of a line commutated, phase controlled thyristor converter which feeds a load commutated thyristor converter through a dc link reactor. A simplified one-line diagram of a single channel LCI is given in Figure 1-1. The transformer provides isolation from the ac system bus and provides the correct voltage at the terminals of the rectifier. Also, the internal impedance of the transformer limits the magnitude of any downstream bus faults.

The rectifier is a thyristor bridge whose gating is controlled to produce a variable dc voltage at its output. The output of the rectifier is fed through the dc link reactor, whose function is to smooth the current and keep it continuous over the operating range of the system.

The dc link reactor output is then fed into the inverter bridge which produces variable frequency ac at the stator terminals of the synchronous motor. The inverter bridge and the rectifier bridge use the same power hardware and are controlled by microprocessor-based electronics. Although the bridges are labelled "rectifier" and "inverter" it is possible for their roles and hence power flow to reverse. In this case the motor would be braked by pumping its energy back into the ac line. A more general nomenclature is to call the line side bridge the source converter, and to call the motor side bridge the load converter.

The synchronous motor field is usually excited by a brushless exciter coupled to the motor shaft. The brushless exciter is a wound rotor induction machine whose rotor voltage is rectified to supply field current to the synchronous motor. The exciter stator phase rotation is connected such that its rotor voltage and thus the synchronous motor field excitation will vary with motor speed.

A more detailed diagram of a single channel LCI is shown in Figure 1-2. As shown in this figure, the electronic control receives the following signal inputs:

1. Attenuated line and load bus voltage signals.
2. Attenuated line and load current signals from current transformers.
3. Speed reference signal.
4. Process commands such as stop, start, etc.

The attenuated bus voltage signals are used by the control to synchronize thyristor firing with the line and load voltages, to provide voltage sensing across the thyristor cells, and to provide electronic over/under voltage detection. The attenuated current signals are used for regulator current feedback, electronic overcurrent detection, and software-implemented fault detection.

The electronic control evaluates the process commands and internal status signals to determine whether the drive should be in a stopped, started, alarmed, or

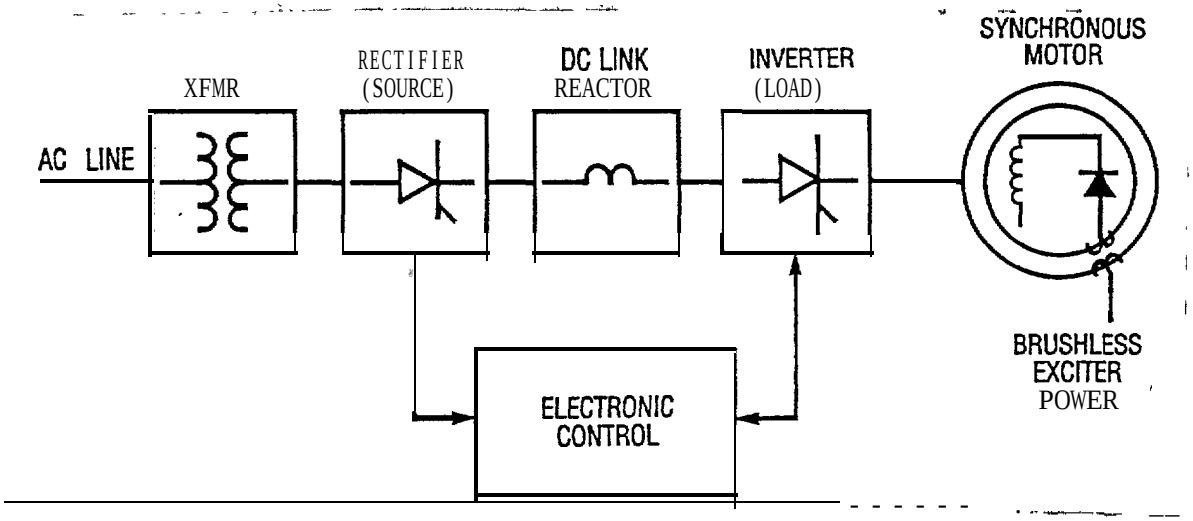


Figure 1-1. Simplified LCI One-Line Diagram

faulted condition. If in a started condition, the electronics provides gate signals to control the thyristor bridges. These signals are low level and are conditioned in the power bridge circuitry to provide the necessary isolation and power level.

The power bridges are of the six pulse, double way type and the thyristor bridge legs "fire" in the order that they are numbered in Figure l-2 and as shown in Figure 1-3.

The source converter current is successfully transferred from one leg to the next by the ac line voltages. Similarly, the inverter current is successfully transferred by the motor stator voltages.

The process of switching motor/inverter current from one leg to the next is illustrated in Figure l-4. The principles illustrated here apply to a rectifier bridge as well as to an inverter bridge. This phase-controlled

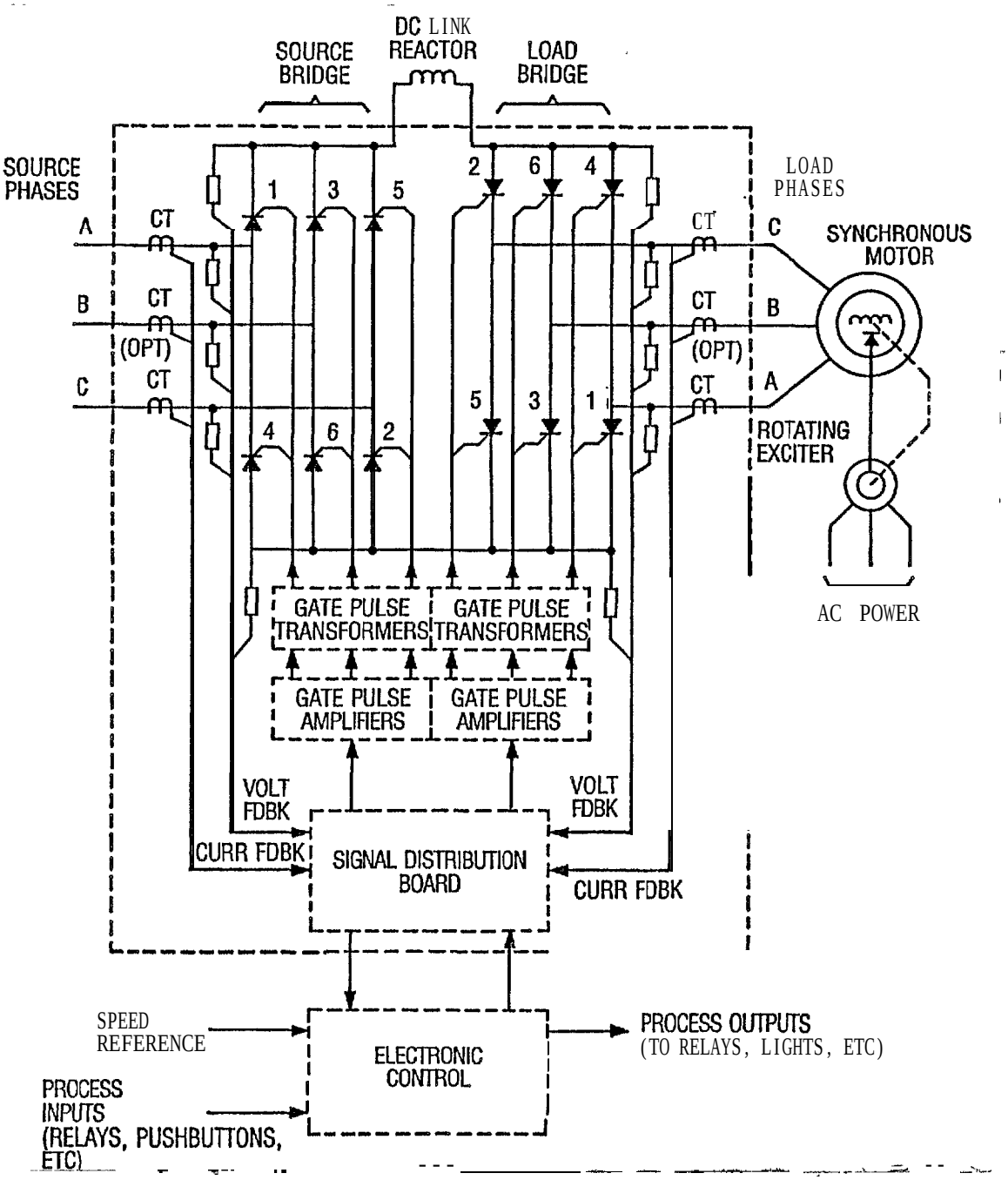


Figure l-2. Single Channel LCI Drive

switching is accomplished by using the following two thyristor characteristics: (1) When the voltage across the thyristor is positive, it can be triggered into conduction, and (2) it will not permit current flow in the reverse direction. Thus, in an alternating voltage circuit, thyristor conduction will cease and reverse voltage will begin to appear when the current becomes zero.

Current transfer must be completed before voltage cross-over, with a margin angle. This angle must be long enough to allow the previously conducting leg thyristors to recover to their blocking state before reverse voltage is applied. This is why the fundamental component of current must lead the voltage, from an inverter/motor viewpoint. From the rectifier/source viewpoint, the fundamental component of current will always lag the voltage; but for successful commutation, the angle = $180° - \beta = 180° - \mu - \gamma$ must always be less than 180"; a practical limit is 1.55"'. Correspondingly, a practical minimum value for , for the inverter bridge, is 25". The motor power-factor angle is always less than, but cannot become zero.

The LCI control system must conform to the characteristics of the synchronous motor operating at leading power factor. A phasor diagram for a synchronous machine operating at leading power factor is shown in Figure 1-5.

The motor voltage characteristic is mainly a function of the rotor field-excitation, E,, and the de-magnetizing action of direct-axis current which produces the $I_D X_{AD}$ voltage in opposition to the voltage produced by field excitation.

Note that an increase in stator current will result in higher direct-axis current, which will increase $I_D X_{AD}$, which in turn decreases the machine voltage E_G available for commutation, thereby increasing displacement angle 0, which further increases stator current, etc. until equilibrium is reached at a new operating point.

Typical voltage, current, and torque characteristics are shown in Figure l-6.

In a typical system, a brushless exciter supplies the motor field excitation. The exciter output voltage varies with motor speed. Field current is exciter output voltage divided by field resistance. Neglecting saturation, field flux is proportional to field current, and flux and volts/hertz are considered synonymous.

The interrelationship of voltage, current, and torque is complex. Typical characteristics for an LCI drive system are depicted in Figure l-7.

The control functions, which conform to the drive characteristics discussed here, will be further described under Control Electronics.

1-2. Power Converters

I-2.1. Basic Operation.

When operating in any mode the electronic control must synchronize firing of the source and load converters to the ac line and motor bus voltages, respectively. The primary feedback for accomplishing this is the attenuated bus-to-ground signals for both converters. Attenuated bus-to-ground voltages are brought into the electronic control and combined to produce line-to-line analogs for both converters. These line-to-line voltages are then integrated to obtain flux signals. The zero crossings of these signals are used in the synchronization of the phase locked loop for the firing control of both source and load converters. At low speed, before

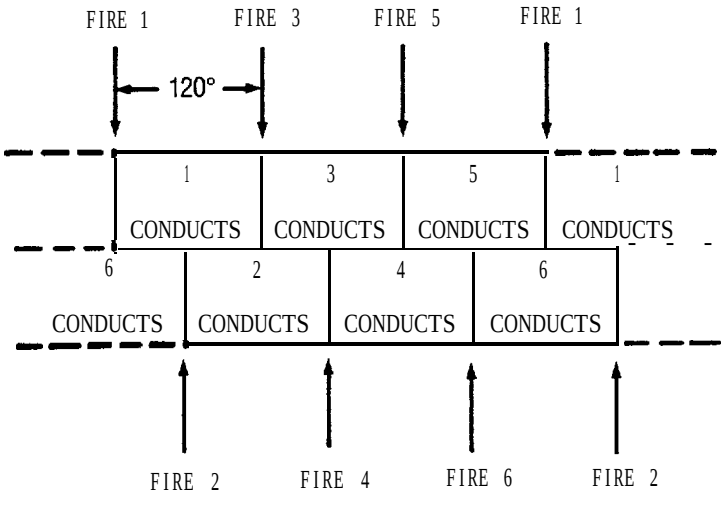


Figure 1-3. Thyristor Bridge Leg Conduction Sequence

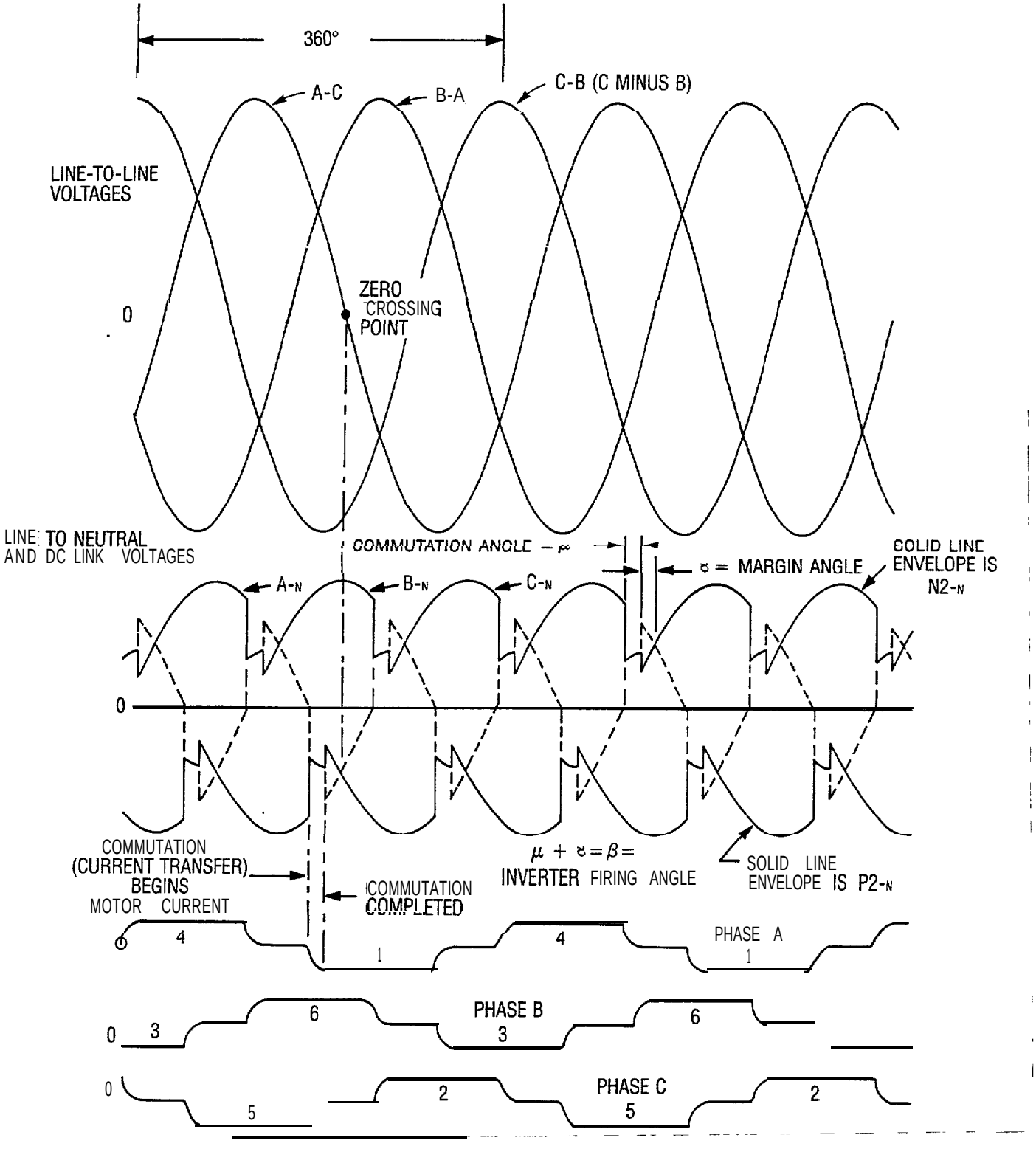


Figure 1-4. Load Converter Voltage and Current

12

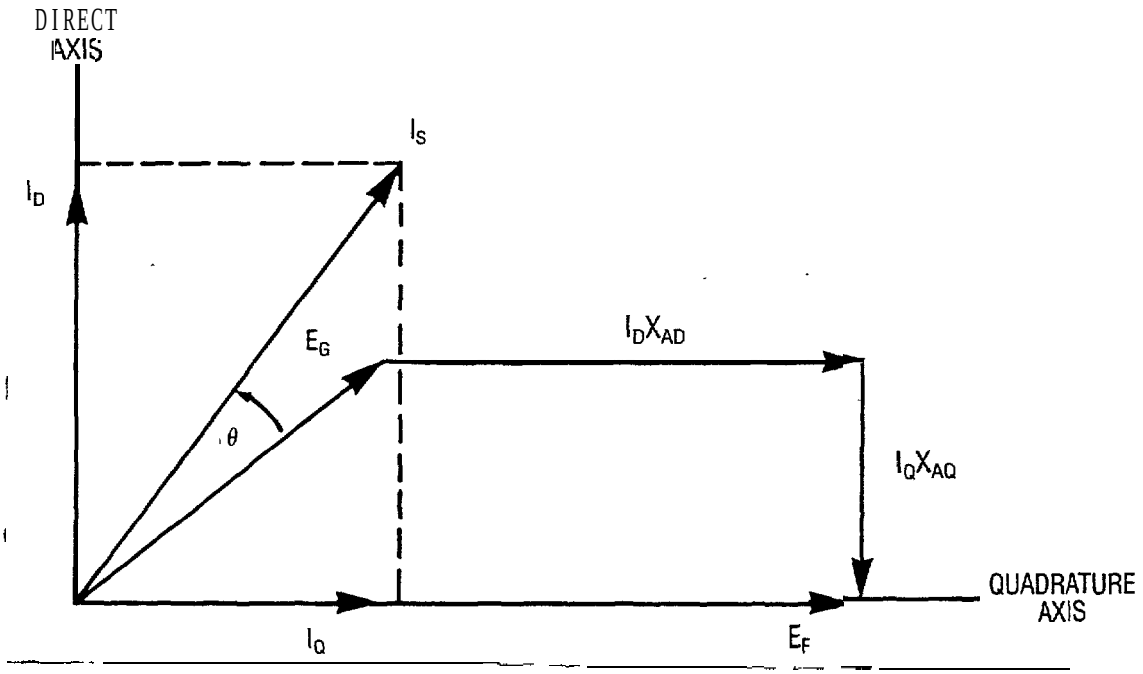


Figure 1-5. LCI-Driven Synchronous Motor Diagram; Leading Power Factor

Where: ID = Direct axis current
IQ = Quadrature axis current
IS = Stator current
E_G = Generated voltage, CEMF
X_{AD} = Armature reactance, Direct axis
X_{AQ} = Armature reactance, Quadrature axis
E_F = Rotor field -excitation voltage
Θ = Displacement angle

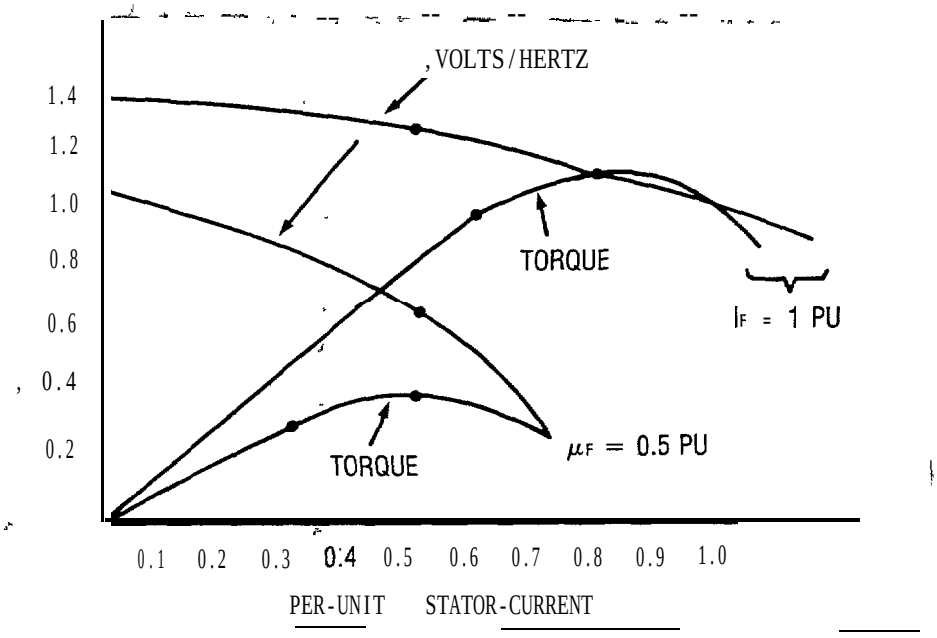


Figure 1-6. Flux and Torque as a Function of Stator Current

the phase locked loop is effective on the load side, the zero crossing marks are used as a timing reference for firing in force-commutated operation.

The source side converter for the LCI always operates line-commutated; i.e., the ac line provides the means for transferring conduction from one thyristor to the next. The load side converter may operate either load (self) commutated or force-commutated, depending on motor speed and flux level.

As the synchronous motor rotor (field) rotates, the near-sinusoidally shaped field flux cuts the stator windings to produce a set of three sinusoidal voltages in the stator which are angularly displaced by 120 electrical degrees. The magnitude of this counter emf is proportional to speed and field strength. At low speeds the induced cemf is insufficient to commutate the thyristors in the load side converter. In this mode, the control must operate in a forced commutated mode.

There are two forced commutated modes: fixed frequency firing and segment firing modes. In fixed frequency operation, the load converter is fired at a fixed low frequency. This frequency is adjustable and optimized at startup for best operation. This mode is used to start the motor and obtain a speed with sufficient cemf to allow the control to lock on to it.

When in forced commutation, conduction of the load converter is stopped by phasing the source converter to inversion limit until the dc link (reactor) current is zero as shown in Figure l-8. Thus the dc link current appears chopped into 60-degree-wide segments of motor frequency (angle). When fixed frequency mode is begun, the motor may initially move backward unless the control is specifically programmed to prevent reverse rotation when starting.

If the field is moving backwards before starting, the control will stop the field (rotor) and accelerate it in the desired direction.

As soon as the control detects sufficient balance and level of the stator flux waves, it transitions to segment firing mode. At this point, the voltages present at the output of the motor voltage integrators (flux) are sufficient to allow the control to synchronize load thyristor firings to the motor voltage. The control compensates for the imperfect integration at low frequency and fires the load thyristors to operate the motor at near unity power factor.

In the segment firing mode, current and load commutation is still controlled by source thyristor bridge shutoff, but now thyristor firings are synchronized to the motor cemf.

Segment firing mode continues until the motor has reached approximately 10% of rated speed, where the synchronous motor cemf is sufficient to commutate the load side converter. At this point, the control transitions to self commutated operation.

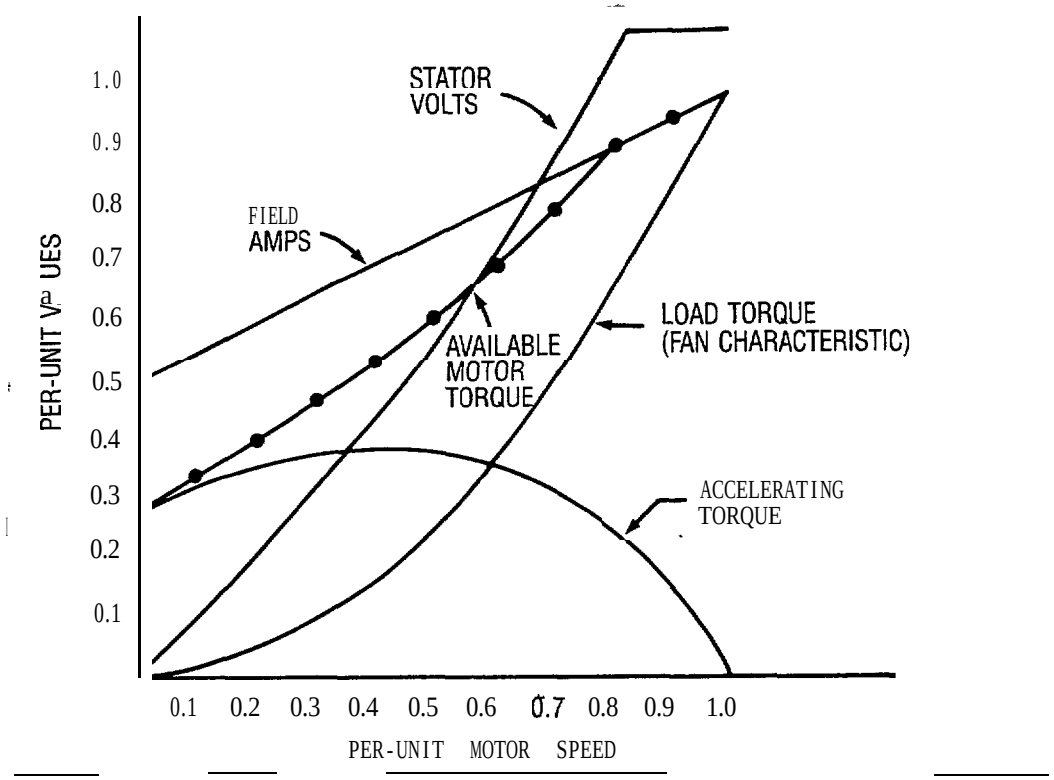
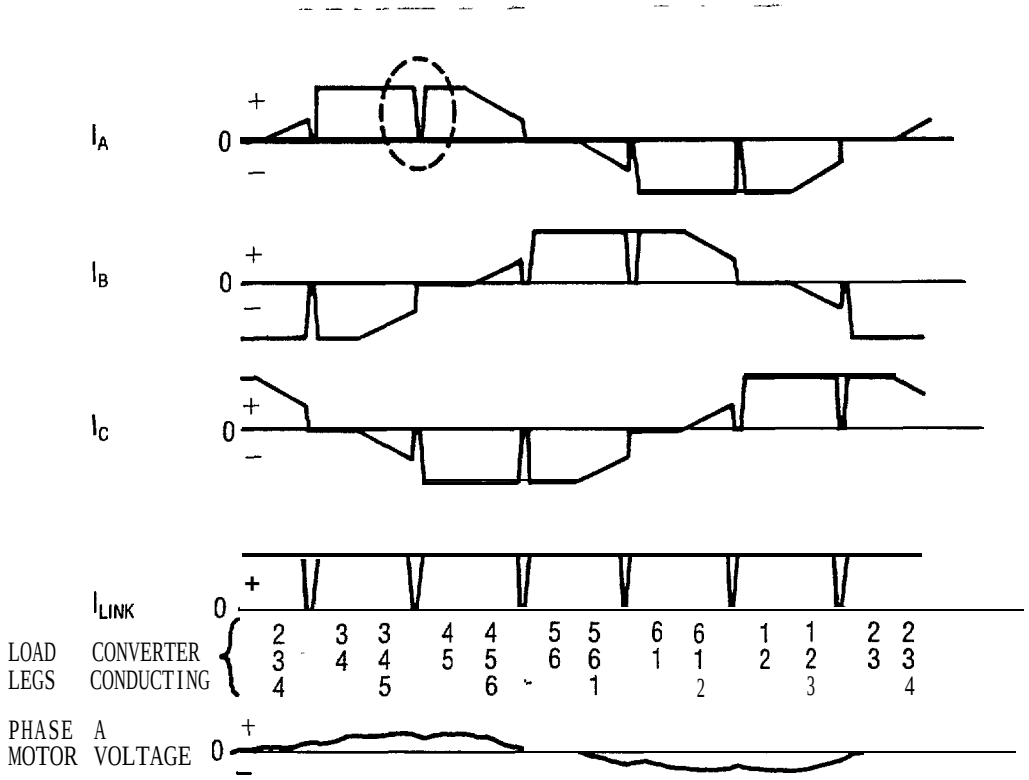


Figure 1-7. Typical LCI Drive Excitation, Voltage, and To&e Variations with Speed



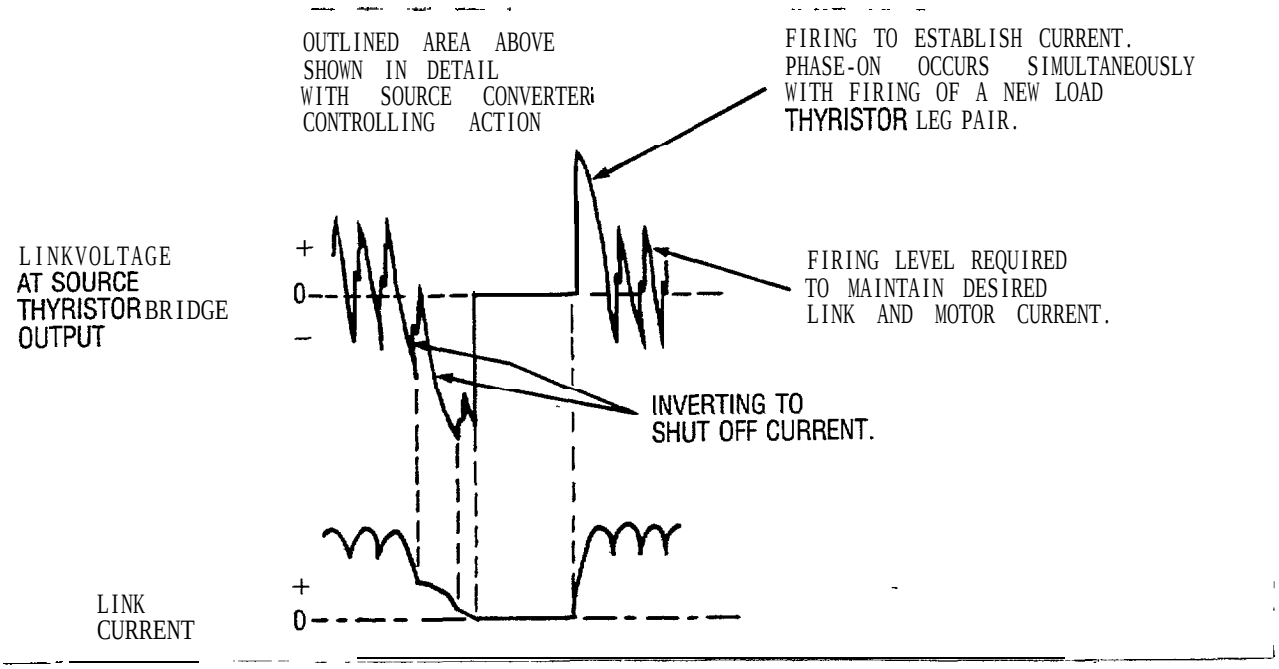


Figure 1-8. Forced Commutation Firing Mode

In the self commutated mode, which is the principal operating mode, the motor must be operated at a leading power factor in order to be able to insure commutation of the load converter. The electronic control acts to keep the motor power factor, and therefore torque per ampere, as high as possible. This is accomplished by firing the load converter as close to inversion limit as possible while maintaining sufficient margin for successful commutation of current from one cell to the next. The commutation time required is a function of motor current, motor (commutating) inductance, and the voltage difference between the lines involved in the attempted commutation.

For a given load current and motor inductance, a corresponding amount of volt-seconds is required for a successful commutation. The control reads the peak volt-seconds of the integrated line-to-line motor voltages and the motor current. The motor commutating inductance is a constant stored in the microprocessor system memory. Using the current and inductance, the amount of commutation volt-seconds required is calculated by the microprocessor. Using this value of volt-seconds and the peak volt-seconds of the previous flux wave, the latest possible time to fire is calculated to give a specified margin after commutation is completed. The relationship of the system voltages, currents, and flux waves are illustrated in Figure l-9.

The commutating "notch" identified in the A-C line-to-line voltage is equal in amplitude to the simultaneous commutating "bump" on the B-C voltage. The corresponding notch in the A-B voltage (A and B are the two lines which are commutating together at this instant) is twice this amplitude; the notch area is twice the commutating inductance per phase times the current. The voltage at the commutating point, where the lines are temporarily connected by the thyristor legs, is practically zero during commutation; the line-to-line voltage is only the forward voltage drops of the conducting thyristor legs.

At high load on the motor, the apparent power factor "seen" by the power source increases. This is because the source converter firing advances (is reduced) to obtain more current. The harmonics in the current and the resultant harmonics in the voltage, due to commutation "notching", will decrease.

The fundamental control strategy is: To increase motor current in response to a load torque increase, the dc link voltage, on the source side, is increased, thereby raising motor current so as to keep motor speed constant.

I-3. Twelve-Pulse Operation

Twelve-pulse operation is like two identical, separate, six-pulse drives operating from a common source at the same current and firing angle, with the firing reference angles shifted 30 degrees by the isolating transformers.

The two motors are combined into one frame. This reduces the motor cost, including installation, and reduces the torque pulsation amplitude while raising the torque pulsation frequency. This is achieved by separating the stator winding into two identical windings, but isolated and phase-shifted 30 degrees.

This construction, utilizing a common magnetic frame, including a common field, causes the load-side converter voltage to be equal in amplitude and frequency between the two drive channels. The transformer design makes the source-side converter voltage equal, but also 30 degrees apart between the two channels. It is not necessary that source-side and load-side voltages be phase-shifted. The motor windings are shifted to obtain smoother torque for equal current. The transformer windings are shifted to reduce the harmonic distortion on the power system and to raise the harmonic frequencies.

Inter-channel communication allows one channel to be master and the other the slave, which takes its torque reference from the master. This communication allows the two motor winding currents to be balanced. Thus the channels deliver equal power, take equal current, and fire at the same relative firing angle, This best utilizes the motor and drive capabilities and also minimizes the harmonic distortion on the power system, as the first two harmonics present in each six-pulse channel cancel (5th and 7th harmonics). Other, higher-order harmonics also cancel, but their amplitudes are much smaller.

A twelve-pulse system also presents the capability of shutting down one channel for maintenance while the motor continues to run on the other channel (with reduced torque and usually reduced speed range). When the out-of-service channel is ready for operation, it may be returned to service without interrupting the drive system.

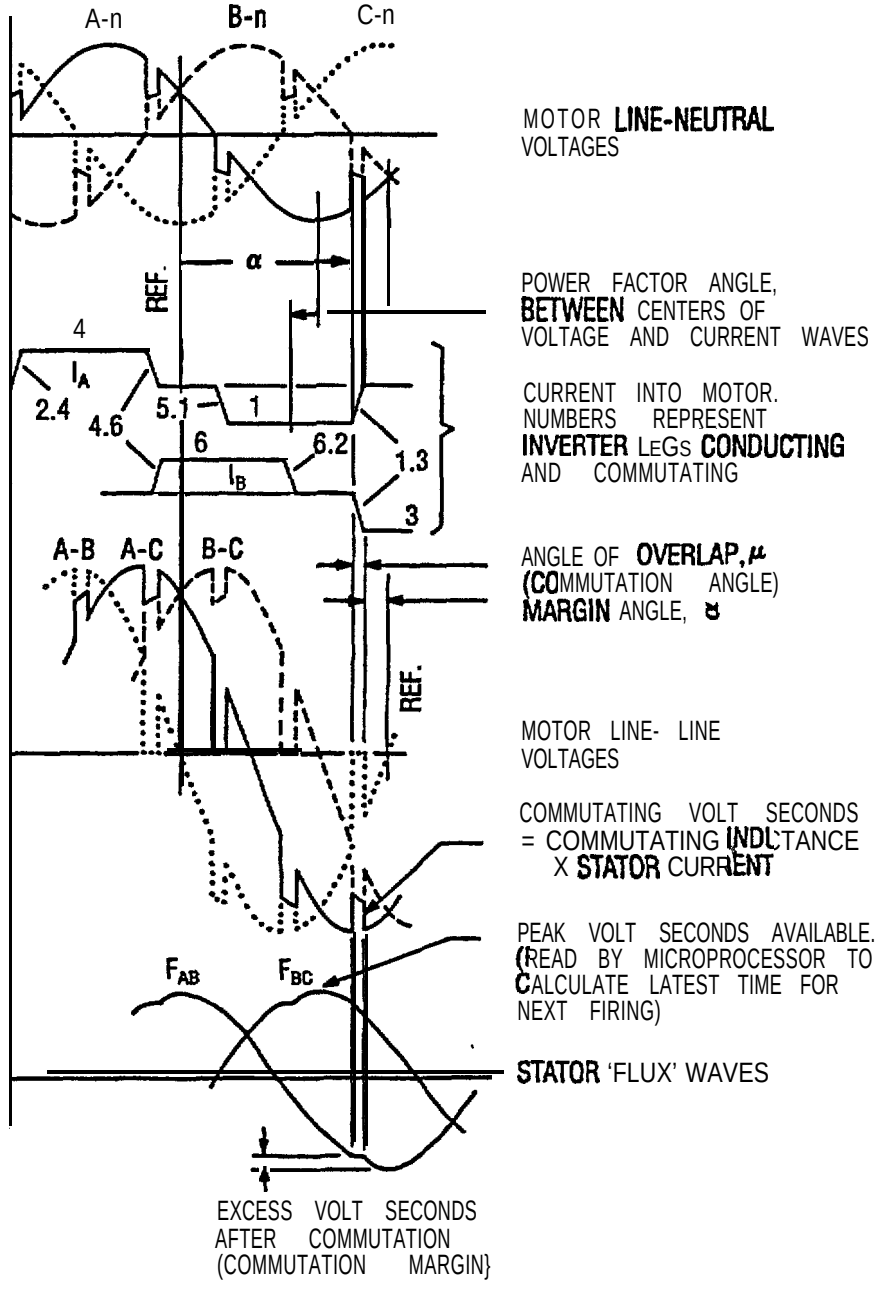


Figure 1-9. Motor Voltage and Current in the Self-Commutated Mode

CHAPTER 2
Power Equipment

2-1. Equipment Options

Two types of bridges may be utilized for power conversion. One type uses three thyristors in series and will be designated the medium voltage bridge (MVB). The other type utilizes one or two thyristors in series and will be designated the SILCOMATIC ® Plus power conversion bridge.

2-1.1. 2KV Medium Voltage Bridge

The arrangement of the power converter silicon controlled rectifier (SCR) stack assemblies are shown in Figure M2-1. This figure and the elementary diagram show the major power components.

SCRs are stacked horizontally; three in series, between four heat sinks. The 18 SCRs between the P-bus (P1, P2) and the AC lines, source and load, are on the left side in the bridge compartment; also CT1, CT3, FU1-4 and FU-6.

The N-bus SCRs are on the right side of the bridge compartment; also CT4, CT6, FU-5, and FU7-10.

The blower compartment contains the blowers, source and load ac and dc filters, MOVs, CT1A, CT3A, CT4A, and CT6A. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay. Figure M4-1 Medium Voltage Bridge Assembly, 4KV

2-1.2. 4KV Medium Voltage Bridge

The arrangement of the power converter SCR stack assemblies is shown in Figure M4-1. This figure and the elementary diagram sheets AC, AD, and AE show the major power components.

Thyristors (SCRs) are stacked horizontally, three in series; between four heat sinks. All 36 SCRs between the P-bus (P1, P2) and the ac lines, source and load, are in the left bridge compartment; also CT1, CT3, FU1-4 and FU-6.
All N-bus SCRs are in the right bridge compartment; also CT4, CT6, FU5 and FU7-10.

Leg modules A are on the left side in each compartment; these are the master stacks containing a full complement of cards in the stack (module) insulating box. Leg modules B are the slave modules, and contain only the XPTL card below the cells.

The Ieft blower compartment contains blower numbers 1 and 2, source ac and dc filters, MOVs, CT1A, CT3A, TB1, and TB2. Also, the hinged base in this compartment contains blower starter numbers 1 and 2, the 70V gating power supply, and thyristor fan-out PWB XTFH.

The right blower compartment contains blower numbers 3 and 4, load ac and dc filters, CT4A and CT6A. The hinged base in this compartment contains blower starters 3 and 4, and the bottom blower selection relays.

2-1.3. SILCOMATIC ® Plus Power Conversion Bridge

Each power converter consists of six thyristor leg modules stacked vertically as shown front-view in Figure S-l.

Each leg consists of two thyristors connected in series or a single thyristor, mounted within a heatsink assembly. An R-C snubber network is also contained in the leg module box.

Figure S-l SILCOMATIC® Plus Power Conversion Bridge Assembly

Five attenuated voltage signals are derived from the bridge as illustrated in Figure S-2 : Note that there is no attenuator connection to leg module 6.

The attenuated voltage signals are combined in the electronics module to produce scaled ac line-to-line and rectified dc bus voltages.

A simplified representation of the firing circuit elements is as indicated in Figure S-3. There may be only one cell per leg.

The pulse transformer delivers a current pulse to the thyristor(s) in the leg. The red light on the card front is illuminated if the HPTF is delivering gate pulses. The neon lamp across the cell goes out if the cell shorts.

The burden resistors for the current transformers are contained on an XTFX board. The auxiliary CT leads plug into the board and small ribbon cables route these signals to the electronics in the control cabinet.

2-2. Thyristor Modules, Medium Voltage Bridge(MVB)

Each power converter leg module consists of three thyristors arranged in series in a heatsink assembly.

The master leg module is mounted in an insulated box with associated printed wiring boards in front as shown in Figure M-2 .

An R-C snubber network is also contained in the leg module box, above the cells.

Each cell stack assembly consists of three cells sandwiched between heatsinks as shown in Figure M-3. The stack assemblies are oriented horizontally in the module as shown. Air flow over the heatsinks is from front to back of the bridge enclosure and the cell modules.

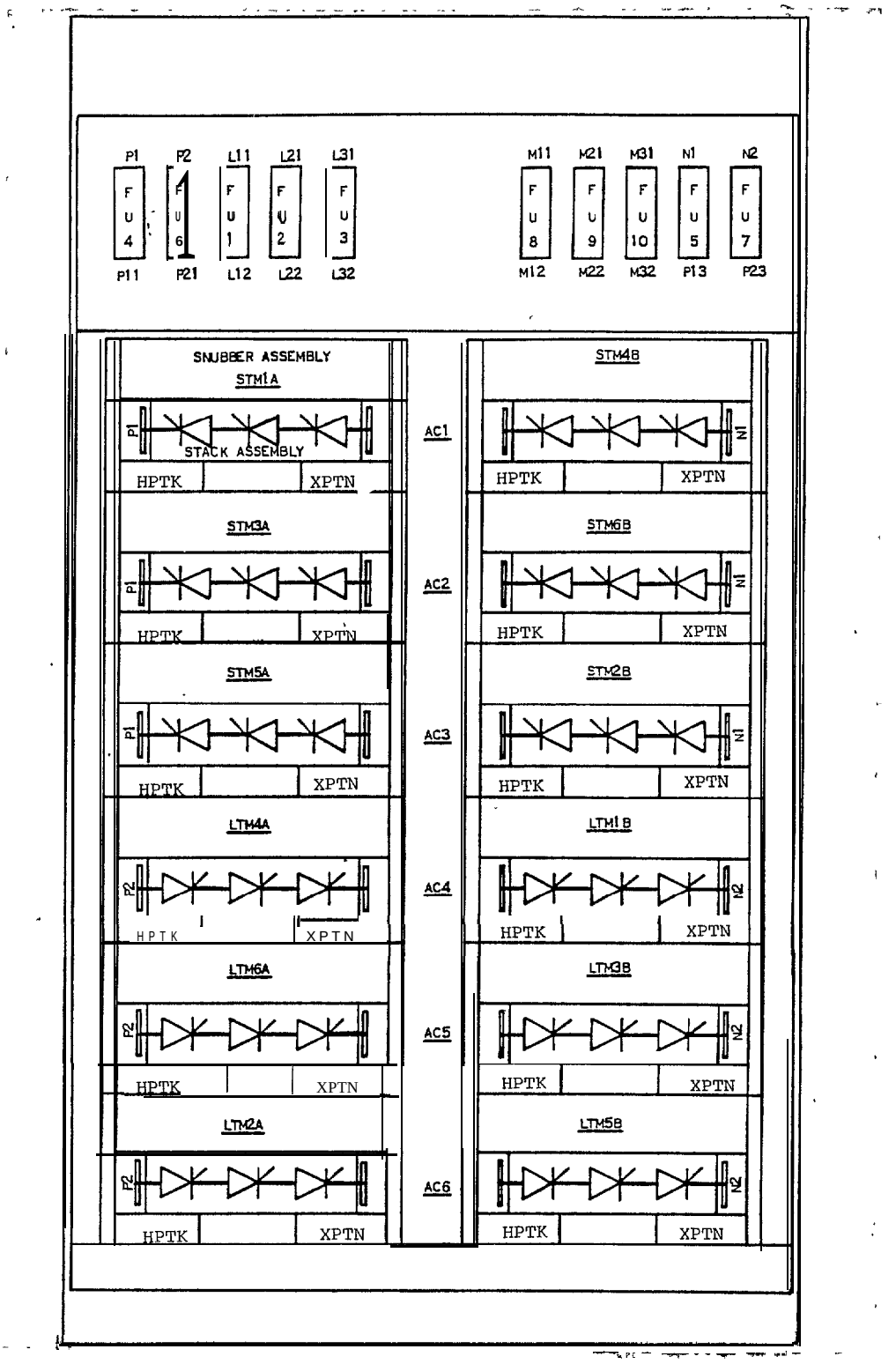


Figure M2-1. Medium Voltage Bridge Assembly

The cells are selected, matched, and assembled at the factory under the correct amount of pressure. For these reasons, it is recommended that the entire stack be replaced if a cell failure occurs. The defective stack may then be sent to the factory for repair. Figure M-3 MVB Cell Stack Assembly

Each cell in the module is paralleled by a voltage-equalizing resistor and by a neon lamp and dropping resistor network which indicates when voltage is present across the cell. These lights are visible through the front doors of the equipment and can be used to detect a shorted cell. The cell light network is mounted on a small card (NTDA) located above each cell on the front of the thyristor modules. The location of these cards is as shown in Figure M-2. The NTDA card also contains thyristor gate interface circuitry which is illustrated in Figure M-4. Thyristor gating signals are routed from the gating transformers through the NTDA cards to the thyristor gates.

2-3. Gate Driver Transformers

The cell gates are driven by current transformers mounted on the front of the module as shown in Figure M-2. The current transformers for each module are wired so that all their primary windings are in series. The primary winding is simply a conductor through the center hole of the transformer. Current pulses are applied in this conductor by the gate driver circuitry. This circuit insures that all thyristors in the string are

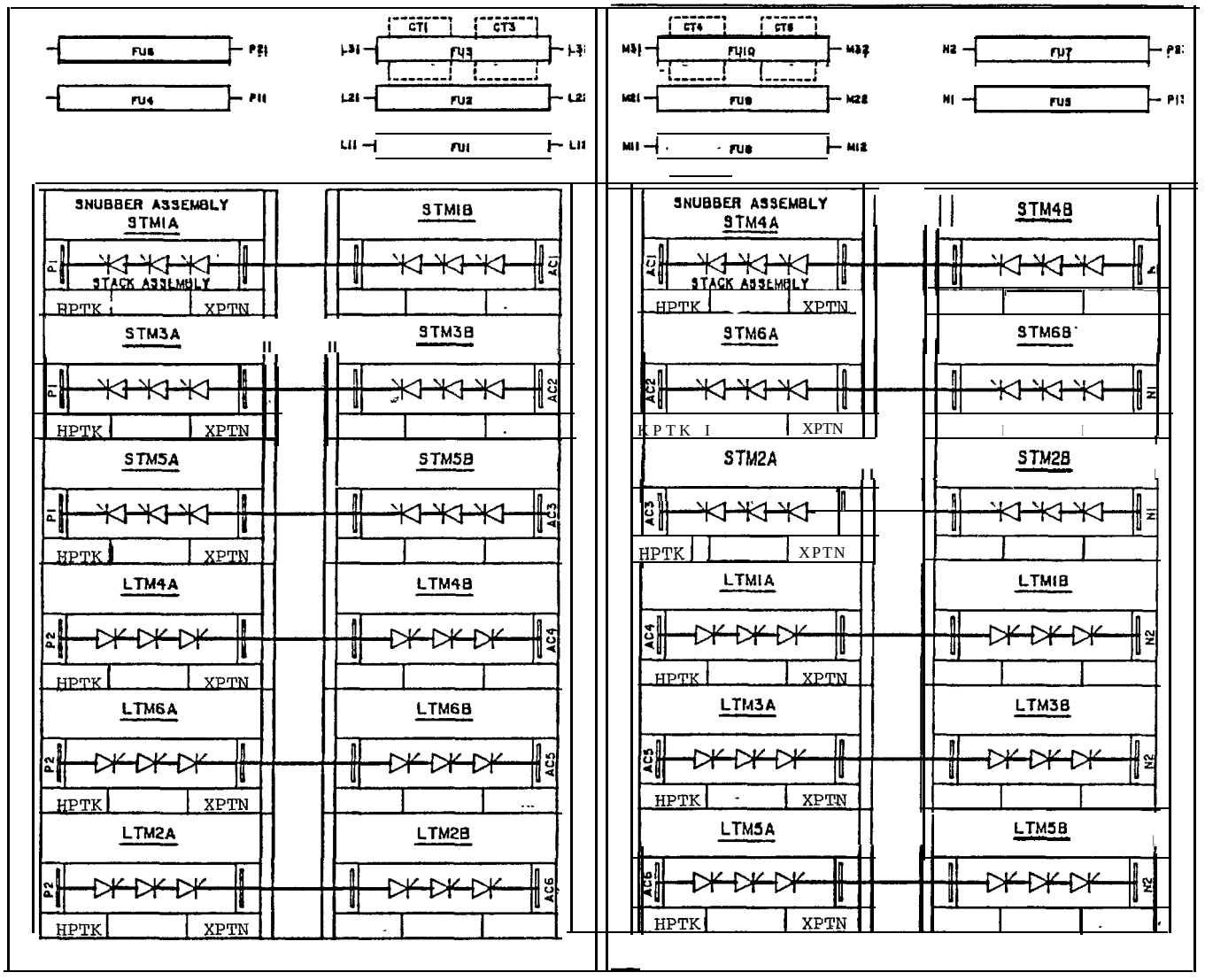


Figure M4-1. Medium Voltage Bridge Assembly, 4KV

gated simultaneously and it further provides voltage isolation between the thyristor gates and the electronic gate pulse generation circuitry.

2-4. Gate Driver Card (HPTK)

Gate pulses of proper duration and timing are developed in an electronics module located in a control enclosure at the left end of the MVB. These signals do not possess the necessary drive and isolation to interface directly with the gate driver current transformers, This interface is provided by the HPTK gate driver card located in the master thyristor module in each leg. The HPTK receives the low level signals and, after conditioning, passes them through a pulse transformer to provide isolation. The pulse transformer output drives a pulse amplifier circuit which is powered by a 70 Vdc gating power supply located in the fan and filter compartment of the MVB. The pulse amplifier output drives all gate pulse transformers in the leg as described above. A red light emitting diode (LED) on the board glows to indicate when gate pulses are present.

A second function of the HPTK card is to detect an undervoltage condition of the 70 Vdc supply. In the event that an undervoltage is detected, the HPTK outputs a logic signal to the electronics module indicating the fault. The electronic control then takes appropriate action to shut down the drive. Since there is an undervoltage detector on each gating card, wiring defects in the gating power will also be detected. A block diagram of the HPTK card is given in Figure M-5. The pulse driver delivers a current pulse simultaneously to all thyristors in the leg module.

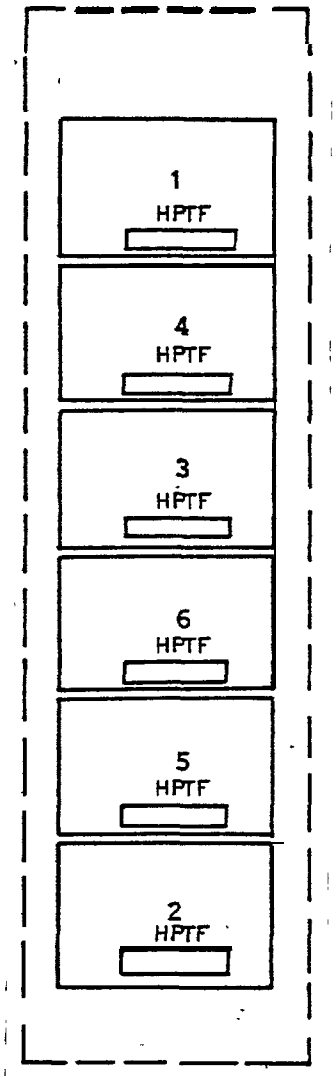


Figure S-l. SILCOMATIC® Plus Power Conversion Bridge Assembly

2-5. Voltage Feedback Attenuator Card (XPTN)

In order to properly control the gating signals to the MVB, the electronics in the control enclosure must have available analogs of all ac and dc bus voltages. The XPTN attenuator cards in the thyristor modules provide bus-to-ground voltage signals which are combined in the electronics module to produce bus-to-bus signals for each converter bridge.

Figure M-6 illustrates the overall attenuator connections for developing bus voltage feedback signals for an entire bridge. Note that only five bus-to-ground signals are required. Consult MVB elementary DS5201 for complete details of attenuator connections.

Five signals are derived from the attenuator units as illustrated below:

In Figure M-6, a series string of thyristors is represented by a single SCR symbol.

Note that the line-to-ground attenuator in leg module 4 is not used.

As explained in the paragraph discussing power converters, the voltage signals are used to control firing of the SCRs and to provide various protective functions.

The XPTN contains a berg jumper that is positioned to provide proper attenuation between the bridge and the electronics. See system elementary for proper setting.

The arrangement of the attenuators for the converter and inverter is identical.

2-6. Signal Distribution Board (XTFH)

All signal interchange between the drive electronic control and the medium voltage bridge is made through the XTFH signal distribution board located in the fan/filter compartment. A small ribbon cable from each HPTK gating card connects to the XTFH board. These twelve ribbons each contain gating control signals, attenuated bus voltage signals, and 70 volt power supply under-voltage signals. The six undervoltage signals from each bridge are combined in the XTFH card to produce a composite under-voltage signal for each bridge. The gating signals originate in the electronics module in the control compartment, while the remaining signals originate in the medium voltage bridge and must terminate in the electronics module. Signal interchange to the electronics module is made via two large ribbons which connect the XTFH board and cards in the source and load electronics.

The XTFH contains the burden resistor circuits for the source and motor line current transformers. The auxiliary ct leads plug into the XTFH board and two small ribbons route these signals to the source and load electronics in the control cabinet.

2-7. Exciter Voltage Controller (EVC)

This installation requires controllable excitation in order to meet requirements such as high performance over a certain speed range, high acceleration/deceleration rates, or a special torque/speed characteristic.

Ordinarily, the excitation is an almost-linear function of speed, where the intercept and slope of the line is determined by the number of poles of the motor,

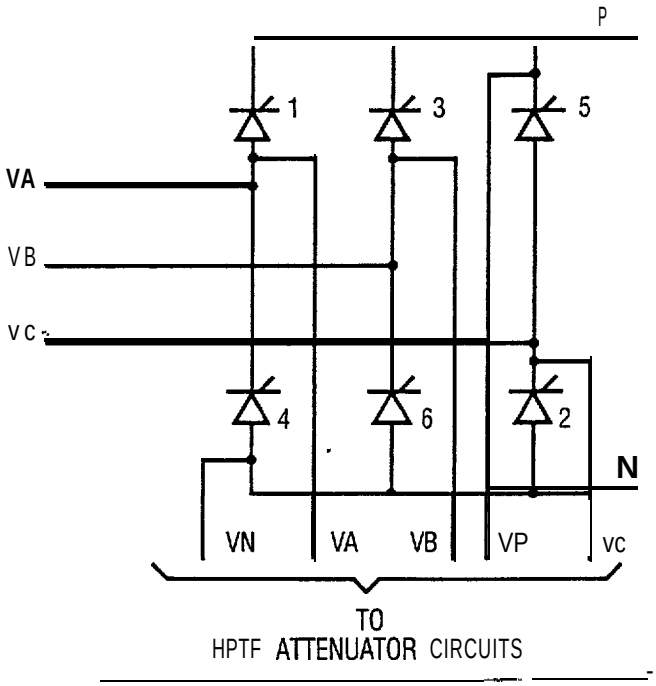


Figure S-2. SILCOMATICB Plus Power Conversion Bridge Attenuator Connections

the number of poles of the brushless exciter, the brushless exciter voltage (usually provided by tapped autotransformers), and the relative phase sequence of motor and brushless exciter stator voltages.

In addition to this, there is a phase-controlled three-phase switch between the source and the brushless exciter. Each of the three legs consists of two parallel thyristors connected in reverse polarity. Refer to Figure CE-1. All six thyristors are phase-delayed identically relative to their own connected source line voltages so that symmetrical, bidirectional current in the three lines is obtained. The effect is to reduce the output three-phase voltage magnitude in response to a reference signal.

The reference signal is programmed to obtain a motor flux vs. speed profile necessary to meet the performance requirements.

The EVC consists of the following major components:

SCR Circuit	DS3800NHVL (PWB)
Six SCR Cells	Six high voltage attenuator circuits
Three line fuses	Six pulse transformer firing circuits
Two Current Transformers	CT burden resistor circuit

The NHVL PWB attenuates the incoming and outgoing line voltages of the SFE, 3-phase rectifies and scales (thru burden resistors) the CT signals, and provides the pulse transformer circuits that fire the SCR gates. The attenuated voltages and the scaled current signal are sent back on the same 50-pin ribbon cable that sent over the pulse signals from the electronics module PWB DS3800NSFE.The NSFE provides signal conditioning and interface to the microprocessor control. The microprocessor fires the SFE SCRs based on a selected phase relation to the source line voltages.

The SFE incoming line must be phased zero degrees, +30°, or -30" to the main power bridge's incoming (source) line voltage.

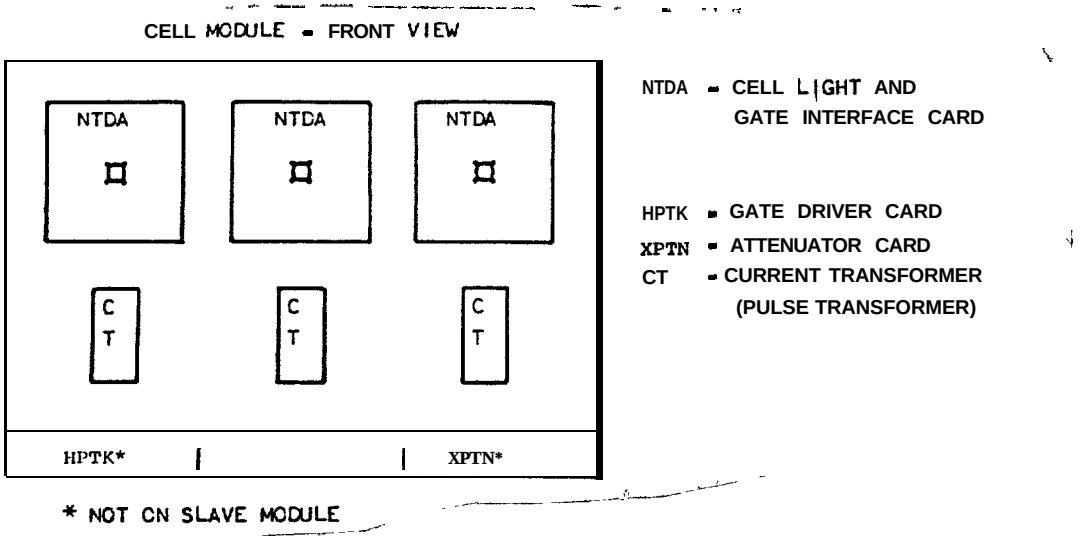


Figure M-2. MVB Thyristor Module

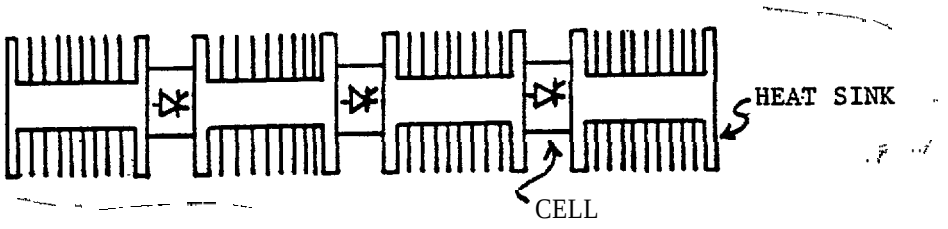


Figure M-3. MVB Cell Stack Assembly

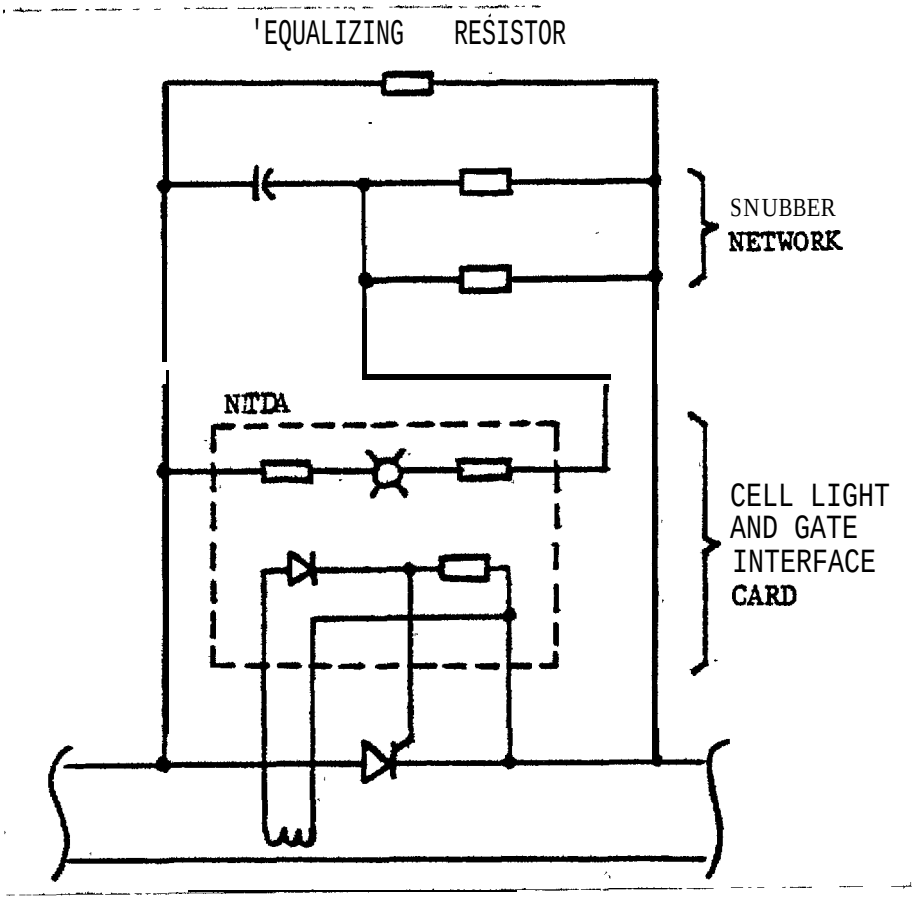


Figure M-4. Cell with Gating, Display, Snubber, and Voltage Equalizing Circuits

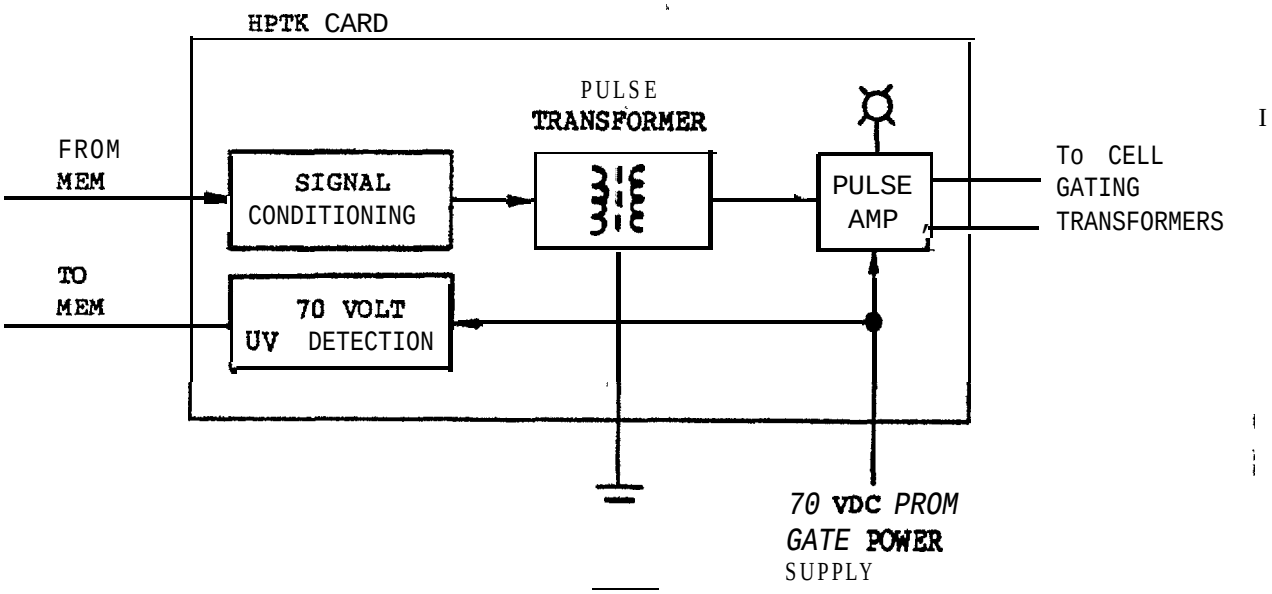


Figure M-5. HPTK Block Diagram

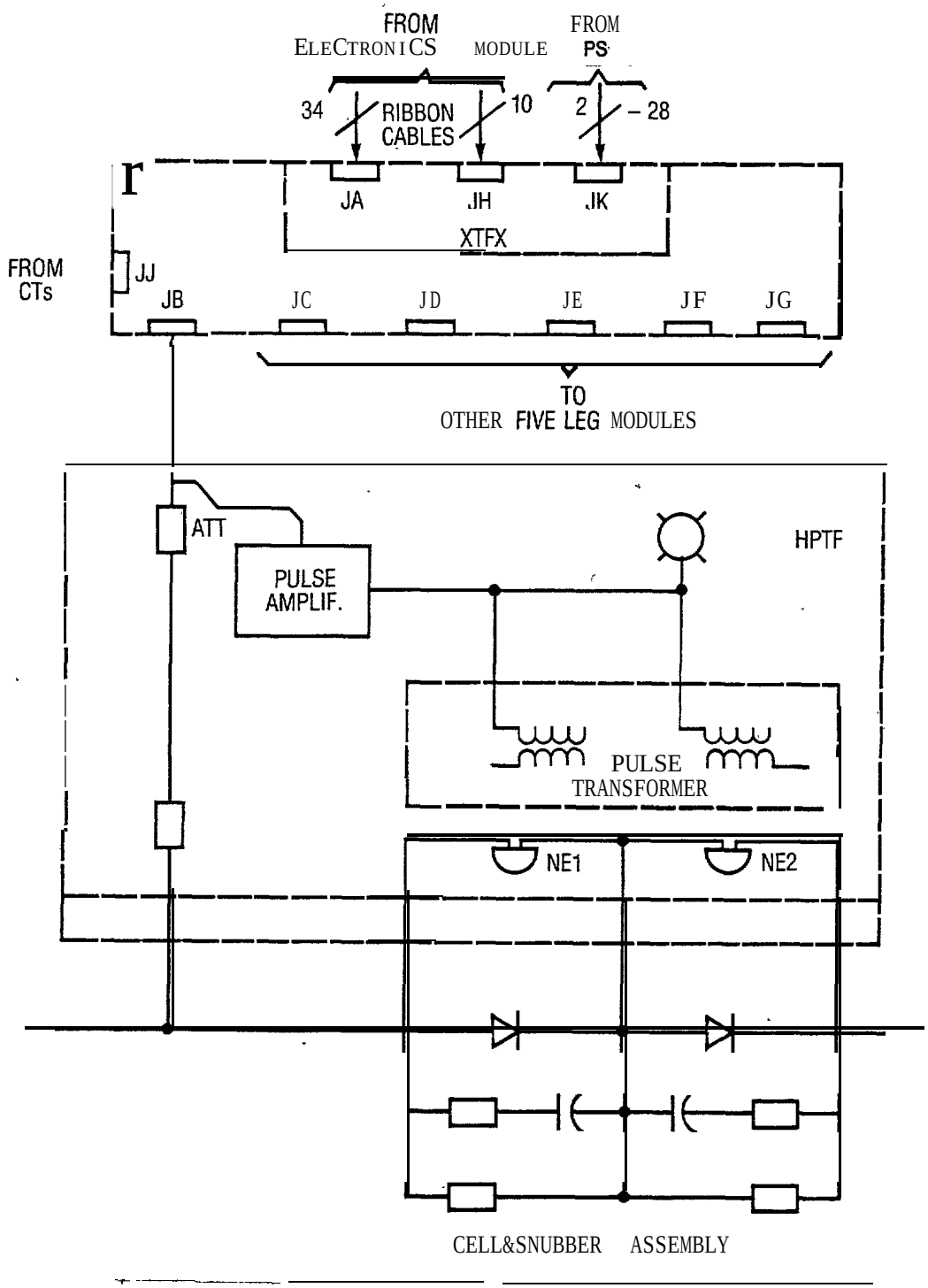


Figure S-3. SILCOMATIC® Plus Power Conversion Bridge Firing Circuit Elements

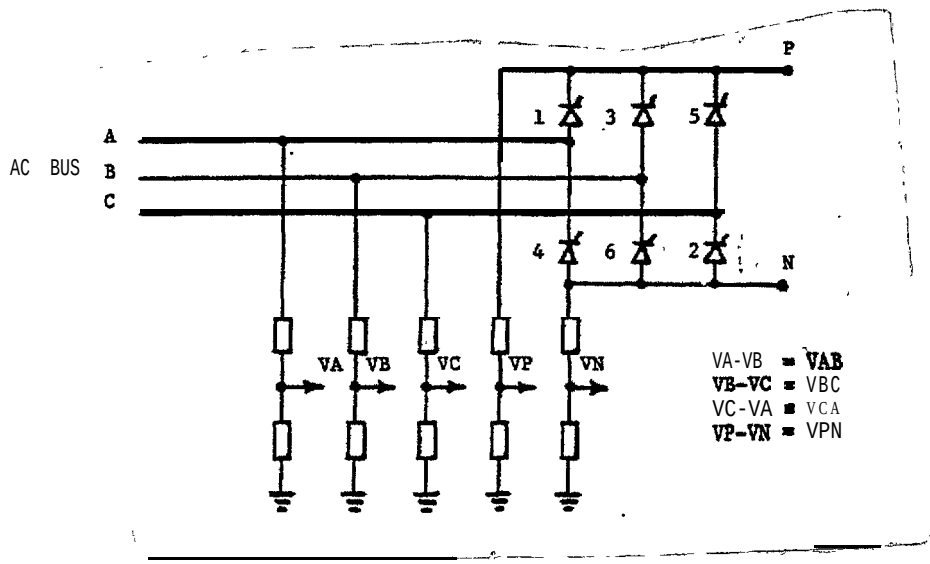


Figure M-6. Bridge Attenuator Circuits

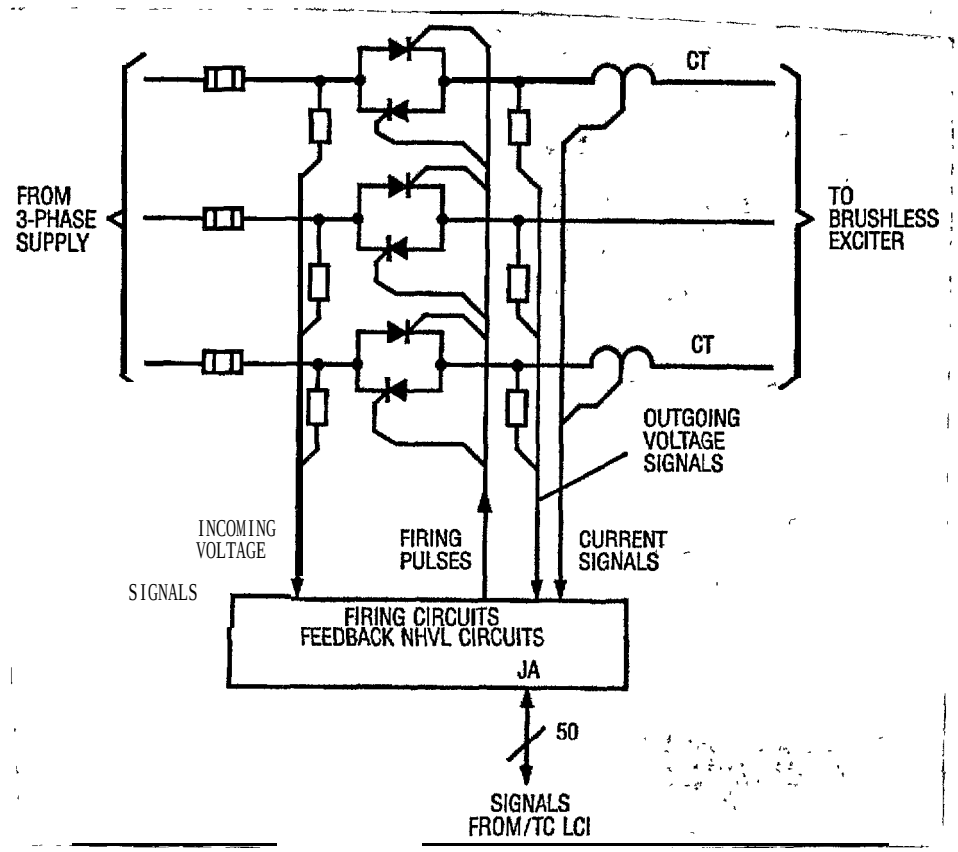


Figure CE-1. Synchronous Field Exciter Module (DS3820SFE)



CHAPTER 3
CONTROLELECTRONICSPANEL

3-1. Introduction

The control electronics panel is made-up mainly by various modules interconnected thru ribbon cables and harnessed wiring. The main modules are:

DS3800LCG	Main Control electronics
DS3820IOM	Inputs/Outputs (I/O)
DS3820SIM	Auxiliary interface electronics (When more than the normal signal are needed)
DS3820SFE	Field Exciter (Discussed in Chapter 2)
DS3820LDM	Display Module

The modules are typically configured as in Figure 3-O. Following are brief descriptions of the functions of the module. For more specific information the corresponding modules elementary and documentation should be consulted.

3-2. Electronics Module (LCG)

The control electronics module consists of cards in 2-rows x 6-slots. Its primary purpose is to control the firing of the power converters and field exciter module SCRs for driving a motor. It also provides various protective functions for the motor and bridges. Interface between the LCG to the customer is thru the IOM and SIM.

See Figure 3-1 for the physical layout of the module. Connection is made to the electronics module (SEM) through the plugs located at the top, and also through card front connectors.

From a functional standpoint, the electronics module processes signals as represented in Figure 3-2.

The signal interfaces for the SEM are as follows: Attenuator signals - Attenuated bus to ground signals from each converter.

SIM module - See SIM section

Line/load current feedback - Signals from line/load bus current transformer burden resistors.

Field exciter module - See SFE section

Power supply - The module requires + 5, ± 15, and + 28 volts which are provided by a base-mounted power supply.

Thyristor gate signals - These are the electronic signals which drive the gate pulse circuits in the medium voltage bridge.

The hardware of the electronics module is represented from a block diagram standpoint in Figure 3-3.

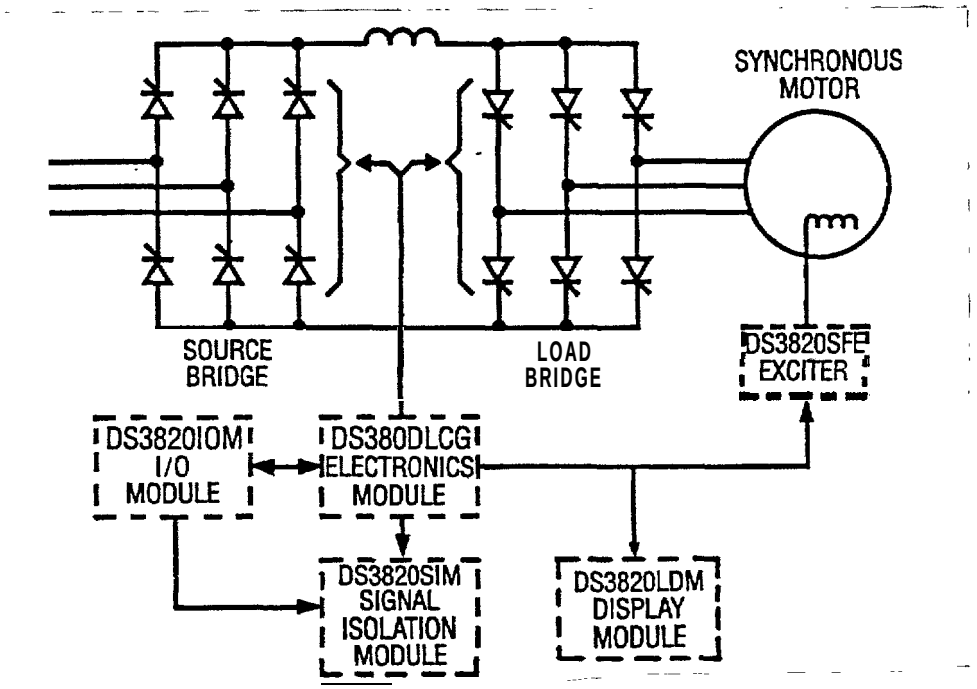


Figure 3-O. Control Electronic Panel

There is one microcomputer controlling the firing of both of the power converters and the field exciter.

The microcomputer establishes a local bus (L-bus) structure on which the input/output for it is interfaced. In addition, it communicates to an Sk byte read/write (RAM) memory and a 2k byte electrically alterable memory (EAROM). This memory combines features of read/write memory (RAM) and read only memory (ROM) in that it is electrically alterable (although slow) and at the same time will not lose its contents on power loss. The primary use of this EAROM memory is to store parameters which must be tailored to each different system. These parameters are field adjustable for drive tuneup.

The local bus has a timer and interrupt card which provides the necessary hardware for control of interrupt priority and timing for the microcomputer.

There are input/output ports on the local bus which interface static logic signals to/from the microcomputer. As an example, thyristor gating commands are written out to one of these ports (on each side) and sent to an analog card to produce the gating signals to the converter bridge.The electronic control contains a number of analog cards which process the voltage feedback, current feedback, meter signals, etc.

Analog to digital converter cards interface the speed reference, voltage feedback, current feedback signals, etc. to the microcomputer bus on each side.

I/O cards interface process command and status signals to the microcomputer. These inputs are combined logically by a sequencing program in the microcomputer to control the relays in the IOM. These outputs are used to control system devices such as panel lights, contactors, etc.

For a dual channel LCI system, each power channel has its own electronic module. To establish load sharing and for communication of other status, it is necessary for information to be interchanged between the two electronic modules. This interchange is accomplished via 12 Bit D/A and A/D converters ($\pm$ 10V) and some I/Os of the IOM.

3-3. Control Operational Description

3-3.1. Basic Principles

Refer to the Control System block diagram in Figure 3-4. The speed reference from the process controller is compared to a speed feedback derived from the integrated motor voltage to develop a speed error input to

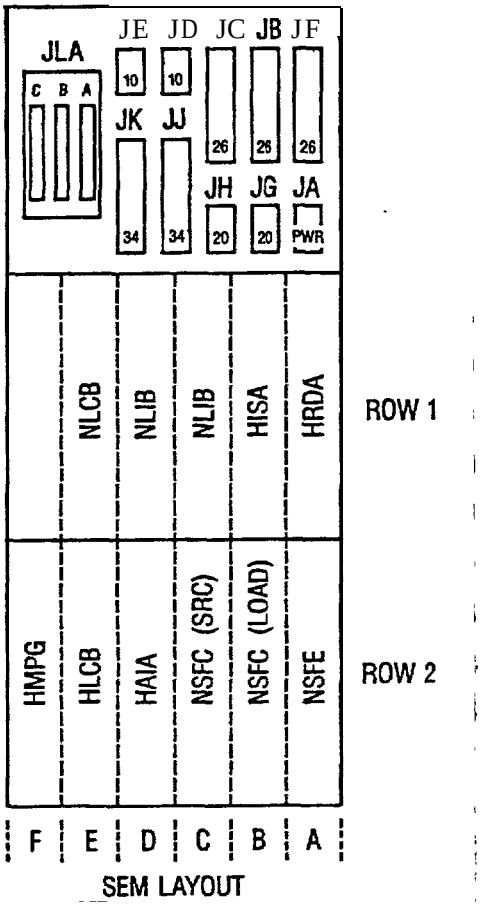


Figure 3-1. Small Electronics Module (SEM) Card and Connector Layout

30

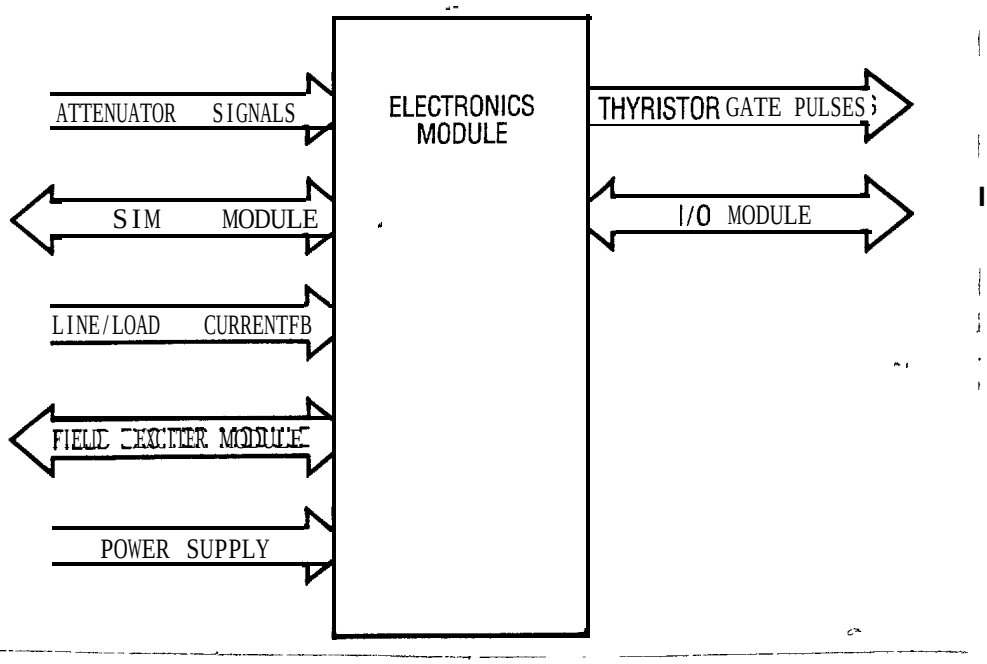


Figure 3-2. SEM **Input/Output**

the speed regulator. The output of the speed regulator is a torque command. The speed regulator program resides in the load side processor.

The torque command signal is applied to both the source and load side control. As illustrated in Figures l-6 and l-7, motor torque is a function of voltage, current, and the angle between them. Therefore, torque can be controlled by adjusting current magnitude at a fixed angle or maintaining a constant current and varying the displacement angle. The current i's controlled from the source side, whereas the angle is varied from the load side . At any instant in time, the torque is actively controlled by only one of these means.

The torque command to the source side control is applied to a maximum and minimum current limiter. The minimum current level is set **to** maintain continuous current in the dc link.

The current level control also affects the load side angle control. When the voltage spillover is not activated, the angle control is in one of its limits. If current is adjusted to control voltage, the angle control gain is decreased and displacement angle θ is increased, which reduces developed torque to that required by the load.

The current command is compared with current feedback, I_{DC}, and the error is applied to the current regulator. The current regulator controls the firing of the thyristors in the source side converter (rectifier). Thus, the source side control adjusts the dc link voltage as required (within its limits) to produce the motor current needed to drive the load.

The load side control is designed to adjust firing angle so as to maintain the displacement angle θ as small as possible. This feature is designated "Fire as late as possible" (FALAP).

Referring to Figure 3-4, it is shown that the angle of advance, β is the sum of the commutation angle, μ and the margin angle, γ. For maximum efficiency, it is desired to keep the margin angle as small as possible without encountering commutation failure. The objective is to determine the latest possible firing point for each oncoming commutation.

When the drive requires regeneration, the load-side thyristors are fired full advance (point "X" in Figure 3-5), and the source side controls current by reversing the dc voltage to match the rectified motor voltage.

3-3.2. Phase Locked Loop Operation

Since the inverter thyristors are fired in synchronism with the integrated motor stator voltages, a phase locked loop (PLL) counter is used to maintain correspondence between the integrated voltage wave zero crossings (Figure 3-5) and the firing pulses.

When the PLL counter output is precisely in synchronism with the "flux" wave zero crossings, the loop is said to be "locked". If the counter output is lagging (slower than) the flux crossings, the PLL controller acts to increase the counter frequency (by decreasing PLLC LOAD). Similarly, if the counter frequency is leading (faster than) the flux crossings, the controller acts to decrease frequency (by increasing PLLC LOAD).

Although the output of a phase locked loop, when "locked", will have the same frequency as the reference signal, there will be a phase difference between the two due to the integrating action of the loop low-pass filters; thus, the name phase locked- loop.

3-3.3. Electronics Module (LCG) Printed Wiring Board (PWB) Descriptions

Following is a list of cards used in the electronics module. The function and condensed theory of operation of these cards follows the list. It is not the intent to describe in great detail the inner workings of each card, but to relate their function in the overall system. For more detail, refer to the appropriate DS3800 drawing, Section 2, for operating notes and Section 4 for elementary diagram. It is also generally helpful to refer to the card symbols in the SEM elementary diagram.

3-3.4. Microprocessor ▪ HMPG/DMPB

The DS3800HMPG is a single board computer with the following features.

1. 80286 microprocessor
2. Programmable communications interface
3. Programmable interval timer
4. On-board RAM memory (16K bytes)
5. On-board PROM memory (64K bytes)
6. Mapped I/O
7. Bus hog timer
8. Stall timer

The configuration chosen gives a powerful 16 bit processor.

Connectors.

A 50 pin ribbon connector is included on the DMPG to interface the EAROMmemory (DMPB).

A 25 pin male connector is present to serve as an RS232C communications port. (See Tables 3-1A and 3-1B) A "null modem" is supplied to allow a standard data terminal interface cable to be used with either communications port.

3-3.5. Electrically-alterable ROM = DS3800DMPB

This is a non-volatile memory board containing 8k bytes of memory used to store drive tuneup parameters which may be customized on a per drive basis. Physically, the board is configured as a daughter board and mounted on the HMPG.

Constants stored in the board's memory are normally changed by first enabling board writes with switch "wren" and then supplying the data through the drive monitor.

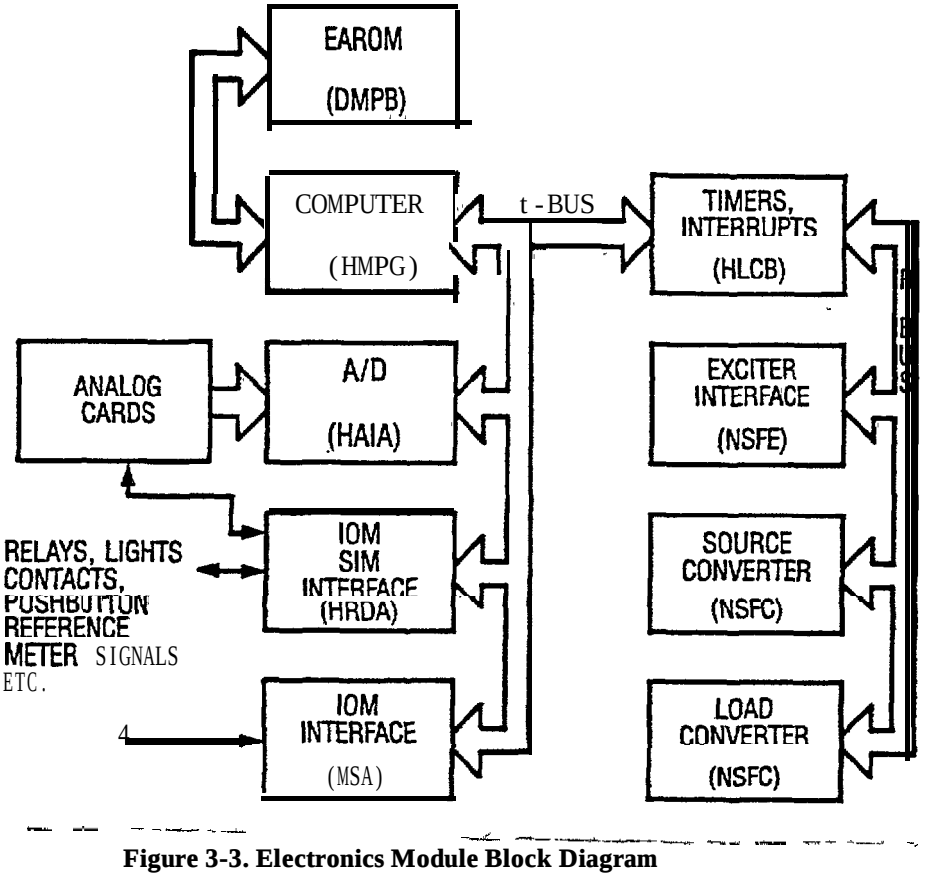


Figure 3-3. Electronics Module Block Diagram

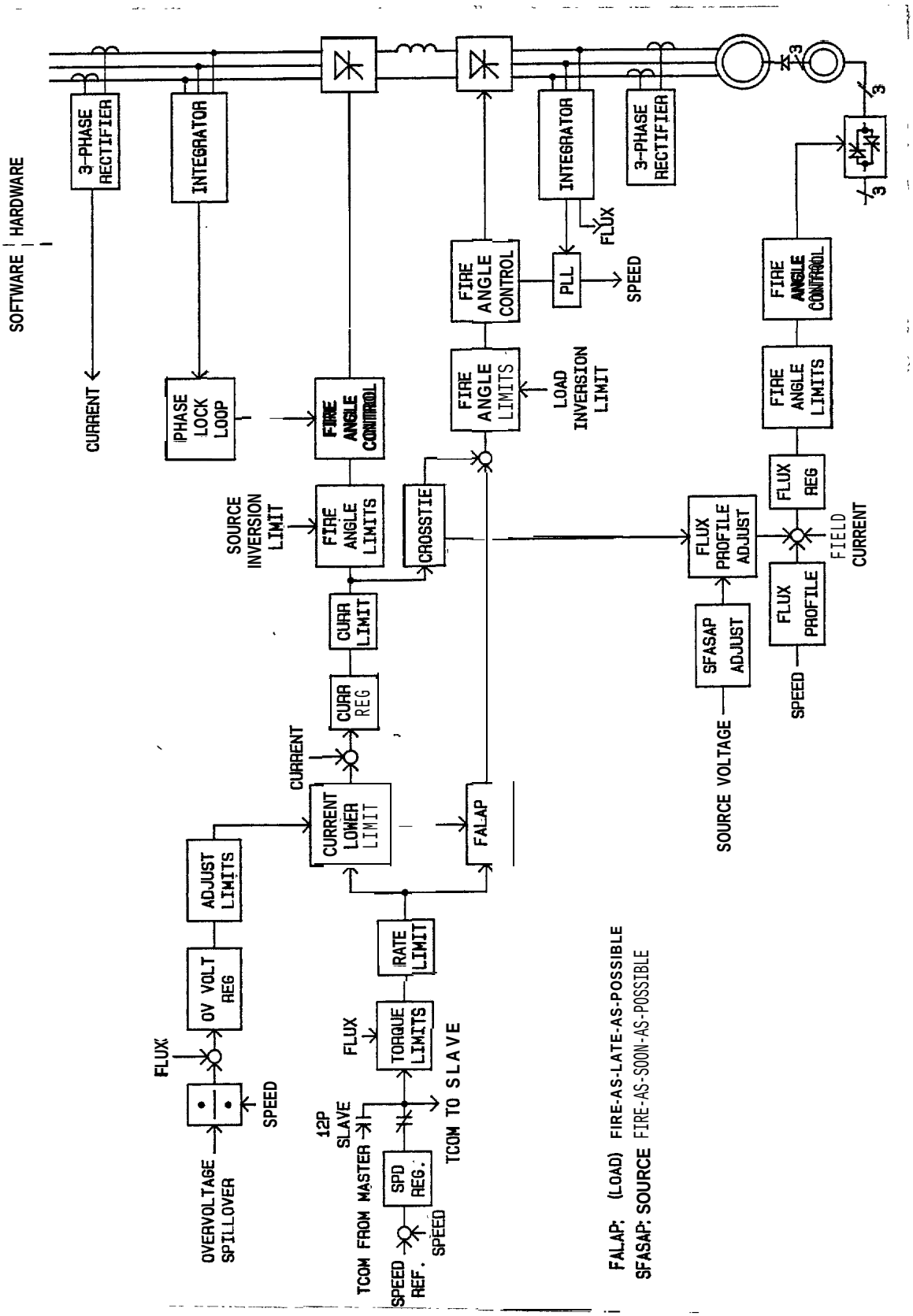


Figure 3-4. LCI System Digital Regulator Block Diagram

In the Drive Systems documentation procedures, DMPB and the two PROMS in sockets U2 and U3 comprise module DS3815PLHA.

DS3815PLHA and HMPG are then combined to form module, DS3815PLGA. (See Table 3-1C)

3-3.6. LCI Processor Extender Board = DS3800HLCB

The HLCB board extends the functions provided by HMPG to include those required for an LCI system. The following functions are provided.

1. Address decoding of on-board functions, NSFC and NSFE
2. Interrupt Controller
   - Source crossover
   - Load crossover
   - Source angle control
   - Load angle control
   - Field excitation control

Table 3-1A. Microprocessor Board Features - RS232C Connector Signals (HMPG)

Pin#	Name	*Signal*	Direction	Function
		RS232C CONNECTOR SIGNALS		
JB-2	BA	TXD	output	transmit data
-3	BB	RXD	input	receive data
-4	CA	RTS	output	request to send
-5	CB	CTS	input	clear to send
-6	CL	DSR	input	data set only
-7	AB	DCOM	input	signal ground
-9	CD	DTR	output	data terminal ready

Table 3-1B. Microprocessor - Berg Jumpers, Testpoints and LED's (HMPG)

Nomen.	Function	Position
	BERG JUMPERS	
J1	Selects type of PROM	8/16
J2	Selects size of PROM	16k
J3	Selects bootup method (on/off bd.)	DGN(not)
J4	Selects 8 or 16 address segments on L bus	8
J5	ground for RS232	GND
	TESTPOINTS	
RST	Reset jumper to common to reset processor	
DCOM	DCOM – Digital common	
I/S	Inhibit stall – Jumper to common to activate	
TNMI	Test nonmaskable interrupt – Jumper to common to activate	
	LED'S	
BHOG	Bushog – On when non-existent address is called by software	
TST	Software controlled and defined – On during initialization self test	

Table 3-1C. Electrically-Alterable ROM - LEDs and Switches (DMPB)

Nomen.	Function
	BERG JUMPERS
None Supplied	
	LED'S
IMOK	Software controlled as a function of self diagnostics — On normally, will be off if selftest IS not passed
WREN	Write enable. On when write capabiltiy to board is enabled with switch, "WREN"
	SWITCHES
WREN	Activates the board to allow alteration of the drive tuneup parameters.

3. 5 bit general purpose input port used for:
   - Ground fault
   - I/O module input ribbon cable continuity
   - I/O module signal cable continuity
   - I/O module output ribbon continuity
   - Door module ribbon cable continuity
4. Zero crossing detection. Circuity is included to derive source and load crossover interrupts from square wave information originating on the NLIB boards. A single clock pulse interrupt is obtained at each crossover.
5. Two programmable timers provide the functions of
   - PLL clock (source and load)
   - Time counter (source and load)
   - Speed calculation

   Cloock for serial communications port
6. RS-232 compatiable port. Only the transmitt function implemented at the port is intended for driving a printer located in the door module. The control line, DSR (data set ready), is also monitored to determine the status fo the remote printer.
7. Digital to analog converters. Three 8-bit D/A converters are supplied. D/A 1 and D/A 2 are 8bit non-committed outputs which are available for transmission to the signal isolation module (SIM) if supplied. The 12 bit D/A is committed to supplying the torque reference between 6 pulse converter controlss when used in a 12 pulse redundant system. System level documentation will define its use (if any) in other systems.

3-3.7. Analog/Digital Interface Card DS3800HAIA

The HAIA board accepts 16 analog input signals, multiplexes them to one line and coverts to a two byte digital word. The A/D provides 11 bits of magnitude solution. Also provided are three, 8-bit digital input ports, three 8-bit output ports and one 4-bit digital port. (See Table 3-3)

3-3.8. Flux Integrators - DS3800NLIB

The NLIB contains various analog signal conditioning and protective functions. Two identical boards are used, one for the source side and one for the loadside. The following functions are provided:

1. Voltage integrators
2. Line/Motor current transformer receivers
3. Voltmeter driver
4. Overvoltage protection
5. Undervoltage sensing
6. Suppression
7. Variable reference comparator
8. Current meter driver

3-3.8.1. Voltage integrators.. A signal proportional to each line-to-line voltage is obtained from the NSFC board. This signal is applied to an analog

Table 3-2. LCI Extender - LED's and Testpoints
(HLCB)

Nomenclature	Function
	BERG JUMPERS
None Used	
	.LED's
IMOK	Software controlled as a function of selftest. On normally, will be off if selftest is not passed.
	TESTPOINTS
ACOM	Analog common.
DA1	Digital to analog output 1. Normally uncommitted.
DA2	Digital to analog output 2. Normally uncommitted.
DA3	Digital to analog output 3. Normally used for horsepower output.
DA4	Digital to analog output 4. Normally used for torque command to other channel (if supplied).
OTESTINO	?
OTESTIN1	?
SFT1	Execution flag for source crossover and angle control routine.
SFT2	Execution flag for load crossover and angle control routine.
SFT3	Execution flag for field angle control interrupt.
SFT4	Execution flag for background program.
VA1	Phase A signal for source power.
VA2	Phase B signal for motor voltage.
DCOM	Digital common.

integrator to obtain a replica of motor flux according to the equation:

$$V = K * dPhi/dt$$

or

$$Phi = V * dt$$

Components of line currents are used to compensate for drops due to motor resistance and commutating impedance so that the flux represents the motor internal counter-emf or the so called voltage behind the commutating reactance. The analog sine wave corresponding to flux *is* then applied to *a* zero crossing comparator where the digitized output is then available to the HLCB for phasing information. The analog flux signals are also read by the microprocessor based software for use in the control algorithms.

Line/motor current transformer receivers.

Two signal conditioning differential receivers are included for line currents A and C. Line current B is calculated from A and C assuming a balanced load. Delta currents are also determined for use in the flux reconstruction.

3-3.8.2. Voltmeter driver. The magnitude of the line-to-line voltage is determined by a 3 phase absolute value network. The absolute value is then scaled by an operational amplifier for use as a line voltage signal.

3-3.8.3. Overvoltage protection.. The absolute value of line/motor volts obtained above is compared with an adjustable reference for a time determined by daughter board component, C37, then the overvoltage latch is set. The signal is wired to an I/O port for fault logging by the control software.

3-3.8.4. Undervoltage sensing.. The absolute value of line volts is compared with an adjustable reference. If the voltage is less than the reference for time delay set by daughter board component, C17, then signal "1LV" becomes true. This signal is used by the software system sequencing to determine low source volts condition. The function is not used on the load side.

3-3.8.5. Suppression.. The absolute value sum of the line currents is taken and applied to a comparator.

If the rectified current exceeds the reference threshold for 26ms, a fault latch is **set.** The time delay is included since the primary protection is in software which includes an on-line recovery. The function implemented here is strictly backup.

3-3.8.8. Current meter driver.. The absolute value of current is scaled and available from either source or load for meter driving. Scaling is 5v = 1pu/ (See Table 3-4)

3-3.9. Gating Reference Board - DS3800NSFC

The NSFC's primary task is to convert information from the microprocessor based control to cell pair and timing for firing 6 thyristors configured into a 3 phase - double way bridge. In performing this task, the following functions are used:

1. R-bus interface
2. Firing mask latch
3. Cell state sensors
4. Line-to-line voltage signal conditioners
5. Pulse train generation
6. Gate drivers
7. Attenuator buffers
8. D-C voltmeter driver

Quantity 2 NSFCs are used, assigned as follows:

1. Source converter
2. Load converter

3-3.9.1. R-bus interface. Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected.
All information exchange with host processor takes place through the R-bus interface.

3-3.9.2. Firing mask latch. The particular cell pair to be fired is selected in software and communicated to

Table 3-3. Analog/Digital Input Signals - Berg Jumpers and Testpoints (HAIA)

Nomen.	Function	Position
	Berg Jumpers	
Nomen.	Function	Position
J1	Selects either a fixed + 7.5v input to channel 15 or a variable input thru pin12. The variable input is selected.	A
J2	Selects either a fixed + 7.5v input to channel 14 or a variable input thru pinl2. The variable input is selected.	A
J3-J8	Selects L bus address for board.	F
J9	Selects filter for analog side of A/Dconverter. Filter is selected.	0
	Testpoints	
ACOM	Power supply analog common.	
FOG	Test input. Tie to common to latch internalcard data bus.	
AN	Analog signal at input to A/D conversion	
DB	Data buffer direction. High transmits to microprocessor and low receives.	
DV	"0" data valid signal	
BSY	"1" busy signal	
ES	"1" enable switch	
CLK	card clock - 1.152 mhz	

the NSFC over the R-bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed. This information is routed to a timer. At the end of the delay an interrupt **is** generated and on the load side the cell mask is clocked into the final latch. On the source side, the generated interrupt calls an "execute firing" program which (if all permissives are met) signals the NSFC to move the firing mask to the final latch.

3-3.9.3. Gate drivers. Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 Vdc levels.

3-3.9.4. Pulse train generation. The gate pulse reference driver signals consist of a "0" true pulse train nominal 120 degrees long during the conduction period. The reference pulse train is generated from a crystal clock and shaped for a duty cycle of 15 microseconds on and 30 microseconds off by a timer and control logic.

3-3.9.5. **Buffers.** Attenuator buffers are provided to receive each attenuator string connected **to** phase voltages 1,2,3 and also the positive anhd negative bus. These buffered signals are then used for analog line-to-line voltage signals as well as cell state sensor inputs.

3-3.9.6. Line-to-line voltage signal conditioners The buffered phase to neutral converter voltage signals are differentially combined to recreate line-to-line voltages. These are used after further processing (on NLIB) as information for the software control algorithms.

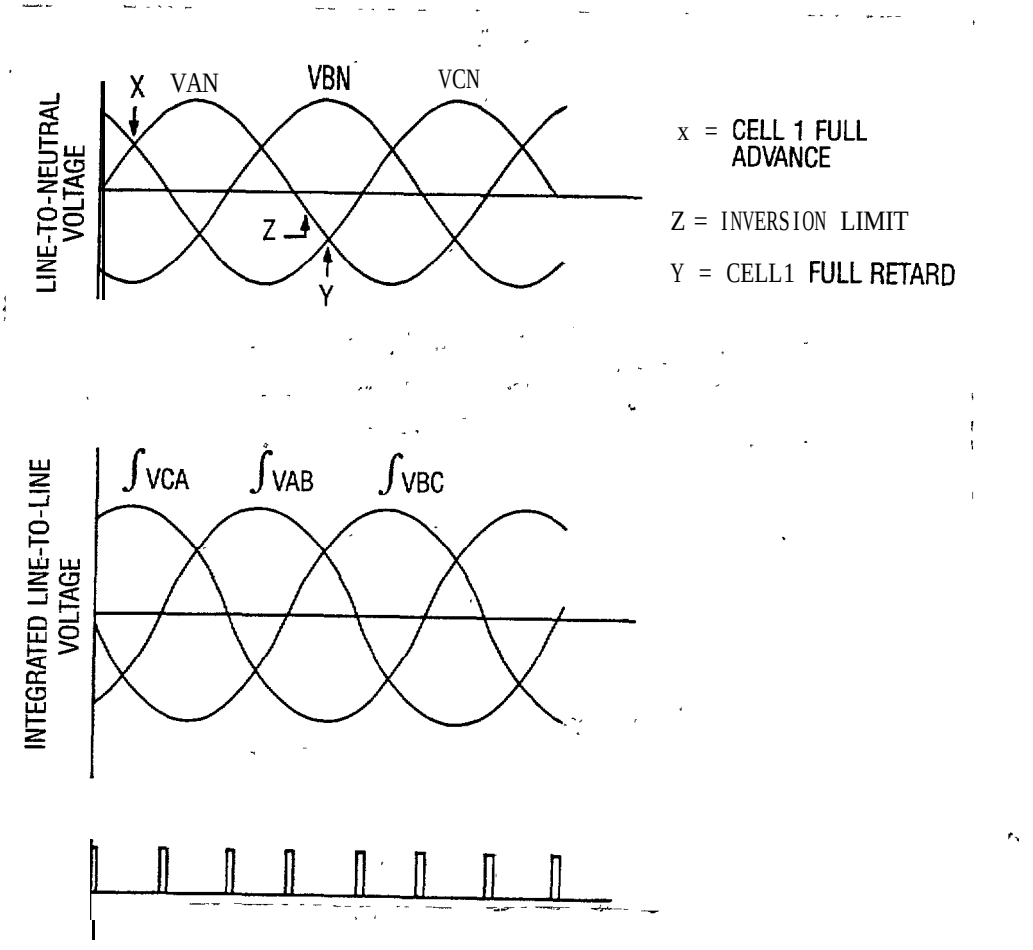


Figure 3-5. Flux Wave Zero-Crossings

Table 3-4. Flux Integrators - Jumpers, Adjustments, LEDs and Testpoints

Nomen.	Function	Position
	BERG JUMPERS	
BJ1-BJ3	Selects voltage integrator scaling for 60Hz or 120Hz	60Hz or 120Hz (var)
	Adjustments	
FAB, FBC, FCA	Scale flux integrators which calculate flux. Scaling is 4.75 volts RMS corresponds to 1 pu.	
o v	Set overvoltage protection trip point.	
u v	Set undervoltage protection trip point.	
SUPP	Set trip point for overcurrent backup fault.	
COMP	Sets field loss level on source side and field overcurrent level of load side.	
VDC	Not used.	
	LED's	
OV	On when overvoltage fault latch set.	
SUPP	On when overcurrent fault latch set.	
IMOK	On if microprocessor based self test is passed and no saturation condition are present.	
	Testpoints	
OVR	Analog D-C voltage proportional to setpoint of over-voltage fault latch.	
UVR	Analog D-C voltage proportional to setpoint of source undervoltage sensor.	
SUPP	Analog D-C voltage proportional to setpoint of overcurrent fault latch.	
FAB,FAC FCA	Analog voltage proportional to motor/line flux reconstructed. See fig. 3-6 for typical wave forms	
FBAR, FCBR, FACR	Analog voltage proportional to motor/line flux reconstructed for IX drop. See fig. 3-6 for typical wave form.	
OIA,OIB OIC	Analog waveform representative of inverted line current. See fig. 3-6 for typical waveforms.	

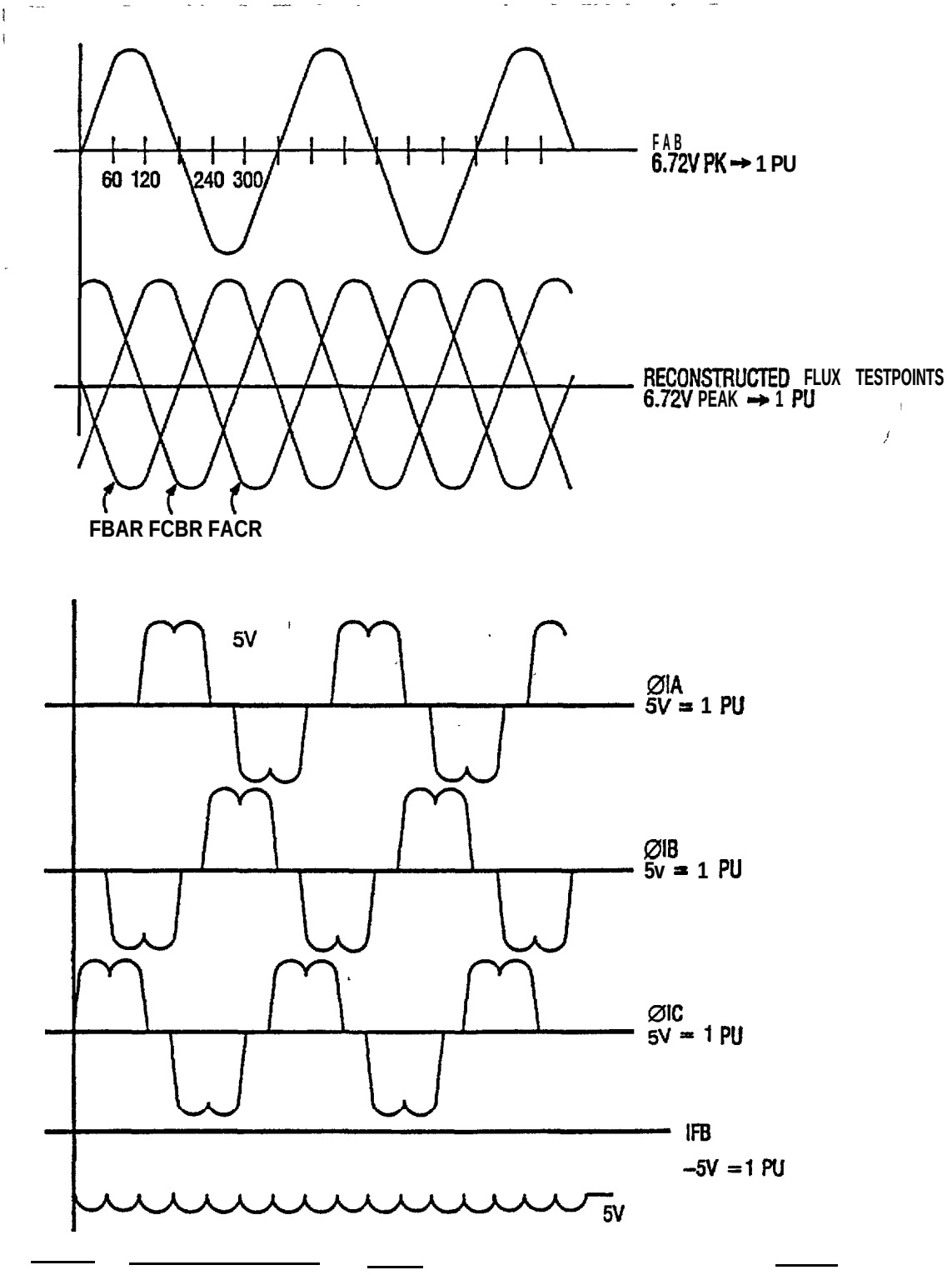


Figure 3-6. Analog Typical Waveforms

3-3.9.7. Cell state sensors . From a cursory check of the standard three phase double way bridge configuration, it can be seen that the voltage across each cell is a combination of one phase voltage and one dc bus voltage. Since each of these voltages is available from an attenuator, the voltage across each cell or "on state" maybe uniquely determined. Circuitry is provided for reading the "state" of all cells at a particular instant and for making the information available to the microprocessor via the L bus.

3-3.9.8. D-C Link volts. The positive and negative dc link bus voltages are combined differentially to obtain a dc link volts signal suitable for driving a remote analog meter with approximately 10 Vdc maximum scale (See Table 3-5).Figure 3-8 – Gating Reference - Berg Jumpers, Adjustments, LEDs and Testpoints

3-3.10. Field Excitation Control Board = DS3800NSFE

The NSFE's primary task is to convert information received from the microprocessor based control to cell pair and timing for firing 6 thyristors configured in a 3 phase switch configuration. It also provides analog interface and diagnostic interface for the external power converter.

1. R-bus interface Firing mask latch Cell state sensors Attenuator signal conditioning Pulse train generation Gate drivers. Current feedback conditioning Current unbalance detector

3-3.10-i. R Bus interface. Control logic is provided for decoding read and write strobes. Eight datalines (either upper or lower byte) are connected. All information exchange with host processor takes place through the R-bus.

3-3.10.2. Firing mask latch. The particular cell pair to be fired is selected in software and communicated to the NSFE over the R-bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed from the present time. This information is routed to a timer. At the end of the delay

Table 3-5. Gating Reference - Berg Jumpers, Adjustments, LEDs and Testpoints

Nomen.	Function	Position
	Berg jumpers	
J1	Determines if new mask should be clocked into final latch at termination of count down timer.	NINT(1d) INT(src)
	Adjustments	
VDC	Sets remote meter full scale to corresponding D-C link volts.	
	LED's	
IMOK	Board hardware status indication. On normally. Off if self diagnostics determines a board hardware failure.	
	Testpoints	
SIG l-SIG 6	Gate pulse command. "0" true. Steady state length of 120 degrees relative to operating frequency. Each gate pulse is shifted 60 degrees relative to lower numbered ones. SIG numbers correspond to thyristor leg being gated.	
VDC	Output of D-C link voltage meter driver.	
VAC,VCB	Analog output proportional to motor/source line-to-line voltage.	
ACOM	Power supply analog common.	

an interrupt is generated and the cell mask is clocked into the final latch to gate the selected cell pair.

3-3.10.3. Gate drivers . Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 Vdc levels.

3-3.10.4. Current feedback conditioning . A differential amplifier is used to receive the field current signal from the field exciter module (remote from SEM). The scaling is arranged for 2V corresponds to 1 per unit field on the input and +5.0 volts equals 1 per unit at EI. An additional signal (EIMTR) is also available for meter driving with a pot adjustable gain range of 2:1. To detect fault conditions within the motor exciter, the field current is applied to a filter tuned tothe excitation fundamental frequency. It can be shown that conditions of unbalance will cause a dramatic increase in current feedback's harmonics. The condition of the sensor is available to the system processor through the data bus. (See Table 3-6)

3-3.11. Analog LCI Board = DS3800NLCB

The NLCB provides various miscellaneous signal conditioning and test functions required for the structured LCI. These features include:

1. Speed feedback generation from motor voltage zero crossing signals with overspeed protection and indication
2. External reference signal conditioning
3. Test reference signal conditioning
4. Torque reference signal conditioning
5. Selftest features
6. 4-20 MA signal drivers
7. Ground fault detector
8. Power applied reset
9. Fault relay drivers
10. Horsepower meter driver
11. Continuity detection circuits
12. Test switch
13. Restart upon power loss

Table 3-6. Field Excitation - Adjustments and Testpoints (NSFE)

Nomen.	Function
	BERG JUMPERS
None Supplied	
	Adjustments
EIMTR	Sets full scale reading on external field current meter (if supplied).
	Testpoints
CIP	Cable in place. When the cable from the NSFE board front is properly installed in the external module the voltage at CIP will be less than .5v. If the cable is not in place, the voltage will be 15v.
Vabs	Signal proportional to the absolute value of exicter output voltage. Always positive and scaled for approximately 10 volts at 1 per unit output.
LVAC	Signal showing the exciter module input waveform on lines A - C. The scaling is approximately 4.5 volts peak corresponds to peak line volts.
FUN	Output of fundamental bandpass filter monitoring field current Will be near zero volts under normal conditions
EIMTR	Output of field current meter driver. Range 0 - 10 volts.
IFB	Field current signal with 5 volts corresponding to 1 per unit.
S1 -S6	Gating signals to external thyristors. Nomenclature corresponds to the firmg order. Each signal is at a 28 volt level and is 0 when true.
ACOM	Analog common.
CONT	Continuity. Signal for checking continuity of ribbon cable to external excitation module.

42

3-3.11 .1. Speed feedback. A pulse rate signal proportional to drive speed is obtained from the zero crossing detection circuitry on the HLCB. A pulse rate to analog converter then produces a corresponding dc level proportional to speed suitable for driving a meter and protective functions.

The overspeed function is implemented by comparing an overspeed setting reference to the analog speed voltage attained above. A latch insures that transient conditions will be sealed. A red LED indicates the overspeed condition until reset.

A differential amplifier committed to accepting the remote speed reference is supplied. Berg jumpers are)used to configure the circuit to accept a 0 to 10 volt signal or a 4 -20 MA signal. In either case, the output voltage is transmitted to channel of the HAIA analog to digital converter for reading by the system processor.

3-3.11.2. Test speed reference.. A differential amplifier is included to condition a voltage source signal from a potentiometer located onthe I/O module. This is also known as the local speed reference.

3-3.11.3. Torque reference signal conditioning . A differential amplifier for receiving the master torque reference from the other six pulse drive (in a 12 pulse system) is also supplied.

3-3.11.4. Selftest features . An array of static switches and voltage sources is included for use in testing the analog circuitry of this and other boards in the module during the module selftest mode, A sinewave oscillator is used to simulate line and motor voltage during "gate test mode". This is to allow the power converter electronics to be tested without the necessity of applying high voltage to the associated bridges.

3-3.11.5. 4 • 20 MA signal drivers . Two voltage to current converters provide capability of remote driving system information to customer interfaces. The signals selected for transmission are motor current and motor speed. In each case, 4 ma corresponds to 0.

3-3.11.6. Ground fault detector . The ground fault detector sums line-to-neutral voltages Va, Vb, and Vc. It operates on the principal that when all impedances from phase to neutral are balanced that the summation of the phase-to-neutral voltages will be zero. Any unbalanced impedance will cause an error to be detected and a ground fault noted by the system processor.

3-3.11.7. Power applied reset . A power applied reset is generated here and distributed throughout the electronics module. It is used to prevent false operations of relay drivers and fault latches during the process of powering up the module.

3-3.11.8. Fault relay drivers . Relay driver no. 1 is generally used as a method of tripping the source side drive switchgear from both hardware and software sensed fault conditions. It is inhibited from false operations during the power up phase by the power applied reset.

Relay driver no. 2 is generally used as a method of tripping the motor side isolator (if used) from both hardware and software sensed fault conditions. It is also inhibited from false operations during the power up phase.

3-3.11.9. Horsepower meter driver . An adjustable meter driver for powering a door mounted meter indicating horsepower is also included. The horsepower signal is calculated in software and converted to analog on the HLCB board.

3-3.11 .10. Continuity detection circuits . Four circuits are included to monitor the continuity of various ribbon cables connected to the electronics module when it is integrated in to a system control panel These signals are only monitored during the panel selftest. An appropriate diagnostic message is printed out if a cable is not in the correct location.

3-3.11.11. Test switch . The test switch on the board front edge is used to initiate the panel selftest mode. When the switch is operated, then the electronics module will begin to execute a selftest routine which extends to other components of the control panel. Upon completion of the selftest, the control will enter gate-test-mode using the three phase test oscillator as a source of synchronous waveforms.

3-3.11.12. Restart upon power loss . If the drive has been previously running and the control power is lost for a time (determined by setting of RTD potentiometer), then the drive will be automatically restarted when power returns by a pulse which appears at signal 1RSRT. Note this is an option and the I/O module must be properly configured. (See Table 3-7)

3-4. Relay driver board = DS3800HRDA

The HRDA is relay driver board with 16 open-collector outputs. Output voltages can range from 5 to 50 volts. In the LCI system, it is applied at a nominal voltage of 28 volt dc. The board may be connected to the computer R-bus for accessibility by the system control.

The relay driver board is used principally as an interface between the system software control and the power devices located in the I/O module. (See Table 3-8)

Table 3-7. Analog LCI Board - Berg Jumpers, Adjustments, Testpoints and LEDs (NLCB)

Nomen.	Function	Position
	BERG JUMPERS	
LOAD	Selects remote speed reference for either current input or voltage input. Should be coordinated with GAIN.	variable
GAIN	Selects gain necessary for remote speed when used as current input or voltage input.	variable
	Adjustments	
SPD	Adjusts analog speed feedback circuit for 10 volts at 60 Hz.	
SPM	Adjusts door mounted speed meter for proper fullscale reading	
OSP	Hardware overspeed reference.	
RTD	Adjusts allowable time for auto restart after loss of control power.	
	Testpoints	
SPD	Output of analog speed feedback calculation	
OSP	Reference to hardware overspeed detection circuit. Scaled same as speed i.e. 10 volts = 60Hz.	
IOA	Output of channel A current source.	
IOB	Output of channel B current source.	
REF	Output of remote speed reference. With a voltage input will be 0 to 10 volts. With a current source input, will be 2 to 10 volts.	
TREF	Output of local test reference. Varies from 0 to 8.3 volts.	
12P	Output of torque reference obtained from other channel.	
NTL	Neutral. Summation of line to neutral voltages.Normally near 0 volts.	
RTD	Run time delay. Sets allowable time for loss of control power while still allowing an auto restart. Scaled 0 to 10 volts.	
PHA	Test oscillator, phase A. Shows a 55Hz. sine wave.	
PHB	Test oscillator, phase B. Shows a 55Hz. sine wave,lagging phase "A" by 120 degrees.	
	LED's	
OSP	Red LED indicating that the hardware overspeed latch is set.	
RSRT	Green LED which is pulsed when the auto restart circuit is activated.	
FT1	Green LED Indicating the status of fault relay driver No. 1. Will be on when the relay driver is energized.	
FT2	Green LED indicating the status of fault relay driver No. 2 Will be on when the relay driver is energized.	
TEST	Green LED which luminous when the test switches closed and the system software indicates gate test mode.	
IMOK	Yellow indicating light will be on normally. An off condition indicates a possible hardware problem with the board. It can be turned off either locally or from the processor as a function of selftest.	

3-5. Input signal conditioning = DS3800HISA

HISA is designed to work in conjunction with HRIA (in I37/O module) to condition thirty high level (105Vdc) signals to logic levels and allow access to them through the processor R-bus. Each signal is separately isolated by a photocoupler and indicated by a bar LED. Two relay drivers are also included which are only utilized in the hardware selftest of the I/O module. (See Table 3-9)

Table 3-8. Relay Driver - Testpoints and LEDs

Nomen.	Function
	BERG JUMPER
None supplied	
	ADJUSTMENTS
None supplied	
	TESTPOINT
TP1	Reset. Connecting testpoint 1 to testpoint 2 resets only this board. This function is not useful in the LCI system.
TP2	DCOM
	LEDs
O - F	A red LED for each of the 16 relay drivers indicates the energized status of each circuit.
ADR	Yellow LED. Turns on when the board is addressed; stays on if the board is addressed at least every 200 microseconds.
WPOK	Parity check. Not used in this system will always be on.

Table 3-9. Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LEDs

Nomen.	Function
	BERG JUMPERS
None Supplied	
	ADJUSTMENTS
None supplied	
	TESTPOINTS
None Supplied	
	LEDs
I0O - I1D	Green bar LEDs. Each indicates the status of the associated input, luminous when the input is energized.
IMOK	Yellow LED. On during normal operation. will be off only as the result of module selftest.

3-6. Display Module (LDM)

The DS3820LDM is the display module for the structured LCI control panel. The display provides fault, alarm and diagnostic information. The display module comes in two basic layouts. Layout Option 1 provides the following features:

- 3 indicator lights (RUN, ALARM and FAULT)
- 2 push buttons (RESET AND DRIVE STATUS)
- 3 1/2 digit motor speed meter (RPM)

-24 column printer with real-time clock

and Layout Option 2 provides the following features:

- 4 push-to-test indicator lights (RUN, ALARM, FAULT and READY)
- 2 push buttons (RESET and DRIVE STATUS)
- 4 DB40 type meters (motor speed, motor current, motor voltage and horsepower)

All lights and push buttons are on a 28v circuit. The functions and color codes for each light are:

Red for the motor RUN light

Red for a system FAULT light

Amber for a system ALARM light

Green for the system READY light

The push buttons are connected to a 28V circuit and direct a 28V signal to the output when the button is depressed. The RESET push button is used to reset latched faults in either software or hardware (NLIB and NLCB latched faults - overspeed, suppression, overvoltage, etc.). The DRIVE STATUS push button; when depressed, provides information on the motor voltage, current, speed, etc. via the printer.

The digital meter is a dc voltage measurement panel type. The conversion factor for the meter is 2 volts correspond to 1999 cnts. The panel meter is powered by the 5 volt bus via the 26-pin ribbon cable.

The four DB40 type analog meters are 10 volt movement meters with +/- 10% adjustment rheostats (back).

The printer included in the display module is a Digitec Model 6610 Panel mount printer. The 24 column printer inputs data and control signals via a serial RS232 connection and uses an impact print head mechanism to output the data. The printer also supplies a real-time clock that will be used to log the time of status information and system failures.

The LDM interfaces to the LCG thru a 26 pin ribbon cable JK.

3-7. Signal Isolation Module (SIM)

The DS3820SIM is used to provide optional functions to an LCI control panel. In particular, it provides four impedance isolated signal outputs for use in a purchaser's external control system. The four signals may be selected as 0 to 10 volts or 4 to 20ma outputs by header connection jumpers at JM. Each circuit may be selected independantly. Inputs to the module (TEM) are received from the LCI control SEM via 26-pin ribbon cable (JM). The four inputs to the isolation circuits may be chosen by header-connector jumpers at JB from the following list of signals: Source volts, source current, SPD ref., motor SPD, horsepower, DA1, DA2, exciter current, motor volts.

A DS3800NVAA provides the functions of voltage to current source transformation and isolation. A differential amplifier accepts the signal from the remote control SEM. The signal is isolated from the NVAA on-board ground by approximately 100 K ohms of resistance. The isolated signal is then applied as a reference to a current source with the gains weighted such that a 0 to 10 volt signal provides a delta output of16 milliamperes. An additonal bias is provided ot give the 4 milliamperes at 0 volts input. A chopper isolated power supply is present on the NVAA board to power the output stages of the signal isolators. The output of each of the four circuits is referenced to XCOM, the on-board power supply common. It is intended that the XCOM connection be tied to the customer's external circuit common. While it is not necessary for the signals used as inputs to the isolators be referenced to the customer's common or XCOM, the difference between ACOM and XCOM must not exceed 3 volts steady-state or transiently. The voltage developed across any external impedance in series with a current source must a not exceed 10 volts.

The output signals are also available as voltage followers (0 to 10V). Maximum output load is 1 ma.

An operational amplifier monitor (or saturation detector) is provided for monitoring the output stages of the current driver circuits an LED, "IMOK", is luminous as long as no fault condition exists. In additon, if FUSE 1 in the on-board power supply is blown, then the LED will be non-luminous. (See Table 4-O)

3-8. I/O Module (IOM)

The purpose fo the DS3820IOMA (I/O) module is to provide signal conditioning and connection for interpanel and purchaser wiring in a load commutated inverter (LCI) control panel. The following functions are provided.

1. Signal conditioning and isolation for interface wiring at the 105Vdc level.
2. Isolated signal interlocks (form C) to remote locations as a function of programmed system requirements. Rated .25A, 115Vac (Resistive).
3. Fused 105Vdc power supply to provide a source of high level power for interface.
4. Power relay control for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10A, 600Vdc (Inductive).
5. Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).
6. Termianl boards for terminating interconnecting wiring.
7. Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits aand feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

Power relay control for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10A, 600Vdc (Inductive).

Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).

Termianl boards for terminating interconnecting wiring.

Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits aand feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

Table 4-O. Signal Isolation Module (SIM)

Nomen.	Function
	TESTPOINTS
TP1	Voltage output of channel 1 relative to XCOM.
TP2	Voltage output of channel 2 relative to XCOM.
TP3	-2.5 Vdc bias for V-l circuits.
TP4	Voltage output of circuit 3 relative to XCOM.
TP5	-2.5 Vdc bias voltage for I-V circuit.
TP6	XCOM, isolated supply ground.
TP7	Voltage output of channel 4 relative to XCOM.
TP8	Voltage output of 4-20ma to voltage converter.
TP9	N15X, isolated supply.
TP10	P15X, isolated supply.
TP11	ACOM, analog power supply ground.

CHAPTER 4
FIELD ADJUSTABLE PARAMETERS AND PUBLIC VARIABLE PARAMETERS

Section A - EEPROM (Field) Variables

EEPROM Memory is a non-volatile memory space used to hold critical drive constants that may be system dependant. The use of EEPROM Memory allows the drive constants to be changed in the field in order to fine tune the system performance. EEPROM Memory space resides on the DMPB card (HMPG daughter board) and is configured to contain 1024 bytes (1K x 8).

Default values for the drive constants are burned into the ROMs residing on the HMPG (the ROMs also contain the executable software to run the drive). These default values are intended to provide a reasonable starting base to from when tuning the drive. They may be loaded into the EEPROM Memory by turning the DMPB write enable switch "on", jumpering testpoint TINO on the HLCB pwb to DCOM and then executing a hardware SEM reset. In order to change a value in the EEPTOM memory, turn the DMPB write enable switch "on" and use the monitor modify command.

After power up or hardware reset, the microprocessor on the HMPG makes a local copy (shadow) in RAM of the EEPROM Memory. During normal running, the microprocessor accesses the drive contants from this local RAM shadow.

Table 4-1. Description of Variables

Name	Location	Function
CRALPHl	3000:20 (CR)	Variable used in current regulator to improve transient response during segment fire mode. The variable sets the initial value of LINE COM (firing angle) such that the desired value of current is obtained after the first firing. The following value is recommended. CRALPHl = 90 degrees Resolution is 1 degree and the range is 60 to 90 degrees. The associated software variable and its relationship to CRALPHl is given below. CUR REG_ FIXED- ALPH1 = 16384 * COS(CRALPH1)
CRALPH2	3000:22 (CR)	Variable used in current regulator to improve transient response during segment fire mode. The action of the variable is to precondition the regulator integrator to command a firing angle necessary to obtain forced commutation current limit value. CRALPH2 = ARCCOS[(4NE+ IR) + (.075FPU*VM) + (XTpu*Ipu/2)]/(1.35*Ell) E = cell drop (1.25V) N = number of cells/leg FPU = per unit field at .O75 per unit speed I = current(amps) corresponding to current limit value Ipu = value of&current in pu R = system resistance (mostly reactor) VM = motor L-L volts (RMS) Ell = transformer sec volts (rms) XTpu = transformer comm reactance Resolution is 1 degree and range is 60 to 90 degrees. The associated software variable and its relation below. CUR REG FIXED ALPH2 = 16384 * COS(CRALPH2)

Table 4-l. Description of Variables

Name	Location	Function
CRGAIN	3000:24 (CR)	Source side current regulator gain. The variable's relationship to system parameters is given below. CRGAIN = (759*IM*WC*L)/(VL-L) IM = fund rms rated motor current V_{L-L} = rated rms L-L source volts WC = desired regulator crossover (250RAD) L = inductance in conducting path = reactor + 2 (motor comm. reactance) + 2 (transf. comm. reactance). F = source side frequency Resolution is units and range is 10 to 1000. The associated software variable and its relationship to CRGAIN is given below. CUR REG G2 = 16*CRGAIN/(6*FRQS)
CRMIN	3000:28 (CR)	Variable setting the minimum value of dc link current. Normally set to maintain continuous current. The link reactor is usually picked to maintain continuous current at .2 per unit. CRMIN = per unit current Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to CRMIN is given below. CUR REG MIN LMT = CRMIN * 1024
CRSTART	3000:2A (CR)	Value of current used in fixed fire forbreaking away the motor. Also the current command in speed regulator open test mode. The value of CRSTART should be that which would give maximum torque at the stand still field excitation. CRSTART = PU FLD (AT STAND STILL) Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to CRSTART is given below. CUR_ REG_REF_ START = CRSTART * 1024
CRTC	3000:26 (CR)	Time constant used to stabilize the source side current regulator. CRTC = time constant (in seconds) Resolution is .001 seconds and the range is .001 to .500 seconds. The associated software variable and its relationship to CRTC is given below. CUR_ REG_ G1 = 4*CRGAIN*CRTC
FANGLAG	3000:E8 (FC)	Variable setting the number of the field voltage degrees leads or variable setting the number of degrees the field voltage leads (-) or lags (+) the source voltage. Resolution is 1 degree and range is ± 30°. Associated software variables and its relationship to FANGLAG is given below. Field, Phase, zero, E = (FANGLAG/360) *1024

Table 4-l. Description of Variables - Continued

Name	Location	Function
FCDELAY	3000:32 (FC)	Timer controlling the delay to allow field to build up before attempting a drive start FCDELAY = delay time in seconds Resolution is .1 and the range is .1 to 10 seconds. The associated software variable and its relationship to FCDELAY given below. FLD_ DELAY- LIMIT = FCDELAY * FRQS
FCFTIME	3000:34 (FC)	Controls number of consecutive field overcurrents allowed. A penalty function is implemented such that 6 good firings following an overcurrent will cancel an overcurrent firing. This allows recovery from non-repetitive faults. FCFTIME = number of consecutive overcurrents. The resolution is units and the range is 1 to 6.The associated software variable and its relationship to FCFTIME is given below. FLD_ FAULT, PERIOD = (FCFTIME - 1) * 6
FCGAIN	3000:2C (FC)	Field current limit function gain. The gainis set to allow a 2% field current limit with 30% overdrivmg flux. A value of 100% is typical. FCGAIN = required regulator gain Resolution is 1 and range is 0 to 200 The associated software variable and its relationship to FCGAIN is given below. FLD_CUR_GAIN = FCGAIN
FCLIM	3000:2E (FC)	Set the trip level for the field overcurrent. The spillover field current limit value is set at 94% of this value. FCLIM = desired per unit field where 1 per unit field is on the base of the field exciter. Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to FCLIM is given below. FLD_CUR_LMT = FCLIM * 1024
FCLOSS	3000:36 (FC)	Field loss setting for field loss fault function. FCLOSS = desired per unit field loss setting Note that one per unit field is on the base of the field exciter rating. Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to FCLOSS is given below. FIELD_LOSS_SETTING = FCLOSS * 1024

Table 4-l. Description of Variables - Continued

Name	Location	Function
FCMAX	3000:E2 (FC)	Field thermal overload function. Scales minimum field thermal rating at zero speed. Must be entered on the base of the field exciter. Resolution is .01 and the range is .05 to 1.2. The associated software variable and its relationship to FCMIN is given below. MAX_FLD_LIM_E = FCMAX * 1024
FCMTN	3000:E4 (FC)	Field thermal overload function. Scales minimum field thermal rating at zero speed. Must be entered on the base of the field exciter. Resolution is .01 and the range is .05 to 1.2. The associated software variable and its relationship to the FCMIN is given below. MAX_FLD_LIM_E = FCMIN * 1024
FCOLEN	3000:174 (FC)	Field thermal overload function. Used to enable or disable the field thermal overload function. Possible values for FCOLEN are "T"rue or "F"alse.
FCSLOP	3000:E6 (FC)	Variable used in field thermal overload function to define cooling effect as a function of speed. FCSLOP = DELTA PU field current. Note the per unit base is the base rating of the field exciter. Resoluton is .01 and the range is .01 to 1.2 per unit. The associated software variable and its relationship to FCSLOP is given below. FLD_SPD_FACTOR_ E = 16 * [FRQM/60]/FCSLOP
FCSTART	3000:30 (FC)	Sets minimum field current for recovering from field overcurrent during forced commutation. Must be coordinated with value obtained from field precondition. FCSTART = per unit field current where 1 per unit field is on the base of the field exciter. Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to FCSTART is given below. FLD_CUR_STRT_LMT = FCSTART * 1024
FLAGAIN	3000:42 (FL)	Variable used with the flux regulator routine to set gain of flux adjust Integrator to allow source tire-as-early-as-possible. FLAGAIN = required gain The recommended value is 1. The range is 0.01 to 2 with a resolution of .01 The associated software variable and its relationship to FLAGAIN is given below. FLUX_REG_ADJ_G = FLAGAIN * 100

Table 4-l. Description of Variables • Continued

Name	Location	Function
FLALEN	3000:184 (FL)	Variable used to control enabling/disabling of the flux adjust integrator to allow fire as early as possible on the source side. FLALEN — REGULATOR CONDITION F — disabled T — enabled Only values of F or T are allowed. The associated software variable and its relationship to FLALEN is given below. FLUX_ADJ_MODE = FLASEN or SHL(FLALEN)
FLALIM	3000:44 (FL)	Variable used with the flux regulator controlled from source fire as early as possible regulator. function is to limit the range of flux adjustment. FLALIM = desired per unit flux Resolution is .01 units and the range is 0 to .50 The associated software variable and its relationship to FLALIM is given below FLUX_REG_ADJ_LMT = FLALIM * 8254
FLASEN	3000:180 (FL)	Variable used to control enabling/disabling ofthe flux adjust as a function of source voltage variations. FLASEN — REGULATOR CONDITION F — disabled T — enabled Only values of F or T are allowed. The associated software variable and its relationship to FLASEN is grven below. FLUX_ADJ_MODE_SRC = FLASEN or SHL(FLALEN)
FLDPU	3000:F0 (PT)	Variable used in the "print status" routine to define the value of 1 per unit exciter current. The value should be entered in terms of actual exciter current rating (in amps). FLDPU = rated exciter current Resolution is 1.0 and the range is 0 to 100.
FLEARLY	3000:46 (FL)	Enables source fire-as-early-as-possible regulator as a function of drive speed. FLEARLY = desired per unit speed Resolution is .001 units range is .1 to 2 per unit speed. The associated software variable and its relation ship to FLEARLY is given below. SFAEAPR_SPD_ENABLE = FLEARLY* (1024/60)*FRQM

Table 4-l. Description of Variables - Continued

Name	Location	Function
FLPBRKl	3000:3E (FL)	Variable used in "FLUX SETPOINT VS SPEED"function in flux regulator. Sets beginning of first slope. See variable FLPMAXl for profile. FLPBRKl = desired per unit speed Resolution is .001 and the range is 0 to 2.00 per unit. The associated software variable and its relationship to FLPBRKl is given below. FLUX_PROFILE_SPD_BRK1 = FLPBRK1*(1024/60)*FRQM
FLPBRK2	3000:40 (FL)	Variable used in "FLUX SETPOINT VS SPEED"function in flux regulator. Sets takeover for second slope See variable FLPMAXI for profile. FLPBRK2 = desired per unit speed Resolution is .001 and the range is 0 to 2.00 per unit The associated software variable and its relationship to FLPBRK2 is given below. FLUX_PROFILE_SPD_BRK2 = FLPBRK2*(1024/60)*FRQM
FLPLIM	3000:38 (FL)	Variable specifying flux level at rated speed. See variable FLPMAXl for profile. FLPLIM = desired per unit flux Resolution is .01 units and range is 0 to 1.5 The associated software variable and its relationship to FLPLIM is given below FLUX _PROFILE_FINAL = FLPLIM * 8254
FLPMAXl	3000:3A (FL)	Variable used m "FLUX SETPOINT VS SPEED "function in flux regulator. Sets initial per unit flux. FLPMAXl = desired per unit flux (Refer to Flux Profile Drawingin System Elementary Sh. 1EA) Resolution is .01 units and the range is 0 to 1.50 The associated software variable and its relationship to FLPMAXl is given below. FLUX_ PROFILE, MAX1 = FLPMAXI * 8254
FLPMAX2	3000:3C (FL)	Variable used in"FLUX SETPOINT VS SPEED "function in flux regulator. Sets peak flux. See variable, FLPMAXl for profile. FLPMAX2 = desired per unit flux Resolution is .01 units and the range is 0 to 1.50 The associated software variable and its relation ship to FLPMAXl is given below.FLUX_PROFILE_MAX2 = FLPMAXl * 8254

Table 4-l. Description of Variables - Continued

Name	Location	Function
FLUXS	3000:102 (MI)	Used in the starting routine from zero speed. Sets the amount of flux required to transition to segment firing mode. FLUXS = desired per unit flux Resolution is .01 and the range is 0 to 2.0 PU. The associated software variable and its relationship to FLUXS is given below. START_FLUX_E = FLUXS * 2752
FRCLMP	3000:4C (FR)	Sets the maximum advance angle of the exciter voltage controller. Generally will be set to 1 per unit exciter volts. When set to some other value the maximum RMS exciter voltage will be approximately proportional to the value of this variable. FRCLMP = desired per unit output volts Resolution is .01 per unit and the range is 0 to 1.2. The associated software variable and its relation ship to FRCLMP is given below. FLUX_ REG_OUT_CLMP = FRCLMP * 255
FRGAIN	3000:48 (FR)	Variable used in the flux regulator to set the regulator integral gain for a nominal 5 radian regulator. The followmg equation holds. FRGAIN = 1/(63.68 * Geq * MEV) where Geq = motor equivalent transfer function from brushless exciter input flux. MEV = maximum exciter volts Resolution is .01 and the range is .01 to 2. The associated software variable and its relationship to FRGAIN is given below. FLUX_REG_G2 = FRGAIN * 16384/ FRQSl
FRQS	3000:60 (MI)	Source side nominal frequency. The variable is used in vanous regulator gain and time constant calculations. This variable only goes into effect after resetting the processor. Resolution is 1 Hz and range is 45 to 90 Hz.
FRQM	3000:62 (MI)	Load *side* nominal frequency. The variable isused in various regulator gain and time constant calculations. This variable will only affect the drive tune-up after resetting the processor. Resolution is 1 Hz and range is 40 to 90 Hz.
FRSTART	3000:4E (FR)	Variable used in the flux regulator used to initialize the flux regulator integrator to call for a value of exciter volts which gives approximately the correct flux value. Also sets the value of flux in forced commutation when the flux regulator is not active. FRSTART = desired per umt output volts Resolution is .01 per unit and the range is 0 to 1.0. The associated software variable and its relation ship to FRSTART is given below. FLUX _REG_PRECONDITION = (FRSTART * 510) - 2.55

Table 4-l. Description of Variables - Continued

Name	Location	Function
FRTC	3000:4A (FR)	Variable selecting the flux regulator stability time constant. Normally the value is selected equal to the motor field time constant. FRTC = desired time constant (seconds) Resolution is .1 and the range is .2 to 10 seconds. The associated software variable and its relation ship to FRTC is given below. FLUX_REG_G1 = 2048 * FRGAIN * FRTC
HPPU	3000:F2 (PT)	Variable used in the "print status" routine to define the value of 1 per unit drive horsepower. Generally, the per channel horsepower is used as the base in 12 pulse systems. HPPU = rated channel horsepower. Resolution is 1.0 and the range is 0 to 10,000.
IDCPU	3000:EE (PT)	Variable used in th "print status" routine to define the value of 1 per unit link current in amperes. 1DCPU = rated link current (amperes) Resolution is 1.0 and the range is 0 to 2,000.
INTOSTOL	3000:5A (MI)	Variable used in test to determine that flux integrators are balanced before enabling the load side phase- lock- loop. INTOSTOL = desired setting (in per unit flux) Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to INTOSTOL is given below. INTEGRATOR OFFSET TOL = INTOSTOL * 1376
LIMARG	3000:54 (MI)	Variable scaling load-side commutating angle to motor commutating impedance. Used to compensate margin angle for commutating volt-seconds. LIMARG = per unit commutating reactance Resolution is .01 per unit and the range is .03 to .30 per unit. The associated software variable and its relationship to LIMARG is given below. ALPH_LD_IMP_MARG = 180.15 * LIMARG
LKLIMD	3000:6C (PR)	Sets allowable difference between source and load side measurements of link current, If the difference is exceeded then Commutation failure fault is declared. LKLIMD = per unit current difference allowed Resolution is .01 per unit and the range is 0 to 2.0 per umt. The associated software variable and its relationship to LKLIMD is given below. IDC_DIFFERENCE_LIMIT = LDLIMD * 1024

Table 4-l. Description of Variables - Continued

Name	Location	Function
LKLIML	3000:70 (PR)	Sets level of load current which is declared an overcurrent fault. LKLIML = desired per unit current setting Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to LKLIMS is given below. IDC_LINK_LMT_LD = LKLIML * 1024
LKLIMS	3000:6E (PR)	Sets level of source current which is declared an overcurrent fault. LKLIMS = desired per unit current setting Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to LKLIMS is given below. IDC_LINK, LMT_SRC = LKLIMS * 1024
LKOCTIM	3000:72	Controls number of consecutive source overcurrents allowed. A penalty function is implemented such that 6 good firings following a failure will cancel a fault firing. This allows recovery from non-repetitive faults. LKOCTIM = number of consecutive overcurrents. The resolution is units and the range is 1 to 6. The associated software variable and its relationship to LKOCTIM is given below. IDC_LINK_LMT_PERIOD = (LKOCTIM - 1) * 6
LMARG	300050 (MI)	Variable used to control the fixed margin angle for firing the load side. A fixed margin angle of 15 degrees at zero load is normally used. LMARG = desired margin angle m degrees Resolution is 1 degree and the range is 0 to 30 degrees. The associated software variable and its relationship to LMARG is given below. ALPH_FIXED_MARG_LD = 1376*[1 + COS(180-LMARG)]
LPLLTIM	3000.78 (PR)	Sets time delay before a load phase lock loopfault is declared following an error determination LPLLTIM = required time in seconds Resolution is .01 seconds and the range is 0 to 1.5 seconds. The time above will be at rated motor frequency with the time delay inversely proportional to speed otherwise. The associated software variable and its relationship to LPPTIM PLL_OUT_OF_TOL_LD_PERIOD = LPLLTIM * 12 * FRQM

Table 4-l. Description of Variables - Continued

Name	Location	Function
MRPM	3000:D6 (PT)	Variable used in the "print status" routine to define the value of 1 per unit motor RPM. MRPM = rated motor RPM Resolution is 1.0 and the range is 0 to 5000.
MTRFMIN	3000:5C (MI)	Variable used in test to determine if themagnitude of motor flux is adequate to enable the load side phase lock loop. MTRFMIN = desired setting (in per umt flux) Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to MTRFMIN is given below. MTR_FLUX_MIN_TOL = 2*1376*MTRFMIN
MVPU	3000:5C (PT)	Variable used in the "print status" routine to define the value of 1 per unit motor volts. MVPU = rated motor volts Resolution is 1.0 and the range is 0 to 5000.
OVCLIM	3000:64 (OV)	Variable controlling the maximum current limit called for by the overvoltage regulator. A setting is 1.5 per unit current. OVCLIM = required per unit stator current Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to OVCLIM is given below. CUR_REG_LMT_OVVOLT = OVCLIM * 1024
OVGAIN	3000:66 (OV)	Variable controlling the gain of the overvoltage regulator. Presently the overvoltage regulator is selected for a crossover of 10 radians. This results in OVGAIN = 2.4 Resolution is .1 units and range is 0 to 20 The associated software variable and its relation ship to OVGAIN is given below. OVVOLT_REG_GAIN = 256*OVGAIN
OVSPIL	3000:68 (OV)	Variable controlling the setpoint of thevoltage regulator which is configured in a spillover mode. OVSPIL = per unit motor volts Resolution is .01 per unit and range is 0 to 2.0 per unit. The associated software variable and its relationship to OVSPIL is given below. OVVOLT_ REG,SPLOVR = 1.15 * FRQM * OVSPIL where F = rated motor frequency Vl-lpu = motor voltage (P.U.) where voltage regulator first cuts in.

Table 4-l. Description of Variables - Continued

Name	Location	Function
OVTC	3000:6A (OV)	Variable controlling an adjustable lag in theforward loop of the overvoltage regulator. The filter is in the forward loop of the voltage regulator and therefore, affects stability. The following table shows the values available OVTC T(SECONDS) (60HZ) T(SECONDS) (50HZ) 0 H .01662 .0199 1H .050 .06 2H .1166 .140 3H .245 .30 4 H .5166 .62 5H 1.05 1.26 6 H 2.116 2.54 7H 4.25 5.07 8H 8.52 10.22 9H 17.05 20.46 The recommended value is OVTC = 6. The associated software variable and its relation ship to OVTC is given below. OVVOLT_REG_TC = OVTC
PLLMODE	3000:58 (MI)	Variable used with the load phase-lock-loop regulator to determme the speed at which the phase-lock-loop switches from 6 to 3 calculations per cycle. PLLMODE = desired speed (HZ) Resolution ıs 1 HZ and the range is 1 to 90 HZ. The associated software variable and its relationship to PLLMODE is given below PLL _SPD_MODE_3 _LD = 7200/PLLMODE
PLLTOL	3000:74 (PR)	Sets maximum allowable error in load phase-lock- loop. If error is exceeded then a phase lock loop fault is declared. PLLTOL = allowable error (in degrees) Resolution is units and the range is 0 to 20 degrees. The associated software variable and its relationship to PLLTOL is given below. PLL_ERROR_TOL_LD = PLLTOL * (1024/360)

Table 4-l. Description of Variables - Continued

Name	Location	Function
PO1 PO2 PO3 PO4 PO5 PO6	3000:F6 (PT) F8 FA FC FE 100	Variables used in the print status routine to select the variables from the following list to be printed on the door mounted printer. The number of the information to be printed is entered as data into each of the variables. If no data for a particular entry is required, then it should be set to zero. Note that the print-out will be formatted in the order of the PO-variables with PO1 first. 0 - No print out 1 - Motor RPM 2 - Speed reference (RPM) 3 - Motor line current 4 - Field current 5 - Motor Horsepower 6 - DC link volts 7 - DC link amperes 8 - Motor line volts 9 - Source amperes 10- Source line volts 11- Source firing angle 12- Load firing angle 13- Motor power factor 14- Source power factor 15- Source KVA 16- Current limit setpoint 17- Voltage limit setpoint
POE	3000:182 (PT)	Variable used in the "print status" routine to enable the auto print feature. In this mode, the printer prints the selected information at time intervals selected by variable POT. POE = "T"rue enables function = "F"alse disables function
RR6PA0 1 2 3 4 5	3000:7E 3ooo:so 3000:82 3000:84 3000:86 3000:88 (RR)	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input range is 0.0001 to .250 per unit/sec. RR6PA() = Per unit speed change/ second The related software variable array is SPD, RATE, ACCL1_ 6P(0-5) and is related to the corresponding tuneup variable according to the following equation: ACCL_ 6P (1) = (RR6PA(1))*5 592 235/ *(FRQM/60)/FRQS

Table 4-1. Description of Variables - Continued

Name	Location	Function
RR6PD0 1 2 3 4 5 (RR)	3000:96 3000:98 3000:9A 3000:9C 3000:9E 3000:A0	Family of variables controlhng the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode./Resolution is 0.01% and the input range is 0.0001 to .250 per unit/sec. RR6PD( ) = Per unit speed change/ second The related software variable array is SPD, RATE_ DECEL, 6P(0-5) and is related to the corresponding tuneup variable according to the following equation: DECEL, 6P( ) = (RR6P( ))*5 592 235 * (FRQM/60)/FRQS
RR12PA0 1 2 3 4 5 (RR)	3000:8A 3000:8C 3000:8E 3000:90 3000:92 3000:94	Family of variables controlling the rate-of-change of the drive specd. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input is 0.0001 to .250 per unit/sec. RR12PA( ) = Per unit speed change/ second The related software variable array is SPD_ RATE, ACCL_12P(0-5) and is related to the corresponding tuneup variable according to the following equation: ACCL_ 12P( ) = (RR12PA( ))*5592235/ * (FRQM/60)/FRQS
RR12PD0 1 2 3 4 5 (RR)	3000:A2 3000:A4 A6 A8 AA AC	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input is 0.0001 to .250 per unit/sec. RR12PD( ) = Per unit speed change/ second The related software variable array is SPD, RATE- DECEL-12P(0-5) and is related to the corresponding tuneup variable according to the following equation: DECEL, 12P( ) = (RR12D( ))*5592235/ * (FRQM/60)/FRQS
RRBPO 1 2 3 4 (RR)	3000:AE 3000:B0 3000:B2 3000:B4 3000:B6	Selects regions of drive speed where different accel/deceleration rates may be selected. The resolution is .1% and the range is 0 to 2.0 per unit speed. RRBP( ) = DESIRED per unit SPEED The related software variable is SPD_ REF_ BREAK_ POINT(O-4) The relationship between the variables is defined below: BREAK POINT( ) = RRBP( )*1024* (FRQM/60)

Table 4-1. Description of Variables - Continued

Name	Location	Function
RRMAX6P	3000:EA (SR)	Maximum speed reference clamp active only in 6 pulse operation. RRMAX6P = Desired per unit speed, where 1 per unit speed is the speed at 60Hz. Resolution is .1% and the range is 0 to 2 per unit. The associated software variable is SPD_ REG_ REF_ MAX_ 6P_ E The relationship between the variables is REF_ MAX_ 6P_ E = (RRMAX6P)* 1024 * (FRQM/60)
RRMAXL	3000:B8 (RR)	Variable controlling maximum speed reference *scaling* in the following speed modes: Mode — Definition 3 — Local volts ref - fwd operation *only* RRMAXL = Desired per unit speed, where 1 per unit speed is drive speed at 60Hz Resolution is .1% and the range is' 0 to 2.0 per unit speed. The related software variable is SPD_ REF_ MAX_ LMT This is related to the subject variable by _MAX_LMT = RRMAXL * 1024 * (FRQM/60)
RRMAXR	300Q:BC (RR)	Variable controlling maximum speed reference scaling in the following speed modes: Mode — Definition 0 — Remote volts ref - fwd operation only 1 — Remote 4-20ma reference 2 — Remote volts ref - reversmg operation 4 — Up-dn contact closure mode RRMAXR = Desired per unit speed, where 1 per unit speed is the drive speed at 60Hz Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD_ REF_ PROCESS, MAX This is related to the subject variable by PROCESS, MAX = RRMAXR * 1024 * (FRQM/60)

Table 4-l. Description of Variables - Continued

Name	Location	Function
RRMINL	3000:BA (RR)	Variable controlling mmimum speed reference scaling in the following speed modes: Mode 3 — Definition: Local volts ref - fwd operation only RRMINL = Desired per unit speed, where 1 per unit speed is the drive speed at 60Hz Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD_ REF_ MIN_ LMT This is related to the subject variable by MIN_ LMT = RRMINL * 1024 * (FRQM/60)
RRMINR	3000:BE (RR)	Variable controlling minimum speed reference scaling in the following speed modes: Mode — Definition 0 — Remote volts ref - fwd operation only 1 — Remote 4-20ma reference 2 — Remote volts ref - reversing operation 4 — Up-dn contact closure mode RRMINR = Desired per unit speed, where 1 per unit speed is the drive speed at 60Hz Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD_ REF_ PROCESS, MIN This is related to the subject variable by PROCESS, MIN = RRMINR * 1024 * (FRQM/60)
RRMODE	3000:C2 (RR)	Variable selecting the speed reference for normal operation. That is , the mode which will be selected if the drive is not in local, turning gear, or in emergency stop. The value of the RRMODE determines the mode according to the following RRMODE = 0 0 TO 10V unipolar reference proportional to speed thru the remote speed reference input - voltage source. RRMODE = 1 4-20ma customer reference through the remote reference input. This is a unidirectional mode. RRMODE = 2 Bipolar 10 volt reference through remote speed reference input. Allows reversing operation. RRMODE = 3 Selects local speed reference and non-reverwng operation only. RRMODE = 04 Allows control of speed reference by raise/ lower interlocks via I/O module. RRMODE = 05 Selects the turning gear speed reference set by RRTG. RRMODE = 06 Selects the reference set by variable RRMINR.

Table 4-l. Description of Variables - Continued

Name	Location	Function
RRTG	3oOO:co (RR)	Variable setting motor fixed speed when speedset point mode 05H is selected. 05H is normally used for turning gear function. RRTG = Desired per unit speed where 1 per unit speed is the drive speed at 60Hz Resolution is .1 % and range is 0 to 1.2 per unit speed. The related software variable is SPD_ REF_ TURNING-GEAR The following relationship holds _TURNING_ GEAR = RRTG * 1024 * (FRQM/60)
SIMARG	3000:56 (MI)	Variable controlling the source-side retard limit as a function of current Compensates the firing as a function of link current for source impedance. SIMARG = per unit commutating impedance Resolution is .01 per unit and the range is .01 to .25 per unit. The associated software variable and its relationship to SIMARG is given below. ALPH_ SRC_ IMP, MARC = 67 * (XPU)
SMARG	3000:52 (MI)	Variable controlling the fixed source side margin when inverting. Normal margin is 20 degrees, SMARG = desired fixed margin in degrees. Resolution is 1 degree and the range is 0 to 30 degrees. The associated software variable and its relationship to SMARG is given below. ALPH _FIXED_ MARG_ SRC = 256[1 + COS(180 degrees-SMARG)]
SPLLTIM	3000:76 (PR)	Sets time delay before a source phase- lockloop fault is declared following an error determination SPLLTIM = required time in seconds Resolution is .01 seconds and the range is 0 to 1.5 seconds. The associated software variable and its relationship to SPLLTIM is given below. PLL_ OUT_ OF, TOL_ SRC_ PERIOD = SPLLTIM * 6 * FRQS
SQCB	3000:178 (SQ)	Test variable to allow operation of the source current regulator by diametrically firing the load side cells in a controlled fashion such that the motor load is not required. SQCB = "F"alse (normal operation) = "T"rue (crowbar mode enabled)
SQETST	3000:172 (SQ)	Test variable used to disable the field exciter selftest feature. This is normally disabled on 12 pulse drives to avoid closing in on field in operation. SQETST = "F"alse disable field exciter test = "T"rue enable field exciter test.
SQFE	3000:17A (SQ)	Test variable to allow operation of the exciter alone. Source power must be available to provide sychronizmg signals. SQFE = "F"alse (normal operation) = "True (exciter only)

Table 4-1. Description of Variables - Continued

Name	Location	Function
SQIOTST	3000:6E (SQ)	Variable controlling the enabling of microprocessor interrupts. Normally all interrupts are enabled. When SQIOTST is set, all interrupt programs are disabled. SQIOTST = "F"alse (normal operation) = "T"rue (interrupts disabled)
SQMCH	3000~170 (SQ)	Selects the master channel when two channels are connected in a 12 pulse master-slave configuration. SQMCH = "False (selects channel as slave) = "T"rue (selects channel as master)
SQOPTl	3000:14E (SQ)	This is a family of flags which control the enabling of various options in the ladder logic sequencing program. The flags are encoded into SQOPTl as shown below. Background Variable / Bit Position / Function SQMOVTEMP 0 -Mask MOVTEMP mput SQBFDIFSW 1 -Mask BFDIFSW input SQMB1ST 2 -Mask MB1ST input SQMB2ST 3 -Mask MB2ST SQMB3ST 4 -Mask MB3ST mput SQMB4ST 5 -Mask MB4ST input SQUSERA 6 -Mask USERA input SQTTRBLE 7 -Mask 'ITRBLE input SQLTRBLE 8 -Mask LTRBLE input SQPCROT 9 -Mask PCROT input SQBFILF 10 -Mask BFILF input SQRUNPM 11 -Mask RUNPM input SQSTARTPM 12 -Mask STARTPM input SQNBLF 13 -Mask NBLF in system fault SQFSP 14 -Enable flying spare logic. 15 -Spare Setting a variable to the "1" state masks out the function so that ladder diagram logic does not affect the sequencing. Setting the "0" allows the function to affect the sequencing. Refer to the ladder diagram logic. The value of SQOPTl is obtained by setting 1s and 0s in the bit positions and connecting the binary number obtained to decimal.
SQOPT2	3000:150 (SQ)	This is a a family of flags which control the enabling of various options in the ladder logic sequencing program. The flags are encoded into SQOPT2 as shown below Background Variable / Bit / Function SQBFNDR 0 -Mask BFNDR mput SQRDY 1 -READY option Settmg a bit to the "I" state masks out the function so that ladder diagram logic does not affect the sequencing. Setting the "0" state allows the function to affect the sequencing. Refer to the ladder diagram logic. The value of SQOPT2 is obtained by setting 1s m bit positions required and converting the binary number obtained to decimal.

Table 4-1. Description of Variables - Continued

Name	Location	Function
SQPLTIM	3000:14C (SQ)	Variable setting the time delay after a loss of source power is sensed and before a power loss fault is declared. The time delay is based on background program executions and is approximately 3 milliseconds per pass. SQPLTIM = (time delay)/3 E-03 The resolution is 1 count and the range is 0 to 20,000. The associated software variable and its relationship to SQPLTIM is given below. POWER- LOSS, TIMER = SQPLTIM
SQRA	3000:17C (SQ)	Selects exciter mode of operation for slave converter in 12 pulse system. SQRA = "F"alse slave exciter does not operate = "T"rue slave exciter operates
SQRT	3000:152 (SQ)	Determines if I/O module outputs arc to be checked during selftest. It is necessary to defect these on 12-pulse and common source breaking type applications SQRT = "F"alse - I/O outputs not checked.
SQTMR1 SQTMR2 SQTMR3 SQTMR4 SQTMR5 SQTMR6	3000:140 3000:142 3000:144 3000:146 3000:148 3000:14A (SQ)	Timers used in the drive sequencing module to implement ladder logic functions. The definition of each timer's use is given in the background program. Note that the time base for the counter is the source.-sidecrossover interrupts. SQTMR- = setting (in seconds) The resolution is .1 seconds and the range is 0 to 20 seconds. The associated software variable and its relationship to SQTMR - is given below. BKGND_ TIMER( ) _LIMIT = SQTMR- * FRQS/10
SRCLFC	3000:C4 (SR)	Speed regulator torque command limit when drive operation is in one of the forced-commutated regions. A typical value is .2 PU torque. SRCLFC = Desired per unit current Resolution is 1% and the range is 0 to 2.0 per unit. The associated software variable is SPD_ REG_ CUR_ LMT_ 1 This is related to the subject variable by CUR, LMT_ 1 = SRCLFC * 1024

Table 4-l. Description of Variables - Continued

Name	Location	Function
SRCLSC	3000:C6 (SR)	Speed regulator torque command limit when drive operation is in self-commutation. A typical value is 1.0 PU torque. SRCLSC = Desired per unit current Resolution is 1% and the range is 0 to 2.0 per unit. The associated software variable is SPD_ REG_ CUR_ LMT_ 2 This is related to the subject variable by CUR_ LMT_ 2 = SRCLSC * 1024
SRERRF	3000.C8 (SR)	Variable controlling the selection of the speed regulator speed error filter. SRERRF: 0H — break frequency 60Hz source: filter is bypassed; break freq 50Hz source: filter is bypassed 1H — 60Hz: double at 17 rad; 50Hz: 14 and 14 rad 2H — 60Hz: double at 40 rad; 50Hz: 28 and 28 rad The related software variable is SPD_ REG_ ERR, FILTER whose value is equivalent.
SRETOL	3000:CA (SR)	Variable setting the allowable speed error (difference between reference and feedback). If this error is exceeded, a signal to initiate an "EXCESSIVE SPEED ERROR" alarm is generated. SRETOL = Desired % speed error, where 1 per unit speed is speed at 60Hz Resolution is .1% and the range is 0 to 1.2 per unit speed. The associated software variable is SPD_ REG_ DELTA- TOL The relationship between the variables is DELTA_ TOL = SRETOL * 1024 * (FRQM/60)
SRGAIN	3ooo:cc (SR)	Variable setting the speed regulator gain. The following equation defines the required gain: $GAIN = \frac{(.0000006195) X (N^2) X (WK2)}{(HP) X (Tr^2)} * \frac{60^2}{FRQM^2}$ N = rated motor RPM Tr = regulator response time WK2 = total drive WK2 at motor shaft HP = drive HP/channel Resolution is .1 and the range is 0 to 200

Table 4-1. Description of Variables - Continued

Name	Location	Function
SRLLADJ	3000:DO (SR)	Variable for selecting the speed feedback time constant. This time constant is normally coordinated with the speed regulator crossover desired to give critical damping.(usually SET FOR .94 RAD.)

spd reg crossover	lead T.C. 60Hz	lead T.C.50Hz	SRLLADJ
	FILTER BY-PASSED		0 H
			1H
80.05	.0292	.0243	2H
16	.0625	.0521	3H
7.74	.1292	.1077	4H
3.80	.2629	.2191	5H
189	. 530	.4417	6H
.94	1.064	.8867	7H

Software variable

SPD, REG_ LEAD, LAG, ADJ

takes on values equivalent to SRLLADJ and is directly in the software

SRMAXR	3000:D2 (SR)	Maximum speed reference clamp. SRMAXR = Desired per unit speed, where 1 per *unit* speed is the speed at 60Hz Resolution is .1% and the range is 0 to 1.2 per unit. The associated software variable is SPD, REG, REF_ MAX, LMT The relationship between the variables is _REF_ MAX_ LMT = SRMAXR * 1024 * (FRQM/60)
SRMINR	3000:D4 (SR)	Minimum speed reference clamp. SRMINR = Desired per unit speed, where 1 per unit speed is the speed at 60 Hz The associated sofhvare variable is SPD, REG, REF_ MIN_ LMT Resolution is .1% and the range is 0 to 1.2 per unit. The relationship between the variables is REF_ MIN_ LMT = SRMINR * 1024 * (FRQM/60)

Table 4-l. Description of Variables - Continued

Name	Location	Function
SRTC	3000:CE (SR)	Variable setting the location of the speed regulator "zero" for stability compensation. SRTC = (Desired time cnst), where the time constant is specified in seconds. Resolution is .01 seconds and the range is .05 to 3 seconds. Software vanables related to SRTC and SRGAIN are: SPD_REG_G1 and SPD_REG_G2 The following relations hold: -G1 = 8 * (SRGAIN)* (SRTC) -G2 = 128 * (SRGAIN)/FRQS where FRQS = source line frequency
STHRUTIM	3000:7A (PR)	Controls number of consecutive commutation failures allowed. A penalty function is implemented such that 6 good firings following a failure will cancel an fault firing. This allows recovery from non-repetitive faults. STHRUTIM = number of consecutive overcurrents. The resolution is units and the range is 1 to 6. The associated software variable and its relationship to FCCOUNT is given below. SHOOT, THRU_CNT = (STHRUTIM - 1) * 6
STIM	3000:104 (MI)	Used in motor starting strategy to set the time for one cycle. Resolution is .1 second and the range is .1 to 6 seconds. The relationship to the software variable is given below. $\text{TRANS_START_FREQ_E} = \frac{181}{\text{STIM}}$ takes on values equivalent to SRLLADJ and is directly in the software
SVPU	3000:F4 (PT)	Defines the value of 1 per unit source voltage. Used in "print status" routine. Resolution is units and the range is 0 to 500 volts.
SWINDTOL	3000:7C (PR)	Controls the allowable tolerance on motor fluxwave unbalance before a shorted winding fault is declared. SWINDTOL = desired per unit flux error Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to SWINDTOL is given below. SHORTED, WINDING, FLUX_TOL = SWINDTOL * 1361

Table 4-l. Description of Variables - Continued

Name	Location	Function
SWINDTOL	3000:7C (PR)	Controls the allowable tolerance on motor fluxwave unbalance before a shorted winding fault is declared. SWINDTOL = desired per unit flux error Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to SWINDTOL is given below. SHORTED, WINDING, FLUX_ TOL = SWINDTOL * 1361
TRSPD	3000:5E (MI)	Sets the transistion speed for entering self-commutation mode. TRSPD = desired per unit speed Resolution is .001 per unit and the range is 0 to 1.2 per unit. Where 1 per unit speed is the drive speed at 60 HZ. The associated software variable and its relationship to TRSPD is given below. LAC _MODE3_ TRAN_ SPD = TRSPD * 1024*(FRQM/60)
QCBTOL QCMAX QCMIN QCVTOL	(MI)	Variables used by Factory QC Personnel.

DESCRIPTION OF PUBLIC VARIABLES

SECTION B

ALARM
Equivalent software variable - ALARM_ WORD
Variable type - WORD
Scaling - BIT CODED

Definition - Variable used to pass alarm states sensed in various software modules to the background (sequencing) programs. The bits of the variable are coded as follows:

Bit	0	Field input volts not present
	1	Rotor position start failed
	2	I/O addressing problem
	3	Division by zero
	4	Improper running mode for UP
	5	Improper running mode for UP
	6	Numerical overflow in calculation
	7	Non-maskable interrupt service
	8	Improper interrupt request on HLCB
	9	System time interrupt running
	A	Interrupt rquest unresolved
	B	Loss of speed control
	C	UV flag
	D	Field thermal model
	E	Loss of speed reference
	F	Spare

The ALARM will remain set until the "RESET" is operated.

ALPHFLD
Equivalent software variable - ALPH_ CMND_ FLD
Variable type - INTEGER
Scaling - +255 counts -> ALPHA = 0 degrees
255 counts -> ALPHA = 180 degrees

Definition - Sets the firing angle of the field exciter voltage controller. It is normally calculated by the Flux regulator but may be set by RAC if the Flux regulator is opened.

ALPHLD
Eauivalent software variable - ALPH_ CMND_ LD
Variable type - INTEGER
Scaling - -t-256 counts -> ALPHA = 0 degrees
-255 counts -> ALPHA = 180 degrees

Definition - Sets the firing angle for the load side power converter. During motoring operation, ALPHA ranges between 90 and 180 degrees. Values between 0 and 90 indicated regenerative mode. The load angle control regulator calculates variable value in self-commutation mode and forced-commutation mode when the phase-lock-loop is locked. In seg-fire mode, the variable is not calculated.

Description of Variables (Continued)

ALPHSRC
Equivalent software variable - ALPH_ CMND_ SRC
Variable type - INTEGER
Scaling - +256 counts -> ALPHA = 0 degrees
-255 counts -> ALPHA = 180 degrees

Definition - Sets the value of the source side power converter firing angle. The variable is calculated in the source current regulator.

ALPHSRCO
Equivalent software variable - ALPH_ SPLOVR _SRC
Variable type - INTEGER
Scaling - 0 TO 97 counts

Definition - When the source side angle reaches full advance, the current regulator can maintain current by advancing the load side firing angle. This is known as the crosstie regulator. The excess angle command is set in ALPHSRCO for transmission to the load angle regulator.

BBFS
Equivalent software variable - BBFS
Variable type - BYTE
Scaling - T or F

Definition - Bottom blower selected. This variable represents the state of the blower switch input from the power conversion bridge. It is used in the blower alarm/fault logic.

BFDIFSW
Equivalent software variable - BFDIFSW
Variable type - BYTE
Scaling - T or F

Definition - Bridge differential pressure switch. Represents the state of the power conversion bridge differential pressure switch. This is an optional function. Used in the background program for fault alarm monitoring.

BFILF
Equivalent software variable - BFILF
Variable type - BYTE
Scaling - 1 '

Definition - Bridge filter fuse indication. Represents the state of the interlocks operated by the ac and dc filter fuses. Used in the background program for fault/alarm monitoring.

BFNDR
Equivalent software variable - BFNDR
Variable type - BYTE
Scaling - T or F

Definition - Bridge fan door compartment open indication. Used in the background program for fault/alarm monitoring. This is an optional function.

Description of Variables (Continued)

CBMODE
Equivalent software variable - CROWBAR MODE
Variable type - BYTE
Scaling - T or F

Definition - This variable indicates when the load side *is* operating in crowbar mode which is a test mode where current is circulated through diametrically fired load side thyristors bypassing the motor. Crowbar mode is obtained through the monitor tuneup mode.

CCMND
Equivalent software variable - CUR REG.. CMND
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU motoring torque
-1024 counts -> 1 PU regenerative torque

Definition - Variable calculated as the output of the speed regulator. It is then separated into two parts to control the drive torque. When above the minimum current level, current is increased to increase torque. Below that level, the load side power factor is modified to control torque.

CFBK
Equivalent software variable - CUR REG_ FDBK
Variable type - INTEGER
Scaling - 1024 counts -> 1 per unit current

Definition - Current feedback is derived in hardware from CT signals and read into the software via A/D converters. During forced commutated modes, current feedback is a function only of source-side current read 30 degrees after source firings. During self commutated modes, current feedback is calculated from the average of source current read 30 degrees from firings and load current read at source firings. This more complex scheme minimizes the effect of operation near beat frequencies.

CFC
Equivalent software variable - CFC
Variable type - BYTE
Scaling - T or F

Definition - Close field contactor. This is the field contactor control signal developed in the background sequencing program.

CHRNG
Equivalent software variable - CHANNEL _RUNNING
Variable type - BYTE
Scaling - T or F

Definition - The channel running variable is passed between master and slave channels to indicate the other's status. It is used in the background sequencing program to determine the mode of operation. Thisvariable serves no function on 6 pulse drives.

Description of Variables (Continued)

CITERM
Equivalent software variable - INTEGRAL_____ TERM
Variable type - INTEGER
Scaling - not applicable

Definition - This variable represents the product of the current regulator integral gain and the current summing junction error.

CPTERM
Equivalent software variable - PROPORTIONAL _TERM
Variable type - INTEGER
Scaling - not applicable

Definition - This variable represents the product of the current regulator proportional gain and the latest current summing junction error.

CRERR
Equivalent software variable - CUR _REG_ ERR
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU error

Definition - Difference between current regulator reference and feedback after being limited by the current regulator error clamp.

CROUT
Equivalent software variable - CUR, REG _OUTPUT
Variable type - INTEGER
Scaling - 22528 counts max
range+ or - 22528 counts

Definition - Current regulator control output. Controls source firing angle to full advance and then advances load side angle to maintain current.

CRSTAR
Equivalent software variable - CUR, REG_ STAR
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU current

Definition - Current regulator reference from speed regulator after being limited by the current lower limit function.

CSTEP
Equivalent software variable - CUR, REG_ STEP
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU current

Definition - Test input for current regulator. Used to step the current regulator to determine its transient response.

Description of Variables (Continued)

CURLL
Scaling
Equivalent software variable - CUR_ LOWER_ LMT
Variable type - INTEGER
Scaling 1024 counts -> 1 PU current

Definition Current mininum limit. The minimum value of current lower limit is fixed by a tuneup variable to maintain continuous link current. The value may be raised by the overvoltage regulator when required to limit motor voltage.

CURRCMD
Equivalent software variable - CUR RATE, LMT_ CMND
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU current

Definition - Current reference after being rate limited. The rate of change of reference is limited to 25 PU per second while in self-commutation mode. No limit is applied in forced-commutated mode.

DCL
Equivalent software variable - DCL
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable whose state is determined by comparing source current against the drive current limit. True when the drive is operating in current limit.

ESPDRQ
Equivalent software variable - ESPDRQ
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing input. When true initiates an emergency speed reduction to minimum operating speed.

ESTOP
Equivalent software variable - ESTOP
Variable type - BYTE
Scaling T or F

Definition - Background sequencing variable. True when an emergency stop input is received from the I/O module.

FLDCN
Equivalent software variable - FLDCN
Variable type - BYTE
Scaling - T or F

Definition - Signal representing the state of the field contactor as determined from an auxiliary interlock input. Used in the background sequencing program.

Description of Variables (Continued)

FLDCUR
Equivalent software variable - FLD _CUR
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU field current

Definition - Ssignal representing the current transferred to the exciter via the exciter voltage controller. Used for field protective functions.

FLDEMODE
Equivalent software variable - FLD_ EXC_ MODE
Variable type - BYTE
Scaling - T or F

Definition - Field exciter test mode. When true the field contactor will be close, the local test reference will be redirected to control the field exciter angle command, and the field interrupt will be enabled.

FLFLT
Equivalent software variable - FIELD _LOSS_ FLT
Variable type - BYTE
Scaling - T or F

Definition - True when field loss fault is present.

FLTWD
Equivalent software variable - FAULT, WORD
Variable type - WORD
Scaling - bit coded

Definition - Faults, sensed by the foreground programs, are encoded into FLTWD for transmission to the background. The bits are defined as follows;

Bit		
	0	Field current unbalance
	1	Field overcurrent
	2	Shorted winding detected
	3	DC link overcurrent SRC
	4	DC link overcurrent LD
	5	IDC source shoot thru
	6	IDC LOAAD shoot thru
	7	Spare
	8	Source PLL fault
	9 -	Load PLL fault
	A	Spare
	B	Stack fault
	C	Failure to Start
	D	EEPROM transfer failure
	E	Spare
	F	Spare

Description of Variables (Continued)

FLUXADJE
Equivalent software variable - FLUX, ADJ, ERR
Variable type - INTEGER
Scaling - 256 counts -> 180 degrees

Definition - Error signal for Flux adjust function.

FLUXADJ
Equivalent software variable - FLUX, ADJ
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Output of flux adjust function used to modify the flux profile.

FLUXERRA
Equivalent software variable - FLUX, REF_ ADJ_ IF
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Variable which reduces the flux setpoint when the field current limit function is active; that is, when the field current limit set point is exceeded, the flux setpoint is reduced to avoid excessive field current.

FLUXERR
Equivalent software variable - FLUX, ERR SCL
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Error signal for flux regulator after being clamped.

FLUXFBK
Equivalent software variable - FLUX, REG_ FDBK
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Feedback signal for flux regulator. Derived by reading the motor phase peak flux.

FLUXREFL
Equivalent software variable - FLUX, REF_ LMT
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Limit variable on flux setpoint. Limits reference as a function of motor voltage to prevent motor overvoltage.

Description of Variables(Continued)

FLXSP
Equivalent software variable - FLUX, SP
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Reference to flux regulator. Calculated as a function of the motor flux profile, source flux programming function, and fire-as-early-as-possible regulator.

FLXSRCT
Eauivalent software variable - SRC_ FLUX _TST
Variable type - INTEGER
Scaling - 1192 counts -> 1 PU flux

Definition - Test input to function which programs the flux setpoint as function of source flux. The input is used to simulate changes in the source flux.

FLXSRC
Equivalent software variable - FLUX_ SRC
Variable type - INTEGER
Scaling - 1192 counts -> 1 PU flux

Definition - Source flux read in source crossover routine. The particular phase read is Fc which corresponds to motor voltage Va-c.

FLXLOV
Equivalent software variable - FLUX JIM, OV
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Variable representing the value of fluxwhich, at a given speed, will give rated motor volts. Used as a limit on the value of flux which can be regulated. The variable is calculated in the overvoltage regulator.

FSTEP
Equivalent software variable - FLUX _REG_ STEP
Variable type - INTEGER
Scaling - 8254 counts -> 1 PU flux

Definition - Variable used to step the flux regulator when tuning the transient response.

GTSTM
Equivalent software variable - GATE-TEST, MODE
Variable type - BYTE
Scaling - T or F

Definition - When true the control is operating in Gate test Mode where the control is synchronized to a 3-phase oscilator and the thyristor gate drive can be analyzed without the need for high voltage application.

Description of Variables (Continued)

HITQ
Equivalent software variable - HI_TQ
Variable type - BYTE
Scaling - T or F

Definition - When true, control is operating in high torque mode. This is a special setup for drives requiring high starting torques.

ILD
Equivalent software variable - IDC_ LINK, LD
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU current

Definition - D-C link current as read by the load side CT's.

INVLIM
Eauivalent software variable - INVLIM
Variable type - WORD
Scaling - bit coded

Definition - When any bit of the variable is non-zero, the source and load side converters will be set to fire at inversion 'limit. The drive will not necessarily fault trip and may recover to normal operation. The bits are coded as follows:

Bit		
	0	Differential current - load
	1	Differential current - source
	2	Gvercurrent source
	3	Gvercurrent load
	4	Spare
	5	Shutdown flag
	6	Spare
	7	UV *flag*
	8	Startup flag
	9	Spare
	A	Spare
	B	SFLAG source
	C	SFLAG load

IRSC
Equivalent software variable - IDC_LINK_SRC
Variable type - INTEGER
Scaling - 1024 counts -> 1 PU current

Definition - D-C link current as measured by source side CT's.

Description of Variables (Continued)

LBKRS
Equivalent software variable - LBKRS
Variable type - BYTE
Scaling - T or F

Definition - Variable representing the status of the source side switchgear. Using in the background sequencing program.

LMODE
Equivalent software variable - LMODE
Variable type - BYTE
Scaling - bit coded

Definition - Variable which represents the load side firing algorithm being used. Values are as follows:

VALUE	FUNCTION
0	lookstart
1	segment firing
2	closed loop forced-commutated mode
3	self-commutated mode

LOCAL
Equivalent software variable - LOCAL
Variable type - BYTE
Scaling - T or F

Definition - Indicates that the drive is under control of the local operator's devices mounted on the I/O module.

LSPD
Equivalent software variable - LOWER_SPD
Variable type - BYTE
Scaling - T or F

Definition - Input from purchaser's lower interlock when raise/lower speed reference is used.

LTRBLE
Equivalent software variable - LTRBLE
Variable type - BYTE
Scaling - T or F

Definition - Input from link reactor interlock.

List of Variables (Continued)

LVLTS
Equivalent software variable - LD, VOLTS, DC
Variable type - INTEGER
Scaling - range -256 to +256
256 counts -> 1 PU load DC volts

Definition - Variable representing load side dc link voltage. It is calculated from speed, flux, and firing angle. The variable is used as a feed forward compensation to the source current regulator.

LVLTS
Equivalent software variable - LD, VOLTS, DC
Variable type - INTEGER
Scaling - range -256 to +256
256 counts -> 1 PU load DC volts

Definition - Variable representing load side DC link voltage. It is calculated from speed, flux, and firing angle. The variable is used as a feed forward compensation to the source current regulator.

MAMS
Equivalent software variable - MAMS
Variable type - BYTE
Scaling - T or F

Definition - Motor above minimum speed. Variable derived from comparison between motor speed and minimum speed reference setting. State controls interlock to the customer.

MB1ST
Equivalent software variable - MB1ST
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable representing the state of an interlock from the power conversion blower number 1 motor starter. Used in fault and alarm logic.

MB2ST
Equivalent software variable - MB2ST
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable representing the state of an interlock from the power conversion blower number 2 motor starter. Used in fault and alarm logic.

Description of Variables (Continued)

MB3ST
Equivalent software variable - MB3ST
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable representing the state of an interlock from the power conversion blower number 3 motor starter. Used in fault and alarm logic.

MB4ST
Equivalent software variable - MB4ST
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable representing the state of an interlock from the power conversion blower number 4 motor starter. Used in fault and alarm logic.

MFPK
Equivalent software variable - MTR_ FLUX, PEAK
Variable type - INTEGER
Scaling - 1376 counts -> 1 PU

Definition - Instantaneous highest of the three motor flux waves. Only calculated in load side firing modes 0 and 1.

MFWD
Equivalent software variable - MTR _ROTATION_ FRWD
Variable type - BYTE
Scaling - T or F

Definition - Motor direction flag. Calculated in load PLL routine. Used to determine load firing mask rotations.

MISOLS
Equivalent software variable - MISOLS
Variable type - BYTE
Scaling - T or F

Definition - Input to background sequencing program as a function of the motor isolator auxiliary interlock.

Description of Variables (Continued)

IMODEFLG
Equivalent software variable - MODE, FLAG
Variable type - WORD
Scaling - bit coded

Definition - Variable used to interface the background sequencing program to the foreground programs which control the firing algorithms. The variable is coded as follows:

Bit		
	0	Run with field exciter
	1	Run without field exciter
	2	Run field exciter only
	3	not used
	4	crowbar
	5	switchgear closed
	6	twelve pulse
	7	master channel
	8	gate test mode
	9	high torque
	A	not used
	B	not used
	C	factory field exciter
	D	factory source side firing
	E	factory loadside firing
	F	factory pumpback

MOVTEMP
Equivalent software variable - MOVTEMP
Variable type - BYTE
Scaling - T or F

Definition - Motor overtemperature device interlock input. Used in background sequencing program.

MTRVLTS
Equivalent software variable - MTR VOLTS
Variable type - INTEGER
Scaling - 128 counts -> 1 PU motor volts

Definition - Motor line-to-line volts. Calculated from the equation

MTRVLTS = K * SPEED * (motor flux)

NALM
Equivalent software variable - NALM
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable which is true when no alarms are present in the control.

Description of Variables (Continued)

NBLA
Equivalent software variable - NBLA
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable which is true when no blower alarms are present. That is , when the power conversion blower selected to operate is the one operating.

NBLF
Equivalent software variable - NBLF
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable which is true when no blower faults are true. That is, the power conversion blower selected to run (or its redundant backup) is running.

NCFLT
Equivalent software variable - NCFLT
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when faults are present as sensed by the control.

NPSUV
Eauivalent software variable - NPSUV
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Monitors the state of the P105V power supply and is true when the power supply is okay.

NSFLT
Equivalent software variable - NSFLT
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when no faults which trip source side switchgear are present.

OVLTE
Equivalent software variable - OVVOLT_ ERROR
Variable type - INTEGER Scaling - 8254 counts -> 1 PU flux

Definition - Overvoltage regulator error signal. The flux level necessary to hold the overvoltage regulator setpoint is continually calculated as a function of speed. A comparison of this flux setpoint and the flux feedback level results in OVLTE.

84

Description of Variables (Continued)

P900C
Equivalent software variable - PORT, 900C
Variable type - BYTE
Scaling - bit coded

Definition - Background sequencing variable representing an input port. The bits represent the following inputs.

Bit	0	Source overvoltage fault
	1	Source undervoltage alarm
	2	Source suppression fault
	3	Source Gating supply undervoltage
	4	Load side overvoltage
	5	Load side suppression fault
	6	Load side gating supply undervoltage
	7	Overspeed fault

PCROT
Equivalent software variable - PCROT
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when input from PCROT interlock is closed.

PLLSERR
Equivalent software variable - PLL_ ERROR SRC
Variable type - INTEGER
Scaling - 1024 counts -> 360 degrees

Definition - Source phase lock loop regulator error signal.

PLLSFLGL
Equivalent software variable - PLL_ SFLAG _LD
Variable type - INTEGER
Scaling - not applicable

Definition - Variable used as a measure of the load phas lock loop reliability. A calculation where the error is excessive will result in the variable being incremented by 3 counts. A good calculation will decrement the variable by 1 count.

PLLSFLGS
Equivalent software variable - PLL _SFLAG_ SRC
Variable type - INTEGER
Scaling - not applicable

Definition - Variable used as a measure of the source phase lock loop reliability. A calculation where the error is excessive will result in the variable being incremented by 3 counts. A good calculation will decrement the variable by 1 count.

Description of Variables (Continued)

PLLSLD
Equivalent software variable - PLLC_ LOAD, SRC
Variable type - INTEGER
Scaling - 120 counts -> 60 HZ

Definition - Value loaded into source PLL counter to maintain synchronizm with source line frequency. The equation controlling its value is

PLLSLD = 7200/(line frequency)

RAC
Equivalent software variable - REG_ ANGLE, CMND
Variable type - INTEGER
Scaling - +255 counts -> full advance
-255 counts -> full retard

Definition - Variable used to set converter firing angle when the corresponding regulator is open looped.

RAMPDEL
Equivalent software variable - ACCL_ LMT
Variable type - INTEGER
Scaling - 155.3 counts -> 1 HZ per second

Definition - Intermediate variable in the ramp reference function. Its value reflects the rate of change of speed.

READY
Equivalent software variable - READY
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when the drive ready to run function is satisfied.

REGOPEN
Equivalent software variable - REG _OPEN _CMND
Variable type - WORD
Scaling - bit coded

Definition - Variable when may be used in tuneup to open loop the various regulators and control firing angles directly. It is code as follows:

Bit		
	0	- opens current regulator
	1	- opens speed regulator
	2	- opens flux regulator
	3	- opens load angle regulator

Descriptions of Variables (Continued)

RETDLSRC
Equivalent software variable - ALPH_ RETARD, LMT_ SRC
Variable type - INTEGER
Scaling - **256** counts -> 180 degrees

Definition - Variable representing the maximum angle of retard for the source power conversion bridge. It is programmed as a function of current.

RSPD
Equivalent software variable - RAISE, SPD
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing input variable. When raise/lower speed reference function is used, this variable reflects the state of the raise speed interlock.

RTONI
Equivalent software variable - RTONI
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Reflects state of RTON input. This input is used for return-to-normal in flying spare configurations and other channel running in twelve pulse configurations. It is not used in other configurations.

RTON
Equivalent software variable - RTON
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Output variable which controls the return-to-normal relay driver. Only used in flying spare configurations.

RUNPM
Equivalent software variable - RUNPM
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Reflects the state of the run permissive interlock input.

RUNP
Equivalent software variable - RUNP
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable which controls the run permissive relay for closing the motor isolator.

Description of Variables (Continued)

RUNX
Equivalent software variable - RUNX
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Reflects the state of motor isolator control relay. True when a run request is latched in.

RUN
Equivalent software variable - RUN
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when per-missives to run and run request are true. Sets mode flag variable for drive to leave lookstart.

SFDBKL
Equivalent software variable - SPD _FDBK
Variable type - INTEGER
Scaling - 1024 counts - > 60 HZ

Definition - Variable used in the speed regulator. This is the speed feedback signal after passing through the lead/lag speed filter.

SFDBK
Eauivalent software variable - SFDBK
Variable type - INTEGER
Scaling - 1024 counts - > 60 HZ

Definition - Variable used in the speed regulator. This is the speed feedback variable which is equal to SPEED below 35 HZ and the average of the last 6 SPEED values above 60 HZ.

SFLUXLIM
Equivalent software variable - SOURCE _FLUX _SCALE
Variable type - INTEGER
Scaling - 256 counts - > 1 PU

Definition - Variable which trims flux profile as a function of source flux. Variable is proportional to source flux after being filtered and limited.

SPDEFIL
Equivalent software variable - SPD_ ERR FIL
Variable type - INTEGER
Scaling - 1024 counts - > 60HZ

Definition - Speed regulator error signal after passing through the speed error filter.

Description of Variables (Continued)

SPDRP
Equivalent software variable - SPD, RAMP
Variable type - BYTE
Scaling - not applicable

Definition - Background sequencing variable. The value of the variable selects which set of corresponding accel/decel/6P/12P tuneup variables controls the speed reference ramp.

SPDSP
Equivalent software variable - SPD_ SP
Variable type - INTEGER
Scaling - 2048 counts - > 10 volts

Definition - Background variable. Variable reflects the value of the presently selected speed reference input.

SPDSTP
Equivalent software variable - SPD, STP
Variable type - INTEGER
Scaling - 1024 counts - > 60 HZ

Definition - Speed reference setpoint before the speed ramp function.

SPDSTR
Equivalent software variable - SPD STR
Variable type - INTEGER
Scaling - 1024 counts - > 60 HZ

Definition - Speed regulator reference (after speed ramp).

SPEED
Equivalent software variable - SPEED
Variable type - INTEGER
Scaling - 1023 counts - > 60 HZ

Definition - Speed feedback variable calculated in load PLL routine from time between flux wave crossovers. It is made bi-polar by use of the motor rotation flag.

Description of Variables (Continued)

SREFMD
Equivalent software variable - SPD, REF_ MODE
Variable type - BYTE
Scaling - not applicable

Definition - Selects the speed setpoint source as selected by the background sequencing and the tuneup variables. The modes are defined as follows:

Value	0	Remote voltage reference
	1	Remote current reference
	2	Remote voltage reference (reversing)
	3	Local voltage reference Raise/lower speed reference
	4	Select turning gear function
	6	Select minimum process speed

SSTEP
Equivalent software variable - SPD_REG_STEP
Variable type - INTEGER
Scaling - 1024 counts -> 60 HZ

Definition - Speed regulator input variable for stepping the regulator and determining its transient response.

STRTPM
Equivalent software variable - STRTPM
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Reflects the state of the start permissive interlock input.

SUVFLT
Equivalent software variable - SUVFLT
Variable type - BYTE
Scaling - T or F

Definition - Source undervoltage fault exists when true.

TGRRQ
Equivalent software variable - TGRRQ
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Shows the state of the turning gear mode request interlock.

TIMESRC
Equivalent software variable - TIME, SRC
Variable type - INTEGER
Scaling - 1024 counts -> 360 degrees

Definition - Value of source time counter reading at time of firing.

Description of Variables (Continued)

VTRNGR
Equivalent software variable - TRNGR
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when control is operating in turning gear mode.

TTRBLE
Equivalent software variable - **TTRBLE**
Variable type - **BYTE**
Scaling - T or F

Definition - Background sequencing variable. Represents the state of the transformer trouble contact input.

TWPLSE
Equivalent software variable - TWELVE, PULSE
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. True when drive is operating in twelve-pulse mode.

USERA
Equivalent software variable - **USERA**
Variable type - BYTE
Scaling - T or F

Definition - Background sequencing variable. Reflects the state of the user alarm contact input.

VLTSECCMD
Equivalent software variable - VOLTSECCMD
Variable type - INTEGER
Scaling - 1376 counts -> 1 PU flux

Definition - This is the load side voltsecond command and is equal to VLTSECSTR except when the crosstie regulator is active.

VLTSECMTR
Eauivalent software variable - VOLTSECMTR
Variable type - INTEGER
Scaling - 1376 counts -> 1 PU flux

Definition - Motor voltseconds available for commutating the load bridge. Calculated from peak motor flux modified by current commutated.

VLTSECSTR
Equivalent software variable - VOLTSECSTR
Variable type - INTEGER
Scaling - 1376 counts -> 1 PU flux

Definition - Load side volt-second command to determine firing angle. When motoring VLTSECSTR is equivalent to VOLTSECMTR; however, when inverting, it is equal to motor peak flux.

CHAPTER 5
SECTION A
STRUCTURED LOAD COMMUTATED INVERTER DIAGNOSTIC MONITOR SPECIFICATIONS

5-1. INTRODUCTION

The diagnostic monitor for the Structured LCI product provides the user with a variety of functions. Via an RS-232 terminal, the user may display regulator and logic variables and system fault and status information. The monitor also provides the ability to modify system variables and regulator constants, which facilitates troubleshooting and tuning procedures.

Communication with the monitor is handled through the RS232 port on the drive control. Hence, the communication device may be any RS232 compatible terminal, including handheld, hard copy and CRT types. Provisions are made to specially format the monitor output when a handheld terminal is being used.

All communication via the RS232 port are 7-bit data, even parity, with one stop bit. Acceptable baud rates are 300, 1200, 2400, 4800, and 9600 bits per second (baud). Upon power up, the monitor searches for carriage return characters to lock onto the baud rate used by the terminal device.

Monitor functions are accessed by entering a valid command (from the monitor's instruction set) at the diagnostic terminal. Figure l-l illustrates the basic command structure for the Digital Drive. Many of the commands are called directly from the MONITOR level, '* >' prompt, and control is returned to this level after execution of the requested process. The ENGINEERING and TUNEUP functions, however, are actually extended modes of operation that have their own instruction sets. After execution of one of the ENGINEERING mode commands, for example, control is returned to the ENGINEERING mode, 'E >' prompt. To return to the MONITOR level again, the QUIT command must be invoked -- this is also true for the TUNEUP mode.

In sections 1-3 and l-4, the purpose and syntax of each monitor command is described in detail. Table 5-l defines the conventions used in specifying the command formats.

TABLE 5-l Defmitions for Command Syntax Notation

Symbol	Definition
:: =	"Is defined to be"
[ ]	Items between brackets are optional
|	Separates items in a list • choose one
{ }	Braces delimit items for the purpose of minimizing ambiguity
( )	Optional characters in a command orparameter are in parentheses
Highlited	User entries (Highlited)

< decimal no > :: = < l-5 decimal digits > T

< hex no > :: = < l-4 hex digits > [H]

< new value > :: = [+ > -] < l-5 decimal digits w/ or w/o dec. pt. >

<pointer > ::= [{CS | DS | < hex no > } :] < hex no >

< register > ::= [CS | DS]

CS ::= Code Segment

DS :: = Data Segment

5-2. OPERATING INSTRUCTIONS

5-2.1. General Information

To initiate communication with the Monitor, the user should enter several "carriage return" characters at the terminal keyboard. The terminal will then respond with a banner message and the monitor prompt character (*>) to indicate that it is ready to accept command inputs.At this point, the operator may enter any of the MONITOR level commands or select one of the extended modes of operation (for example TUNEUP) as described in detail in sections 3 and 4.

When finished using the diagnostic monitor, the terminal session may be terminated by entering QUIT at the terminal keyboard (see section l-3.9). The logoff message is then printed at the terminal, indicating that the session is over, and the monitor will idle, waiting for the carriage return characters that indicate the start of another terminal session.

Terminal Format:

The Diagnostic Monitor is designed to work with either a handheld terminal or a standard keyboard type. The default (at the start of the terminal session) is the handheld format. In this mode of operation, the output messages are sent one line at a time to accommodate the limited display length of the handheld terminal (16 characters max per line). Entering a carriage return causes the next line of output to be sent to the terminal. The TERMINAL command (section l-3.4) may be used to switch to an 80 column listing format with no pauses for use with a standard CRT or hardcopy terminal.

Power-up Status Messages:

When power is first applied to the drive control, a

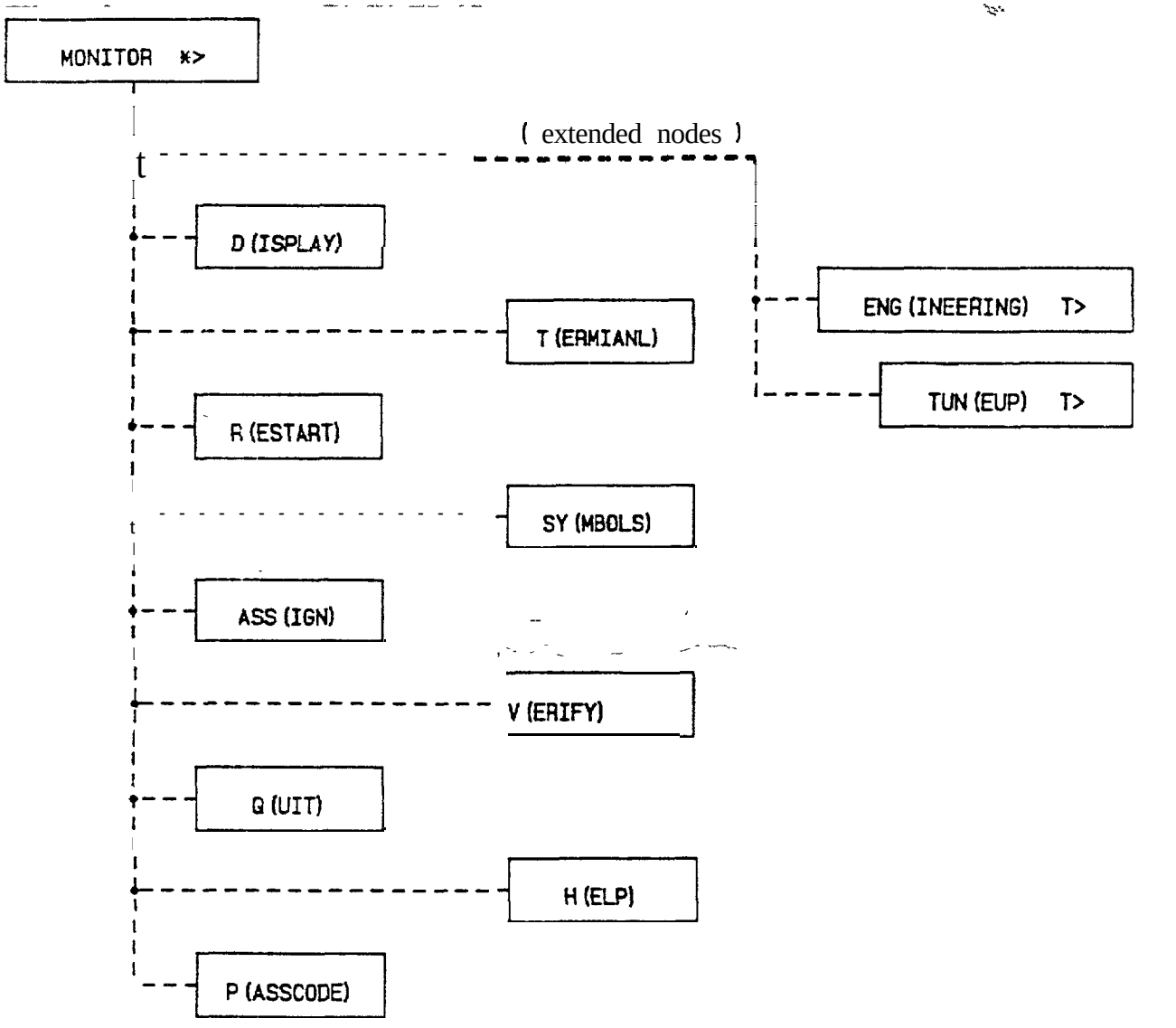


Figure 5A-1. Monitor Command Structure

sequence of self-tests and initialization procedures is performed. Any failure status of these tests is automatically sent to the panel printer. Should the printer fail, diagnostic messages will be rerouted to the diagnostic terminal.

5-2.2. Editing Functions

The diagnostic monitor provides text editing functions for correcting errors entered within a command string. Each function is listed below with a brief description.

NOTE: In the examples below, the symbol * > is the diagnostic monitor prompt (not typed by the user), and the symbol [CUR] represents the cursor position at the end of the sequence shown.

DEL/RUBOUT -

Removes the previous character from the message and echoes the deleted character to the terminal.

Example: Typed - * > HEKP < DEL > < DEL > LP

Echoed - * > HEKP /K P LP [CUR]

CONTROL-R

Causes all characters currently in the message to be transmitted to the terminal on a new line. This function is convenient after using DEL/RUBOUT because it shows the state of the message after the editing.

Example:

Typed - * > HEK < DEL > LP < CONTROL-R >

Echoed - * > HEK /K LP
HELP [CUR]

NOTE: Typing subsequent consecutive CONTROL-R keys will invoke the monitor recall function. This feature calls up previous commands that were typed at the terminal, starting with the most recent and progressing to the oldest command remembered. A total of 20 command lines are remembered. Using the recall feature helps the user avoid retyping a command that he wishes to execute repetitively.

CONTROL-X

Deletes the entire message that has been entered so far.

NOTE: This has the same effect as using the DEL/RUBOUT function to remove all the characters from the message.

Example: Typed - * > HEKL < CONTROL-X > HELP
Echoed - * > HEKL
HELP [CUR]

CONTROL-Y

Aborts the current command sequence and returns the monitor to the current mode of operation (e.g., ENGINEERING, TUNEUP or the main level).

NOTE: CONTROL-Y or CONTROL-C may be entered at any time to abort the current process--even during the listing of a message at the terminal.

Example:

Typed - T > HE < CONTROL-Y >

Echoed - T> HE^ Y

T> [CUR]

CONTROL-C

Aborts the current command sequence and returns the monitor to the outermost (main) level of operation. When a CONTROLC is received, the monitor also displays a message identifying the product as a LCI Drive as in the login sequence.

Example:

Typed - T > HE < CONTROL-C >
Echoed - T> HE^ C
-- LCI DRIVE --
*> [CUR]

CONTROL-S

Causes the current output to be halted until it is restarted using the CONTROL-Q switch.

CONTROGQ

Causes the current output to be restarted after having been stopped using the CONTROL-S switch.

5-2.3. Command Syntax

The general syntax for a monitor command is shown below:

< command > :: = < instruction >
[< delimiter > <parameter > . ..] < terminator >

where <instruction> is one of the valid monitor instructions (defined in section l-3) and <terminator > is a carriage return (<cr >). The <delimiter > and <parameter > entries, if necessary, depend on the particular instruction being called. Each instruction has its own set of valid delimiters and parameters. Also, some commands may require several < delimiter > <parameter > pairs, while some require no parameters at all.

As an aid to the operator, and to facilitate handheld terminal use, all the <delimiter > <parameter > pairs need not be entered on the same line as the instruction -- each needed parameter will be requested by the monitor. The syntax for this type of sequence is:

< command > :: = c instruction > < terminator >
<parameter > < terminator > .

After each <terminator > entry, the monitor displays an appropriate prompt that asks the user for the necessary parameter.

The use of abbreviated commands is another feature designed to aid the handheld user. Rather than typing the complete instruction word, only the first one or two letters need to be entered (see the syntax definition tables in sections 3 and 4 for the minimum acceptable abbreviations).

5-2.4. Protected Commands

Some monitor commands allow the user to modify system variables and change tuning parameters. These functions, because of their potentially destructive nature, are available only after executing a privilege enabling sequence. This is so the user will not inadvertantly make modifications that could inhibit the proper operation of the LCI Drive control.

The privileges are enabled by entering a password at the monitor terminal. Privileges are automatically disabled when: (1) the monitor is logged off (QUIT).

5-2.5. Help

The user may print out a list of available commands by asking for HELP. From either the MONITOR mode, or one of the extended modes, a HELP request will print the list of commands available at that mode only.For example, if in the ENGINEERING mode, only the ENGINEERING mode commands, and not the MONITOR mode commands, will be listed.

5-3. MONITOR COMMANDS

The MONITOR level commands are intended to be the primary tools used in interrogating the drive control and monitoring its status. The instruction set provides display and modify functions, the ability to reset the control, and the power to direct variables to analog outputs for strip chart monitoring, among other things. Table 5-2 shows the complete list of instructions and their correct syntax. The commands are defined in their single line entry format, although the "prompt-through" type sequence (as discussed in section 2.3) is also valid.(See Table 5-2)

TABLE 5-2. Syntax Definitions for MONITOR Level Commands

ASSIGN: A(SSIGN) {LIST | DA}

DISPLAY: {D(ISPLAY) { <varname> [C] | |STATUS | LIST | FAULTS }
DS |
DL |
DF } <cr>

HELP: H(ELP) < cr >

PASSCODE: P(ASSCODE) < cr >

QUIT: {Q(UIT) EX(IT) <cr>

RESTART: R(ESTART) < cr >

SYMBOLS: SY(MBOLS) [<symbolname>] <cr>

TERMINAL: T(ERMINAL) < cr >
<current format > CHANGE ? [Y| N] <cr>

VERIFY: V(ERIFY) [<symbol group name>] <cr>

5-3.1. ASSIGN

Function: To designate that a particular variable be assigned to a circular list and/or D/A converter.

Using the ASSIGN function, the user may modify the configuration of the internal circular lists by changing the number of entries (maximum of 8) to be stored at each sample interval and/or selecting which entries are being captured. A total of 2K bytes of memory are available for storage of the diagnostic list information. This memory is evenly divided among the parameters within the list. As an example, if 4 word parameters are monitored in the circular a total of 256 events will be recorded.

The list may be triggered either by a level trigger or mask trigger. In either case the trigger is compared against the variable selected in list position no. 1.. In a level (>, <, =). When the bit trigger mode is selected, any bit of variable 1 becoming true causes the list to be taken. Bits of variable 1 may be masked using the mask variable option of the list. The list can only be triggered by the level trigger or the bit trigger. When more than one trigger has been activated only the last will be accepted by the list function.

The user may also direct drive variable contents to a D/A converter for observation on an oscilloscope or chart recorder. The analog outputs are available from test points on the front edge of the printed circuit board HLCB. When assigning a variable to a D/A output, the operator selects the two parameters, SCALE and OFFSET. Since the possible range of a variable often exceeds the range of the D/A converter, a function is provided to scale the variable contents before sending data to the D/A converter. The following table gives the possible SCALE parameters; beside each one is the value by which the variable contents are multiplied before being converted if that SCALE factor is selected.

SCALE PARAMETERS

SCALE	MULTIPLIER	SCALE	MULTIPLIER
15	1/32768	0	1
14	1/16384	-1	2
13	1/8192	-2	4
12	1/4096	-3	8
11	1/2048	-4	16
10	1/1024	-5	32
9	1/512	-6	64
8	1/256	-7	128
7	1/128	-8	
6	1/64	-9	256
5	1/16	-10	1024
4	1/8	-11	2048
3		-12	4096
2	1/4	-13	8192
1	1/2	-14	16384
0	1	-15	32768

An OFFSET may also be selected during the ASSIGN process -- this value is subtracted from the variable contents before the scaling process. This function may be used when the slight variations around a fixed value need to be examined (see example below).

Examples:

```
* c ASSIGN LIST
Settings;
1 – samples                    ; selects variables to be sampled
2 – trig                       ; sets up level triggered list
3 – bit trig                   ; sets up bit triggered list
4 – site                       ; selects site where data is taken
Choice? 1
NO OF ENTRIES= 1               ; previously was set to capture1 parameter
NEW VALUE ? 3 <cr>             ; set to capture 3 parameters
ENTRY 1 = FLUXFB               ; previously set to capture FLUXFB
NEW ENTRY? SPDREF < cr >       ; speed reference (no. 1)
ENTRY 2 = OOOOH                ; null value
NEW ENTRY? SPDFB < cr >        ; speed feedback (no. 2)
ENTRY 3 = OOOOH                ; null value
NEW ENTRY? IAFB < cr >         ; armature current feedback (no. 3)
* > ASSIGN LIST

SETTINGS:
  1 – samples
  2 – trig
  3 – bit trig
  4 – site

Choice? 2
TRIGGER LEVEL = 0
NEW VALUE? 200                 ; level of trigger in counts
TRIGGER CONDITION? OFF
NEW VALUE? <                   ; choices are c, >, =
* > ASSIGN LIST

Setting:
1 – samples
2 – trig
3 – bit trig
4 – site
```

```
Choice? 4
TRIGGER SOURCE = OFF          ; present data site
CHOICES(1-5)                  ; available data sites
SAC, SPLL, LAC, LPLL, BG
NEW VALUE?1                   ; selects SAC (0 disables list
*ASSIGN LIST
Settings:
  1 - samples
  2 - trig
  3 - bit trig
  4 - site

Choice? 3
PRESENT MASK FF               ;you must turn off level trigger
NEW VALUE? OF                 ;value must be in hexadecimal
*>
* > ASSIGN LIST                go to choice 4 to select site)
*>


*>ASSIGN DA1<cr>               ; choose DA1 (TP2) on HLCB (slot2E)
ENTRY = SPDREF                ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = IAFB <cr>             ; set new parameters -- 1 feedback
SCALE = 6<cr>                 ; divide by 64 to get in range
OFFSET = <cr>                 ; leave the offset the same (at 0)
*>

*>AS DA2<cr>                   ; choose the DA2 (TP3) on HLCB (slot 2E)
ENTRY = SPDFB                 ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = <cr>                  ; still look at speed feedback
SCALE = -2 <cr>               ; multiply 4 to accent variations
OFFSET = 1000<cr>             ; look at fluctuations around 1000
*>
```

The following notes on using the diagnostic list apply.

1. When using the bit trigger option the mask must always be entered in hexadecimal. The bit (s) to trigger the list must be set to a "a" in the appropriate bit location.
2. In either the bit trigger or magnitude trigger mode, entry "1" in the variable selections is compared against either the magnitude or bit trigger word.
3. When using the bit trigger option, the word must be re-entered, wherever selection 3 of "ASSIGN LIST" is accessed, otherwise the mask is reset to "0".
4. The list has completed taking data when the trigger source has returned to "OFF" after entering another selection.
5. List sites must be selected with care. The list may not necessarily fill up if a site is selected which doesn't operate during the event to be captured. For example, SAC is disabled when the drive "RUN" command is removed.

S-3.2. DISPLAY

Function: to display the contents of select drive control variables and circular lists.

Several options are allowed in the DISPLAY command:

1. DISPLAY LIST, to display the circular list information.

2. DISPLAY FAULTS, to display the status of the fault and alarm logic.

3. DISPLAY STATUS, to display a concise list of the drive's status.

The DISPLAY FAULTS command will produce a list

of the most recent STATUS messages (stored in a circular "fault log"). This list includes current status indications (shown also with the DISPLAY STATUS command).

Drive variables are displayed in their respective engineering units. Generally they are in per unit (of flux, current, speed,etc...). The general format in which variable contents are displayed is:
<varname > = <value >
A particular variable may be displayed repetitively by simply typing a "C" (for continuous) following the variable name on the command line.

Some DISPLAY command options may be abbreviated to two letters. These abbreviations and their equivalents are listed below:

DF ::= DISPLAY FAULTS
DL :: = DISPLAY LIST
DS ::= DISPLAY STATUS

Examples:
*** > DISP SPDSP < cr >**
SPDSP = 472
***> D<cr>**
NAME? **SPDSP < cr >**
SPDSP = 472
*>

***> D STATUS<cr>**
<drive status messages >
*>

*** > DISPLAY LIST < cr >**
< listing of circular list >
*>

*** > DISPLAY FAULTS < cr >**
< listing of circular fault log >
*>

5-3.3. HELP

Function: to display at the terminal a list of the commands available in the current mode of monitor operation.

Examples:

*> H <cr>
< list of available commands >
*>

*> HELP <cr>
< list of available commands >
*>

5-3.4. PASSCODE

Function: to enable the monitor's privileged functions.

NOTE: For security, the characters are not echoed when entering the password at the terminal. The present passcode is a carriage return. [GESALEM9]

Examples:

***> PASSCODE <cr>**
PASSWORD **:** <password> ; password not echoed
PRIVILEGES ENABLED
*>
To change the passcode, see SET password under TUNEUP

5-3.5. QUIT

Function: to terminiate the current terminal session. Upon execution of the QUIT command, the monitor will print a logout message and then begin idling, waiting for the carriage return inputs that indicate the start of a new terminal session.

Examples:
***> Q <cr>**
MONITOR OFF

***> EX <cr>**
MONITOR OFF

5-3.6. RESTART

Function: to execute a software reset of the microprocessor.

Executing the RESTART command is essentially a soft reset. The diagnostic checks are run the same as during power-up, and the drive control is re-initialized to restore the hardware (counters,etc.) to its normal running state.

**** NOTE ****

To execute the RESTART command, the drive must be stopped. If the drive is running, the RESTART routines will not run, and a warning will be printed at the terminal

Examples:

***> RES <cr>**
< selftest results messages >

* > RESTART < cr > < selftest results messages >

5-3.7. SYMBOLS

Function: to display at the diagnostic terminal the symbolic names that may be used to access drive control variables. If no qualifier is given, the complete list will

be printed; if a partial name is specified, however, all symbols that match that partial name are listed. See Table 5A-1 for list of symbolic names.

Examples:

***> SY <cr>**
<complete symbol list >
*>

* > SYMBOLS VR **<cr>**
VRGAIN
VRTIME
*>

*> SYM SPDREF **<cr>**
SPDREF
***>**

List of Available Symbols:

ALARM	ALPHFLD	ALPHLD	ALPHSRCO
ALPHSRC			
BBFS	BFDIFSW	BFILF	BFNDR
CBMODE	CCMND	CFBK	CFC
CHRNG	CITERM	CRALPHI	CPTERM
CRALPH2	CRERR	CRGAIN	CRMIN
CROUT	CRSTAR	CRSTART	CRTC
CSTEP	CURLL	CURRCMD	
DCL			
ESPDRQ	ESTOP		
FANGLAG	FCDELAY	FCFTIME	FCGAIN
FCLIM	FCLOSS	FCMAX	FCMIN
FCOLEN	FCSLOP	FCSTART	FLAGAIN
FLALEN	FLALIM	FLASEN	FLDCN
FLDCUR	FLDEMODE	FLDPU	
FLEARLY	FLFLT	FLPBRK1	FLPBRK2
FLPLIM	FLPMAX1	FLPMAX2	FLTWD
FLUXADJ	FLUXADJE	FLUXERR	FLUXERRA
FLUXFBK	FLUXREFL	FLUXS	FLXSP
FLXSRCT	FLXSRC	FLXLOV	
FRCLMP	FRQM	FRQS	FRGAIN
FRSTART	FRTC	FSTEP	
GTSTM			
GTSTM			

List of Available Symbols: (Continued)

HITQ	HPPLJ		
IDCPU	ILD	INTOSTOL	INVLIM
ISRC			
LBKRS	LIMARG	LKLIMD	LKLIML
LKLIMS	LKOCTIM	LMARG	LMODE
LOCAL	LPLLTIM	LSPD	LTRBLE
LVLTS			
MAMS	MB1ST	MB2ST	MB3ST
MB4ST	MFPK	MFWD	MISOLS
MODEFLG	MOVTEMP	MRPM	
MTRFMIN	MTRVLTS	MVPU	
NALM	NBLA	NBLF	NCFLT
NPSW	NSFLT		
OVCLIM	OVGAIN	OVLTE	OVSPIL
OVTC			
P900C	PCROT		
PLLMODE	PLLSERR	PLLSFLGL	PLLSFLGS
PLLSLD	PLLTOL	PO1	PO2
PO3	PO4	PO5	PO6
POE			
QCBTOL	QCMAX	QCMIN	QCVTOL
RAC	RAMPDEL	READY	REGOPEN
RETDLSRC			
RR12PA0	RR12PA1	RR12PA2	RR12PA3
RR12PA4	RR12PA5	RR12PD0	RR12PD1
RR12PD2	RR12PD3	RR12PD4	RR12PD5
RR6PA0	RR6PA1	RR6PA2	RR6PA3
RR6PA4	RR6PA5	RR6PD0	RR6PD1
RR6PD2	RR6PD3	RR6PD4	RR6PD5
RRBPO	RRBP1	RRBP2	RRBP3
RRBP4	RRMAXL	RRMAXR	RRMAX6P
RRMINL	RRMINR	RRMODE	RRTG
RSPD	RTONI	RTON	RUN
RUNP	RUNPM	RUNX	

List of Available Symbols: (Continued)

SFDBKL	SFDBK	SFLUXLIM	SIMARG
SMARG			
SPDEFIL	SPDRP	SPDSP	SPDSTP
SPDSTR	SPEED	SPLLTIM	
SQCB	SQFE	SQETST	SQIOTST
SQMCH	SQOPT1	SQOPT2	SQPLTIM
SQRA	SQRT	SQTMR1	SQTMR2
SQTMR3	SQTMR4	SQTMR5	SQTMR6
SRCLFC	SRCLSC		
SREFMD	SRERRF	SRETOL	SRGAIN
SRLLADJ	SRMAXR	SRMINR	SRTC
SSTEP	STRTPM	STHRUTIM	STIM
SUVFLT	SVPU	SWINDTOL	

TGRRQ	TIMESRC	TRNGR	TRSPD
TTRBLE	TWPLSE		

USERA

VLTSECCMD	VLTSECMTR	VLTSECSTR

5-3.8. TERMINAL

Function: to examine and possibly modify the terminal format. The monitor will display the current format (either "handheld" or "standard") and then ask the operator whether a change is required. Entering Y(ES) will cause the monitor to toggle to the other terminal format. Any other entry will cause no change.

Examples:

*> TERMINAL <cr>
HNDHLD-CHANGE? Y <cr>
STANDARD FORMAT
*>

*> TER <cr>
STD - CHANGE? <cr>
STANDARD FORMAT
*>

53.9. VERIFY

Function: displays the current state of all drive EEPROM settings and adjustments to allow the user to verify factory settings. Individual *groups* of variables may be displayed for verification by appending one or more group names to the VERIFY command (see example). The groups will be listed in the order specified. Following are the possible group names:

CR - Current Regulator
FI - Field Regulator
FL - Flux Reference
FR - Flux Regulator
MI - Misc. Variables
OV - Overvoltage Regulator
PR - Protection Variables
RR - Speed Ramp Rate Conditioning
SQ - General Sequencing
SR - Speed Regulator

Examples:

*> VERIFY <cr>
<grouped list of all software jumpers and adjustments >
*>

*> V SR, FR, SQ <cr>
(list of speed reg. software jumpers and adjustments>
(list of flux reg. software jumpers and adjustments >
(list of software sequencing jumpers >
*>

5-4. EXTENDED MODES

5-4.1. ENGINEERING

The ENGINEERING mode provides primitive functions like examining and modifying memory and input and output ports. Absolute addresses, rather than symbolic names, are used to access memory locations and I/O ports. Table 5-3 summarizes the command syntax for this mode of monitor operation.

Table 53 Syntax Definitions for ENGINEERING Level Commands

DISPLAY:	D[W] <start addr>[,<end addr>] <cr>
HELP:	H <cr>
INPUT:	I[W] <port addr>,[, . ..] < cr >
PUT:	P[W] <port addr>,[, . ..] < cr >
MOVE:	M < start addr > , < end addr > , < dest addr > < cr >
OUTPUT:	OIW] <port addr>, <data> [, <data> . ..] <cr>
QUIT:	{Q (EX} <cr>
SUBSTITUTE:	S[W] < addr >,[< new value >, . ..] < cr >

< start addr > :: = <pointer >
<port addr>:: = < hex no>
<end addr> :: = <pointer>
< new value>:: = <hex no>
< dest addr > :: = <pointer >
<data> :: = < hex no>
< addr> :: = <pointer >

5-4.2. DISPLAY

Function: to display the contents of a memory location or a block of memory. The current value of each byte (or word) is listed in hexadecimal form. Additionally, the ASCII equivalent of the memory shown on a line is displayed in a field to the right of the standard display. Only printable ASCII characters are shown --others are displayed as a period (.). The ASCII display function is suppressed when operating in the handheld terminal format.

Format:

D[W] c start addr >[, < end addr >] < cr >

Examples:

E> D DS:800 <cr>
F800:0800 41 A
E>

E> DW 200, 210 <cr>

0000:0200 55AA OOFF 1233 5256 4147 4E49
2020 0020 .U..3.VRGAIN .
0000:0210 4243 CB
E>

5-4.3. HELP ▪

Function: to list the available command set for the ENGINEERINGmode.

Examples:

E> H <cr>
< list of available commands >
E>

5-4.4. INPUT

Function: to display a byte or word at an input port.

Format:

IIW] <port addr> <cr>

Examples:

E> 19008 <cr>
7F ,
7F ,
7F <cr>
E>

E> IW 9008 <cr>
007F <cr>
E>

5-4.5. PUT

Function: to modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the PUT mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

P[W] <addr> <cr>

Examples:

E> P DS:A40 <cr>
FC00:0A40 A5- ,
FC00:0A41 5A- AA <cr>
E>

E> PW DS:50 <cr>
FC00:0050 OFFF- 7FF,
FC00:0051 5A5A- <cr>
E>

5-4.6. MOVE

Function: to move a block of data from one area of memory to another. Memory contents are transferred sequentially from low memory to high.

Format:

M <start addr> , <end addr> , <dest addr> <cr>

Examples:

E > M DS:50,59,CS:00 < cr >
E>

E > M 800,900,DS:42 < cr >
E>

5-4.7. OUTPUT

Function: to output a byte or word to an output port.

Format:

OIW] c port addr > , <data> [, <data> . ..] <cr>

Examples:

E> 0 8010, FF <cr>
• F8,
-FO <cr>
E>

5-4.8. QUIT

Function: to return from the ENGINEERING mode to the MONITOR mode.

Examples:

E> Q <cr>
*>
E> EX <cr>
*>

5-4.9. SUBSTITUTE

Function: to exam and modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the SUBSTITUTE mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

S[W] <addr> <cr>

Examples:

E> S DS:A40 <cr>
FC00:0A40 A5- ,
FC00:0A41 5A- AA <cr>
E>

E> SW DS:50 <cr>
FC00:0050 OFFF- 7FF,
FC00:0051 5A5A- <cr>
E>

5-4.10. TUNEUP

In the TUNEUP mode of monitor operation, the user is able to adjust the performance of the drive by modifying regulator parameters and configurations. A step-by-step tuneup procedure is contained in Chapter 6. Following are descriptions of the commands that are used as part of this procedure. Table 5-4 summarizes the command syntax for the TUNEUP mode.

For on -line tuneup of any regulator, the background sequencing RUN logic must be satisfied. Monitor commands alone will not allow the drive to run. This mode is intended to be used primarily when the drive is initially installed.

Table 5-4 Syntax Definitions for TUNEUP Level Commands

APPLY:	**AP(PLY)** < cr >
ASSIGN: *	**AS(SIGN)** {LIST | DA]
DISPLAY: *	**{D(ISPLAY)** { <varname > [C]| STATUS | LIST <**listname** > | } | **DS** I DL <**listname**>} <**cr**>
HELP:	H <**cr**>
MODIFY:	M(ODIFY) < varname > < cr > <**old** value message > < new value prompt > [I | D | <new value >] < cr >
MODIFY REFERENCE:	**M(ODIFY) REF(ERENCE)** < cr > or MREF
QUIT:	{Q | EX} <**cr**>
REMOVE:	**REM(OVE)** < cr >
RESTART	set password
SAVE:	**SAV(E)** < cr > <**verify** query> [Y | N] <**cr**>
SAVE D/A	**(SDA)** <**verify query**> [Y | N] <**cr**>
SYMBOLS *	**SY(MBOLS)** [< symbolname qualifier >] < cr >
TERMINAL: *	**T(ERMINAL)** < cr > <**current** format> CHANGE ? [Y | N] <**cr**>
TUNEUP SYMBOLS:	TSY(MBOLS) c cr >
TUNEUP:	**TUN(EUP) {F(IELD) | A(RMATURE)}** < cr >
VERIFY: *	**V(ERIFY)** [<symbol group name >] < cr >

* - Also available in MONITOR mode -- see section 5-3 for descriptions.

5-4.11. APPLY -

Function: to apply the test generator reference to the regulator being tuned up.

Examples:

T> AP <cr>
T>

5-4.12. HELP -

Function: to list the available command set for the TUNEUP mode.
Examples:

T> H <cr>
< list of available commands >
T>

5-4.13. MODIFY -

Function: to allow the operator to modify the contents of select drive control variables. A new value may be entered, or the current value may be incremented or decremented repeatedly. Also,the increment (or decrement) interval may be selected by placing a new interval after the I or D entry (see example below). If none is specified, the increment interval will be one unit of the least significant digit. Values are displayed and entered in appropriate engineering units.

**** NOTE ****

Because modifying certain memory locations could have adverse effects on the drive control operation, the MODIFY command may only be executed after completing the privilege enabling sequence.

Examples:

T> MODVRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? **15 <cr>**
VRGAIN = 15.0
NEW VALUE ? < cr >
T>

```
T> M VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? I <cr>
VRGAIN = 1.2
NEW VALUE ? I .3 <cr>
VRGAIN = 1.5
NEW VALUE ? I <cr>
VRGAIN = 1.8
NEW VALUE ? D <cr>
VRGAIN = 1.5
NEW VALUE ? <cr>
T>
```

**** NOTE ****

When a tuneup case is selected (section 5-4.20), a numbered list of symbols associated with that case is displayed at the terminal. While running that particular test, a symbol's NUMBER (from the list) rather than its name may be used as the argument in the DISPLAY and MODIFY commands. This eliminates the need to laboriously type out long symbol names at the terminal during a tuneup session.

Examples:

```
T> MOD3 <cr>
FRKGAIN = 1.1
NEW VALUE ? 1.3 <cr>;change gain from 1.1 to 1.3
FRKGAIN = 1.3
NEW VALUE ? <cr>
T>
```

5-4.14. MODIFY REFERENCE

Function: to modify (customize) an internal reference generator. When a tuneup case is selected, its corresponding reference generator is automatically selected (see Section 5-4.20). The user then has the option to modify the characteristics of the reference generator as shown in the following examples. A reference generator is a series of two steps, each with selectable amplitude and duration. Step durations are entered in seconds (i.e., .05 for 50 milliseconds) and reference levels are specified in per unit levels (i.e., for a current test, the level is in per unit current and for a speed test in per unit speed, etc.).

After the steps are fully defined and the test generator is enabled by the APPLY command, the regulator specified by the TUNEUP command will be excited with a periodic signal corresponding to the test reference levels defined. A fixed constant level may be obtained by setting the amplitudes of both steps to the same value.The REMOVE command disables the test generator and removes all excitation to the specified regulator.

Examples:

(Fixed Reference)

```
T> MOD REF <cr>
STEP        LEVEL(PU)        TIME(SEC)
  1          0.100             1.00
  2          0.000             1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.10 <cr>  ;make levels agree
STEP 2 DURATION = <cr>
T>
```

(Stepped Reference)

```
T> MODREF <cr>
STEP        LEVEL(PU)        TIME(SEC)
  1          0.100             1.00
  2          0.100             1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.00 <cr>  ;set for 10% step
STEP 2 DURATION = <cr>
T>
```

5-4.15. QUIT

Function: to return from the TUNEUP mode to the MONITOR mode. On leaving the TUNEUP mode, the monitor asks the user whether hewould like to have any new constant values transferred into the non-volatile memory (i.e., execute a SAVE command). This is so the operator will not inadvertantly forget to save the new drive status after finishing the TUNEUP sequence.

Examples:

```
T> Q <cr>
DO YOU WANT TO SAVE NEW CONSTANTS?
? Y <cr>
CONSTANTS SAVED
*>

T> EX <cr>
DO YOU WANT TO SAVE NEW CONSTANTS ?
? N <cr>
CONSTANTS NOT SAVED
*>
```

5-4.16. REMOVE

Function: to remove the test generator signal from the regulator being tuned up.

Examples:

```
T> REM <cr>
T>
```

5-4.17. SET PASSCODE ▪

Function: to to allow the user to put in his own password – up to 6 characters long. For this to be a permanent change it must be saved.

** NOTE **

For security passcodes are not
echoed
:Password address

Example:

OLD PASSCODE =: < OLD PASSCODE > < cr >
; password not echoed
NEW PASSCODE =: <NEW PASSCODE> <cr>
; password not echoes
NEW PASSCODE =: <NEW PASSCODE > <cr>
; verify to enable
T > ; can now save if want permanent change.

5-4.18. SAVE ▪

Function: to store the present values of the regulator variables in the system's nonvolatile memory. After modifying system parameters (for example, gains and time constants during a tuneup session), the new values may be transferred into permanent storageusing the SAVE command. The previously stored values, of course, are lost during this process; hence, the monitor will prompt the user for confirmation before actually executing the command.

NOTE: Privileges must be enabled to execute the SAVE command. Also, the hardware write enable switches on the DMPB cards must be in the "store enable" position before values may be stored in the nonvolatile memory.

Examples:

*> SAVE <cr>
ARE YOU SURE? Y < cr >
CONSTANTS
SAVED ;
saved if DMPB switch on

*>

*> SAVE <cr>
ARE YOU SURE? < cr >
CONSTANTS NOT SAVED
*>

5-4.19. SAVE D TO A (SDA)

Function: To store the present values of the D to A assignment variables in the system's nonvolotile memory. During tuneup mode, the D to A converters (1 and 2) are continually reassigned according to the default values required for the regulator being turned. If the D to A values were included in the global "SAVE" command, the intended setup values would be lost. A separate 'SAVE "SAVE" command is therefore included.

When it is required to modify, the D to A assignments permanently, the "ASSIGN DA" command used first be used to setup the required parameters.

The hardware write enable switches on the DMPB card must be in the "write enable" position before values may be stored in the nonvolatile memory.

Examples:

*> SDA <cr>
ARE YOU SURE? Y < cr >
CONSTANTS SAVED
* >

*> SDA <cr>
ARE YOU SURE? <cr >
CONSTANTS NOT SAVED
*>

5-4.20. TUNEUP SYMBOLS ▪

Function: to list all tuneup variables associated with the currenttuneup case.Examples:

T> TSY <cr>
TUNEUP VARIABLES ARE:
<numbered symbol list > or NONE
T>

5-4.21. TUNEUP ▪

Function: to initiate a particular regulator tuneup sequence. The user must specify which of several regulator is to be tuned. He is then allowed to select the desired tuneup case. Examples of valid commands are:

CR ▪ Current Regulator
FI - Field Regulator
FL ▪ Flux Reference
FR ▪ Flux Regulator
MI - Misc. Variables
OV ▪ Overvoltage Regulator
PR - Protection Variables
RR ▪ Speed Ramp Rate Conditioning
SQ - General Sequencing
SR - Speed Regulator

Example:

(Current Regulator Tuneup)
T> TUNE CR <cr>
AVAILABLE VARIABLES ARE:

1-CRALPH1 = 90
2-CRALPH2 = 86
3-CRGAIN = 338
4-CRTC = 0.020
5-CRMIN = 0.20
6-CRSTART = 0.25
T>

5-4.22. Monitor Commands -

The following main-level monitor commands are also available in the TUNEUP mode:

ASSIGN
DISPLAY
SYMBOLS
TERMINAL
VERIFY

The syntax definitions and functional descriptions for these commands are contained in section 5-3.

5-5. LOADING DISPLAY MESSAGES HEADERS

Customized display messafges can be loadeed into EEPROM to identify the user, drive, personnel to **contact,** actions to be taken, etc. Five (5) different messages can be loadeed - starting at the locations below. Each message can be up to 64 characters long (minus 1 for end of message = 63).

Starting Address	Message type
3000:200	Header - message printed at power-up
3000:240	Repetition - message printed at beginning of "data logging"
3000:280	Requisition - message printed when "Status" P.B. used
3000:2C0	Fault - message printed at beginning of "Fault" messages
3000:300	Alarm - message printed at beginning of "Alarm" messages

Procedure (see example 5B-1):

1. Write the message out on Form 5B-1 (remember to leave 2 spaces) for < cr > < LF > (by word) if you don't need to force a new line.
2. Convert the characters to their ASCII equivalent using Table 5B-1. Add after the last conversion 00 to signal end-of-message.
3. Using the monitor Put command, load the ASCII equivalents.

** NOTE **

You can use a instead of a < cr > after each ASCII code to increment to the next location - rather than reloading the address (faster).

Suggestions

Always include user's name and the drive/channel #. For the fault, alarm, and repetition messages, a reminder such as "Log items in maint book' may be useful. The default messages should be okay. It should only be necessary to add the user's name and drive I.D. to the beginning (if sufficient space) or to the end (will need to move "00" of the default message.

```
MONITOR OFF < cr >     ; Form 5B-1 or 5B-2 should
                         be filled out.

LCI MONITOR
* > PASSWORD
PASSWORD ? <cr>

PRIV. ENABLED
*> ENG<cr>

E > PUT 3000:200 < cr >   ; Enable EEPROM write
                            switch

3000:200 ??- 53,
3000;201 ?? 43,           ; Continue till all characters done
   :

3000:2??  ??- 00 < cr >
```

Example 5B-1 : MESSAGE HEADERS

Message Type _____

Starting Address - 3000: _____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01___	___	33___	___
02___	___	3 4 -	___
03___	___	3 5 -	___
04___	___	36___	___
05___	___	3 7 -	___
06___	___	3 8 , ,	___
07___	___	3 9 -	___
O S , ,	___	4 0 , ,	___
09___	___	41___	___
l o -	___	42___	___
l l -	___	43___	___
12___	___	44___	___
13___	___	45___	___
14___	___	46___	___
15___	___	47___	___
16___	___	48___	___
17___	___	49___	___
18___	___	50___	___
19___	___	51___	___
20___	___	52___	___
21___	___	53___	___
22___	___	54___	___
23___	___	55___	___
24___	___	56___	___
25___	___	57___	___
26___	___	58___	___
27___	___	59___	___
28___	___	60___	___
29___	___	61___	___
30___	___	62___	___
31___	___	63___	___
32___	___	64 < EOM >	00

FORM 5B-1: For Header Message

Message Type _____

Starting Address - 3000: _____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01	20	33 d	64
02	20	34	20
03	20	35 i	69
04	20	36 t	74
05	20	37 e	65
06	20	38 m	6D
07	20	39	20
08	20	40 i	61
09	20	41 n	6E
10	20	42	20
11	20	43 L	4C
12	20	44 O	4F
13	20	45 G	47
14	20	46	47
15	20	47 b	20
16	20	48 o	6F
17	20	49	6F
18	20	50 k	6B
19	20	51	00
20	20	52	
21	20	53	
22	20	54	
23	20	55	
24	20	56	
25	20	57	
26 <cr>	0D	58	
27 <lf>	0A	59	
28 R	52	60	
29 e	65,	61	
30 c	63	62	
31 o	6F	63	
32 r	72	64 <EOM>	00

FORM 5B-2: Default for Header Message

Character	ASCII	Character	ASCII
A	41	a	61
B	42	b	62
C	43	c	63
D	44	d	64
E	45	e	65
F	46	f	66
G	47	g	67
H	48	h	68
I	49	i	69
J	4A	j	6A
K	4B	k	6B
L	4c	l	6C
M	4D	m	6D
N	4E	n	6E
0	4F	o	6F
P	50	P	70
Q	51	q	71
R	52	r	72
S	53	s	73
T	54	t	74
U	55	u	75
V	56	v	76
W	57	w	77
X	58	x	78
Y	59	Y	79
Z	60	z	80
0	30	< space >	20
1	31	<cr>	OD
3	32	<1f>	OA
3	33	#	23
4	34	&	26
5	35	'	2c
6	36	-	2D
7	37		2E
8	38		3A
9	39	@	40

TABLE 5B-1: ASCII Character Set

Message Type / Y_____

Starting Address - 3000:___ANY

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01_S_	53,	33_d_	64_
02x-	43,	34___	20,
03-0,	4F_	35_i_	69_
04_T_	54_	36_t_	74_
0.5_T_	54_	37- e-	65,
06___	20_	38_m_	6D_
07_P_	50_	39___	20_
08_A_	41_	40_i_	61_
09_P_	50_	41_n_	6E_
10_E_	45,	42___	20_
11_R_	52_	43_L_	4C_
12 < cr >	OD_	44_O_	4F_
13	OA_	45_G_	47_
14_D_	44_	46___	20_
15_r_	72_	47_b_	62_
16_i_	69_	48_o_	6F_
17_v_	76, ,	49- 0-	6F_
18_e_	65,	50_k_	6B_
19_#_	23_	51	00_
20_1_	31_	52___	___
21___	20_	53___	___
22_c_	43_	54___	___
23_H_	48,	55___	___
24_._	2E_	56___	___
25_A_	41_	57___	___
26 < cr >	0D_	58___	___
27 < 1f >	0A_	59___	___
28_R_	52_	60___	___
29- e-	65_	61___	___
30_c_	63_	62___	___
31_0_	6F_	63___	___
32_r_	72_	64 < EOM >	00

FORM 5B-2: Default for Header Message

EXAMPLE 5B-1: MESSAGE HEADERS

112

CHAPTER 6
INSTALLATION AND STARTUP GUIDE (OUTLINE)

6-1. INTRODUCTION

1. Keep detailed installation log for each drive
2. Install Equipment
   (Align motor: Set up: coupling to run uncoupled from driven device; Motor protectives - lubrication, thermocouples, RTDs, vibration detectors; Transformers - fault relays, liquid alarm, overtemperature alarm, CTs, RTDs; Source and motor breakers - close and trip circuits, manual trip, IOC/LOR)
3. Interconnect equipment
4. Check Wiring
5. Checkout source switchgear and load isolators
6. Preliminary Checkout
   Check Berg jumpers, daughter board components, burden resistors
7. Apply control power
   Check out starters and blowers (Rotation)
   Check power supplies - PSFF, P105
   Check relays and lights
   Check/adjust electronics calibration voltages
8. Gate test mode Operation
   Check source and load side converter firing
   Check process inputs
   Check process outputs
   Adjust dc offset for flux waveform at NLIB cards
9. Apply source voltage
   Check attenuator signals
   Magnitude and scaling
   Ph sequence
   DC to ground
   Flux waves
   Zero crossings
   VABS, OV, UV
10. Exciter voltage control
    Measure voltage, Ph sequence, current
    Measure field signals at SEM
11. Perform crowbar test
12. Run motor uncoupled - check lube, cooling vibration
    Local
    Remote
    Accelerate, decelerate, stop, regenerate
    Check input, output signals and contacts
    Repeat all of the above for other channel (B) – if 12 pulse or standby system
13. Couple motor and tune-up the drive evaluate performance

6-1.1. DETAILED INSTALLATION INSTRUCTIONS

It is assumed that all interconnecting wire and cables are in place and connected. The erection and alignment of the motors is not described here, nor is the routine checkout of the circuit interrupters. It is recommended that a detailed installation log be maintained for each drive.

Preliminary Considerations

PWB Variable Components Refer to system elementary. Extract each PWB listed on elementary sheet 1DA and confirm that Berg jumpers are in the position indicated and that variable components are the values shown in the tables.

Megger Tests

Preparation - Temporarily disconnect ribbon cable and chassis connections as follows:

MVB:
Disconnect the ribbon cable to each HPTK card in the converter leg modules. Also disconnect the 'GND' connection to the HPTK and the XPTN in each leg module. Disconnect the load-side line filters at the fuses.

Silco - Disconnect the attenuattor (JE-4) cable to each HPTF card in the leg modules. Disconnect the load-side line filters at the fuses.

Megger **Power Busses** - Confirm that both the source-side circuit interrupter and the load-side circuit interrupter are open.

Temporarily jumper the converter source-side terminals (AC1, AC2, AC3), load-side terminals (Ml, M2, M3), and the dc buses (P1, N1, P2, N2) together. Megger from any bus to ground (they are all tied together} preferably using at least a 1000V megger. Indicated resistance should be greater than 100 megohms.

Remove jumpers and reconnect the ribbon and chassis cables. Reconnect the load-side line filter.

Jumper the load-side circuit interrupter load-side terminals (T1, T2, T3) together and megger to ground. Indicated resistance should exceed 100 megohms. Remove jumpers.

Check continuity across cells
Using a multimeter, check across each cell in the bridge. Measurement should indicate approximately 10K ohms. If greater, check the equalizing resistors in the snubber boxes. If **less** than 100 ohms, check for defective snubber capacitor.

Apply Auxiliary Power
Open the electronics power circuit breaker at the power supply. Close-in the 120Vac control power (May be off 480Vac).

Close-in the 480Vac breaker at the motor control center for drive blowers and confirm that the blower(s) bring air through the converters front grill and out the top of the fan assembly. Also check direction of rotation.

Signal Input Power Supply Measure output of P105 volt power supply on the I/O module and confirm that voltage output is approximately 105Vdc.

Electronics Power Supplies Pull all the cards in the SEM. Turn on the power supply and measure the bus voltages at the test rings on the small electronics modules (SEMs) header (top). Voltages should be as follows:

(See elementary sheet 1BB for trimpot adjustments)

P5 = + 5.0V
P28 = +28.0V (+2V)
P15 = + 15.0V
N15 = -15.0V

Turn off electronics power supply, check that PROMS are in the HMPG and DMPB and reinsert the cards into the SEM. Turn on power supply and check the bus voltages again.

Set-up Panel Printer - Consult the panel printer's manual to enable the battery backup (clock). Then set the time and date. Also set the internal time greater than 60 minutes.

Run LCI Selftest:
Turn-on (up) NLCB test switch. Reset HMPG by jumpering TP1(RST) to TP2(DCOM). Follow the indicated instruction and investigate any fault messages. Repeat till unit passes all tests.

Operating Set-up Adjustments and EEPROM Memory Initialization:
Measure and set the voltages for all the OV, SUPP, RTD, UVR, OSP test rings as tabulated in the system elementary sheet 1DC.

Reset SEM, using the HLCB RESET switch; the HMPG IMOK LED should come on. Jumper testpoint TINO on HLCB-2E to DCOM, turn-on(up) WREN switch on DMPB(HMPG), WREN LED on the DMPB card should come on and reset SEM . Remove the jumper and turn off WREN switch. Dump EEPROM memory (using VERIFY) on a printing terminal and compare to values shown on system elementary (sheet 1DE, 1DF, 1DG).

Bridge Current Feedback Calibration (MVB):

Verify resistors and Berg jumpers on the XTFH against system elementary (sheet DA). Remove CT lead plugs JR and JT from the XTFH. Measure the resistance across each of the burden resistors on the XTFH; it should equal the "R BURDEN SETTING" value per the elementary. Adjust the corresponding pot to fine tune. Re-connect plugs JR and JT.

Bridge Current Feedback Calibration (Silco) -Verify resistors and Berg jumpers on the XTFX-S and the XTFX-L against system elementary (sheet DA). Remove CT lead plugs JJ from the XTFX. Measure the resistance acrosss each of the burden resistors on the XTFX; it should equal the "R BURDEN SETTING" value per the elementary. Adjust the corresponding pot to fine tune. Re-connect plugs JJ.

Gate Test Mode Operation

Refer to the paragraph entitled GATING TESTS in Chapter 7 for a description of procedure.

Verify that the source-side circuit interrupt and load-side circuit interrupter are open.

Follow the procedure described in GATING TESTS.

Check Process Inputs

Verify that the source-side circuit interrupter and load side circuit interrupter are open and locked out.

The purpose of the tests in this section is to operate each external input shown on system elementary sheets BD, BE into the SEM. Operate each of these inputs by the most convenient means and confirm that the correct LED lights with a closed circuit in each case.

Check Process Outputs

Verify that the source contactor and load isolator are open and locked out.

Connect the hand terminal (or equivalent) to the HMPG and MODIFY EEPROMs variable (SQIOTST) to "T" to temporarily disable the interrupts. (See Chapter 5, section MONITOR - section EXTENDED MODES for procedure.)

Using the OUTPUT command, operate the output relays shown on system elementary sheets BG, BJ in accordance with the tables below. Confirm that the relays operate the proper device(s) per the system elementary.

Relays may be operated in any desired combination by adding the appropriate DATA values from Table 6-l.

Adjust dc offset for flux waveform at NLIB cards.

This procedure should be performed whenever the NLIB card is replaced by a new one.

When the source power is off and the control is not in gate test mode, measure TP4, TP5 and TP6 at NLIB with a DUM in millivolts dc range. Adjust offset pots R1, R2 and R3 respectively on the NLIB mother board so that TP4, TP5 and TP6 read zero millvolt dc.

If a DS3820SIM module is provided, and one or more isolated output from the module are from the D to A port then use the "ASSIGN DA" command to assign the correct variable, then use "SDA" command to save the variable in the nonvolatile memory. See elementary sheet CQ.

Re-MODIFY SQUITST to original value (F).

No-Load Full-Voltage Tests

The tests in this section require that the source power be connected and available. Open and lock-out the load side circuit interrupter. (If no interrupter is available disconnect leads to motor.)

Apply the control power, reset the SEM and check that all the IMOK LEDs are on. Check for any fault indications. Investigate and clear any indicated faults. Null all meters.

Close source side circuit interrupter. Observe that all source side bridge neon lamps are illuminated.

Voltage Phasing and Balancing

A. Connect an oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2C. Confirm that the phase sequence is l-2-3 and that all these are appromimately equal.

The actual system voltage = (ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet 1EA.

Note: Must measure between TP and ACOM testpoint on these cards.

Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:

$$FAB = \frac{\text{Actual System Voltage}}{\text{Nominal System Voltage}} \times 4.75 \text{ Vac rms}$$

(See elementary sheet 1EA for nominal system voltage).

Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for dc offset at TP4, TP5 and TP6 of NLIB (< 100 millivolts).

C. Check for presence of zero crossing pulses at TP8 (SFT1) on the HLCB-2E. These are narrow 2.5V pulses evenly spaced at 2.78 millisecond intervals.

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the dc voltage level is approximately 2.229 X Vac at TP9 of NSFC-2C.

Set Overvoltage:

Connect thevoltmeter to TP18 (OVR) on the NLIB-1C. Turn R8 (OV) slowly counterclockwise until the "OV" LED comes on. The source side circuit interrupter should trip.

Observe the dc voltage on TP-18 and re-adjust R8 so that the voltage is per unit source overvoltage times the trip value. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See elementary sheet dc for per unit source overvoltage.

Set Undervoltage.

Clear any faults and close the line contactor. Connect voltmeter to TP19 (UVR) of NLIB-1C.

TABLE 6-l. SLOT 1A

ADDRESS	DATA	LED ON
8040 :	01 :	0
8040 :	02 :	1
8040 :	04 :	2
8040 :	08 :	3
8040 :	10 :	4
8040 :	20 :	5
8040 :	40 :	6
8040 :	80 :	7
8044 :	01 :	8
8044 :	02 :	9
8044 :	04 :	A
8044 :	08 :	B
8044 :	10 :	C
8044 :	20 :	D
8044 :	40 :	E
8044 :	80 :	F

Connect an oscilloscope to pin 55-1C; this should be a logic zero. Turn R9 (UV) slowly CW until pin 5.5 just transitions high (may alternate high and low).

Observe TP19-1C and adjust UV(R9)-1C so that the measured value is per unit source undervoltage X trip level. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See system elementary sheet 1DC for per unit source under-voltage.

Load Side Bridge Voltage Tests with signals from Source Side Bridge

Trip source circuit interrupter. Disconnect ribbon cable JA from NSFC-2B and move the ribbon cable JA from NSFC-2C to NSFC-2B. Connect a test jumper from backplane pins 1C-55 to 1C-1 (disables source under-voltage fault).

Voltage Signal Phasing and Balancing

Close source circuit interrupter.

A. Connect oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2B. Confirm that the phase sequence is l-2-3 and all balanced. The testpoints should measure the same as for the source-side. Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1D. Adjust R1 (FLXAB) to obtain voltage as calculated ine Equation 1.

Rated frequency and nominal motor volts from elementary sheet 1EA.

Now connect the voltmeter to TP2 (FBC) and adjust R2 (FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for dc offset at TP4, TP5 and TP6 of NLIB (< 100 millivolts). potentiometer.

C. Check for the presence of zero-crossing pulses at TP9 (SFT2) on the HLCB-2E. These are narrow 5Vdc pulses evenly spaced at 2.78 millisecond intervals.

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1D and check that the dc voltage level is the same as for the source-side 2.229 X VAC at TP9 of NSFC-2B

Set Overvoltage Connect voltmeter to TP18 (OVR) on the NLIB-1D. Turn R8 (OV) slowly CCW until the "OVA" LED comes on. The motor contactor should trip. Observe the dc voltage on TP18 and re-adjust R8 so that the voltage is as calculated in Equation 2.

Set Speed and Voltage Indicators

Since speed is derived from the motor voltage frequency, the source 60 Hz can be used to set the overspeed.

Set OSP -lE to + 11V dc. Close-in line contactor. Connect a voltmeter to TP1-1E (SPD). Adjust R1 (SPD) until TP1 = 10.0 V dc.

Speed Indication:

Adjust R4 (SPM)-1E so that the speed indicator on the control panel reads rated motor speed for 60 Hz.

Motor Voltage Indication:

Adjust VM (R7) -lD on DLIB so that the motor voltmeter reads actual line voltage.

Set Overspeed:

Connect voltmeter to testpoint OSP (TP6)-1E and slowly turn OSP (R2)-1E CCW until the OSP LED comes on. The voltage on TP6 should be + 10 V dc at the trip point. See elementary sheet DC for per unit overspeed.

Adjust OSP until TP6 = [per unit overspeed] X trip level.

Trip the source circuit interrupter.

Restore NSFC ribbon cables to correct configuration.

Remove jumpers.

Eauation 1

$$FAB = \frac{\text{Actual System Voltage}}{\text{Nominal Motor Voltage}} X 4.75 Vacrms X \frac{\text{Rated Freq.}}{60}$$

Equation 2

$$\text{"OVA" volts} = 1.2 \text{ X trip value X } \frac{\text{Nominal Motor Voltage}}{\text{Actual System voltage}}$$

Equation 3

$$TP\ IFB\text{-}2A = \frac{\text{Rated exciter current}}{\text{Rated EVC output}} 5V\ dc.$$

(See elementary Sh. 1EA)

Exciter Voltage Control

Preliminary Checkout:

Field Exciter Current Feedback Calibration

Remove CT lead plugs JB and JC from the NHVL. Measure the resistance across the burden resistance; It should equal the "Field R burden setting" value given on system elementary sheet 1EA. Adjust R (IFB) to fine tune. Reconnect JB and JC.

Phase Sequence Check:

Apply source voltage (and 480Vac). Connect an oscilloscope to simultaneouly observe the waveforms on testpoints ATTEN-AI(TP1), ATTEN-BI(TP3), ATTEN-CI(TP5) on the NHVL and confirm that the phase sequence is l-2-3 and are balanced. Close-in the source switchgear and using a oscilloscope, compare the TP9-2C(VAC-NSFC) against TP9-2A (LVAC-NSFE). Measure # of degrees LVAC-2A leads (-) or lags (+) VAC-2C.

Record, modify and save FANGLAG to value. (Typical values of -30, 0, +30).

Power Check:

Attach a voltmeter across one phase of the exciter output. Attach a clamp-on ammeter around one of the output lines.

Control From SEM

1. Set local speed reference to minimum (drive control in local).
2. Open the flux regulator by setting SQFE to "T"rue (Example 6-l). (Set SQETST to "T" if not already).
3. Using the local speed command reference, increase the exciter current. Stop when rated exciter current is achieved as measured on the clamp-on. (Verify Equation 3.)
4. Verify all 3 phase currents and voltages are equal
5. Reduce exciter current to zero and SQFE to "F"alse. (Reset SQETST to "F" if changed in step two).

MONITOR OFF <cr> ; Adjust local speed reference pot to minimum

LCI MONITOR

* >PASSWORD <cr>
PASSWORD ? < cr>
PRIV. ENABLED
* > T U N <cr>

T > MOD SQFE <cr>
SQFE = F
VALUE? = T <cr>
SQFE = T ; Adjust local reference to get desired
VALUE ? = F < cr> EVC current ; when finished.
SQFE = F
VALUE? = <cr>

Example 6-l : EVC TEST

Crowbar Current Tests

Crowbar Test Description The crowbar test permits controllable dc link current to be circulated without involving the motor windings. On the load side, the current circulates through one or more diametric cell pairs and thus bypasses the motor. The signals to be observed on the oscilloscope will indicate whether the source-side CTs are phased correctly. For the tests in this section, the load-side circuit interrupter should be open.

Test Setup:

A. Connect one channel of the oscilloscope to firing pulse #1 (SIG1-2C) on NSFC. Connect second channel to testpont OIA (TP9)-1C on NLIB.

Establish Initial Conditions:

Connect a hand terminal to the HMPG. Set-up the following EEPROM variables (Example 6-2).

A. Reset SEM and apply primary power.

B. Connect the hand terminal to the HMPG.

C. Enable crowbar mode;
Modify variable SQCB to "T"rue

D. Set minimum current limit to zero;
Modify CRMIN to 0

E, Set starting current to zero.
Modify CRSTART to 0

Actual Test:

A. Initiate a run sequence; i.e., sequence drive to pick-up the software logic ladder's RUNFLG. The current regulator should now be active but regulating zero current. Check that word MODEFLAG - Bit 0 or 1 is set.

B. Using the modify command, set integer CRSTART to .1 PU (10% current **command). Check the** IFB(TP7)-1C on the NLIB for -0.5Vdc + 0.2Vdc. Current IFB should appear stable as viewed on an oscilloscope.

C. Increase CRSTART to .2 (20% current command) using the MOD command. Current should now be continuous (not going to zero). Observe the testpoint pairs given in Figure 6-1, and check for proper phasing.

Note: When circulating current in the crowbar mode, current should be applied only for short periods of time (about 15 second duration and 25% duty) when operating above .2 PU current.

Take scope traces of all three currents as indicated in Figure 6-1 ; record peak volts. This also checks source-side CT phasing.

D. Ammeter Check

1. Apply .5 P.U. (CRSTART) and check testpoint IFB-1C for .5 P.U. current (approximately 2.5Vdc +20%). Adjust ammeter to read 50% motor current (see motor nameplate).

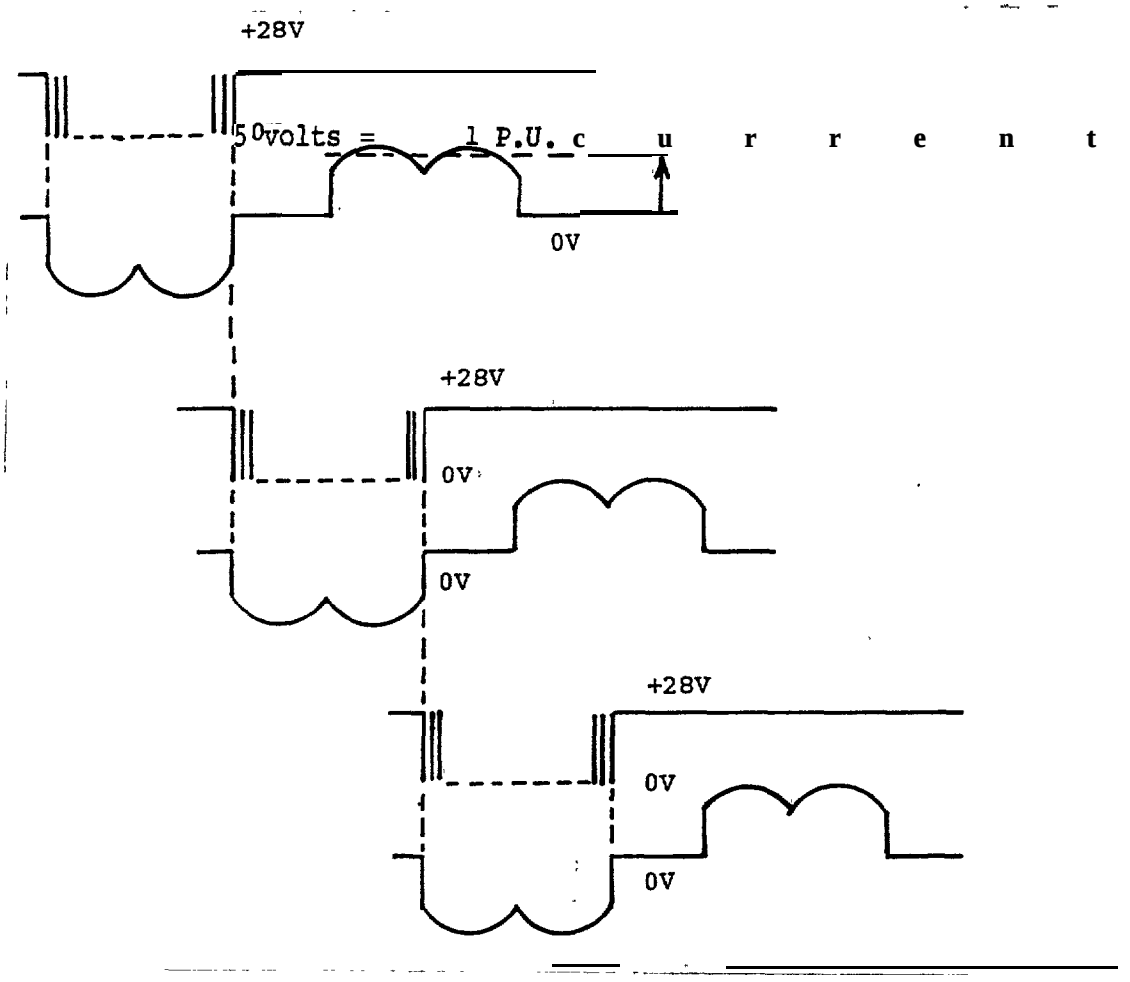


Figure **6-1. Firing Signals and Current Signals**

E. Source Hardware Suppression

1. Adjust SUPP (R6)-1C on NLIB to minimum (fully CW).
2. Initiate current flow and adjust CRSTART up from .5 P.U. At .6 P.U. ± 30%, the SUPPression LED on the NLIB-1C should come on and the line contactor should trip.
3. Readjust SUPP (R6)-1G for the value shown on system elementary sheet DC.

F. Modify variable SQCB, CRSTART and CRMIN to original value and reset SEM.

```
MONITOR OFF <cr>

LCI MONITOR
* > PASSWORD
PASSWORD ? <cr>
PRIV. ENABLED
* >TUN <cr>
T >MOD SQCB <cr>
SQCB = F
VALUE ? T <cr>
T> MOD CRMIN <cr>
CRMIN = 0.20
VALUE? 0 <cr>
CRMIN = 0.0
VALUE ? <cr>
VALUE ? <cr>
T >MOD CRSTART <cr>
CRSTART = 0.5
VALUE ? 0 <cr>
CRSTART = 0.0
VALUE ? < cr >
T >MOD CRSTART <cr>
CRSTART = 0.0          ;Start drive
VALUE ? 0 < cr>
CRSTART = 0.00         ;Check IFB
VALUE I .1 < cr >      ;Increase by 0.1
CRSTART = 0.1  ;Check IFB level and if continous
VALUE ? I <cr> ;Continue increasing I till
                       IFB = 0.5 pu


CCMD = 0.5       ;Check IFB and calibrate ammeter
VALUE ? I <cr>  ;Adjust SUPP to min and
                   increase I till trip
                ;Approximately 0.6 pu
```

Example 6-2:

Run Motor Uncoupled

A. Test Connections

Considerations:
Motor bearing protection must be functioning at this time to protect the motor. Once the motor is started, it should be run-up to full speed and magnetic center checked, bearing temperature rise checked, bearing oil (sling) rings checked, and vibration checked.

1. Make the following connections to a six channel chart recorder:
   CH1 - DA1 TP1-2E
   CH2 - OIA TP11-1D
   CH3 - FCBR TP5-1D
   CH4 - OIC TP9-1D
   CHS - IFB TP7-1C
   CH6 - SPD TP5-1E

2. Temporarily modify the following EEPROM variables:
   SRGAIN to 4.5
   SRTC to 1.0
   MTRFMIN to .25pu
   CRSTART to .125pu
   FLALEN to "F'
   FLASEN to "F"

   FLPBRK1 to 1.1
   FLPBRK2 to 1.1
   FLPMAX1 to 0.5
   FLPMAX2 to 0.5

3. Assign LMODE to D/A #1
   (Scale = -5)
   (offset = 0

**** NOTE ****

Scale is minus 5 which means multiply LMODE by 32 before output to D/A port. (See S-6 for this procedure)

B. Start and Run Motor

Check that auxiliary and control power is applied.

1. Reset processor and clear any faults and/or alarms.
   Close source side circuit interrupter.
2. Select local speed reference and set that reference to minimum.
3. Initiate run. Motor should start in made "0"; switch to mode "1", switch to mode "2" at about 2 to 6 Hz and then switch to mode "3" at about 7.5% speed. Check motor rotation. If motor runs backward, check LCI-to-motor wiring; swap two motor leads if necessary.

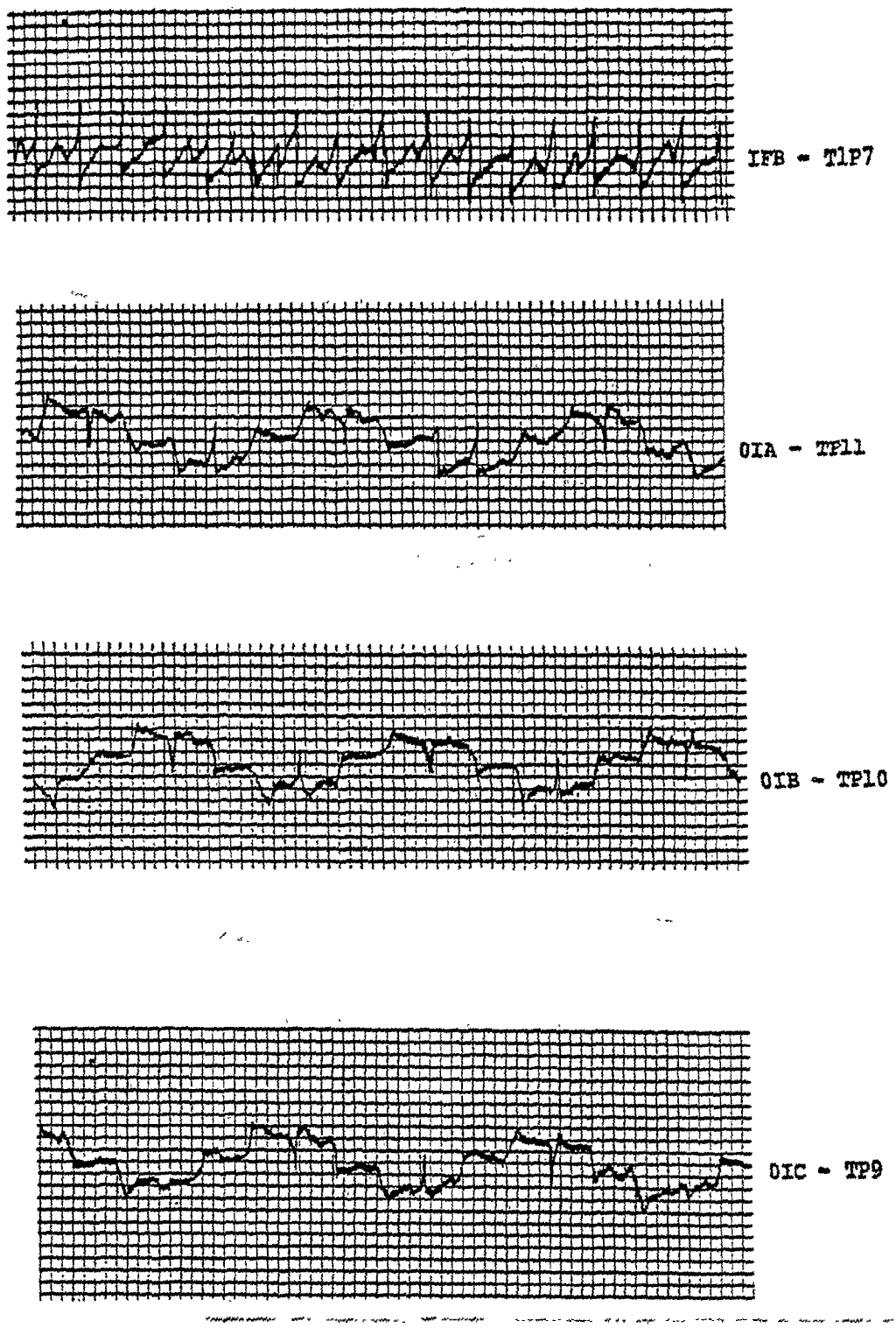


Figure 6-2. Phase **Current** Relationships

If motor fails to start, observe the memory' variable FLUXFDBK. Then adjust the EEPROM variable MTRFMIN down until a start is obtained, or see Section B of "O"perate with motor coupled.

Running mode is determined by the voltage level out of the DALO testpoint of the DA1-2E (channel 1 of recorder) as follows:

Mode		
0	0 Vdc	IDLE (No firing)
1	2.5 Vdc	Forced Commutated (PLL open-fire 30" segments).
2	5.0 Vdc	Forced Commutated with PLL closed.
3	7.5 Vdc	Self Commutated using PLL.

4. Initiate stop and motor should coast to a stop.
5. Restart and check the load-side NSFC(2B) and NLIB(1D) are like figure 6-1.

C. Speed Range Test

Start the drive and gradually increase the drive speed to rated value. If motor voltage fails to increase, reverse two field leads.

Monitor TP5 (SPD)-2E. Voltage should be -10 Vdc. Panel speed meter should read rated rpm.

D. Speed Ramp Test

Initiate a run from minimum speed to top speed by changing the reference as fast as possible, while recording the speed signal. Verify that the acceleration rate is correct.

Reduce reference to minimum as fast as possible while recording the speed signal. Verify that the deceleration rate is correct.

E. Operation from Remote Speed Control

Stop the drive and remove any test jumpers. Place drive in "remote" mode. Reduce the remote ref. signal to minimum. Start drive from "remote". Drive should run up to minimum speed. Increase the speed, verifying that the drive follows the reference up to maximum speed. Decrease reference while verifying that the speed decreases.

F. 12-Pulse Checkout

Preliminary: Once one channel of the 12-pulse drive has been run, the EEPROM in the idle channel should be updated to that of the previously run channel. Both channel's EEPROM must be identical before the drive can be run as a 12-pulse - except for SQMCH (Channel A should be "T" and channel B should be "F").

Uncoupled: After checking out and running the motor uncoupled from both channels (one-at-a-time &pulse), test the 12 pulse link. Start Channel A (Channel B remains stopped) and run to minimum speed. Verify DA4-2E (TP5-HLCB) on Channel A equals 12P-1E (TP2-NLCB) on Channel B.

Verify that channel B is slave and 12 pulse. Start channel B and see that the currents balance. Run the drive up near full speed, checking that currents, voltages, and flux are balanced. Reduce speed, stop A, then restart A.

Stop both channels and then start both channels simultaneously. If starting is now too fast, reduce CRSTART, SRCLFC, CRALPHA1, and CRALPHA2 on both channels.

Z. Modify EEPROM

Variables changed for the uncoupled tests back to their default values. These variables include:

RRMINL, SRMINR, CRSTART, SRGAIN, SRTC, MTRFMIN, CRMIN, CRALPH1, CRALPH2.

**** NOTE ****

Leave (SAVE), FLALEN and FLASEN as "F"alse till told otherwise.

Operate with Motor Coupled (6-Pulse)

A. Test Connections

Make the following connections to a six channel chart recorder.

CH1 - DA1 TP1-2E
CH2 - OIA TP11-1D
CH3 - FCBR TP5-1D
CH4 - OIC TP9-1D
CH5 - IFB TP7-1C
CH6 - SPD TP5-1E

B. Starting Tune-up

The start-up of the LCI should be optimized by varying EEPROM variables MTRFMIN. The procedure is as follows:

1. Connect terminal to the HMPG.

2. Attempt to start drive and monitor flux FCBR (TP5-1D) on chart recorder. If motor fails to start, stop and start drive several times - adjusting STIM to try to maximize flux during the first few firings. If motor still fails to start then modify MTRFMIN down until drive starts. Modify and save .8 x the value of MTRFMIN into EEPROM. Modify and save FLUXs to .9x and value of MTRFMIN. Record.

3. Tune up current regulator in SEG FIRE.

   Modify EEPROM variable as follows:

   SRMINR to .05pu
   RRMINL to 0.05pu

12/

This will allow the drive to operate continuously in SEG FIRE with the speed reference at zero. The purpose of this tune-up is to minimize rise time and overshoot of the current regulator during segment firing. Figure 6-3 shows an idealized trace of IFB (TP7)-1C.

4. Stop and start the drive several times from a dead stop. If it fails to start quickly every time, monitor flux feedback signal FCBR (TP5-1D), on a chart recorder. Then stop the drive and lower MTRFMIN by .01 steps to a minimum of .04. Repeat as necessary, trying to maximize the flux signal during the first few pulses when the motor is trying to start. Save and record the final value used.

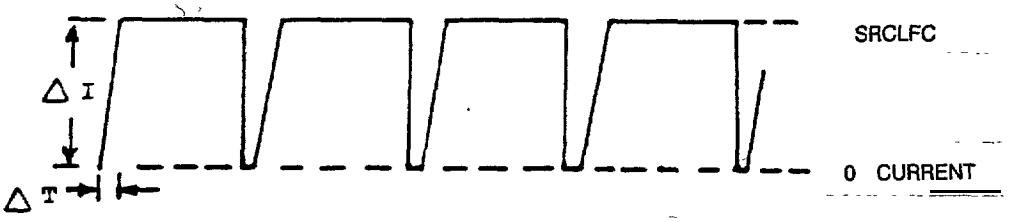


Figure 6-3. IFB during SEGFIRE

The variables to be set up are CRALPHl and CRALPH2. CRALPHl initializes the first source-side firing after a current zero to a value that should force link current to equal the setting of CRMIN. CRALPH2 then initializes the current regulator to call for a value of firing which will sustain this current. On the third and subsequent firings, the current regulator becomes active.

Start-up the drive and run at 5% speed. Confirm that the drive is in SEG FIRE. Run the chart recorder at 125mm/sec and obtain a trace of IFB. Compare the trace obtained with the cases shown in Figure 6-4 and take appropriate tune-up action. Zap final values of CRALPHl and CRALPH2 into EEPROM. Record for factory records.

C. Motor Flux Check

1. Start the drive and run up to some speed between 50% and 75%. Connect oscilloscope to testpoint"VAC"-2B. The voltage at testpoints FBAR, FCBR, and FACR should appear as in the waveform shown in Figure 6-6.

It is important that these be balanced as closely as possible.

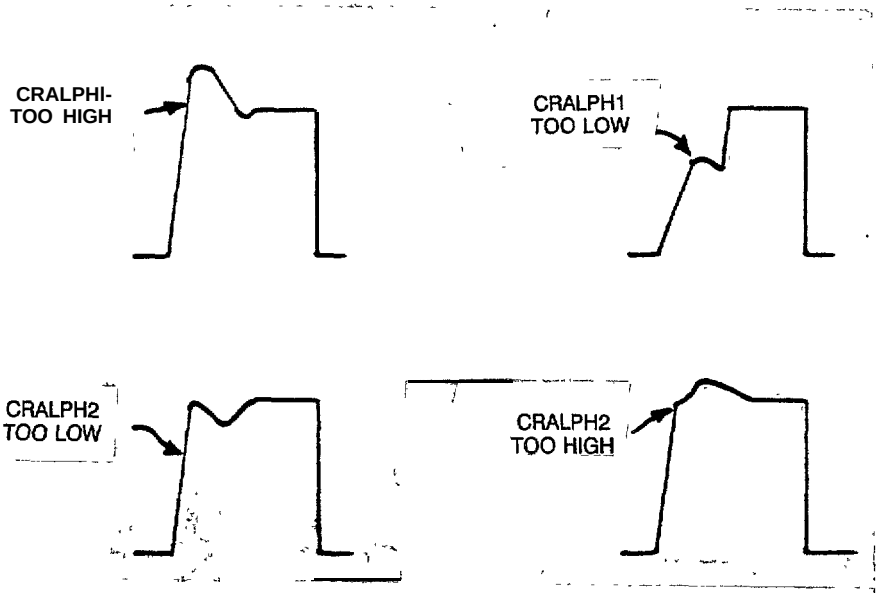


Figure 6-4. CRALPHA1/2 TUNE-UP

122

Calculate motor volts/Hz (flux).

$$\overline{V/Hz} = \frac{\text{(attn ratio) (Vpk/1.414)/f}}{\text{(rated motor volts)/60Hz}}$$

The flux integrators on 1D should then have peak voltages of:

Vf = $(\overline{V/Hz})$ * 6.718 = _____ Vpk

D. Current Regulator

The current regulator for the LCI is calculated for a 250 (± 50) radian response. The parameters involved in this calculation are fairly well known and measurement of the response is impracatical; however; the stability of the regulator can be checked by the following procedure:

Run drive to 55%speed and connect the brush recorder to IFB-1C (TP7).

Apply a 2-5% step for .1-.5 seconds to the current regulator using the monitor commands - MODIFY REFERENCE, APPLY and REMOVE (TUNEUP MODE-CURRENT REGULATOR). See Example 6-3.

The brush recorder should be running at its fastest speed.

It is important that these be balanced as closely as possible

Verify IFB-1C does not overshoot or oscillate on both the up-step to a higher current and speed and the return step to the lower current and speed.

```
*>TUN <cr>
T >TUN CR <cr>

Available Variables are:
(List of Variables)

T >MOD REF <cr>
Step          Level (PU)          Time(sec)
1
2
Step 1.       Level. =            0.02 <cr>
Step 1        Time =              .1 <cr>
Step 2        Level =             0 <cr>
Step 2        Time =              .1 <cr>
T > APPLY <cr>                    ; take brush recording of
                                    up + down response
T > REMOVE <cr>                   ; remove step function
```

Example 6-3

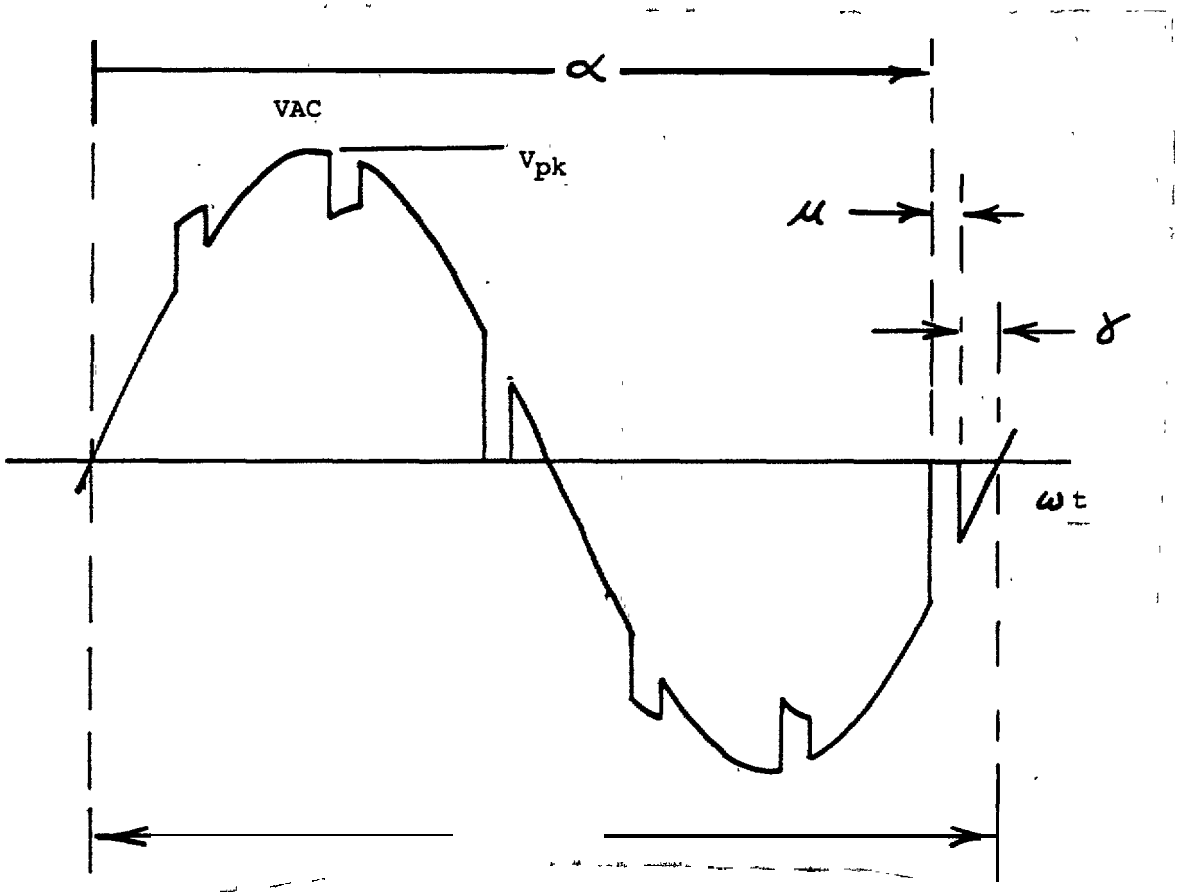


Figure 6-5. MOTOR LINE TO LINE VOLTAGE WAVE FORM

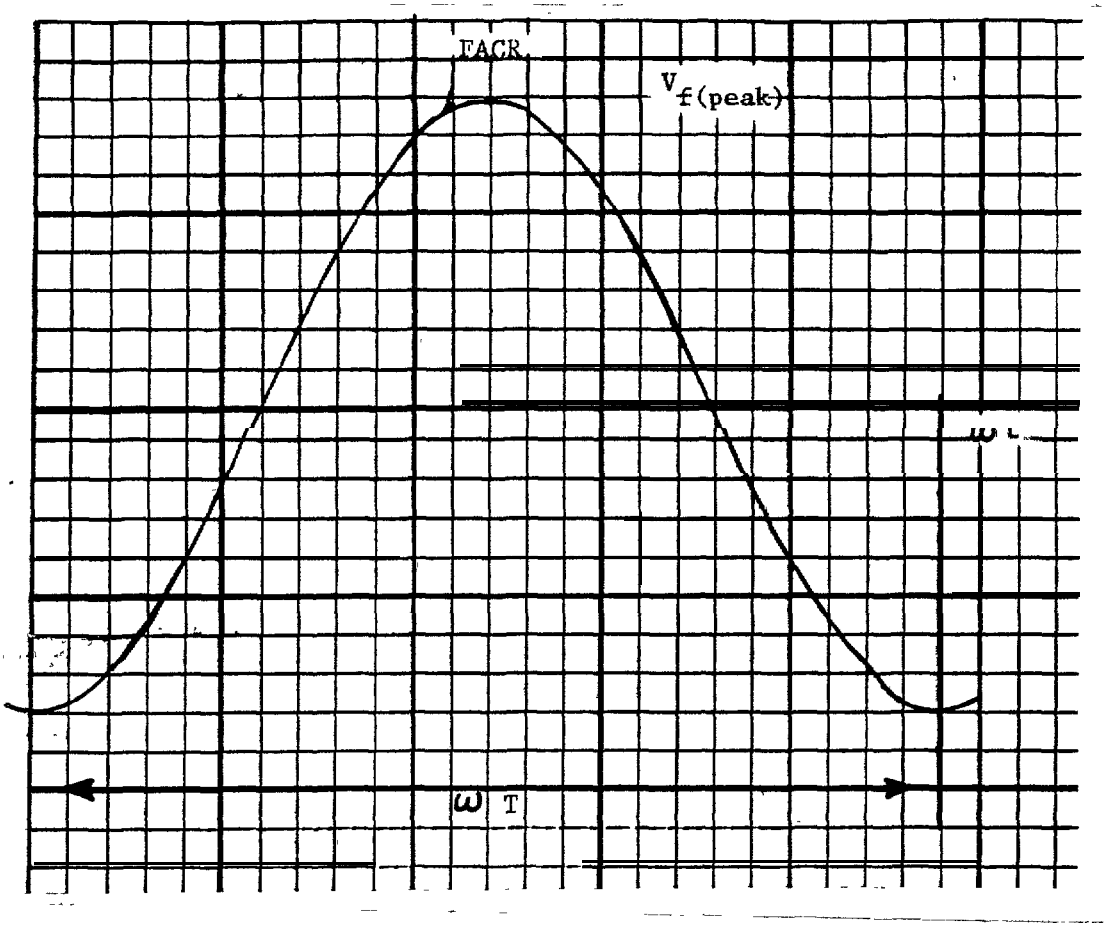


Figure 6-6. Ideal Flux **Wave**

**** NOTE ****

DA1 has been temporarily assigned to current feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.

DA2 has been temporarily assigned to source side firing angle, for reference only.

100

E. Flux Regulator

1. Check flux vs. speed profile over the full speed range. Connect a DVM to testpoint "FCBR" on NLIB 1D. Operate the drive from minimum speed to top speed, measuring the flux amplitude at selected speeds to verify the flux is following the profile shown on the Drive Data sheet of the system elementary diagram. Rated flux = 4.75 vrms at "FCBR".
2. Connect "VABS" on the NLIB in slot 1D to a chart recorder. This is a fairly clean dc signal which will be proportional **to** flux if the motor speed does not change during the following test.
3. Start the drive and run it up to approximately 55% speed. Apply a 5-10% step for 3-5 seconds to the flux regulator using the monitor (in the TUNEUP MODE FLUX REGULATOR case) MODIFY REFERENCE, APPLY and REMOVE commands to step the flux regulator reference.
4. Observe the response on the chart recorder-. Response should be critically damped at approximately 5 radians crossover frequency as shown in Figure 6-7.
5. If the response is too fast (or too slow), decrease (or increase) EEPROM variable FRTC until the nearest desired response is otained.
6. If the response is overdamped or underdamped, increase or decrease the integral gain FRGAIN to obtain critical damping. If damping is now correct but response is not, return to step 5.

F. Torque Regulator (NOT USED)

**** NOTE ****

DA1 has been temporarily assigned to flux feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.

```
*TUN <cr>
T> TUN <cr>

Available Variables Are:
(List of Variables)

T> MOD REF <cr>

Step          Level(PU)          Time(sec)
1
2
Step 1.       Level(pu)  =       0.02 <cr>
Step 1        Time  =            5 <cr>
Step 2        Level  =           0 <cr>
Step 2        Time  =            5 <cr>
t> APPLY <cr>      :Take brush of up + down
                                 response
T> REMOVE < cr >   ;Remove step function
```

Example 6-4

G. Speed Regulator

Run the drive speed up to 75% speed. Verify that CCMD is greater than CURLL and that ALPHSRCO = 0. Apply a step size (2-5% for 5 seconds) to obtain as large a step as possible without hitting current limit (observe IFB on chart recorder). (Using TUNEUP mode-SPDREG: commands MODIFY REFERENCE, APPLY, REMOVE) See Example 6-5.

**** NOTE ****

It may be necessary to buck-out the speed signal with a battery voltage to obtain enough amplification. If-step down response is different from step up, use step down response.

Observe the step response and adjust SRGAIN and SRTC for a 1 radian critically damped response as shown in Figure 6-8.. It should only be necessary to modify SRGAIN at all times:

Save these values for SRGAIN, SRTC, into EEPROM. Record.

**** NOTE ****

DA1 has been temporarily assigned to speed feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.

DA2 has been temporarily assigned to current feedback, monitor the signal so that during the step to make sure the current is not in current limit.

List of Variables are:

```
*> TUN <cr>
T> TUNSR <cr>

T> MOD REF <cr>

Step          Level(PU)          Time(sec)
1
2
Step 1.       Level(pu)  =       .05 <cr>
Step 1        Time  =            2 <cr>
Step 2        Level  =           0 <cr>
Step 2        Time  =            2 <cr>
t> APPLY <cr>      :Take brush of up + down
                                 response
T> REMOVE <cr>     ;Remove step function
```

Example 6-5

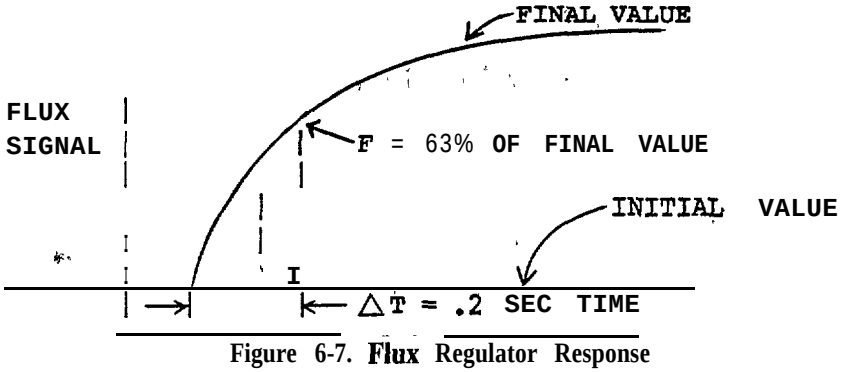


Figure 6-7. Flux Regulator Response

H. Load-side Fire-As-Late-As-Possible (FALAP)

The load-side converter ordinarily operates (when under load) so that firing occurs at a time which permits a 20 degree commutation margin. Since the commutation time is a function of current and volts/Hz, it is necessary to adapt the firing angle to these parameters. It is the purpose of the load-side fire-as-late-as-possible routine to adjust the firing as a function of load current such that commutation is complete at alpha = 160 degrees. It should be noted that other routines may affect the load-side firing, especially when operating at very light loads.

It is desirable to tune-up under the heaviest steady load possible. Run the drive to any convenient speed and check variables CURLL, CCMD, ALPHSRCO. If CCMD is greater than CURLL and ALPHSRCO equals zero, then the drive should be operating in the fire-as-late-as-possible mode. Verify FLALEN and FLASEN are "F".

Connect a scope to "VAC"-2B testpoint on load NSFC. The waveform obtained should be similar to that shown in Figure 6-9.

γ = MARGIN ANGLE. Proper drive adjustment keeps this angle constant, at a minimum value (after transients have died down), over the entire process range.

μ = COMMUTATION ANGLE

β = ANGLE OF ADVANCE

$$X_c = \frac{3}{\frac{\pi}{I_D}} * \overline{V/H} * (\cos\gamma - \cos a);$$

Xc = per-unit commutating reactance,

I, = per-unit current = I/5V.

LIMARG = $\overline{Xc}$

EEPROM variable LIMARG controls the amount of advance of the firing angle with load current. If the margin angle is less than 18 degrees, increase the value of LIMARG. If the margin is greater than 22 degrees, decrease LIMARG. The margin angle should remain constant regardless of current if LIMARG is tuned correctly. If change in original value SAVE into EEPROM.

This waveform also shows certain performance parameters. The angle (90" - ß) shows the transition from motoring to regenerating. When the angle is positive (as illustrated) the motor is driving the load in the normal manner. When the angle goes to zero and beyond (negative), the system is regenerating, with the load driving the motor as a generator. The current waveform and the flux wave can be observed also, but the voltage wave by itself shows all the information. The voltage wave is preferred, for accurate firing angle measurement.

If it can be assumed that the voltage behind commutating reactance is an undistorted sine wave, then a reasonably accurate calculation of power can be made by multiplying the line-to-line voltage behind commutating fundamental inverter power factor, by $\sqrt{3}$, and by the motor efficiency. The RMS line-to-line voltage is Vpk * Attenuation ratio divided by $\sqrt{2}$. The current can be read from the panel ammeter; it is also I_D * motor nameplate current (per channel).

The fundamental inverter power factor is:

$$\frac{\cos(\beta - \mu) + \cos\beta}{2}$$

I. Source-side Line Impedance

Normally the source-side operates in the rectifying quadrant. However, when the drive is forced to slow down, the source converter may have to regenerate (invert). To avoid commutation failures, the angle of retard must be programmed as a function of current. Since a regenerating condition is difficult to hold steady-state, a measurement of system impedance will be made and this value will be programmed into EEPROM.

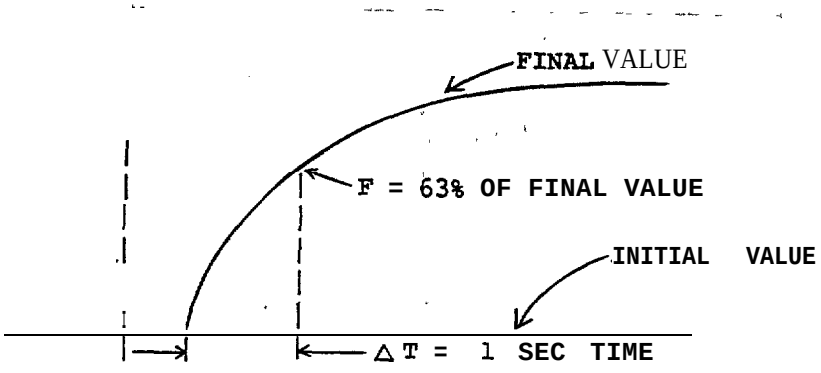


Figure 6-8. Speed Regulator

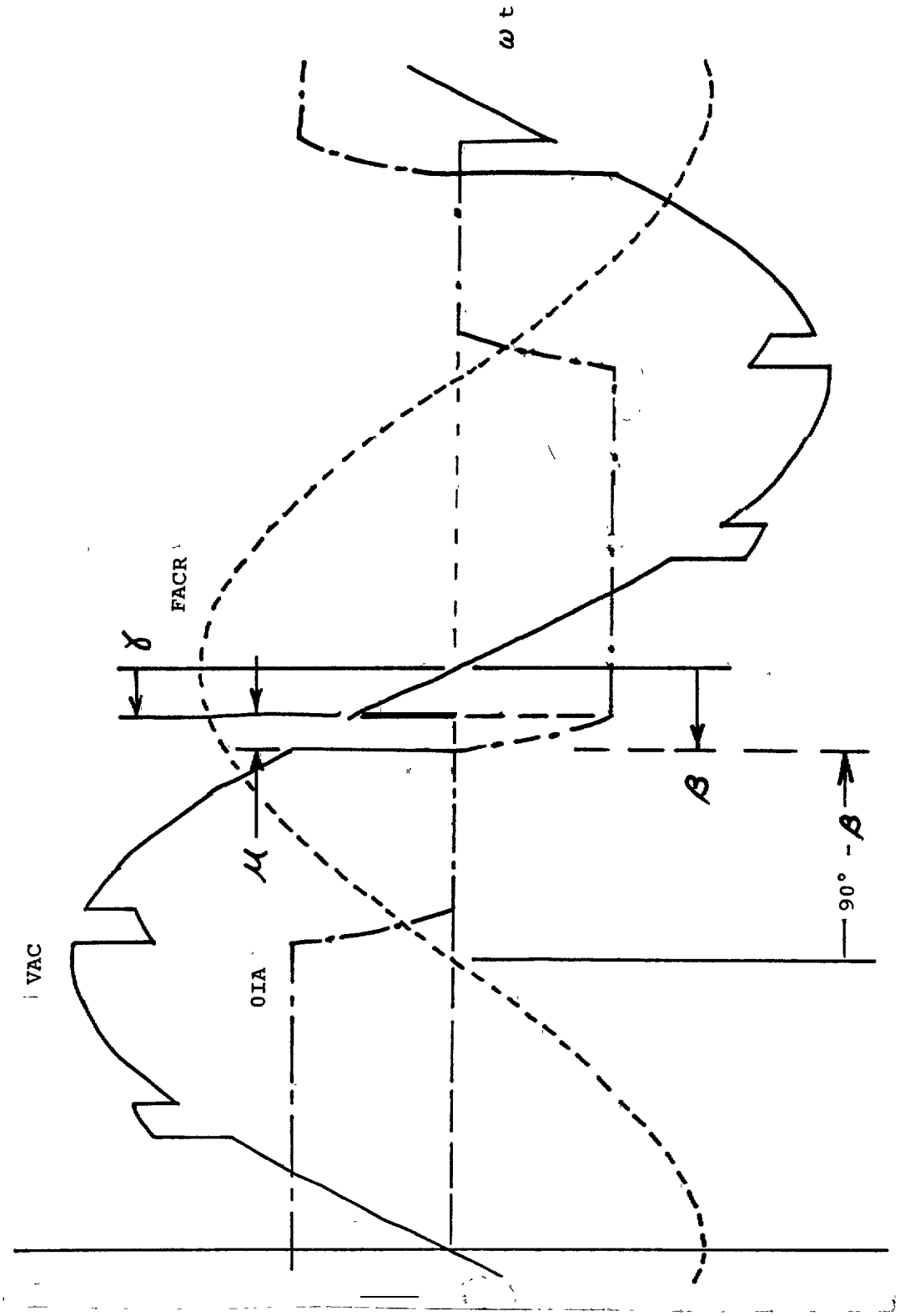


Figure **6-9.** Load-side FALAP

Run the drive to some convenient speed (the higher the better). Connect channel 1 of the oscilloscope to testpoint "VAC"-1C. Connect channel 2 to testpoint "IA"-1C. The signal should be similar to those shown in figure 6-10.

$\overline{VI}$-f = (attn ratio) (Vpk/1.414)/Vnom

where Vnom = source-side nominal voltage

$\overline{I_d}$ = I/5

$\overline{X_c}$ = ($\overline{V1}$-1/$\overline{Id}$)[cos ∝ • cos(∝ + μ)]

SIMARG = $\overline{(Xc)}$ = _____

Round-up to nearest integer and SAVE into EEPROM. Take drive to top speed then decelerate as fast as possible. If a software overcurrent occurs, then increase SIMARG.

J. 12-Pulse Coupled Tests

Start channel A and run up to maximum 6-pulse speed. Load channel A to rated current, if possible.

Start channel B and observe that current balances in 3-5 seconds. Run drive up and down checking current, voltage, and flux balance. Run the drive to maximum 6-pulse speed, stop A, and then restart A.

Reduce load and run the drive up to near rated speed. Drop off channel A and see that channel B stays on.

Repeat for channel B.

Stop the motor, then start both channels at once. Verify simultaneous starting current, following rotor position determination at standstill.

If starting is too rapid, reduce SRCLSC, SRCLFC, and retune CRALPH1 and CRALPH2 as necessary.

** NOTE **

When in 12-pulse operation, additional commutation notches are present in the voltage waveforms. These notches are not compensated for in the flux wave reconstruction, and cause a phase shift in the direction which increases the margin angle. This 12-pulse waveform is shown below. Also, the current waveform in 6-pulse and 12-pulse operation is not the ideal flat-topped waveshape. It contains a "double-hump" ripple at both the power system frequency; and the motor frequency, in addition to other, smaller irregularities.

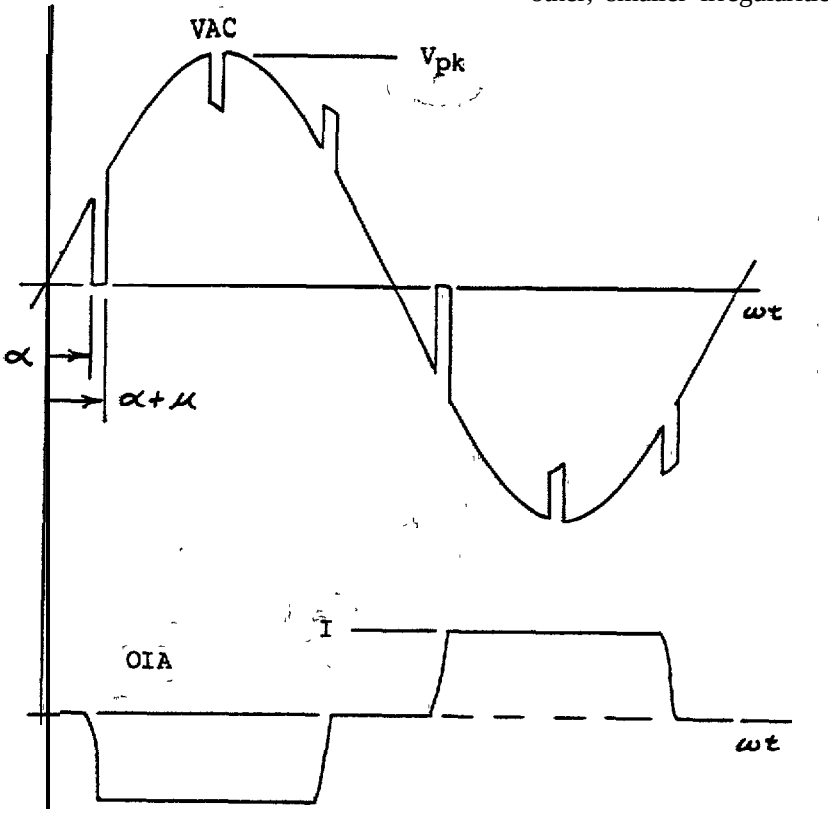


Figure 6-10. Source-side Firing Angle

K. Source-side Fire-As-Soon-As-Possible

1. Modify and save FLALEN and FLASEN to value listed on drive system elementary sheets 1DE - 1DG.
2. Rerun drive - verifying that there are no performance problems (trips, excessive swings in current, etc.).

1. Motor power factor and power readings (per channel) use Figure 6-9

1. Power factor (PF)
   a. Measure μ
   b. Measure β

$$\text{Power factor} = \frac{\cos(\beta - \mu) + \cos \beta}{2}$$

2.1 Kilowatts (KW)

a. Measure power factor (PF).

b. Measure OIA (peak average) (OIA).

c. Measure peak voltage of VAC (Vacpk).

$$\text{Kilowatts} = \frac{\text{Vacpk} * \text{attn ratio}}{1000 * 2} * \frac{\text{OIA}}{5\text{V/unit}}$$

*rated current per winding * PF.

2.11 Alternative to 2 (eliminates P.F reading)

a. If VDC (NSFC) not being used, turn to min (full CCW).

b. Measure VDC

c. Measure IFB

$$\text{Kilowatts} = \frac{\text{IFB}}{5\text{V/volt}} * \text{rated current} * 1.2825 * \text{per winding}$$

$$\frac{\text{VDC} * \text{ATTN RATIO}}{1000} * 2/3$$

Eff-efficiency can be estimated at .97 for large motors See motor data sheets for accurate data).

3. Horsepower
   a. Measure kilowatts

$$\text{Horsepower} = \frac{\text{kilowatts}}{.7457} \cdot \text{Eff.}$$

4. Torque
   a. Measure horsepower
   b. Measure SPD (RPM)

Put oscilloscope's time base on calibrated for this entire measurement. Use as much as possible of the screen to display a half-cycle. Measure the time of this half-cycle (T).

SPD = [(l/rated frequency) * rated RPM at rated frequency)/(2*t)]

$$\text{Torque} = \frac{5250 * \text{HP}}{\text{SPD}}$$

CHAPTER 7
DIAGNOSTICS and TROUBLESHOOTING

7-1. SECTION A - SELFTEST

7-1 .1. BASIC STRUCTURE

The Selftest routines are PLM86 procedures which may be executed by the HMPG resident in the Structured LCI's control Small Electronics Module (SEM) to test individual PWBs and functions. These routines are executed upon power-up if the TEST switch on the NLCB is in the up position. These routines may also be run when the drive is in a non-RUN condition by operator commands received via a terminal connected to the HMPG's RS-232 port or by depressing 'a test switch. The function of the Selftest is to test most of the control panel for the Structured LCI, with great emphasis placed on the testing of the PWBs in the SEM.

Testing of the panel begins by performing tests of the hardware elements within a neighborhood of the HMPG's microprocessor and continues to tests of increasingly remote hardwares. All of the tests incorporated within the selftest may be classed into 3 groups:

1. PWB-Specific tests
2. Paired PWB tests
3. Panel-level tests

7-1.1.1. PWB-Specific Tests. : A great amount of effort has been placed on performing tests that will pinpoint problems on specific PWBs first. In this manner, if any easily diagnosed problem exists, it will be clearly spelled out to the operator, instead of vaguely defined by a list of possible solutions. Tests such as these comprise the PWB-Specific test group.

7-1.1.2. Paired PWB Tests:. Tests which depend on the operation of several PWBs are classed within the Paired-PWB group. Diagnostic messages for failures of these tests will indicate a functional problem and also a list of possible solutions which may be general in nature.

7-1.1.3. Panel-level Tests. : Panel-level tests also require correct operation of several PWBs or modules, however, these tests are segregated to allow the user to perform either a module-level test (involving only the control SEM) or a complete panel-level test which would include this last catagory of procedures.
From a practical point of view, the sequence of testing within the control SEM progresses in the following manner:

1. HMPG
2. HLCB
3. HAIA
4. NSFC (source)
5. NSFC (load)
6. NSFE
7. NLCB
8. NLIB (source)
9. NLIB (load)
10. HISA
11. HRDA

Further tests which include PWBs outside the SEM, progress in the following sequence:

1. Cables leading from the SEM.
2. the I/O Module.
3. the Field Exciter Module.

7-1.2. EXECUTING THE TESTS

The self test routine may be initiated by reseting the HMPG with the TEST switch (on the NLCB) in the UP position. The entire test is completed in approx 20 seconds. At its completion, status information will be printed out at the door-mounted printer. In the present Structured LCI product, the entire self test routine is executed; however, provisions for executing small sections of the code are in place.

Upon completion of the self test, the drive will operate in the gate test mode. To return to normal operation, the NLCB's test switch must be returned to the normal position and the HMPG must be reset.

CAUTION

Selftest excercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped (if 12-pulse).

7-1.3. DIAGNOSING FAILURES

As mentioned above, failure of any of the selftest procedures will produce a diagnostic message on the door-mounted printer.

The format of the diagnostic message is:

```
DATE TIME FAILED
(aabb) DURING TEST OF:        aa = TEST NUMBER
description of test           bb = SUB-TEST NUMBER

PROBLEM:problem statement

POSSIBLE SOLUTIONS:
solution 1
solution 2
solution 3
```

Usually, a failure can easily be diagnosed by the problem statement. At least one solution will be suggested in the print out. It is suggested that troubleshooting be performed at the PWB level (Replace the PWB not its component). Although the diagnostic messages may indicate more specific failures, it is not recommended that PWB components be replaced or repaired (always write or attach the diagnosis to the card).

The TEST NUMBER and SUB-TEST NUMBER uniquely define a failure within the Selftest procedure. The TEST NUMBER is incremented from 1 to 31 (hex) as the tests progress. For different test numbers, the SUB-TEST parameter will vary form 1 up to the number of sub-tests within that major test. The Description of Tests section that follows details the complete Selftest procedure according to the TEST NUMBER and SUB-TEST NUMBER parameters.

7-1.4. DESCRIPTION OF TESTS

The following lists all tests available with sub-test information:

TEST DESCRIPTION

0. HMPGROMTST: Reads all of ROM memory (F000:0 to F000:FFFF). Possible failures are:
   - .O - BusHog error
   - .1 - Read error

1. HMPGRAMTST: Reads all of RAM memory (0:0 to 0:3FFF). Possible failures are:
   - .O - BusHog error
   - .1 - Read error
   - .2 - Write error

2. HMPGINTRTST
   - .1 - Verifies control and data lines to HMPG's 8259

3. HMPG,TMR
   - .1 - Verifies addr lines & data bits 6 & 7
   - .2 - Verifies addr lines & data bits 6 & 7
   - .3 - Verifies addr lines & data bits 6 & 7
   - .4 -Verifies data bits 5-O with walking 1 pattern
   - .5 - Verifies that all outputs are free to toggle.
   - .6 - FOR TMR0: Checks for null count = 1
   - .7 - FOR TMR0: Loads TMR; checks for null count = 0
   - .8 - FOR TMRO: Verify TMR time out
   - .9 - FOR TMRO: Verify TMR reloads
   - .A - FOR TMR1: Checks for null count = 1
   - .B - FOR TMR1: Loads TMR; checks for null count = 0
   - .C - FOR TMR1: Verify TMR time out
   - .D - FOR TMR1: Verify TMR reloads
   - .E - FOR TMR2: Checks for null count = 1
   - .F - FOR TMR2: Loads TMR; checks for null count = 0
   - .10 - FOR TMR2: Verify TMR time out
   - .11 - FOR TMR2: Verify TMR reloads

4. HMPG_SIO_PRES
   - .1 - Check that trans and rec buffers are empty

5. HMPG,STALL
   - .1 - Verifies software reset of Stall circuit.
   - .2 - Initiate stall timer; checks for no STALL after 100
   - .3 - Verifies STALL asserted after 1.5 seconds.

6. HMPGJNTR
   - .1 - Verify operation of BusHog interrupt.
   - .2 - Verify that Bus Hog does not occur for legitimate read.
   - .3 - Verify operation of Time Tic interrupt.

7. DMPB
   - .1 - Checks that Write Enable Switch is not asserted.
   - .2 - Checks for no BusHog during EEPROM reads.
   - .3 - Checks for expected pattern in EEPROM.

8. HLCB_PRES
   - .1 - Enables IMOK, checks for 0IMOK=0.
   - .2 - Disables IMOK; checks for OIMOK= 1.

9. HLCB_DAK
   - .1 - Checks for no BusHog during reads of HLCB.

A. HLCB_TMR_U22
   - .1 - Verifies addr lines & data bits 6 & 7
   - .2 - Verifies addr lines & data bits 6 & 7
   - .3 - Verifies addr lines & data bits 6 & 7
   - .4 - Verifies data bits 5-O with walking 1 pattern
   - .5 - Verifies that all outputs are free to toggle.
   - .6 - FOR TMRO: Checks for null count = 1

.7 - FOR TMRO: Loads TMR; checks for null count = 0
.8 - FOR TMR0: Verify TMR time out
.9 - FOR TMRO: Verify TMR reloads
.A - FOR TMR1: Checks for null count = 1
.B - FOR TMR1: Loads TMR; checks for null count = 0
.C - FOR TMR1: Verify TMR time out
.D - FOR TMR1: Verify TMR reloads
.E - FOR TMR2: Checks for null count = 1
.F - FOR TMR2: Loads TMR; checks for null count = 0
.10 - FOR TMR2: Verify TMR time out
.11 - FOR TMR2: Verify TMR reloads

B. HLCB_TMR_U23
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines.& data bits 6 & 7
.4 - Verifies data bits 5-O with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.
.6 - FOR TMRO: Checks for null count = 1
.7 - FOR TMRO: Loads TMR; checks for null count = 0
.8 - FOR TMRO: Verify TMR time out
.9 - FOR TMRO: Verify TMR reloads
.A - FOR TMR1: Checks for null count = 1
.B - FOR TMR1: Loads TMR; checks for null count = 0
.C - FOR TMR1: Verify TMR time out
.D - FOR TMR1: Verify TMR reloads
.E - FOR TMR2: Checks for null count = 1
.F - FOR TMR2: Loads TMR; checks for null count = 0
.10 - FOR TMR2: Verify TMR time out
.11 - FOR TMR2: Verify TMR reloads
.12 - FOR TMR2: Initialize to Mode 0; verify status. .13 - FOR TMR2: Load with 10 ms delay; verify status.
.14 - FOR TMR2: After 4 ms, verify status.
.15 - FOR TMR2: After 15 ms, verify time out.
.16 - FOR TMR2: Check that count = 0.

C. HLCB_INTR_PRES
.1 - Verify addr and data lines to HLCB's 8259.

D. HLCB,SIO
.1 - Initialize and check status of HLCB's 8251.
.2 - Verify that single byte of data is send, by sensing status.
.3 - Verify transfer of data pattern.

E. HAIA_PRES
.1 - Checks that HAIA is present by reading single bit.

F HAIA_DAK
.1 - Checks for no BusHog during reads of HAIA.
.2 - Checks for no BusHog during writes to HAIA.

10. HAIA_TMR
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 - Verifies data bits 5-O with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.
.6 - FOR TMRO: Checks for null count = 1
.7 - FOR TMRO: Loads TMR; checks for null count = 0
.8 - FOR TMRO: Verify TMR time out
.9 - FOR TMRO: Verify TMR reloads
.A - FOR TMR1: Checks for null count = 1
.B - FOR TMR1: Loads TMR; checks for null count = 0
.C - FOR TMR1: Verify TMR time out
.D - FOR TMR1: Verify TMR reloads
.E - FOR TMR2: Checks for null count = 1
.F - FOR TMR2: Loads TMR; checks for null count = 0
.10 - FOR TMR2: Verify TMR time out
.11 - FOR TMR2: Verify TMR reloads

11. HAIA_CONVERSION
.1 - Start A/D conversion; check that BUSY = 1.
.2 - Check that 1DAVAL = 0.
.3 - Wait for conversion to complete; check that BUSY = 0.
.4 - Check that 1DAVAL = 1.

12. HAIA_PSUPPLY
.1 - Check that N15 supply is read correctly.
.2 - Check that P15 supply is read correctly.

13. NSFC_S_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

14. NSFC_S_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7

.4 -Verifies data bits 5-O with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.

15. NSFC- S - TMODE
.1 - FOR TMRO: Checks for null count = 1
.2 - FOR TMRO: Loads TMR; checks for null count = 0
.3 - FOR TMRO: Verify TMR time out
.4 - FOR TMRO: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

16. NSFC_L_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

17. NSFC_L_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 - Verifies data bits 5-O with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.

18. NSFC_L_TMODE
.1 - FOR TMRO: Checks for null count = 1
.2 - FOR TMRO: Loads TMR; checks for null count = 0
.3 - FOR TMRO: Verify TMR time out
.4 - FOR TMRO: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same.
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

19. NSFE_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

1A NSFE_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 -Verifies data bits 5-O with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.

1B NSFE_TMODE
.1 - FOR TMRO: Checks for null count = 1
.2 - FOR TMRO: Loads TMR; checks for null count = 0
.3 - FOR TMR0: Verify TMR time out
.4 - FOR TMRO: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same.
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

1C NSFE_DIAG
.1 - Tests for dormant field status information.

1D NSFC_S_PLL
.1 - FOR TMR2: Enables PLL; Checks for null count = 1
.2 - FOR TMR2: Loads TMR; checks for null count = 0
.3 - FOR TMR2: Verify TMR time out
.4 - FOR TMR2: Verify TMR reloads

1E NSFC_L_PLL
- .1 - FOR TMR2: Enables PLL, Checks for null count = 1
- .2 - FOR TMR2: Loads TMR; checks for null count = 0
- .3 - FOR TMR2: Verify TMR time out
- .4 - FOR TMR2: Verify TMR reloads

1F NSFE_PLL
- .1 - FOR TMR2: Enables PLL; Checks for null count = I
- .2 - FOR TMR2: Loads TMR; checks for null count = 0
- .3 - FOR TMR2: Verify TMR time out
- .4 - FOR TMR2: Verify TMR reloads

20. ROLL-CALL
- .1 - Test for ribbon cable connecting HAIA(JA) to SEM(JG).
- .2 - Enable NLCB IMOK; check for IMOK true.
- .3 - Disable NLCB IMOK; check for IMOK false.
- .4 - Enable NLIBS IMOK; check for IMOK true.
- .5 - Disable NLIBS IMOK; check for IMOK false.
- .6 - Enable NLIBL IMOK; check for IMOK true.
- .7 - Disable NLIBL IMOK; check for IMOK false.

21. NLCB_DIFT_AMPS
- .1 - Test for 0 volts at REFO.
- .2 - Test for 0 volts at TREFO.
- .3 - Test for 0 volts at 12PREF0.
- .4 - Inject current in REFO circuit; test for change in output.
- .5 - Inject current in TREFO circuit; test for change in output.
- .6 - Inject current in 12PREF0 circuit; test for change in output.
- .7 - Test for 0 volts at SP (speed).
- .8 - Reset exixting faults; test for 1OSP (overspeed) = 0.

22. NSFE_ANALOG
- .1 - Test for EI (exciter current) = 0 volts.
- .2 - Test for no current imbalance
- .3 - Inject signal in current circuit; test for output change.
- .4 - Again, test for no current imbalance.
- .5 - Excite current circuit with 55 Hz inputs; verify that current imbalance is detected.
- .6 - Wait 100 ms; verify no current imbalance condition.

23. HLCB
- .1 - Test for absence of SQ waves with oscillator off.
- .2 - Test for any activity (source or load) with oscillator on.
- .3 - Check for activity on each of the 3 source channels
- .4 - Verify source side VA channel active
- .5 - Verify source side VB channel active
- .6 - Verify source side VC channel active
- .7 - Test phase sequence of source SQ waves
- .8 - Check for activity on each of the 3 load channels
- .9 - Verify load side VA chananel active
- .10 - Verify load side VB channel active
- .11 - Verify load side VC channel active
- .12 - Test phase sequence of load SQ waves

24. NLIB_S_ANALOG
- .1 - Reset existing faults; verify UV & not OV condition.
- .2 - Verify not SUPPression conditon.
- .3 - Verify 1UV70=0.
- .4 - Test Cl7 for time delay. (Overvoltage circuit)
- .5 - Test Cl7 for time delay. (Overvoltage circuit)
- .6 - Test for 0 volts at IFBS.
- .7 - Test for 0 volts at FLUX BAS.
- .8 - Inject current in source current condition circuit. Detect change in IFBS.
- .9 - Detect change in FLUX BAS.

- .A - For FLUX BAS: Check for 0 volts dc offset.
- .B - For FLUX BAS: Check level of positive peak.
- .C - For FLUX BAS: Check level of negative peak.

25. NLIB_L_ANALOG
- .1 - Reset existing faults; verify UV & not OV condition.
- .2 - Verify not SUPPression conditon.
- .3 - Verify 1UV70=0.

.4 - Test Cl7 for time delay. (Overvoltage circuit)
.5 - Test Cl7 for time delay. (Overvoltage circuit)
.6 - Test for 0 volts at IFBL.
.7 - Test for 0 volts at FLUX BAL.
.8 - Inject current in source current condition circuit.Detect change in IFBL.
.9 - Detect change in FLUX BAL.
.A - For FLUX BAS: Check for 0 volts dc offset.
.B - For FLUX BAS: Check level of positive peak.
.C - For FLUX BAS: Check level of negative peak.
.D -For FLUX CBS: Check for 0 volts dc offset.
.E - For FLUX CBS: Check level of positive peak.
.F - For FLUX CBS: Check level of negative peak.
.10 -For FLUX ACS: Check for 0 volts dc offset.
.11 - For FLUX ACS: Check level of positive peak.
.12 - For FLUX ACS: Check level of negative peak.

26. HLCBJNTR
.1 - Enable test oscillator; verily occurance of source zero crossing interrupt (slave intrerrupt 0).
.2 - Verify frequency of source zero crossing interrupt.
.3 - Enable test oscillator; verify occurance of loadzero crossing interrupt (slave intr. 1).
.4 - Verify frequency of load zero crossing intr.
.5 - Verify occurance of source firing interrupt.
.6 - Verify occurance of load firing interrupt.
.7 - Verify occurance of field firing interrupt.

27. NLCB,SPEED
.1 - Enable test oscillator; verify speed circuits reads 55 Hz.
.2 - Verify no overspeed condition.

28. HISA_PRES
.1 - Disable IMOK; verify by reading back bit.
.2 - Enable IMOK; verify by reading back bit.

29. CABLE,CONT
.1 - Tests for continuity in ribbon cable from SEM, JA to I/O Module.
.2 - Tests for continuity in ribbon cable from SEM, JK to I/O Module.
.3 - Tests for continuity in ribbon cable from SEM, JH to I/O Module.
.4 - Tests for continuity in ribbon cable from SEM, JB to Door Module.

2A. HISA_IO
.1 - Set all inputs from I/O Module high; verify by reading back inputs to HISA.
.2 - Set all inputs from I/O Module low; verify by reading back inputs to HISA.

2 B P105_SUPPLY_CHECK
.1 - Check that 105 volt dc power supply is OK.

2C HROA_1A_IO:
In sequence, pick-up single relays on the HROA in slot 1A of theI/O module. Expect feedback signal to indicate change.
.1 -Force all HROA(1A) relays to drop-out; verify by feedback.
.2 -Force 105 Vdc relays to pick-up; verify by feedback.
.3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
.4 -Pick-up single 28 V relay; verify by feedback.
.5 -Drop-out all relays; verify by feedback.
.6 -Pick-up single 28 V relay; verify by feedback.
.7 -Drop-out all relays; verify by feedback.
.8 -Pick-up single 28 V relay; verify by feedback.
.9 -Drop-out all relays; verify by feedback.
.A -Pick-up single 28 V relay; verify by feedback.
.B -Drop-out all relays; verify by feedback.

2 D HROA_1B_IO:
In sequence, pick-up single relays on t he HROA inslot 1B of the I/O module. Expect feedback signal to indicate change.
.1 -Force all HROA(1A) relays to drop-out; verify by feedback.
.2 -Force 105 Vdc relays to pick-up; verify by feedback.
.3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
.4 -Pick-up single 28 V relay; verify by feedback.
.5 -Drop-out all relays; verify by feedback.

.6 -Pick-up single 28 V relay; verify by feedback.
.7 -Drop-out all relays; verify by feedback.
.8 -Pick-up single 28 V relay; verify by feedback.
.9 -Drop-out all relays; verify by feedback.
.A -Pick-up single 28 V relay; verify by feedback.
.B -Drop-out all relays; verify by feedback.
.C -Pick-up single 28 V relay; verify by feedback.
.D -Drop-out all relays; verify by feedback.
.E -Pick-up single 28 V relay; verify by feedback.
.F -Drop-out all relays; verify by feedback.

2 E NLCB_R_DVR
.1 - Verily presence of HROA in slot 1B of I/O Module.
.2 - Verify presence of HROA in slot 1A of I/O Module.
.3 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.
.4 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.

2 F DOOR_MOD_PRES
.1 - Test for CTS asserted.
.2 - Test for DSR asserted.

30. FLD_EXC_MOD_ATTEN
.1 - Check for no absence of ac on the SFE Module, via thecell state sensors.
.2 - Verify presence of ac on phase A, via the cell state sensors.
.3 - Verify presence of ac on phase B, via the cell state sensors.
.4 - Verify presence of ac on phase C, via the cell state sensors.
.5 - Check level of the ac input voltage.
.6 - Check level of the ac input voltage.
.7 - Check dc offset of the ac input voltage.
.8 - Check that exciter output voltage is equal to 0.
.9 - Check that exciter output voltage is equal to 0.
.A - Check that exciter current is equal to 0.
.B - Check that exciter current is equal to 0.
.C - While sampling cell state sensor patterns, verify that cell state sensors are operational.
.D - Verify correct cell state pattern is in sampled data.
.E - Verify phase sequence of SFE Module input voltage by finding correct second cell state pattern in sampled data.
.F - Verify phase sequence of SFE Module input voltage by finding correct third cell state pattern in sampled data. (Completion of phase reversal check.)

31. FLD_EXC_MOD_1PHASE
** Single phase operation of SFE module, A-B. **
.1 - Waiting for negative level of field exciter sync signal.
.2 - Waiting for positive edge of field exciter sync signal.
.3 - After delay, fire correct SCR pair; verify via cell state sensors.
.4 - Waiting for positive level of field exciter sync signal.
.5 - Waiting for negative edge of field exciter sync signal.
.6 - After delay, fire correct SCR pair; verify via cell state sensors.
.7 -After 5 cycles, verify that exciter current < 0.5 per unit.
** Single phase operation of SFE module, B-C. **
.8 - Waiting for negative level of field exciter sync signal.
.9 - Waiting for positive edge of field exciter sync signal.
.A - After delay, fire correct SCR pair; verify via cell state sensors.
.B - Waiting for positive level of field exciter sync signal.
.C - Waiting for negative edge of field exciter sync signal.
.D - After delay, fire correct SCR pair; verify via cell state sensors.
.E - After 5 cycles, verify that exciter current < 0.5 per unit.
** Single phase operation of SFE module, C-A. **
.F - Waiting for negative level of field exciter sync signal.
.10 - Waiting for positive edge of field exciter sync signal.
.11 - After delay, fire correct SCR pair; verify via cell state sensors.

.12 - Waiting for positive level of field exciter sync signal.
.13 - Waiting for negative edge of field exciter sync signal.
.14 - After delay, fire correct SCR pair; verify via cell state sensors.
.15 - After 5 cycles, verify that exciter current < 0.5 per unit.

7-2. SECTION B - GATE-TEST-MODE

7-2.1. Gate-Test-Mode

The gate test mode allows off-line testing of much of the analog signal conditioning, microprocessor firing control, and gate pulse generation circuitry. In the gate-test-mode, a local test oscillator supplies 3 phase, approximately 60 Hz signals in place of the attenuated bus signals from the source and load bridges. The electronic control can then lock onto these signals and generate signals to "dry fire" the thyristors in both bridges. Thyristor gate pulses may then be examined for proper magnitude and phasing. Firing may be adjusted between advance and retard limit for each bridge. In the gate-test-mode, the control produces pulse train firing, thus it is necessary to run the converter ventilating fans for this test.To perform the following tests, a dual channel oscilloscope with voltage probes and a current probe is required.

7-2.2. Getting Into Gate-Test-Mode

A. The drive should be stopped, primary switchgear and/or transformer isolation switch (if any), and load isolation device (if any) should be open. Check the software ladder diagram in the system elementary for any other conditions that must be met.

B. Put the NLCB (1E) "TEST" switch (SW1) in the test position.

C. Reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E or 2J).

** NOTE **

Always use dc coupling on the oscilloscope channels.

7-2.3. Gate-Test-Mode Checkout

A. Connect channel one of the oscilloscope to NLIB (1C) "FCBR" (TP5) and channel two of HLCB (2E) "VA1" (TP12). Trigger the scope on channel one and verify the waveforms look like Figure 7B-1.

B. Connect channel one of the oscilloscope to NLIB (1D) "FCBR" (TP5) and channel two to HLCB (2E) (TP13). Trigger the scope on channel one and verify the waveforms look like Figure 7B-1.

C. Connect channel one of the oscilloscope to HLCB (2E) "VA1" (TP12) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e, 60 degrees per major division or 12 degrees per minor division).

D. One at a time, connect channel two to SIG1 thru SIG6 on the source side "A" NSFC (2C) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.
Note: The firing angle can be adjusted by varying the speed demand reference (place in "LOCAL" and adjust "LOCAL REF").

F. Connect channel one of the oscilloscope to HLCB (2E) "VA2" (TP13) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e., 60 degrees per major division or 12 degrees per minor division).

I. One at a time, connect channel two to SIG1 thru SIG6 on the load side NSFC (2B) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.

K. Verify that each of the red gate pulse LEDS on the HPTK cards in the leg cell modules is illuminated indicating that the cell is getting gate pulses.

L. With a current probe, look at the gate pulse of each thyristor while simultaneously monitoring the corresponding gate pulses on the NSFC card. The two signals should line up with each other as shown in Figure 7B-4A.

Table7B-1

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR	CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2C)	STM1	SIG1 (TP1, 2B)	LTM1
SIG1(TP2, 2C)	sTM2	SIG1 (TP2, 2B)	LTM2
SIG1(TP3, 2C)	STM3	SIG1 (TP3, 2B)	LTM3
SIG1(TP4, 2C)	sTM4	SIG1 (TP4, 2B)	LTM4
SIG1(TP5, 2C)	STMS	SIG1 (TP5, 2B)	LTM5
SIG1(TP6, 2C)	STM6	SIG1 (TP6, 2B)	LTM6

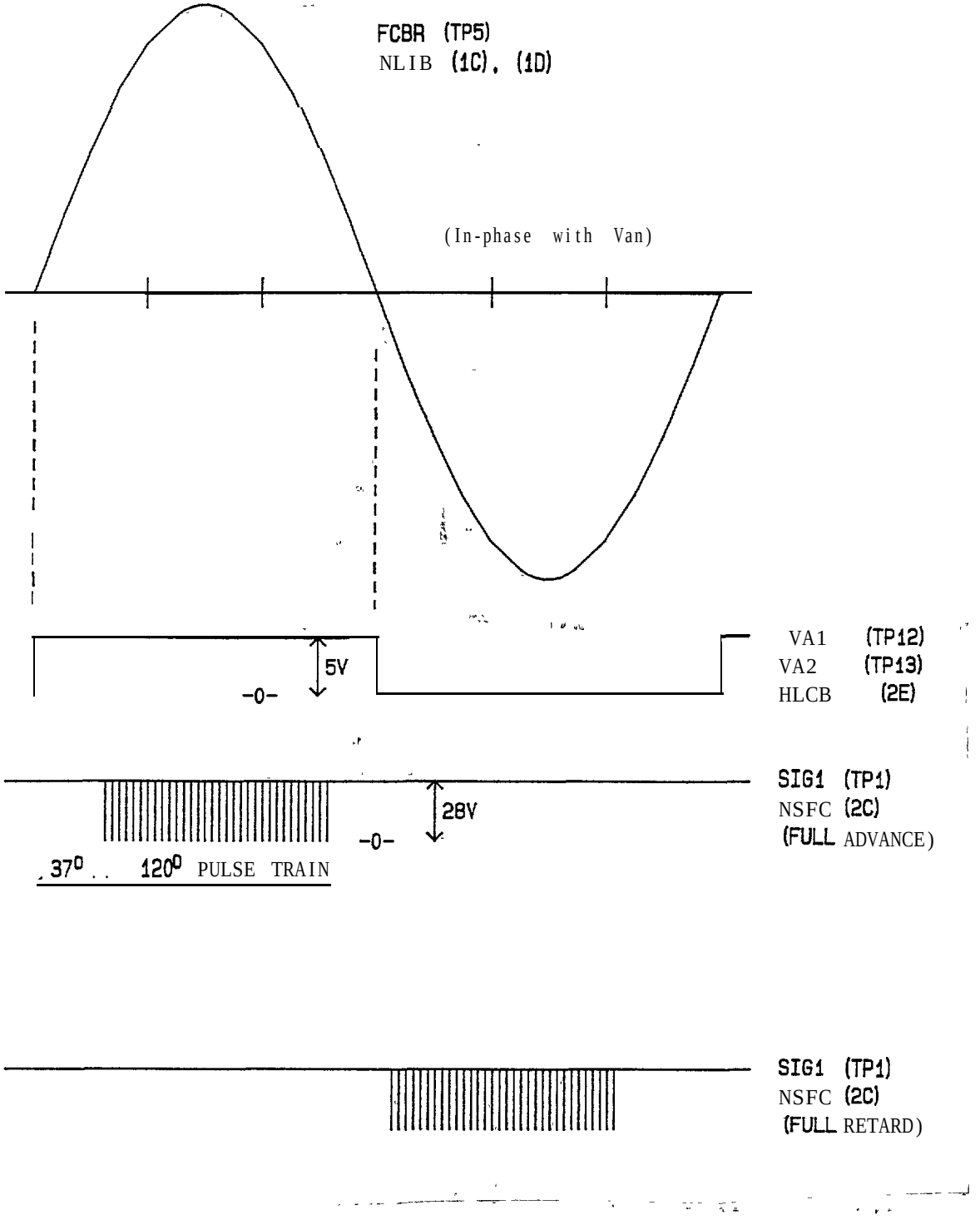


Figure 7B-1. Firing Angle Limits

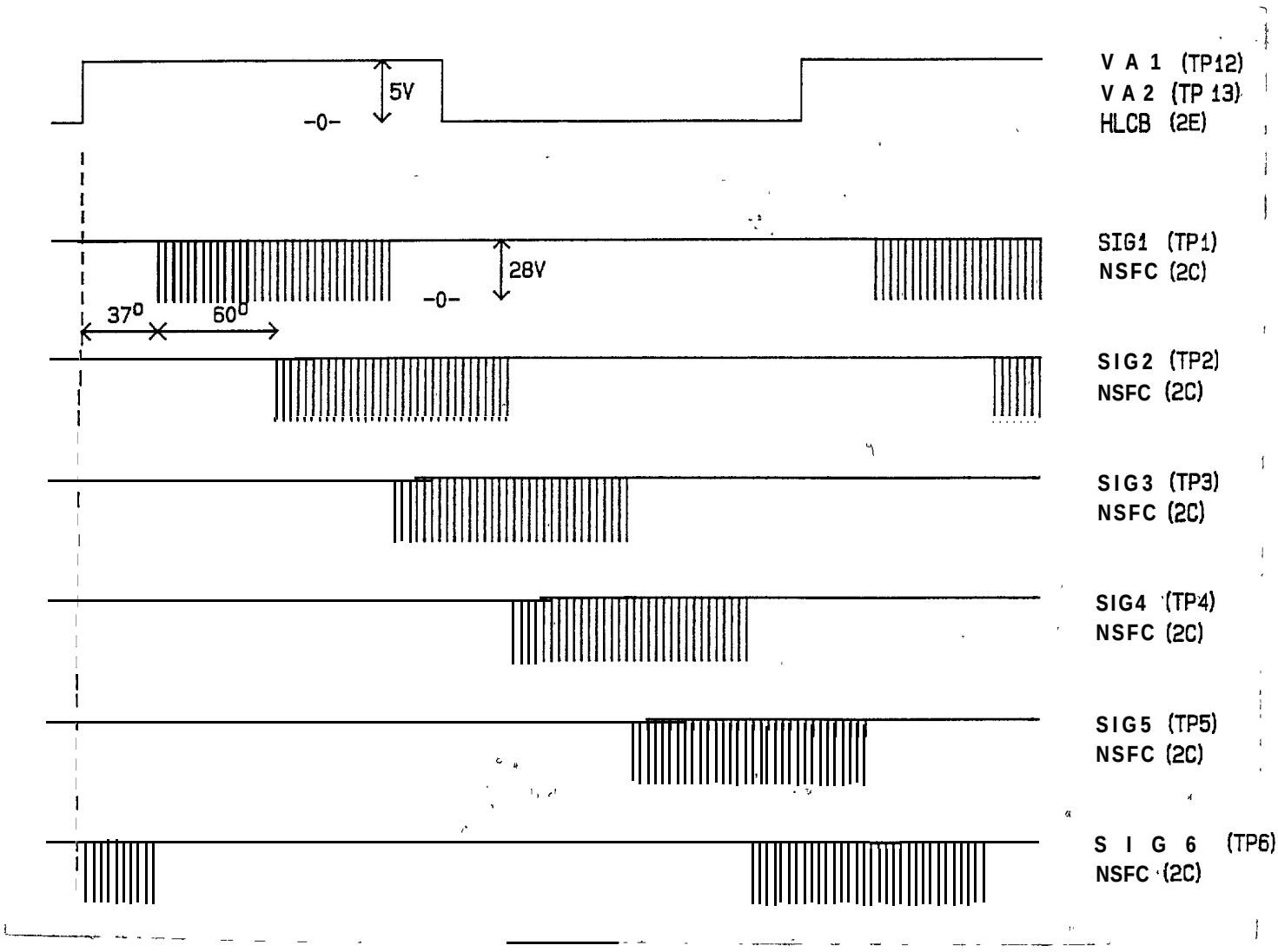


Figure 7B-2A. Source Firing - Full Advance

M. Connect a current probe to the scope and trigger on this channel. Hook the current probe to each of the gate leads (white wire) of the SCRs with the arrow on the current probe pointing away from the gate lead termination point. Verify that the waveform has the same characteristics as Figure 7B-4B. If the pulse fails the criterion, the HPTKN card in that leg cell module is most likely the cause, or the NTDA connections may be reversed.

Z. Put the NLCB (2E) "TEST" switch (SW1) in the OFF position and reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB(2E).

**** NOTE ****

For silco bridge, HPTF will be used instead of HPTK.

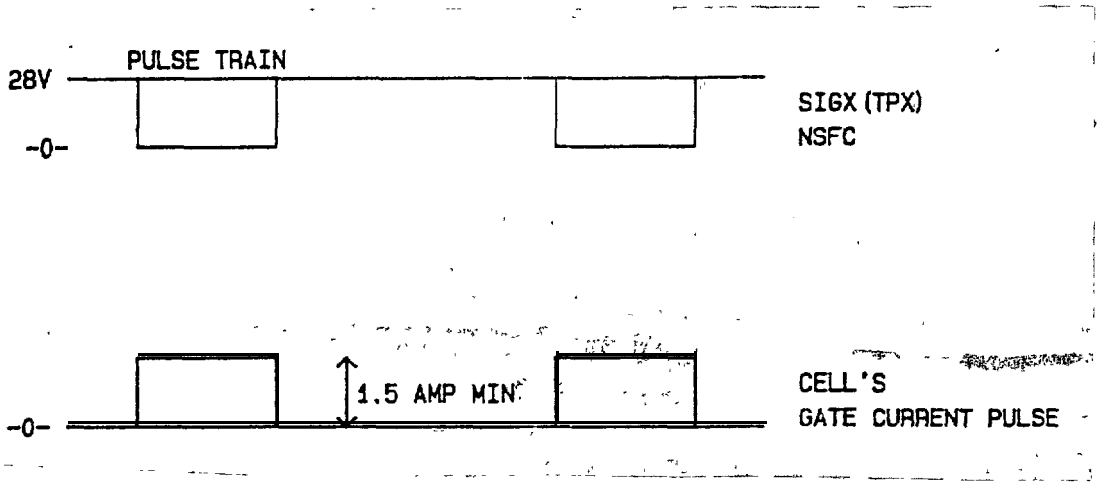


Figure 7B-4A. Typical Gate Pulse Trains

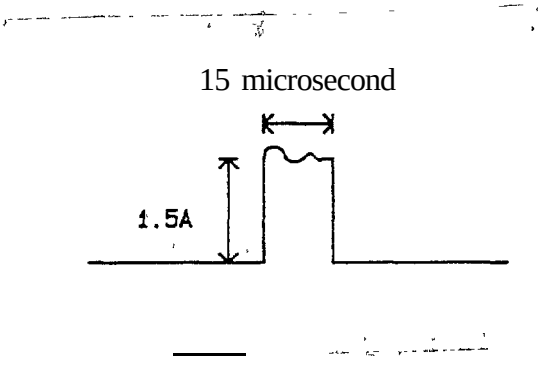


Figure 7B4B. Typical Firing Pulse (Cell Gate Current Pulse)

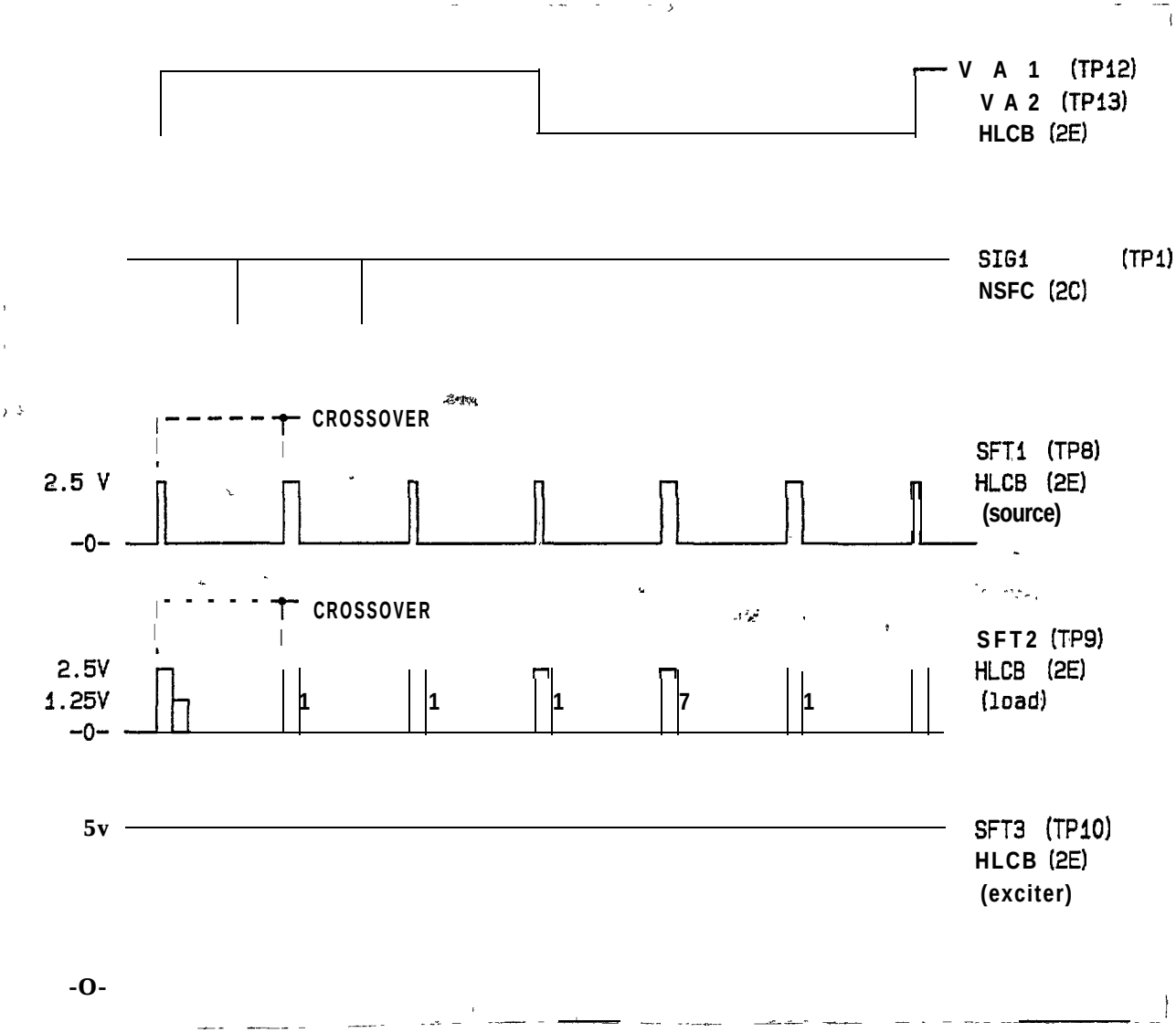


Figure 7B-6. Software Interrupts - Full Advance (Source)

7-3. Section C - Software Regulator Block Diagrams

Following are symbolic block diagrams of various regulators and functions found in the restructured LCI software.

Note: Most of the regulator diagrams are presented in both "S" domain (analog) and "Z" domain (digital). Also, most of the variables listed (in "S" domain) are available by name through the monitor (Review Chapter 5) either as EEPROM variable (Chapter 4, Section A) or public variables (Chapter 4, Section B).

The diagrams should be useful for understanding the overall control of the drive and for troubleshooting by identifying variables that could be checked or used on the diagnostic list or D/A function.

7-4. SECTION D - TERMINALS

7-4.1. Serial Monitor and Hand Terminal (or other Diagnositic Terminal)

Integrated into the firmware for both source and load electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of each microcomputer card. Communicaion with the microcomputer can be established by connecting the hand-held terminal supplied with the drive to RS-232 port. This connection must be made using the HNMB null modem (supplied) with the electronics.

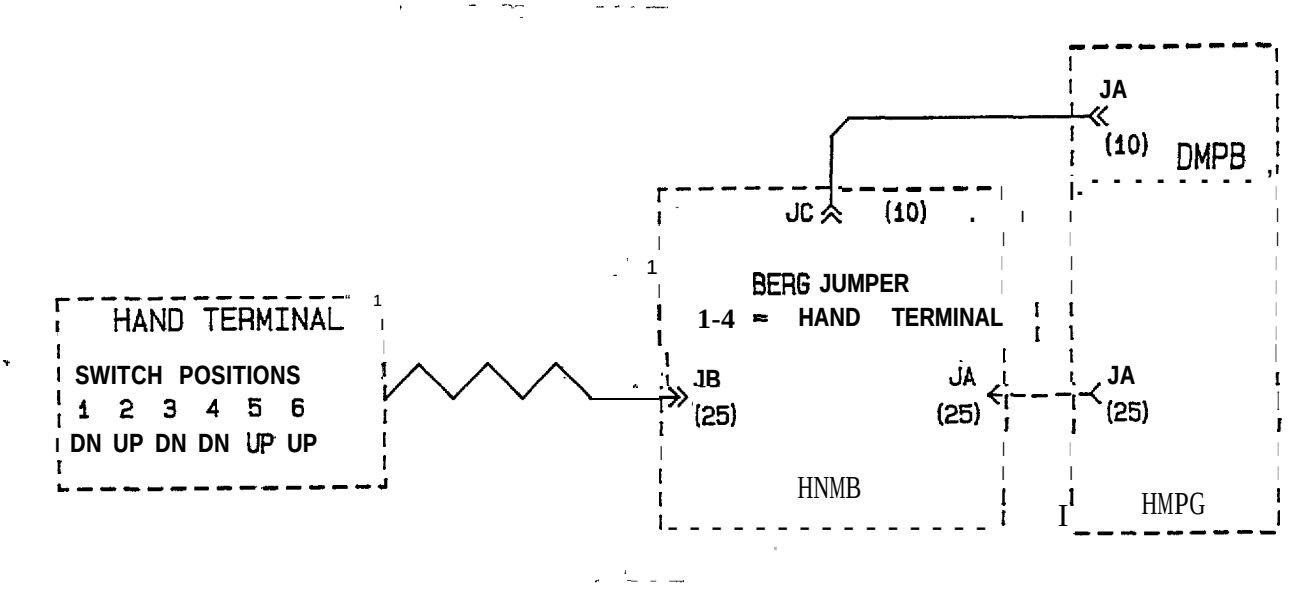


Figure 7D-1. Hand Terminal Connection

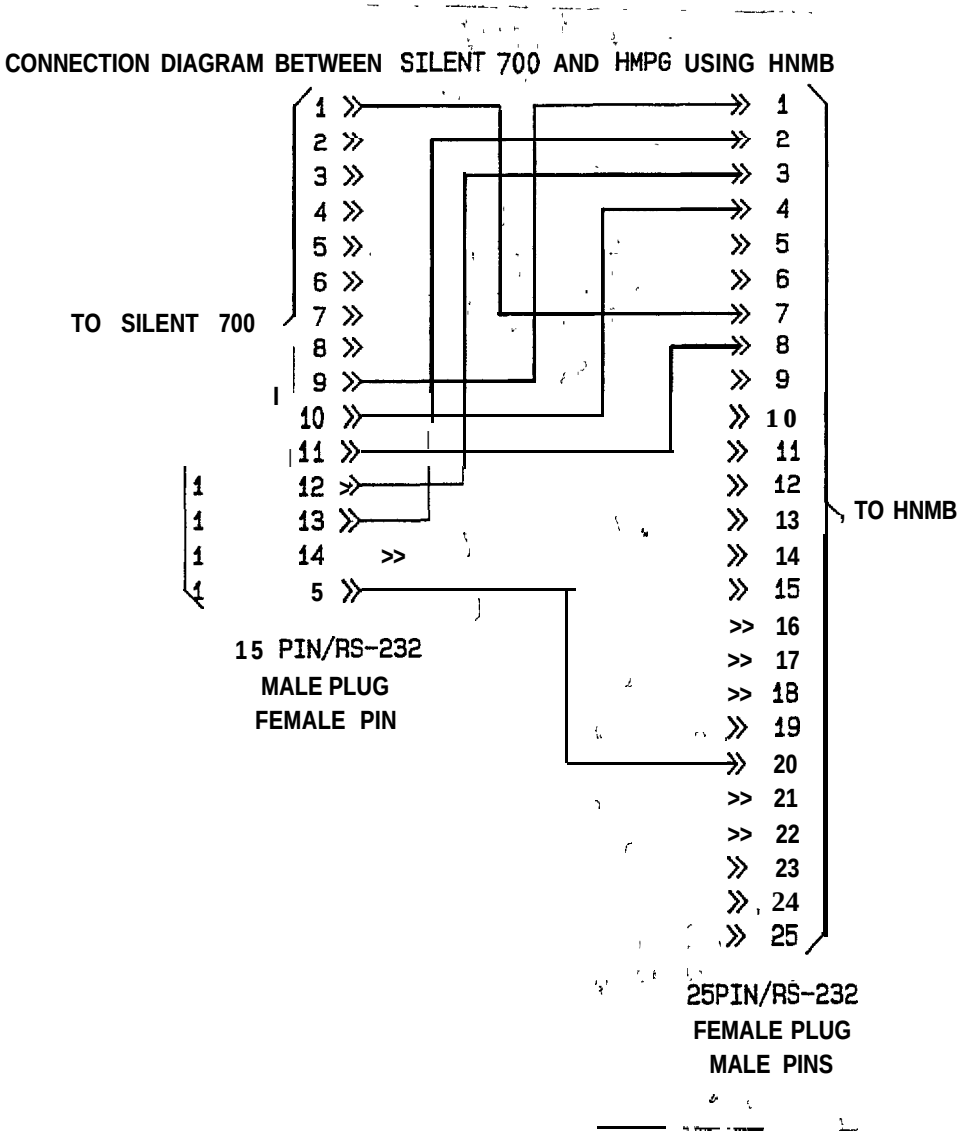

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG USING HNMB
TO SILENT 700
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
TO HNMB
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
25PIN/RS-232
FEMALE PLUG
MALE PINS

Figure 7D-2.

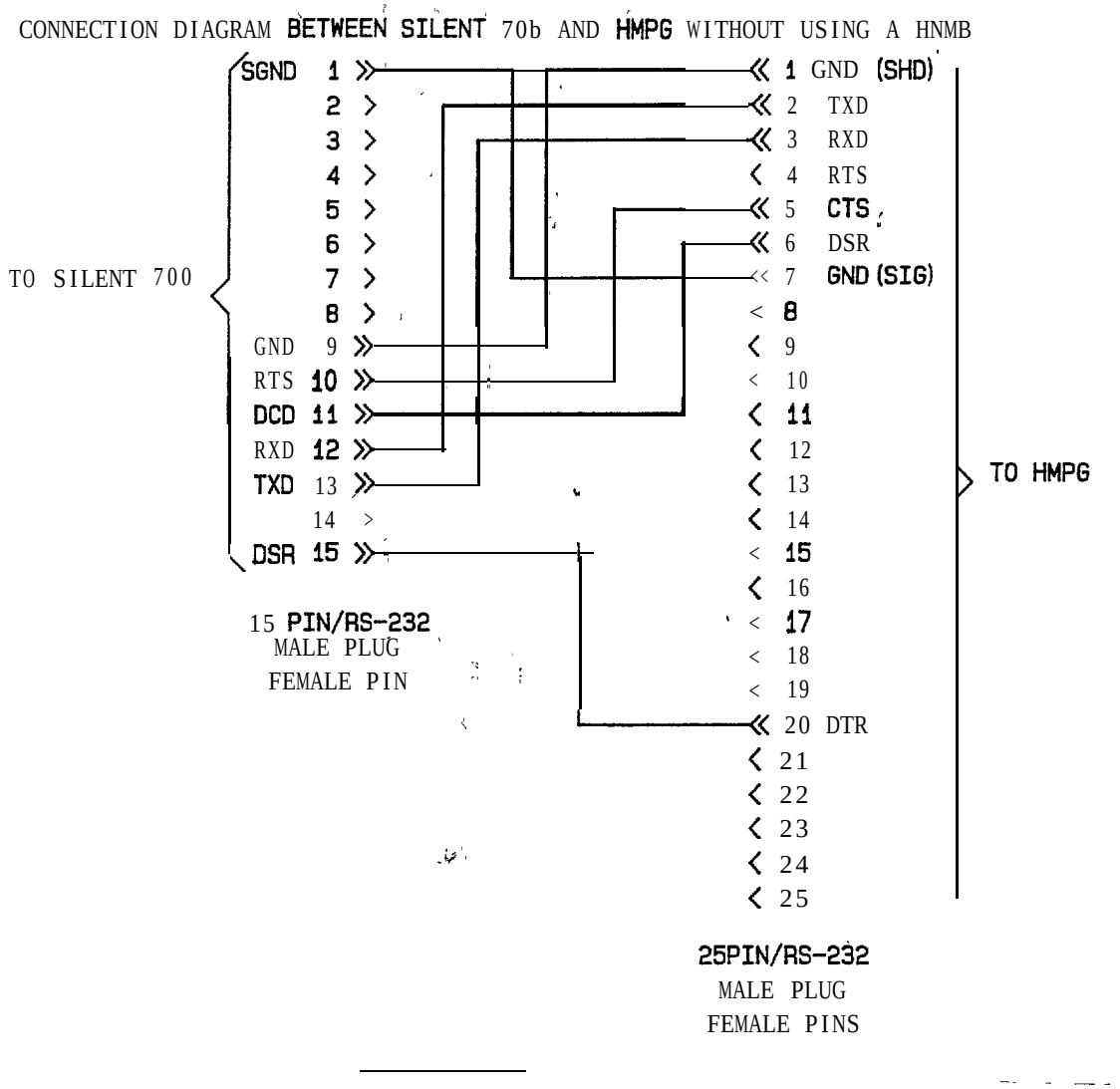

CONNECTION DIAGRAM BETWEEN SILENT 70b AND HMPG WITHOUT USING A HNMB
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-3.

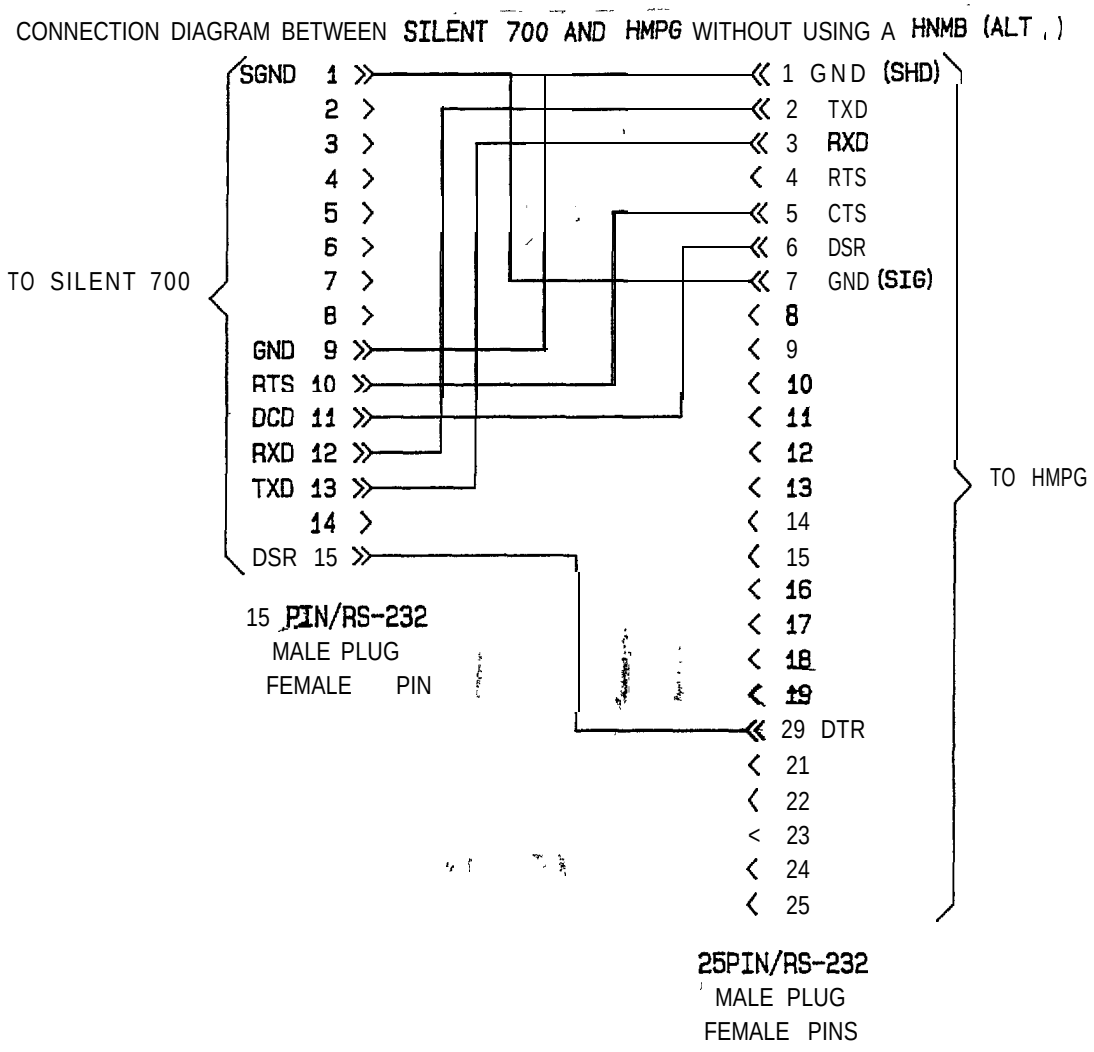
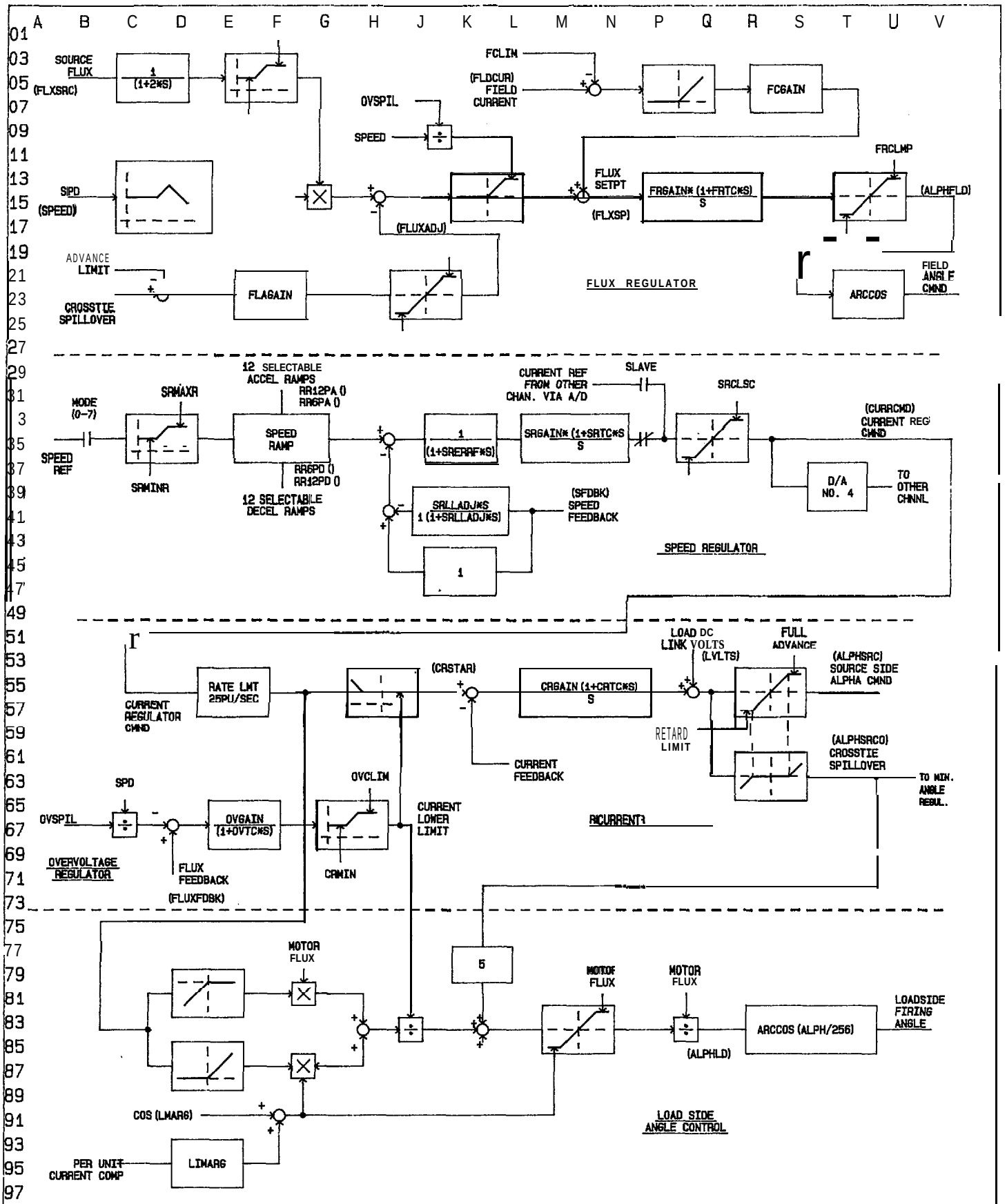

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB (ALT.)
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
29 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-4

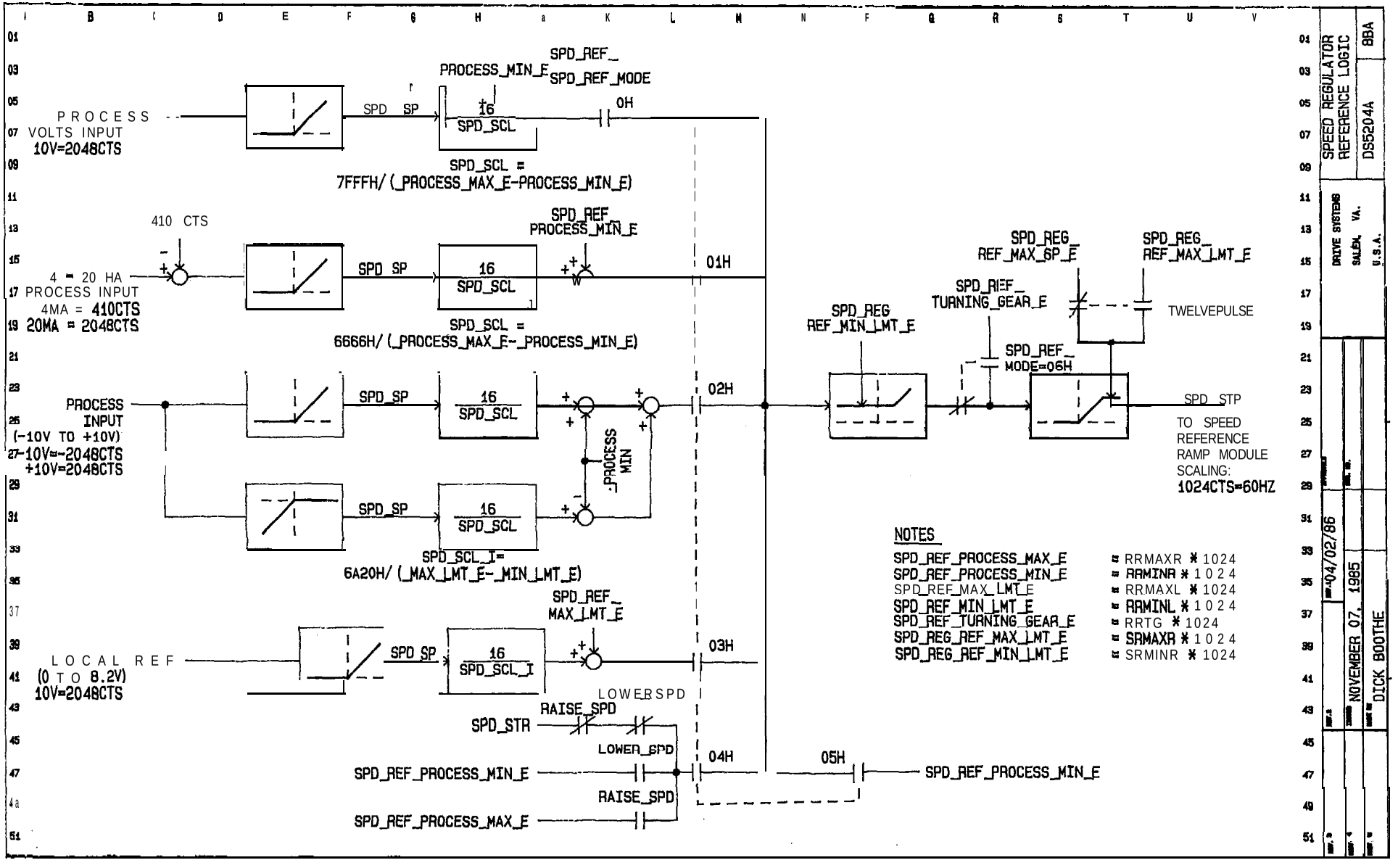


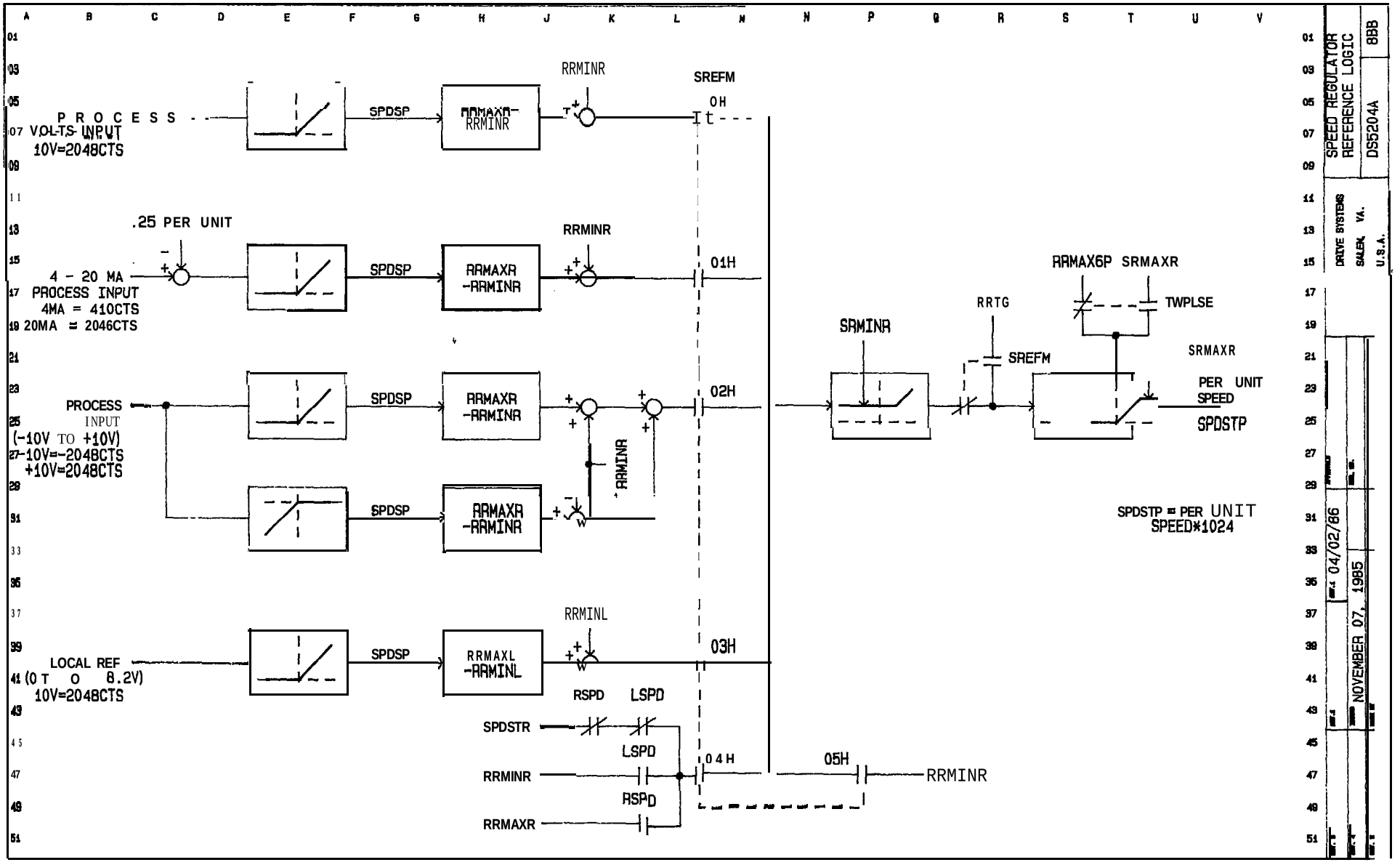
REV.1 04/03/86	REV.4	REV.5	PRINTS TO	APPROVALS -TE PLOTTED ( )	GENERAL ELECTRIC DRIVE SYSTEMS DEPT
REV.2	ISSUED NOVEMBER 18, 1985			1ST MADE FOR	
REV.3	MADE BY DICK DOOTIE			IC. NO.	

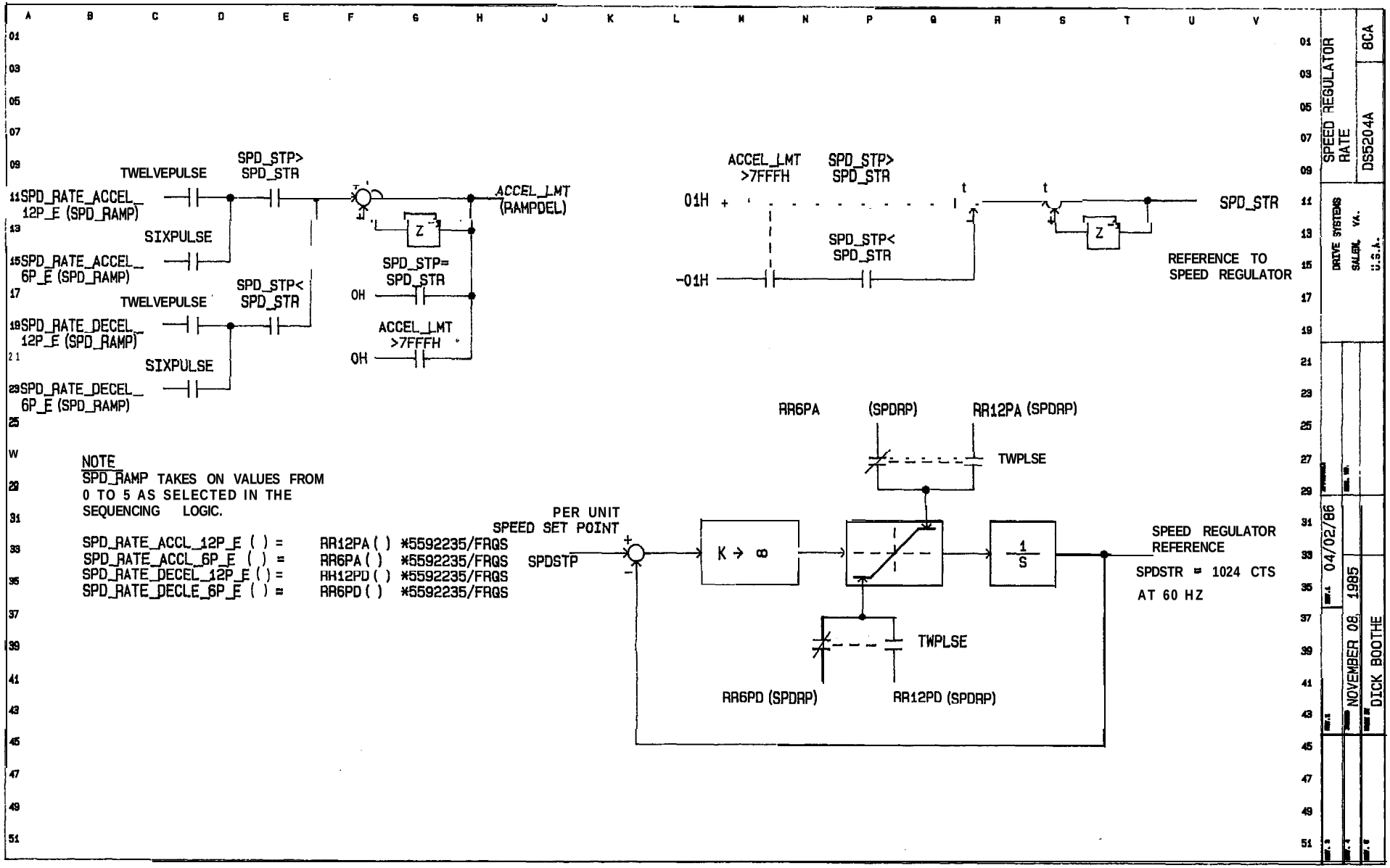
ELEMENTARY DIAGRAM
LCI REGULATOR

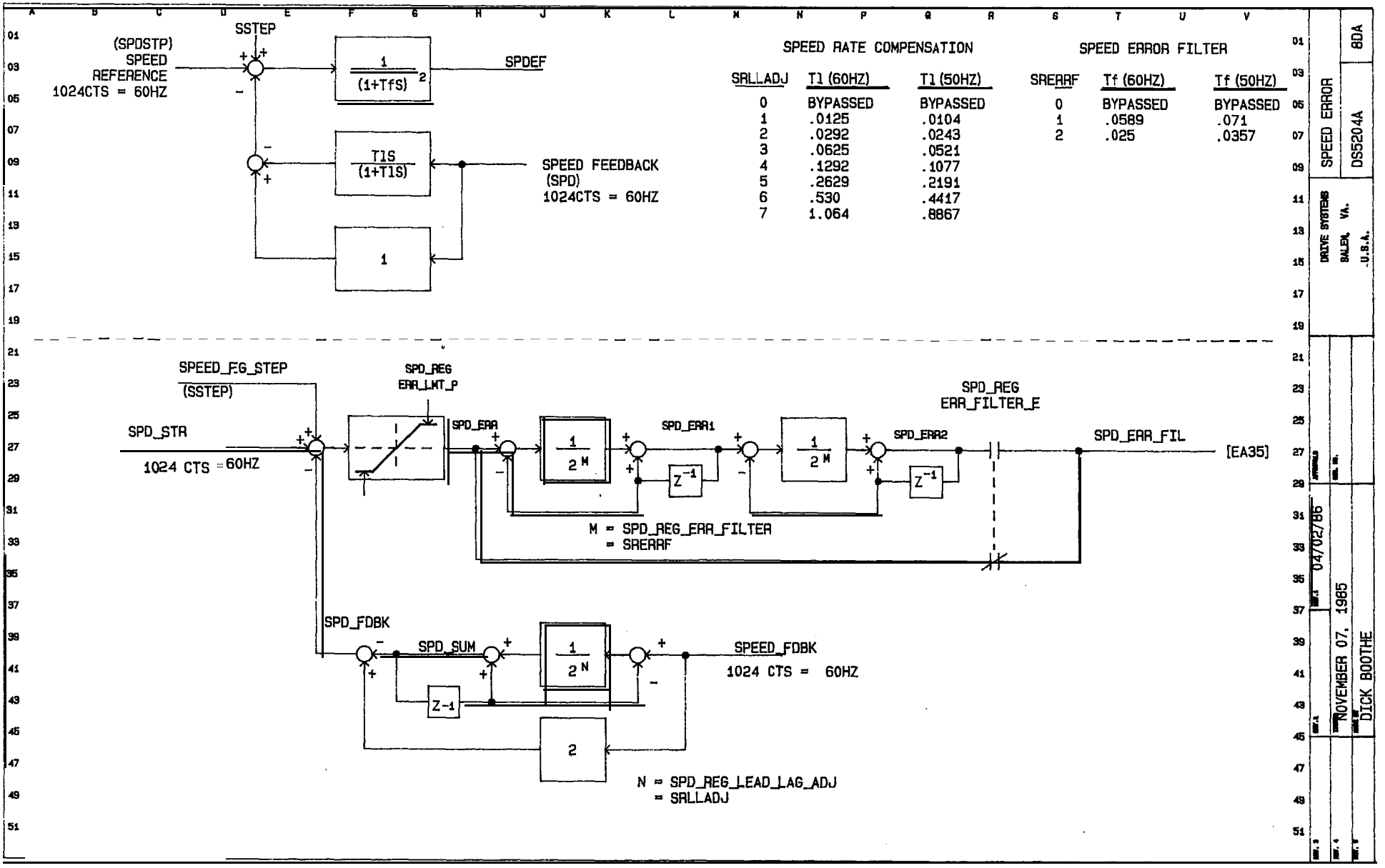
DS5204A

CONT.ON SH. 8BA SH.NO. 8AA


SPEED REGULATOR REFERENCE LOGIC
BBA
DS5204A
DRIVE SYSTEMS SALEM, VA. U.S.A.
04/02/86
NOVEMBER 07, 1985
DICK BOOTHE
PROCESS VOLTS INPUT 10V=2048CTS
SPD_REF_PROCESS_MIN_E
SPD_REF_MODE
0H
SPD SP
16
SPD_SCL
SPD_SCL = 7FFFH/(_PROCESS_MAX_E-PROCESS_MIN_E)
410 CTS
4 - 20 HA PROCESS INPUT
4MA = 410CTS
20MA = 2048CTS
SPD_REF PROCESS_MIN_E
01H
SPD_SCL = 6666H/(_PROCESS_MAX_E-_PROCESS_MIN_E)
PROCESS INPUT (-10V TO +10V)
-10V=-2048CTS
+10V=2048CTS
02H
-PROCESS MIN
SPD_SCL_I= 6A20H/(_MAX_LMT_E-_MIN_LMT_E)
SPD_REF_MAX_LMT_E
03H
LOCAL REF (0 TO 8.2V)
10V=2048CTS
LOWERSPD
RAISE_SPD
SPD_STR
LOWER_SPD
04H
05H
SPD_REF_PROCESS_MIN_E
RAISE_SPD
SPD_REF_PROCESS_MAX_E
SPD_REG_REF_MAX_6P_E
SPD_REG_REF_MAX_LMT_E
SPD_REF_TURNING_GEAR_E
TWELVEPULSE
SPD_REG REF_MIN_LMT_E
SPD_REF_MODE=06H
SPD_STP
TO SPEED REFERENCE RAMP MODULE SCALING:
1024CTS=60HZ
NOTES
SPD_REF_PROCESS_MAX_E = RRMAXR * 1024
SPD_REF_PROCESS_MIN_E = RRMINR * 1024
SPD_REF_MAX_LMT_E = RRMAXL * 1024
SPD_REF_MIN_LMT_E = RRMINL * 1024
SPD_REF_TURNING_GEAR_E = RRTG * 1024
SPD_REG_REF_MAX_LMT_E = SRMAXR * 1024
SPD_REG_REF_MIN_LMT_E = SRMINR * 1024


PROCESS
VOLTS INPUT
10V=2048CTS
SPDSP
RRMAXR-
RRMINR
RRMINR
SREFM
0H
.25 PER UNIT
4 - 20 MA
PROCESS INPUT
4MA = 410CTS
20MA = 2046CTS
SPDSP
RRMAXR
-RRMINR
RRMINR
01H
PROCESS
INPUT
(-10V TO +10V)
-10V=-2048CTS
+10V=2048CTS
SPDSP
RRMAXR
-RRMINR
02H
RRMINR
SPDSP
RRMAXR
-RRMINR
LOCAL REF
(0 T O 8.2V)
10V=2048CTS
SPDSP
RRMAXL
-RRMINL
RRMINL
03H
RSPD
LSPD
SPDSTR
LSPD
RRMINR
04H
05H
RRMINR
RSPD
RRMAXR
SRMINR
RRTG
SREFM
RRMAX6P SRMAXR
TWPLSE
SRMAXR
PER UNIT
SPEED
SPDSTP
SPDSTP = PER UNIT
SPEED*1024
SPEED REGULATOR
REFERENCE LOGIC
DS5204A
8BB
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 07, 1985


SPEED REGULATOR RATE
DS5204A
8CA
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 08, 1985
DICK BOOTHE
TWELVEPULSE
SPD_STP> SPD_STR
SPD_RATE_ACCEL_ 12P_E (SPD_RAMP)
SIXPULSE
SPD_RATE_ACCEL_ 6P_E (SPD_RAMP)
SPD_STP< SPD_STR
TWELVEPULSE
SPD_RATE_DECEL_ 12P_E (SPD_RAMP)
SIXPULSE
SPD_RATE_DECEL_ 6P_E (SPD_RAMP)
SPD_STP= SPD_STR
0H
ACCEL_LMT >7FFFH
0H
ACCEL_LMT (RAMPDEL)
Z
ACCEL_LMT >7FFFH
SPD_STP> SPD_STR
01H
SPD_STP< SPD_STR
-01H
SPD_STR
REFERENCE TO SPEED REGULATOR
NOTE
SPD_RAMP TAKES ON VALUES FROM 0 TO 5 AS SELECTED IN THE SEQUENCING LOGIC.
SPD_RATE_ACCL_12P_E ( ) = RR12PA( ) *5592235/FRQS
SPD_RATE_ACCL_6P_E ( ) = RR6PA( ) *5592235/FRQS
SPD_RATE_DECEL_12P_E ( ) = RR12PD( ) *5592235/FRQS
SPD_RATE_DECLE_6P_E ( ) = RR6PD( ) *5592235/FRQS
PER UNIT SPEED SET POINT
SPDSTP
K → ∞
RR6PA (SPDRP)
RR12PA (SPDRP)
TWPLSE
1/S
SPEED REGULATOR REFERENCE
SPDSTR = 1024 CTS AT 60 HZ
TWPLSE
RR6PD (SPDRP)
RR12PD (SPDRP)

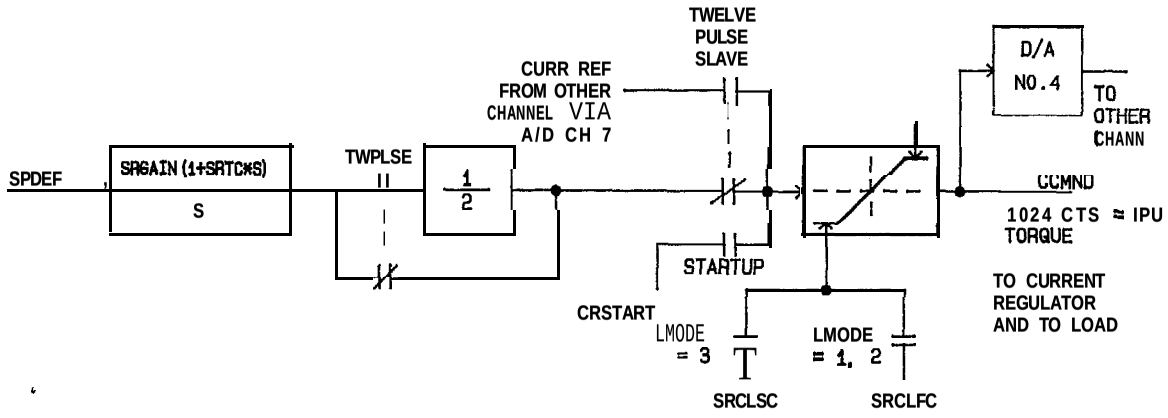

SSTEP
(SPDSTP)
SPEED
REFERENCE
1024CTS = 60HZ
1 / (1+TfS)2
SPDEF
T1S / (1+T1S)
SPEED FEEDBACK
(SPD)
1024CTS = 60HZ
1
SPEED RATE COMPENSATION
SRLLADJ | T1 (60HZ) | T1 (50HZ)
0 | BYPASSED | BYPASSED
1 | .0125 | .0104
2 | .0292 | .0243
3 | .0625 | .0521
4 | .1292 | .1077
5 | .2629 | .2191
6 | .530 | .4417
7 | 1.064 | .8867
SPEED ERROR FILTER
SRERRF | Tf (60HZ) | Tf (50HZ)
0 | BYPASSED | BYPASSED
1 | .0589 | .071
2 | .025 | .0357
SPEED_FG_STEP
(SSTEP)
SPD_REG
ERR_LMT_P
SPD_STR
1024 CTS = 60HZ
SPD_ERR
1 / 2M
SPD_ERR1
Z-1
SPD_ERR2
SPD_REG
ERR_FILTER_F
SPD_ERR_FIL
[EA35]
M = SPD_REG_ERR_FILTER
= SRERRF
SPD_FDBK
SPD_SUM
1 / 2N
SPEED_FDBK
1024 CTS = 60HZ
Z-1
2
N = SPD_REG_LEAD_LAG_ADJ
= SRLLADJ
SPEED ERROR
8DA
DS5204A
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04702/86
NOVEMBER 07, 1985
DICK BOOTHE

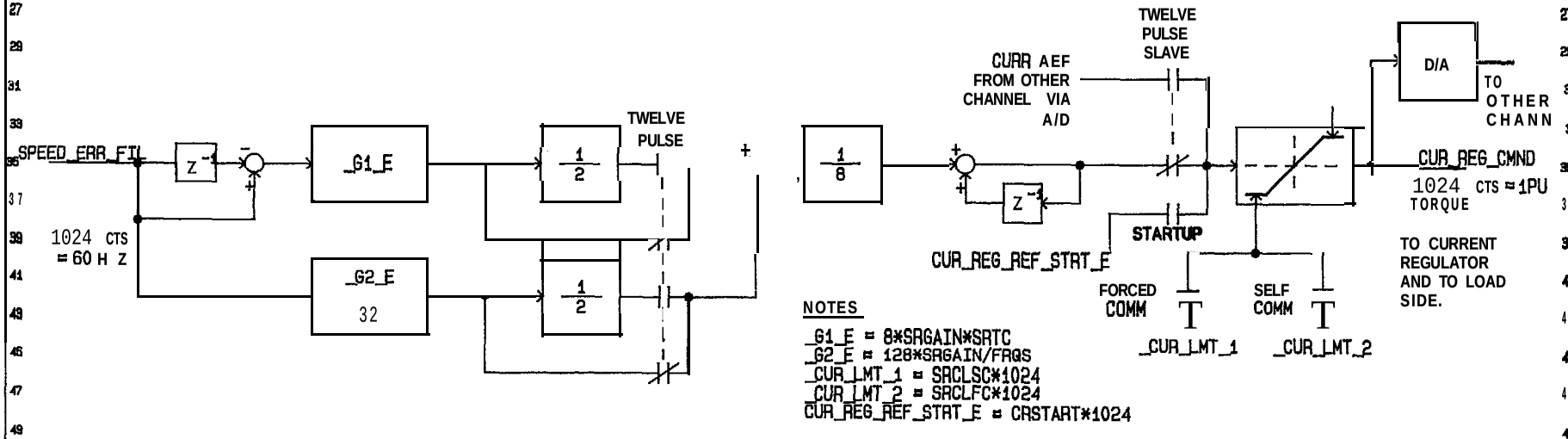
NOTES

THE SPEED REGULATOR CONFIGURATION IS THE STANDARD INTEGRAL PLUS PROPORTIONAL A RATE FEEDBACK IS INCLUDED TO ALLOW A CRITICALLY DAMPED TUNEUP. THE SPEED ERROR FILTER MINIMIZES CURRENT JITTER WHEN LARGE GAINS ARE NEEDED.

AN AUTOMATIC GAIN CHANGE OCCURS WHEN OPERATING IN TWELVE PULSE SUCH THAT THE SPEED REGULATOR RESPONSE REMAINS UNCHARGED.

THE SPEED REGULATOR IS LOCATED IN THE SOURCE PHASE LOCK LOOP INTERRUPT AND IS EXECUTED TWICE PER SOURCE SIDE CYCLE

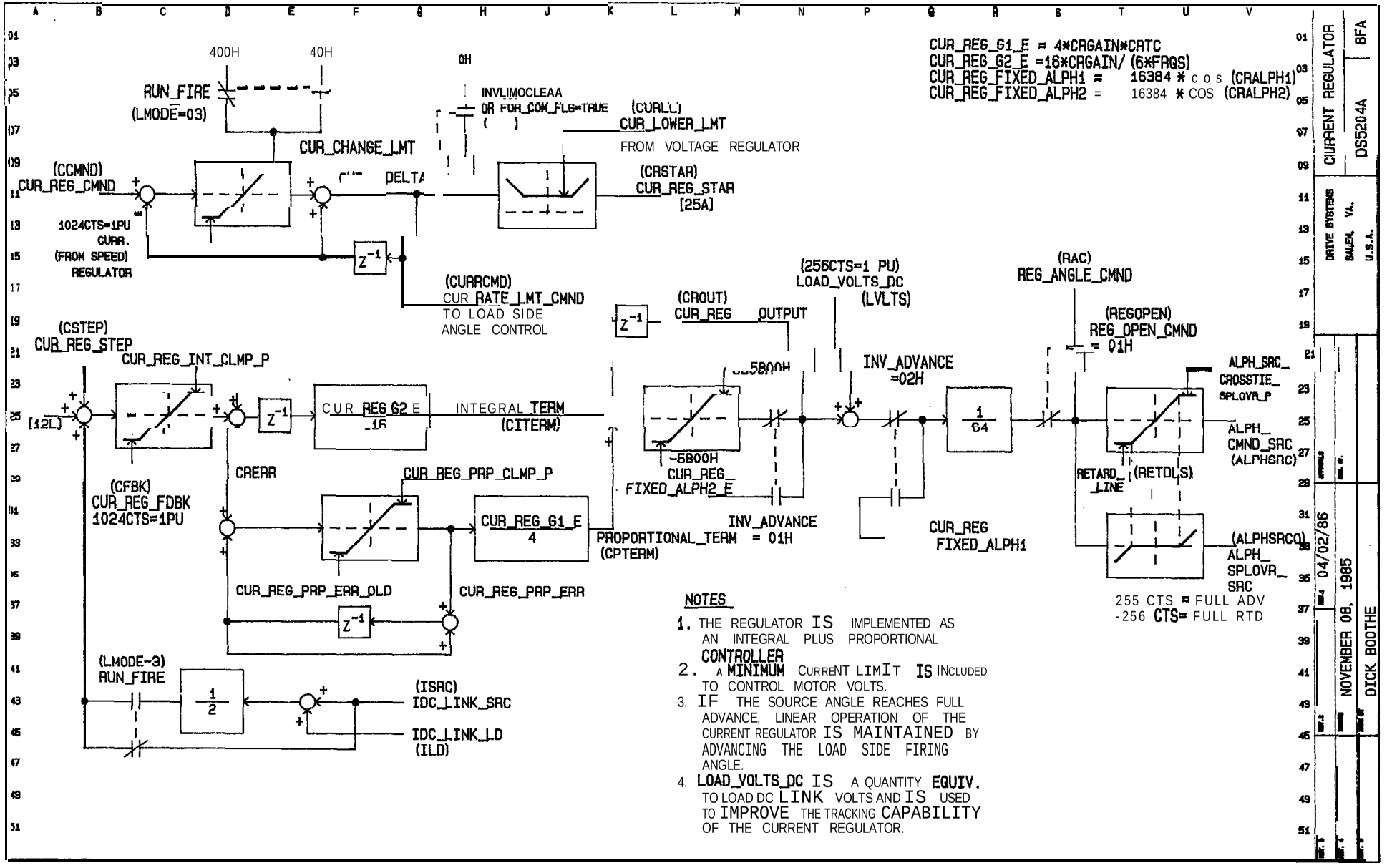


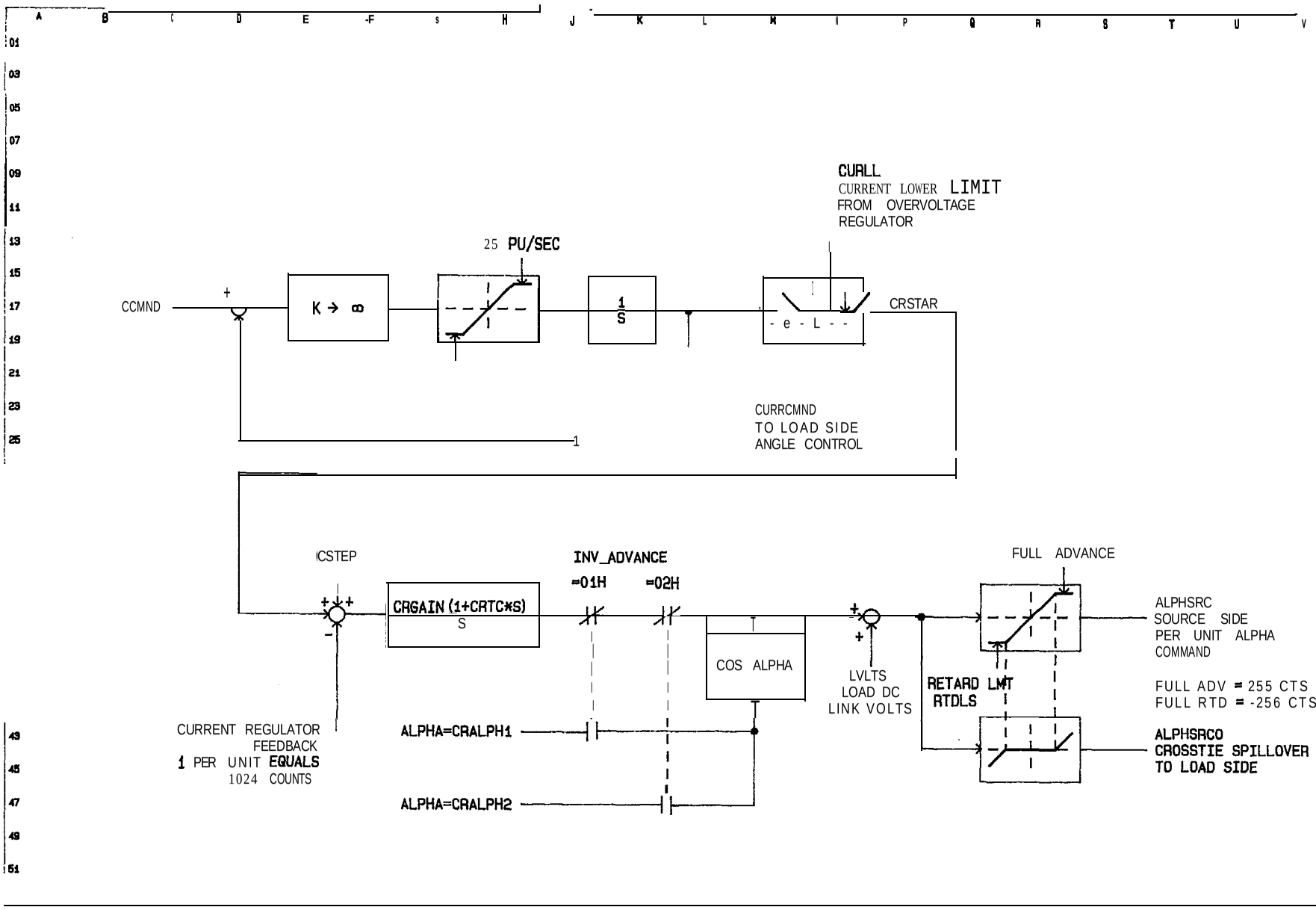


NOTES

_G1_E = 8*SRGAIN*SRTC
_G2_E = 128*SRGAIN/FRQS
_CUR_LMT_1 = SRCLSC*1024
_CUR_LMT_2 = SRCLFC*1024
CUR_REG_REF_STRT_E = CRSTART*1024

SPEED REGULATOR TORQUE CLAMPS | DS5204A | 8EA
DRIVE SYSTEMS SALEM, VA.
NOVEMBER 07, 1985 | 04/02/86
DICK BOOTHE


CUR_REG_G1_E = 4*CRGAIN*CRTC
CUR_REG_G2_E =16*CRGAIN/ (6*FRQS)
CUR_REG_FIXED_ALPH1 = 16384 * cos (CRALPH1)
CUR_REG_FIXED_ALPH2 = 16384 * COS (CRALPH2)
400H
40H
0H
RUN_FIRE
(LMODE=03)
CUR_CHANGE_LMT
INVLIMOCLEAA
OR FOR_COM_FLG=TRUE
( )
(CURLL)
CUR_LOWER_LMT
FROM VOLTAGE REGULATOR
(CCMND)
CUR_REG_CMND
1024CTS=1PU
CURR.
(FROM SPEED)
REGULATOR
DELTA
(CRSTAR)
CUR_REG_STAR
[25A]
(CURRCMD)
CUR_RATE_LMT_CMND
TO LOAD SIDE
ANGLE CONTROL
(CROUT)
CUR_REG OUTPUT
(256CTS=1 PU)
LOAD_VOLTS_DC
(LVLTS)
(RAC)
REG_ANGLE_CMND
(REGOPEN)
REG_OPEN_CMND
= 01H
(CSTEP)
CUR_REG_STEP
CUR_REG_INT_CLMP_P
[12L]
CUR_REG_G2_E
16
INTEGRAL_TERM
(CITERM)
-5800H
CUR_REG_
FIXED_ALPH2_E
INV_ADVANCE
=02H
1
64
ALPH_SRC_
CROSSTIE_
SPLOVR_P
ALPH_
CMND_SRC
(ALPHSRC)
RETARD_
LINE
(RETDLS)
(CFBK)
CUR_REG_FDBK
1024CTS=1PU
CREAR
CUR_REG_PRP_CLMP_P
CUR_REG_G1_E
4
PROPORTIONAL_TERM
(CPTERM)
INV_ADVANCE = 01H
CUR_REG
FIXED_ALPH1
CUR_REG_PRP_ERR_OLD
CUR_REG_PRP_ERR
(LMODE=3)
RUN_FIRE
1
2
(ISRC)
IDC_LINK_SRC
IDC_LINK_LD
(ILD)
(ALPHSRC0)
ALPH_
SPLOVR_
SRC
255 CTS = FULL ADV
-256 CTS= FULL RTD
NOTES
1. THE REGULATOR IS IMPLEMENTED AS AN INTEGRAL PLUS PROPORTIONAL CONTROLLER
2. A MINIMUM CURRENT LIMIT IS INCLUDED TO CONTROL MOTOR VOLTS.
3. IF THE SOURCE ANGLE REACHES FULL ADVANCE, LINEAR OPERATION OF THE CURRENT REGULATOR IS MAINTAINED BY ADVANCING THE LOAD SIDE FIRING ANGLE.
4. LOAD_VOLTS_DC IS A QUANTITY EQUIV. TO LOAD DC LINK VOLTS AND IS USED TO IMPROVE THE TRACKING CAPABILITY OF THE CURRENT REGULATOR.
CURRENT REGULATOR
DSS204A
8FA
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 08, 1985
DICK BOOTHE



CURRENT REGULATOR | 8FB
DS5204A
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 08, 1985
DICK ROUTHE

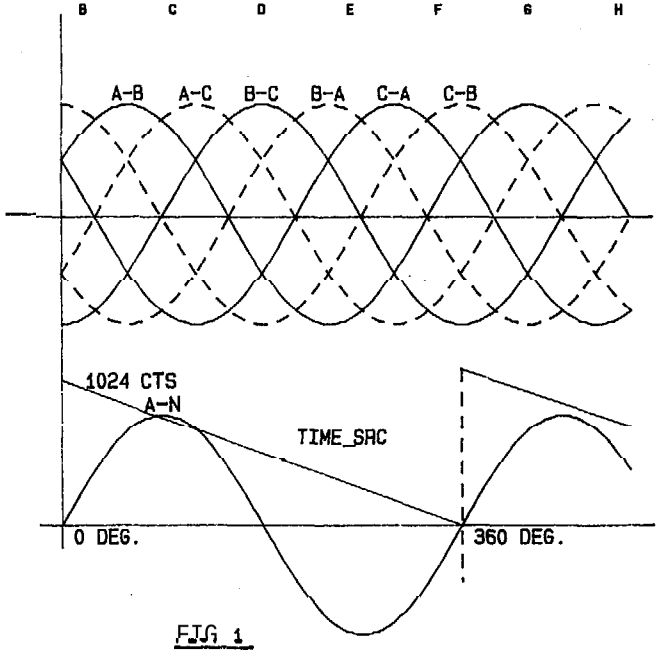


FIG 1

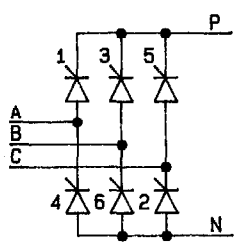


FIG 2

PH TABLE		
L-L	PH	OFFSET (CTS)
A-B	1	55H
A-C	2	OFFH
B-C	3	1AAH
B-A	4	255H
C-A	5	2FFH
C-B	6	3AAH

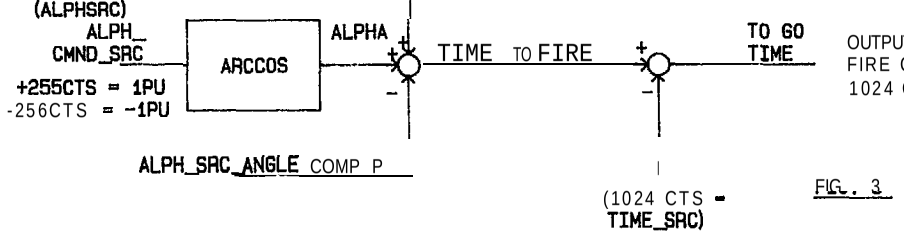
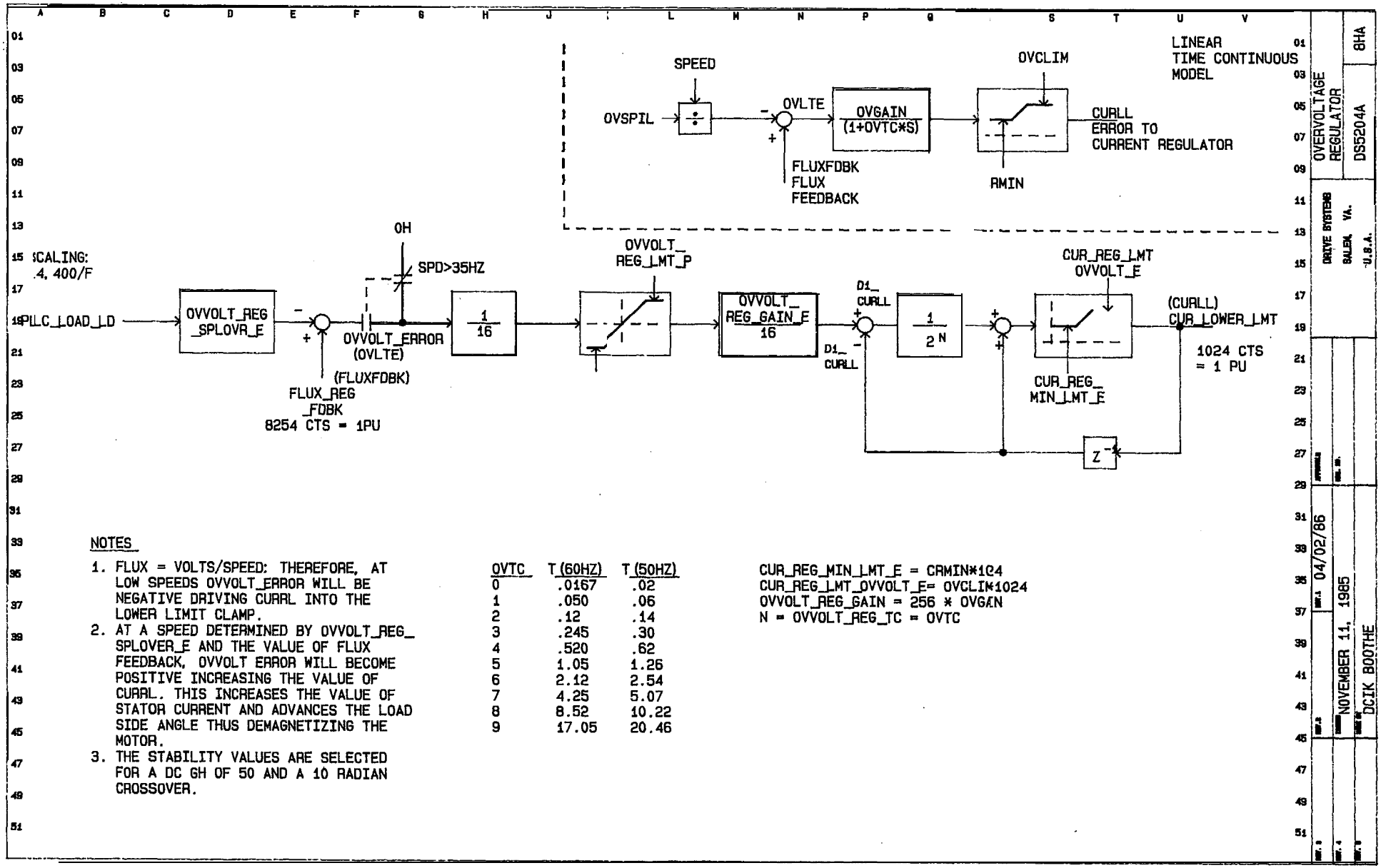


FIG. 3

TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE SOURCE PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE BRIDGE VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 2 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 0 DEGREES (FULL ADVANCE) FOR A GIVEN PH.
3. THE VALUE OF ALPH_CMND_SRC CALCULATED BY THE STATDR CURRENT REGULATOR IS CONVERTEO TO A FIRING ANGLE ACCORDING TO THE EQUATION ALPHA = ARC COS (ALPH_CMND_SRC). THIS IMPLEMENTED IN THE SOFTWARE WITH AN INDEXED LOOKUP TABLE.
4. THE VALUE OF ALPHA IS ADDED TO THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED. A SMALL COMPENSATION IS INCLUDED TO ACCOUNT FOR THE DELAY BETWEEN WHEN THE FIRING INTERRUPT OCCURS AND THE CODE WHICH IS EXECUTED BEFORE THE ACTUAL FIRING. THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME TO FIRE - PRESENT TIME) IS THE DIFFERENCE BETWEEN TIME TO_GO AND PRESENT TIME. THE SOFTWARE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC..

SOURCE SIDE TIME TO GO CALCULATION
8GA
DS5204A
DRIVE SYSTEMS
SALEM, VI.
U.S.A.
04/02/86
JUNE 27, 1985
DICK BOOTHE

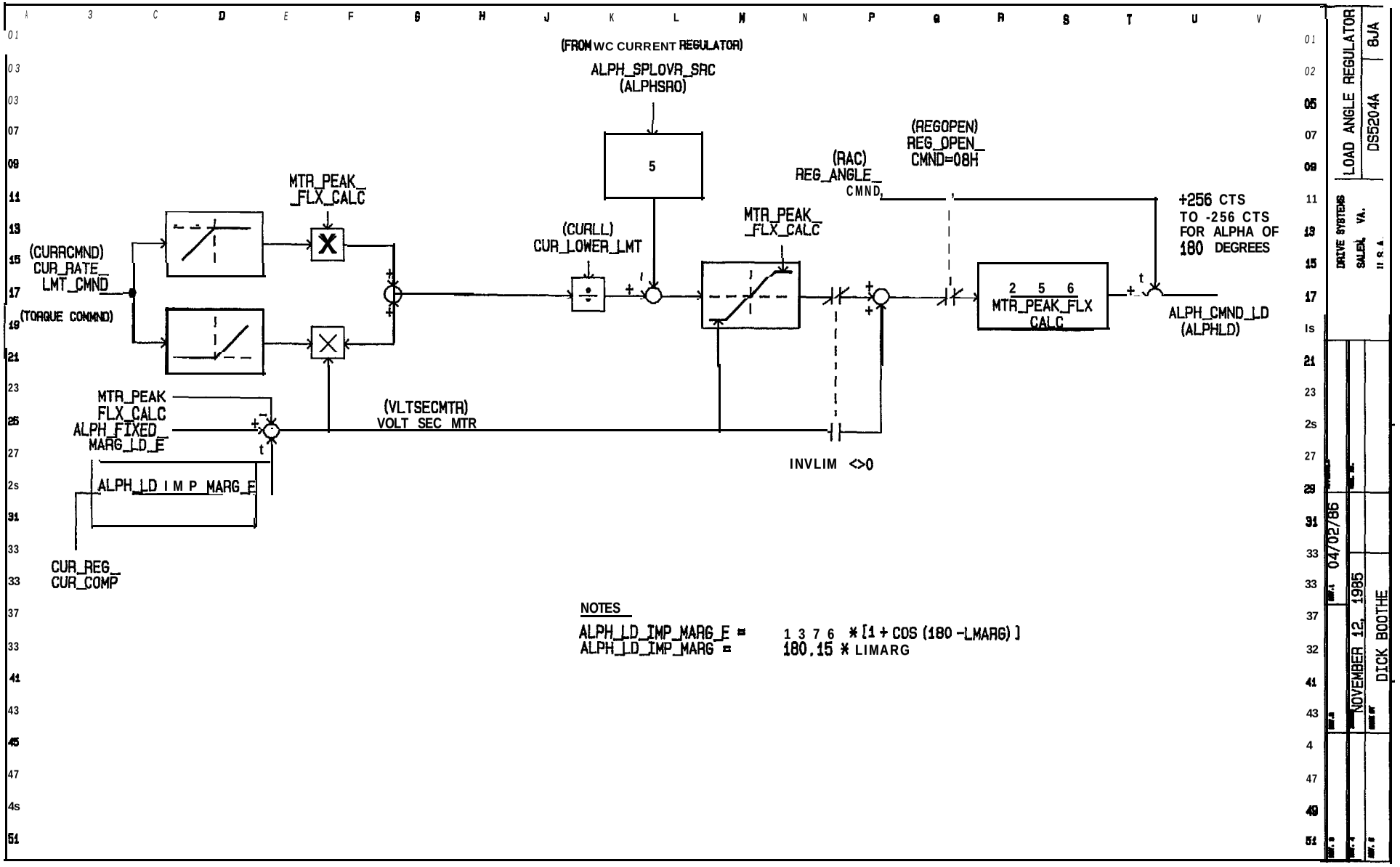


NOTES

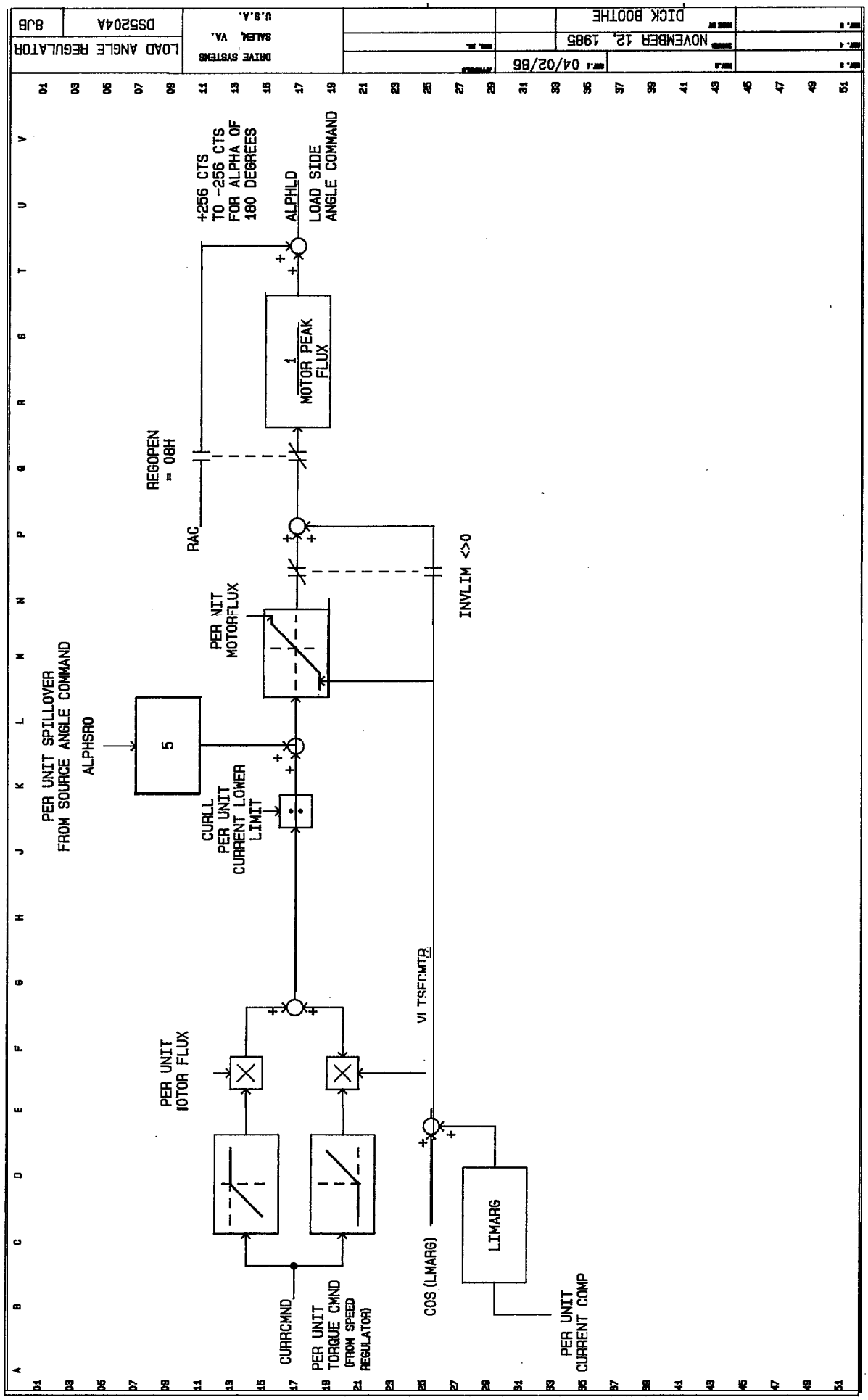
1. FLUX = VOLTS/SPEED: THEREFORE, AT LOW SPEEDS OVVOLT_ERROR WILL BE NEGATIVE DRIVING CURRL INTO THE LOWER LIMIT CLAMP.
2. AT A SPEED DETERMINED BY OVVOLT_REG_ SPLOVER_E AND THE VALUE OF FLUX FEEDBACK, OVVOLT ERROR WILL BECOME POSITIVE INCREASING THE VALUE OF CURRL. THIS INCREASES THE VALUE OF STATOR CURRENT AND ADVANCES THE LOAD SIDE ANGLE THUS DEMAGNETIZING THE MOTOR.
3. THE STABILITY VALUES ARE SELECTED FOR A DC GH OF 50 AND A 10 RADIAN CROSSOVER.

OVTC	T (60HZ)	T (50HZ)
0	.0167	.02
1	.050	.06
2	.12	.14
3	.245	.30
4	.520	.62
5	1.05	1.26
6	2.12	2.54
7	4.25	5.07
8	8.52	10.22
9	17.05	20.46

CUR_REG_MIN_LMT_E = CRMIN*1024
CUR_REG_LMT_OVVOLT_E= OVCLIM*1024
OVVOLT_REG_GAIN = 256 * OVGAIN
N = OVVOLT_REG_TC = OVTC


(FROM WC CURRENT REGULATOR)
ALPH_SPLOVR_SRC
(ALPHSRO)
5
(REGOPEN)
REG_OPEN_
CMND=08H
(RAC)
REG_ANGLE_
CMND,
MTR_PEAK_
_FLX_CALC
(CURRCMND)
CUR_RATE_
LMT_CMND
(TORQUE COMMND)
(CURLL)
CUR_LOWER_LMT
MTR_PEAK_
_FLX_CALC
+256 CTS
TO -256 CTS
FOR ALPHA OF
180 DEGREES
2 5 6
MTR_PEAK_FLX
CALC
ALPH_CMND_LD
(ALPHLD)
MTR_PEAK
FLX_CALC
ALPH_FIXED_
MARG_LD_E
(VLTSECMTR)
VOLT SEC MTR
INVLIM <>0
ALPH_LD IMP MARG_E
CUR_REG_
CUR_COMP
NOTES
ALPH_LD_IMP_MARG_E = 1 3 7 6 * [1 + COS (180 -LMARG)]
ALPH_LD_IMP_MARG = 180.15 * LIMARG
LOAD ANGLE REGULATOR
DS5204A
8JA
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 12, 1985
DICK BOOTHE

157


LOAD ANGLE REGULATOR
DS52044
8JB
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/02/86
NOVEMBER 12, 1985
DICK BOOTHE
+256 CTS
TO -256 CTS
FOR ALPHA OF
180 DEGREES
ALPHLD
LOAD SIDE
ANGLE COMMAND
1
MOTOR PEAK
FLUX
REGOPEN
= 08H
RAC
INVLIM <>0
PER NIT
MOTORFLUX
PER UNIT SPILLOVER
FROM SOURCE ANGLE COMMAND
ALPHSRO
5
CURLL
PER UNIT
CURRENT LOWER
LIMIT
PER UNIT
MOTOR FLUX
VI TSFCMTR
CURRCMND
PER UNIT
TORQUE CMND
(FROM SPEED
REGULATOR)
COS (LMARG)
LIMARG
PER UNIT
CURRENT COMP

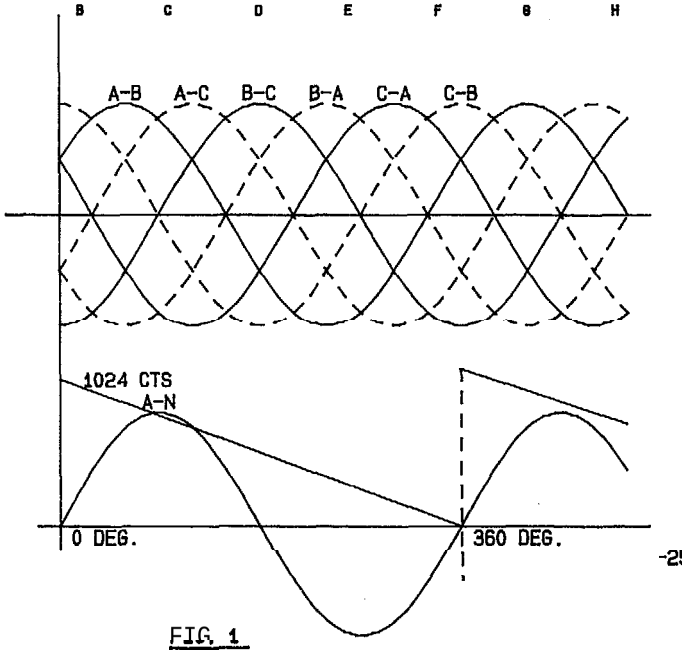


FIG. 1

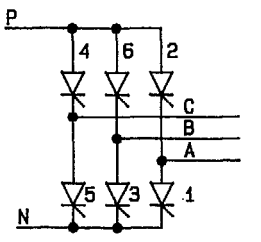


FIG. 2

PH TABLE		
L - L		PH OFFSET (CTS)
A-B	1	3AAH
A-C	2	2FFH
B-C	3	255H
B-A	4	1AAH
C-A	5	0FFH
C-B	6	55H

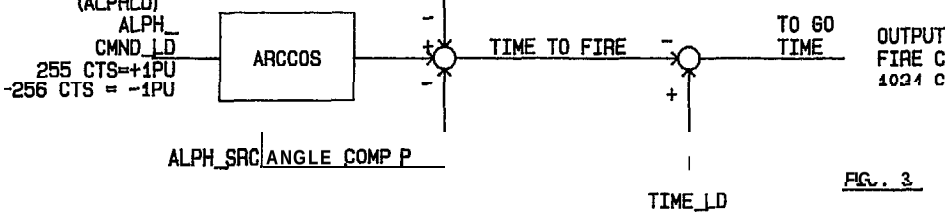
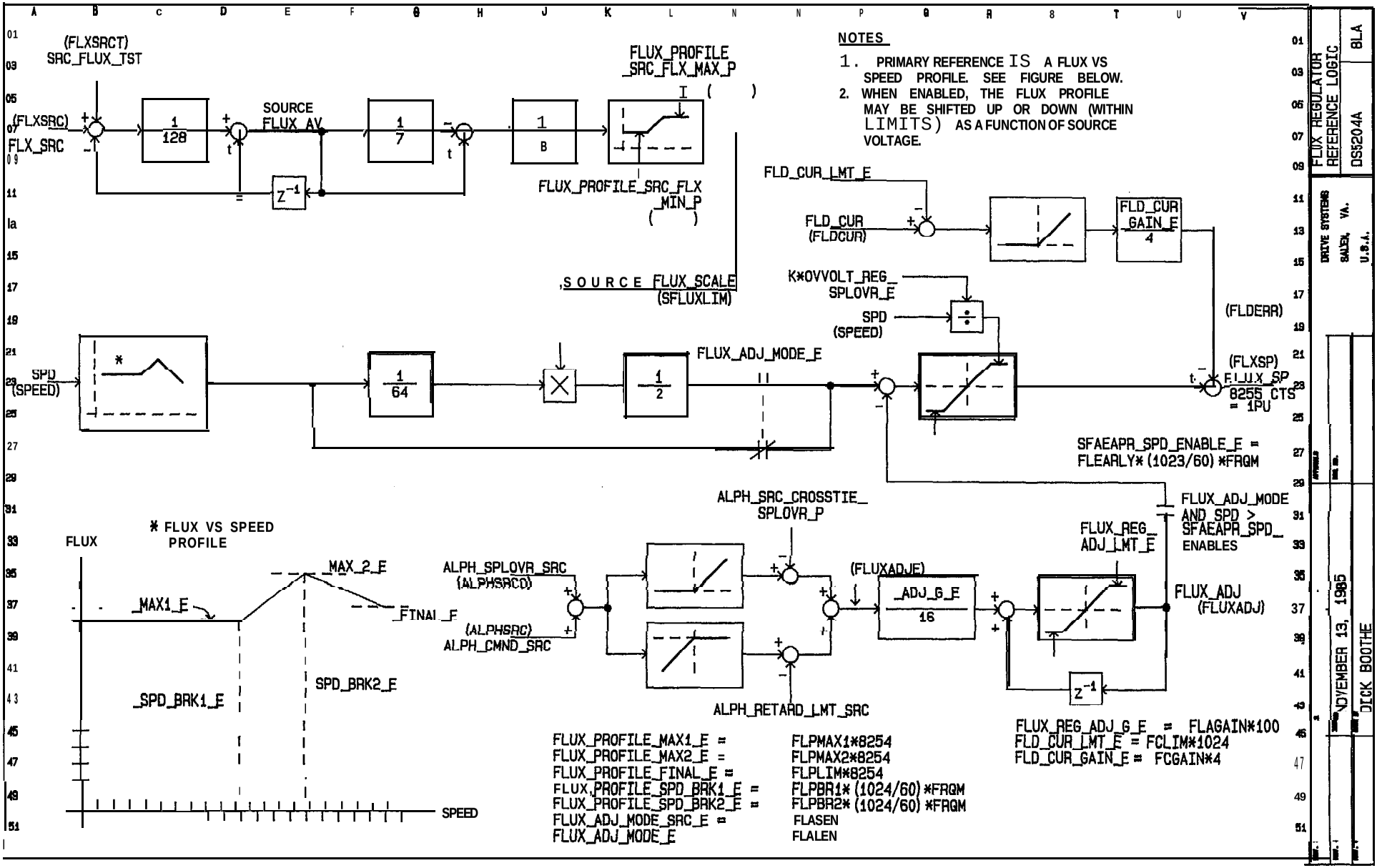


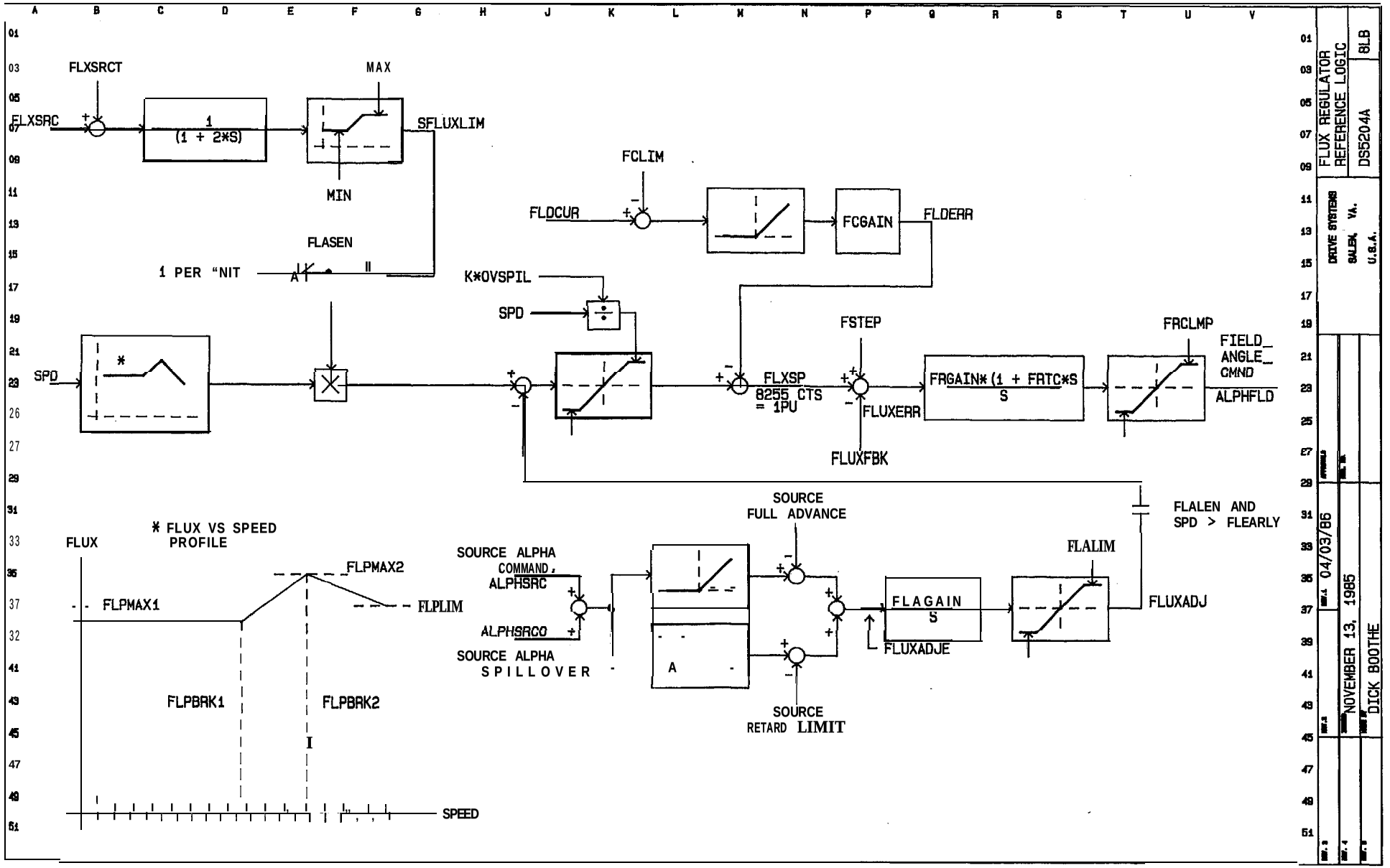
FIG. 3

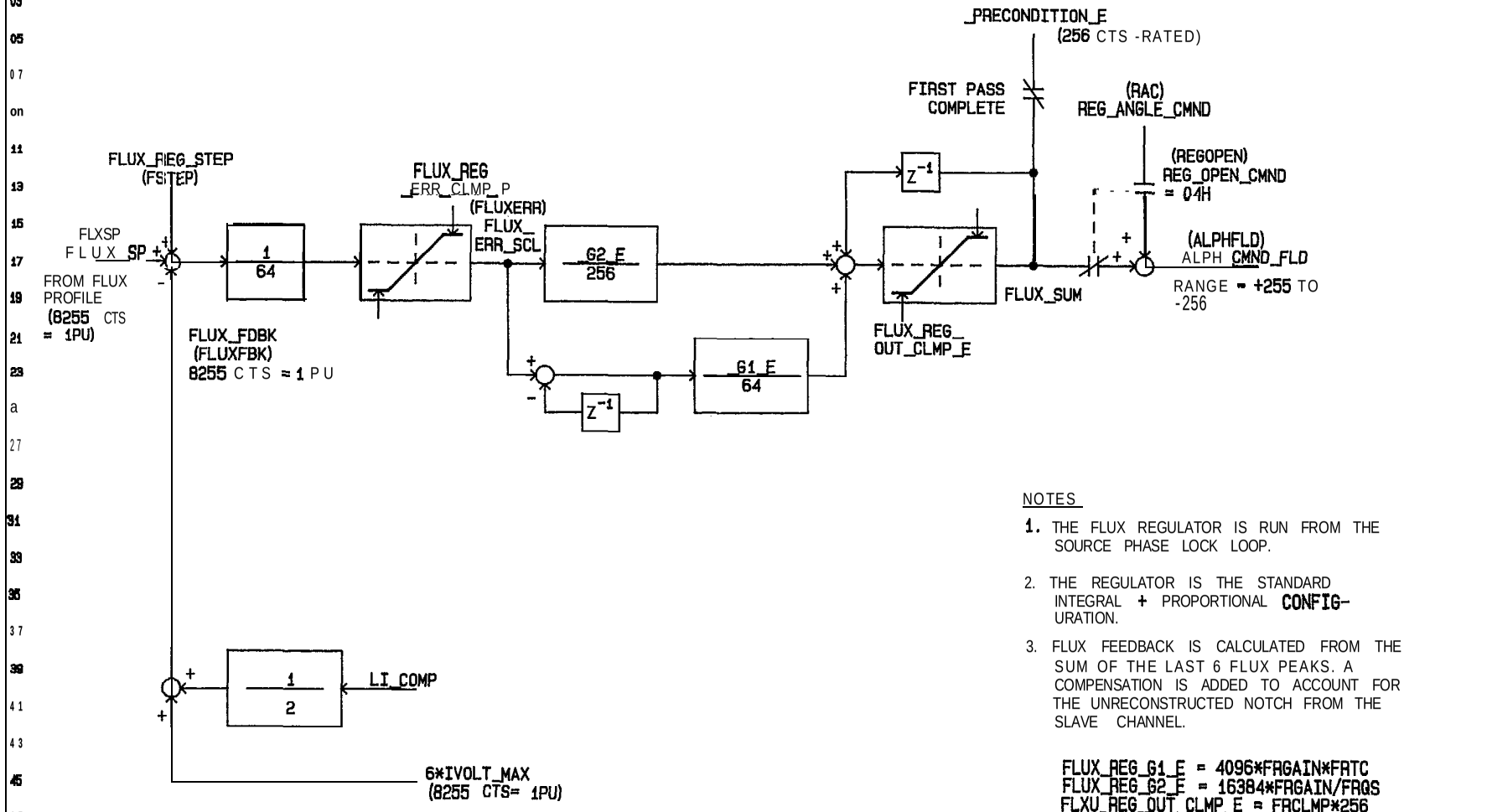
TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE SOURCE PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE BRIDGE VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1) . NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 2 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 0 DEGREES (FULL ADVANCE) FOR A GIVEN PH.
3. THE VALUE OF ALPH_CMND_LD CALCULATED BY THE STATOR CURRENT REGULATOR IS CONVERTED TO A FIRING ANGLE ACCORDING TO THE EQUATION ALPHA = ARC COS (ALPH_CMND_LD). THIS IS IMPLEMENTED IN THE SOFTWARE WITH AN INDEXED LOOKUP TABLE.
4. THE VALUE OF ALPHA IS ADDED TO THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED. A SMALL COMPENSATION IS INCLUDED TO ACCOUNT FOR THE DELAY BETWEEN WHEN THE FIRING INTERRUPT OCCURS AND THE CODE WHICH IS EXECUTED BEFORE THE ACTUAL FIRING. THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME TO FIRE - PRESENT TIME) IS THE DIFFERENCE BETWEEN TIME TO_GO AND PRESENT TIME. THE SOFTWARE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC.

LOAD SIDE TIME TO GO CALCULATION
DS5204A
8KA
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/03/86
NOVEMBER 12, 1985


(FLXSRCT)
SRC_FLUX_TST
(FLXSRC)
FLX_SRC
SOURCE
FLUX_AV
FLUX_PROFILE
_SRC_FLX_MAX_P
I ( )
FLUX_PROFILE_SRC_FLX
_MIN_P
( )
NOTES
1. PRIMARY REFERENCE IS A FLUX VS SPEED PROFILE. SEE FIGURE BELOW.
2. WHEN ENABLED, THE FLUX PROFILE MAY BE SHIFTED UP OR DOWN (WITHIN LIMITS) AS A FUNCTION OF SOURCE VOLTAGE.
FLD_CUR_LMT_E
FLD_CUR
(FLDCUR)
FLD_CUR
GAIN_E
SOURCE FLUX_SCALE
(SFLUXLIM)
K*OVVOLT_REG_
SPLOVR_E
SPD
(SPEED)
(FLDERR)
SPD
(SPEED)
FLUX_ADJ_MODE_E
(FLXSP)
FLUX_SP
8255 CTS
= 1PU
SFAEAPR_SPD_ENABLE_E =
FLEARLY* (1023/60) *FRQM
ALPH_SRC_CROSSTIE_
SPLOVR_P
FLUX_ADJ_MODE
AND SPD >
SFAEAPR_SPD_
ENABLES
FLUX_REG_
ADJ_LMT_E
* FLUX VS SPEED
PROFILE
FLUX
MAX_2_E
MAX1_E
FINAL_E
SPD_BRK1_E
SPD_BRK2_E
SPEED
ALPH_SPLOVR_SRC
(ALPHSRCD)
(ALPHSRC)
ALPH_CMND_SRC
(FLUXADJE)
ADJ_G_E
16
FLUX_ADJ
(FLUXADJ)
ALPH_RETARD_LMT_SRC
FLUX_REG_ADJ_G_E = FLAGAIN*100
FLD_CUR_LMT_E = FCLIM*1024
FLD_CUR_GAIN_E = FCGAIN*4
FLUX_PROFILE_MAX1_E = FLPMAX1*8254
FLUX_PROFILE_MAX2_E = FLPMAX2*8254
FLUX_PROFILE_FINAL_E = FLPLIM*8254
FLUX_PROFILE_SPD_BRK1_E = FLPBR1* (1024/60) *FRQM
FLUX_PROFILE_SPD_BRK2_E = FLPBR2* (1024/60) *FRQM
FLUX_ADJ_MODE_SRC_E = FLASEN
FLUX_ADJ_MODE_E FLALEN
FLUX REGULATOR
REFERENCE LOGIC
8LA
DS5204A
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
NOVEMBER 13, 1985
DICK BOOTHE


FLXSRCT
MAX
FLXSRC
1
(1 + 2*S)
SFLUXLIM
MIN
FCLIM
FLDCUR
FCGAIN
FLDERR
FLASEN
1 PER "NIT
K*OVSPIL
SPD
FSTEP
FRCLMP
FIELD_
ANGLE_
CMND
SPD
FLXSP
8255 CTS
= 1PU
FRGAIN* (1 + FRTC*S
S
ALPHFLD
FLUXERR
FLUXFBK
FLALEN AND
SPD > FLEARLY
* FLUX VS SPEED
PROFILE
FLUX
FLPMAX2
FLPMAX1
FLPLIM
FLPBRK1
FLPBRK2
SPEED
SOURCE
FULL ADVANCE
SOURCE ALPHA
COMMAND,
ALPHSRC
ALPHSRCO
SOURCE ALPHA
SPILLOVER
A
FLAGAIN
S
FLUXADJE
FLALIM
FLUXADJ
SOURCE
RETARD LIMIT
FLUX REGULATOR
REFERENCE LOGIC
DS52044A
8LB
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/03/86
NOVEMBER 13, 1985
DICK BOOTHE



NOTES

1. THE FLUX REGULATOR IS RUN FROM THE SOURCE PHASE LOCK LOOP.
2. THE REGULATOR IS THE STANDARD INTEGRAL + PROPORTIONAL CONFIGURATION.
3. FLUX FEEDBACK IS CALCULATED FROM THE SUM OF THE LAST 6 FLUX PEAKS. A COMPENSATION IS ADDED TO ACCOUNT FOR THE UNRECONSTRUCTED NOTCH FROM THE SLAVE CHANNEL.

```
FLUX_REG_G1_F = 4096*FRGAIN*FRTC
FLUX_REG_G2_F = 16384*FRGAIN/FRQS
FLXU_REG_OUT_CLMP_F = FRCLMP*256
FLUX_REG_PRECONDITION_F = FRSTART*256
```

FLUX REGULATOR
DS5204A
BMA
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
DWG. # 04/03/86
NOVEMBER 13, 1985
DCIK BOOTHE

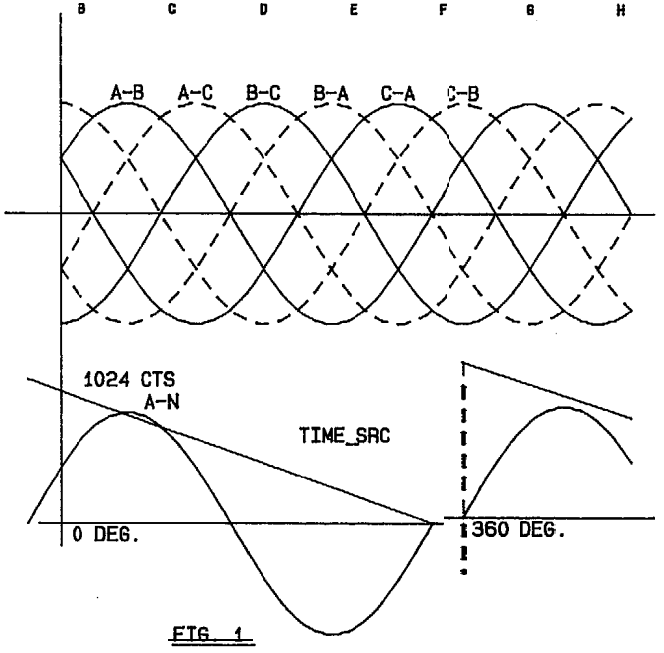


FIG. 1

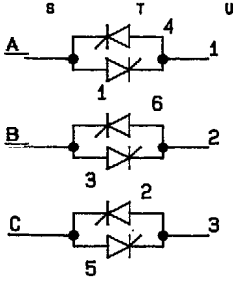


FIG. 2

PH TABLE		
L-L	PH	OFFSET (CTS)
A-B	1	2A
A-C	2	37F
B-C	3	2D5
B-A	4	22A
C-A	5	17F
C-B	6	0DF

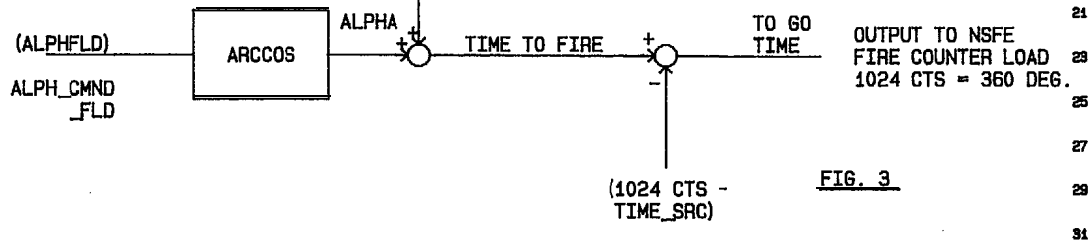


FIG. 3

TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE SOURCE PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE BRIDGE VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 2 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1
3. THE VALUE OF ALPH_CMND_FLD CALCULATED BY THE FLUX REGULATOR IS CONVERTED TO A FIRING ANGLE ACCORDING TO THE EQUATION ALPHA = ARC COS (ALPH_CMND_FLD). THIS IS IMPLEMENTED IN THE SOFTWARE WITH AN INDEXED LOOKUP TABLE.
4. THE VALUE OF ALPHA IS ADDED TO THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED. A SMALL COMPENSATION IS INCLUDED TO ACCOUNT FOR THE DELAY BETWEEN WHEN THE FIRING INTERRUPT OCCURS AND THE CODE WHICH IS EXECUTED BEFORE THE ACTUAL FIRING. THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME TO FIRE - PRESENT TIME) IS THE DIFFERENCE BETWEEN TIME TO_GO AND PRESENT TIME. THE SOFTWARE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFE. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFE.

FIELD REGULATOR TO GO CALCULATION
8NA
DS5204A
DRIVE SYSTEMS
SALEM, VA.
U.S.A.
04/03/86
NOVEMBER 14, 1985
DICK BOOTHE

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC